



**JADARD**

**JD9165BA**

*Rev. 0.0.3*

PRELIMINARY DATA SHEET

**1538CH TFT LCD Single Chip Driver with TCON**

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## 1. General Description

JD9165 is a one-chip solution for Amorphous Si TFT LCD panel. The source driver integrates 1538 source channels, a timing controller, and supports GOA control circuits for the middle size panel application. In addition, the chip also provides Gamma reference voltage and VCOM voltage.

JD9165 also supports various types of peripheral interface such as MIPI interface without frame memory, LVDS interface, 24 parallel RGB interface, and provide 60Hz frame rate. The serial communication interface is also embedded for function setting.

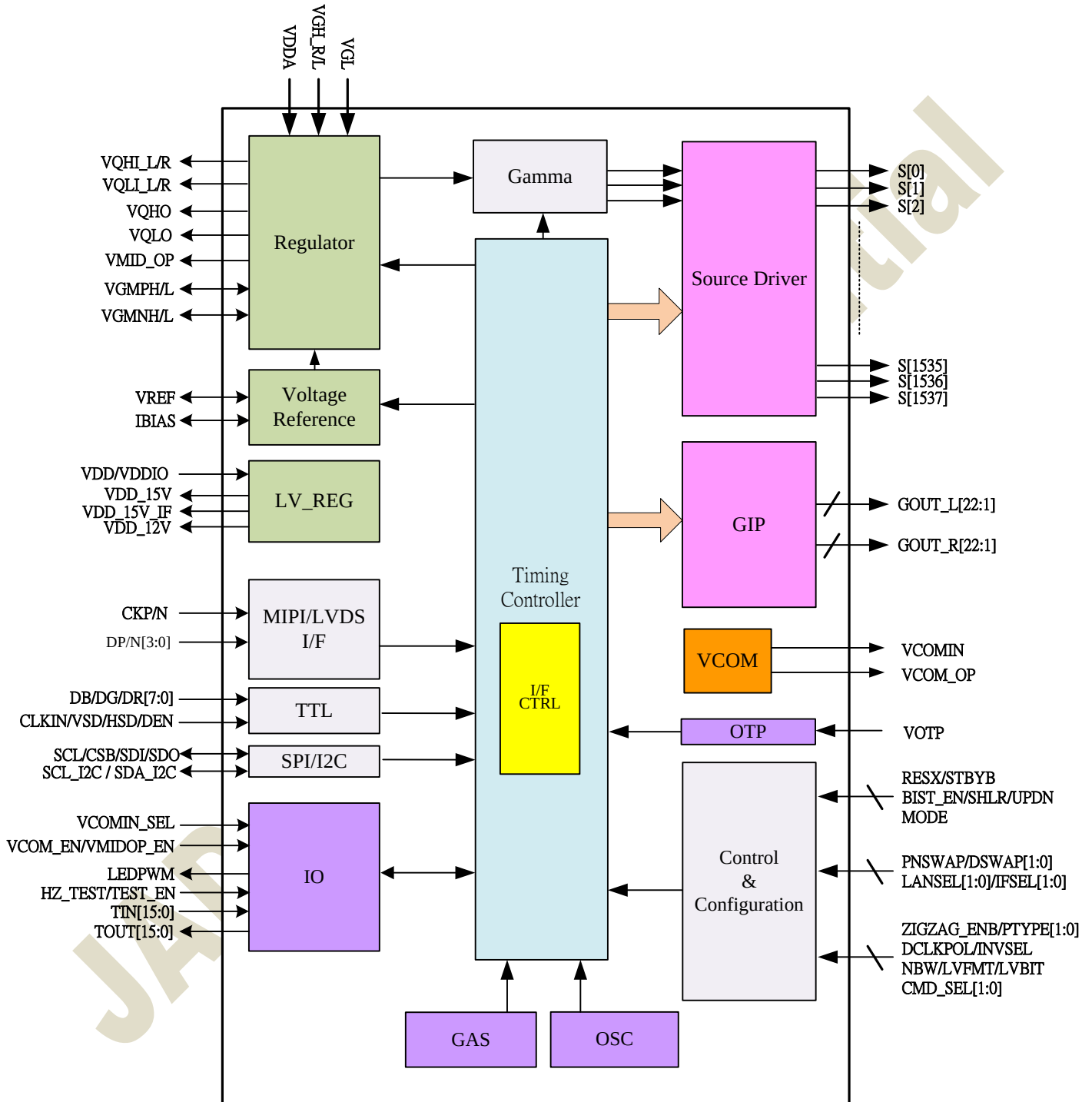
JADARD Confidential

## 2. Features

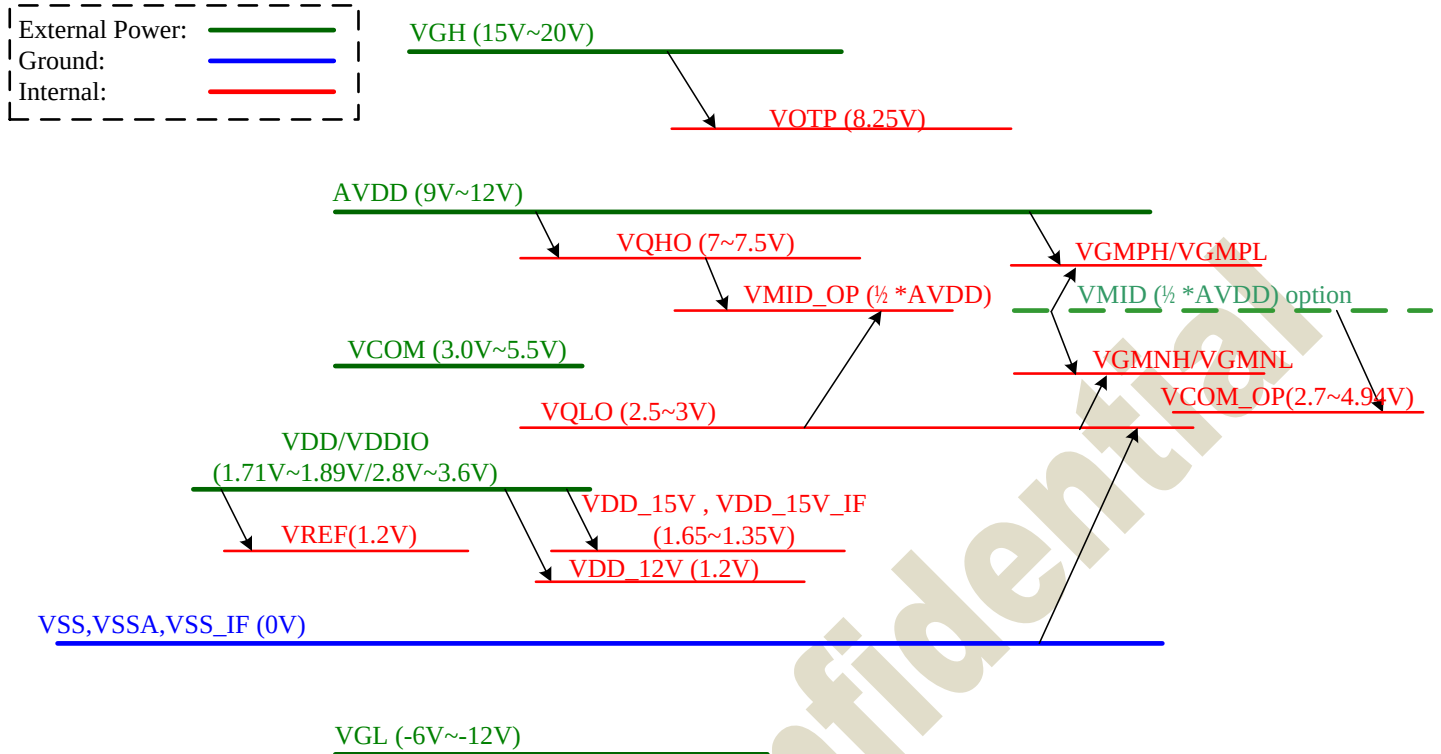
- Single chip solution for Amorphous Si TFT LCD display
- Integrated 1538 channel source driver
- Support panel resolution (HxV):
  - 1024RGBx768, (256RGB+4\*n)xV
- Embedded GOA circuit, support 22 GOUT pins each on R-side/L-side
- Panel type:
  - Single chip: Dual gate panel/Dual gate+Zig-zag panel/Single gate panel
- Inversion architecture:
  - Column inversion
  - 1+2 dot inversion
  - 2 dot inversion
  - Dot inversion
- Support Normally black and white panel
- Provide total 28 register value for gamma correction adjustment
- Support BIST mode
- Support GAS function for abnormal power off
- Support SPI/I2C interface
- Source driver output with 8-bit DAC
- Embedded VCOM = 2.7V to 4.94V
- Embedded VCOM = 2.7V to 5.4V
- OTP memory to store initialization register settings
  - 1 times OTP for GOA setting
  - 2 times OTP for Gamma setting
  - 6 times OTP for VCOM setting
  - 1 times OTP for Other Register setting
- Input power supply:
  - VDD=1.71V to 1.89V / 2.8V to 3.6V (power supply for digital circuit)
  - AVDD=9V to 12V (power supply for analog circuit)
  - VCOM=3.0V to 5.5V (power supply for analog circuit)
  - VGH=15V to 20V (power supply for GOA circuit)
  - VGL= -6V to -12V (power supply for GOA circuit)
- Support MIPI DSI version 1.02
- Support MIPI D-PHY version 1.00
- Support MIPI 1port interface
- Support LVDS 1 port interface
- Support TTL 24-bit parallel RGB interface
- COG package
- Chip size = 28500um x 742um
- Chip thickness = 230um
- Operating temperature ( $T_A$ ): -30°C to +85°C
- Storage temperature ( $T_{STG}$ ): -55°C to +125°C

### 3. BLOCK DIAGRAM

#### 3.1 Function block diagram

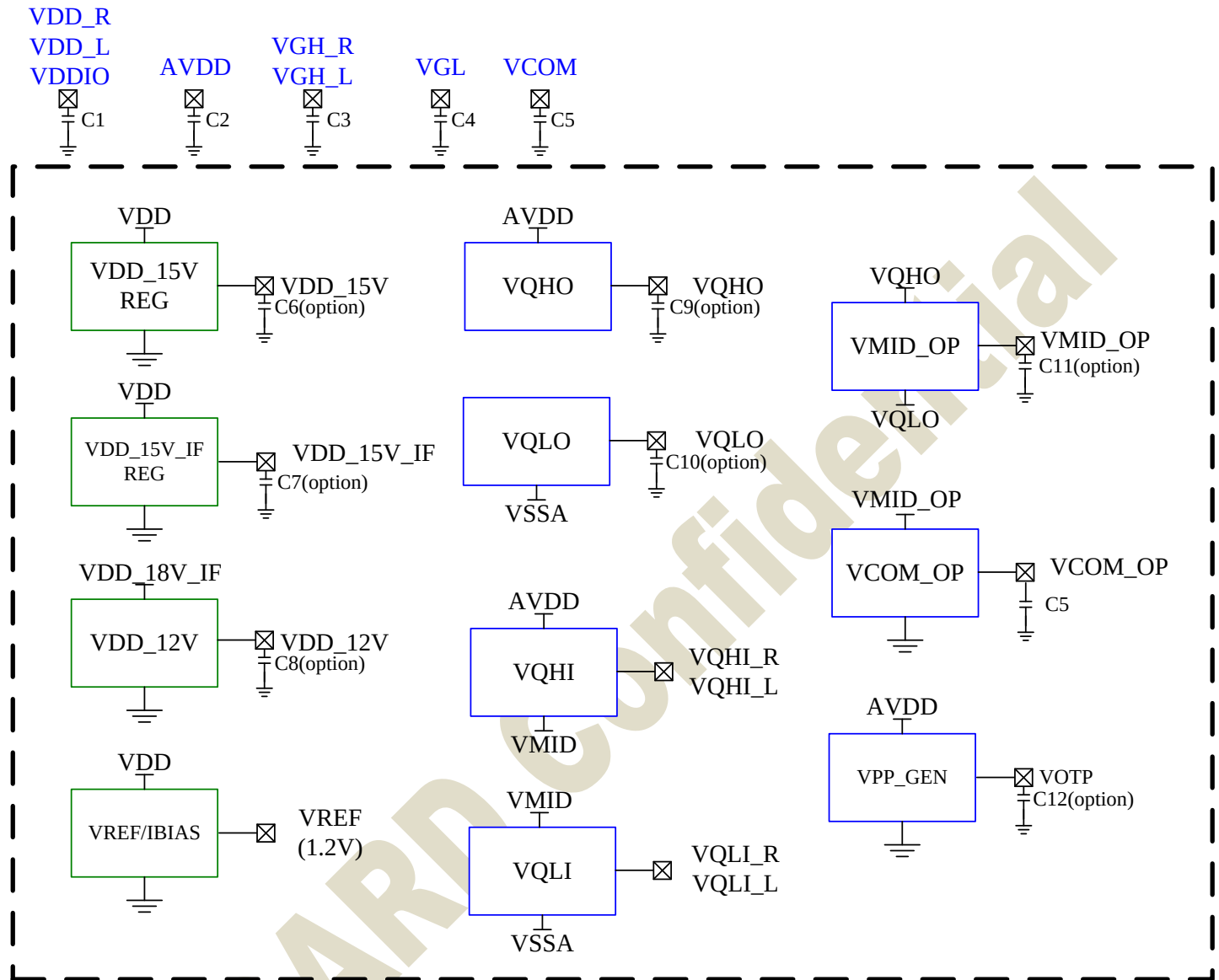


### 3.2 Power structure





## 3.3 DC/DC circuit reference design



| Ref. no | Typical       |
|---------|---------------|
| C1      | 4.7~10uF/6.3V |
| C2      | 4.7~10uF/25V  |
| C3      | 2.2~4.7uF/25V |
| C4      | 2.2~4.7uF/25V |
| C5      | 2.2~4.7uF/16V |
| C6      | 2.2uF/6.3V    |
| C7      | 2.2uF/6.3V    |
| C8      | 1uF/6.3V      |
| C9      | 4.7uF/10V     |
| C10     | 4.7uF/10V     |
| C11     | 4.7uF/10V     |
| C12     | 1uF/25V       |

## 4. PIN DESCRIPTION

### 4.1 Pin description

| Pin name                                                 | I/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|----------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--------------------------|--|--|--|--|--|--|--|--|--|--|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Interface pins                                           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| DP[0]/DN[0]<br>DP[1]/DN[1]<br>DP[2]/DN[2]<br>DP[3]/DN[3] | In  | MIPI and LVDS data input pin.<br>If not used, let it open.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| CKP/CKN                                                  | In  | MIPI and LVDS clock input pin.<br>If not used, let it open.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| DR[7:0]<br>DG[7:0]<br>DB[7:0]<br>(VDD)                   | In  | TTL data input. Select by “IFSEL” pin.<br>If not used, please fix this pin at VSS level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| CLKIN<br>(VDD)                                           | In  | TTL clock input pin. Select by “IFSEL” pin.<br>If not used, please fix this pin at VSS level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| HSD<br>(VDD)                                             | In  | Horizontal sync input. Internally pulled high.<br>If not used, please fix this pin at VSS level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| VSD<br>(VDD)                                             | In  | Vertical sync input. Internally pulled high.<br>If not used, please fix this pin at VSS level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| DEN<br>(VDD)                                             | In  | DE signal when DE mode. Internally pulled low.<br>If not used, please fix this pin at VSS level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| Global pins                                              |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| RESX<br>(VDD)                                            | In  | Global reset pin, active low. <b>(Default pull High)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| STBYB<br>(VDD)                                           | In  | Standby mode selection. <b>(Default pull High)</b><br>(only for TTL and LVDS, MIPI mode = dummy function)<br>STBYB=H, Normal mode.<br>STBYB=L, Standby mode.<br>If not used, please fix this pin at VDD level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| DSWAP[1:0]<br>(VDD)                                      |     | These pins must be connected to VDD or VSSD. DSWAP[1:0] are used for the combination of data lane swap. <b>(Default DSWAP[1:0]=11b)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|                                                          |     | <table><tr><th>DSWAP[1:0]</th><th>DP0</th><th>DN0</th><th>DP1</th><th>DN1</th><th>CKP</th><th>CKN</th><th>DP2</th><th>DN2</th><th>DP3</th><th>DN3</th></tr><tr><td colspan="11">MIPI lanes mapping table</td></tr><tr><td>00</td><td>DP3</td><td>DN3</td><td>DP2</td><td>DN2</td><td>CKP</td><td>CKN</td><td>DP1</td><td>DN1</td><td>DP0</td><td>DN0</td></tr><tr><td>01</td><td>DP0</td><td>DN0</td><td>DP1</td><td>DN1</td><td>CKP</td><td>CKN</td><td>DP2</td><td>DN2</td><td>DP3</td><td>DN3</td></tr><tr><td>10</td><td>DP3</td><td>DN3</td><td>DP2</td><td>DN2</td><td>CKP</td><td>CKN</td><td>DP1</td><td>DN1</td><td>DP0</td><td>DN0</td></tr><tr><td>11</td><td>DP0</td><td>DN0</td><td>DP1</td><td>DN1</td><td>CKP</td><td>CKN</td><td>DP2</td><td>DN2</td><td>DP3</td><td>DN3</td></tr></table> | DSWAP[1:0] | DP0 | DN0 | DP1 | DN1 | CKP | CKN | DP2 | DN2 | DP3 | DN3 | MIPI lanes mapping table |  |  |  |  |  |  |  |  |  |  | 00 | DP3 | DN3 | DP2 | DN2 | CKP | CKN | DP1 | DN1 | DP0 | DN0 | 01 | DP0 | DN0 | DP1 | DN1 | CKP | CKN | DP2 | DN2 | DP3 | DN3 | 10 | DP3 | DN3 | DP2 | DN2 | CKP | CKN | DP1 | DN1 | DP0 | DN0 | 11 | DP0 | DN0 | DP1 | DN1 | CKP | CKN | DP2 | DN2 | DP3 | DN3 |
|                                                          |     | DSWAP[1:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DP0        | DN0 | DP1 | DN1 | CKP | CKN | DP2 | DN2 | DP3 | DN3 |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|                                                          |     | MIPI lanes mapping table                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|                                                          |     | 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DP3        | DN3 | DP2 | DN2 | CKP | CKN | DP1 | DN1 | DP0 | DN0 |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|                                                          |     | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DP0        | DN0 | DP1 | DN1 | CKP | CKN | DP2 | DN2 | DP3 | DN3 |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|                                                          |     | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DP3        | DN3 | DP2 | DN2 | CKP | CKN | DP1 | DN1 | DP0 | DN0 |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|                                                          |     | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DP0        | DN0 | DP1 | DN1 | CKP | CKN | DP2 | DN2 | DP3 | DN3 |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|                                                          |     | <table><tr><th>DSWAP[1:0]</th><th>DP0</th><th>DN0</th><th>DP1</th><th>DN1</th><th>CKP</th><th>CKN</th><th>DP2</th><th>DN2</th><th>DP3</th><th>DN3</th></tr><tr><td colspan="11">LVDS lanes mapping table</td></tr><tr><td>00</td><td>DP3</td><td>DN3</td><td>DP2</td><td>DN2</td><td>DP1</td><td>DN1</td><td>DP0</td><td>DN0</td><td>CKP</td><td>CKN</td></tr><tr><td>01</td><td>CKP</td><td>CKN</td><td>DP0</td><td>DN0</td><td>DP1</td><td>DN1</td><td>DP2</td><td>DN2</td><td>DP3</td><td>DN3</td></tr><tr><td>10</td><td>DP3</td><td>DN3</td><td>DP2</td><td>DN2</td><td>CKP</td><td>CKN</td><td>DP1</td><td>DN1</td><td>DP0</td><td>DN0</td></tr><tr><td>11</td><td>DP0</td><td>DN0</td><td>DP1</td><td>DN1</td><td>CKP</td><td>CKN</td><td>DP2</td><td>DN2</td><td>DP3</td><td>DN3</td></tr></table> | DSWAP[1:0] | DP0 | DN0 | DP1 | DN1 | CKP | CKN | DP2 | DN2 | DP3 | DN3 | LVDS lanes mapping table |  |  |  |  |  |  |  |  |  |  | 00 | DP3 | DN3 | DP2 | DN2 | DP1 | DN1 | DP0 | DN0 | CKP | CKN | 01 | CKP | CKN | DP0 | DN0 | DP1 | DN1 | DP2 | DN2 | DP3 | DN3 | 10 | DP3 | DN3 | DP2 | DN2 | CKP | CKN | DP1 | DN1 | DP0 | DN0 | 11 | DP0 | DN0 | DP1 | DN1 | CKP | CKN | DP2 | DN2 | DP3 | DN3 |
|                                                          |     | DSWAP[1:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DP0        | DN0 | DP1 | DN1 | CKP | CKN | DP2 | DN2 | DP3 | DN3 |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|                                                          |     | LVDS lanes mapping table                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|                                                          |     | 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DP3        | DN3 | DP2 | DN2 | DP1 | DN1 | DP0 | DN0 | CKP | CKN |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|                                                          |     | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CKP        | CKN | DP0 | DN0 | DP1 | DN1 | DP2 | DN2 | DP3 | DN3 |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
|                                                          |     | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DP3        | DN3 | DP2 | DN2 | CKP | CKN | DP1 | DN1 | DP0 | DN0 |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| 11                                                       | DP0 | DN0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | DP1        | DN1 | CKP | CKN | DP2 | DN2 | DP3 | DN3 |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |
| If not used, please fix this pin at VDD level.           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |     |     |     |     |     |     |     |     |     |     |                          |  |  |  |  |  |  |  |  |  |  |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |    |     |     |     |     |     |     |     |     |     |     |

|                      |    |                                                                                                                                                                   |                                 |       |                      |       |       |
|----------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------|----------------------|-------|-------|
| PNSWAP<br>(VDD)      | In | MIPI and LVDS polarity selection. <b>(Default pull Low)</b>                                                                                                       |                                 |       |                      |       |       |
|                      |    | PNSWAP                                                                                                                                                            | D0P/N                           | D1P/N | CKP/N                | D2P/N | D3P/N |
|                      |    | MIPI/LVDS lanes mapping table                                                                                                                                     |                                 |       |                      |       |       |
|                      |    | 1                                                                                                                                                                 | D0N/P                           | D1N/P | CKN/P                | D2N/P | D3N/P |
|                      |    | 0                                                                                                                                                                 | D0P/N                           | D1P/N | CKP/N                | D2P/N | D3P/N |
|                      |    | If not used, please fix this pin at VSS level.                                                                                                                    |                                 |       |                      |       |       |
| LANSEL[1:0]<br>(VDD) | In | MIPI lane number configuration. <b>(Default LANSEL[1:0]=11b)</b>                                                                                                  |                                 |       |                      |       |       |
|                      |    | LANSEL[1:0]                                                                                                                                                       | MIPI lane number                |       |                      |       |       |
|                      |    | 00                                                                                                                                                                | 1 lane                          |       |                      |       |       |
|                      |    | 01                                                                                                                                                                | 2 lanes                         |       |                      |       |       |
|                      |    | 10                                                                                                                                                                | 3 lanes                         |       |                      |       |       |
|                      |    | 11                                                                                                                                                                | 4 lanes                         |       |                      |       |       |
|                      |    | If not used, please fix this pin at VDD level.                                                                                                                    |                                 |       |                      |       |       |
| IFSEL[1:0]<br>(VDD)  | In | Interface selection. <b>(Default IFSEL[1:0]=00b)</b>                                                                                                              |                                 |       |                      |       |       |
|                      |    | IFSEL[1:0]                                                                                                                                                        | Interface                       |       |                      |       |       |
|                      |    | 00                                                                                                                                                                | LVDS                            |       |                      |       |       |
|                      |    | 01                                                                                                                                                                | MIPI                            |       |                      |       |       |
|                      |    | 10                                                                                                                                                                | TTL                             |       |                      |       |       |
|                      |    | 11                                                                                                                                                                | T.B.D.                          |       |                      |       |       |
| BIST_EN<br>(VDD)     | In | BIST mode enable, high active. <b>(Default pull Low)</b><br>If not used, please fix this pin at VSS level.                                                        |                                 |       |                      |       |       |
| SHLR<br>(VDD)        | In | Horizontal scan direction. <b>(Default pull Low)</b>                                                                                                              |                                 |       |                      |       |       |
|                      |    | SHLR                                                                                                                                                              | Function                        |       |                      |       |       |
|                      |    | 1                                                                                                                                                                 | S1537→S1536→.....S1→S0          |       |                      |       |       |
|                      |    | 0                                                                                                                                                                 | S0→S1→.....S1536→S1537          |       |                      |       |       |
| UPDN<br>(VDD)        | In | Vertical scan direction. <b>(Default pull Low)</b>                                                                                                                |                                 |       |                      |       |       |
|                      |    | UPDN                                                                                                                                                              | Function                        |       |                      |       |       |
|                      |    | 1                                                                                                                                                                 | Bottom → Top                    |       |                      |       |       |
|                      |    | 0                                                                                                                                                                 | Top → Bottom                    |       |                      |       |       |
| MODE<br>(VDD)        | In | DE mode and Sync mode selection. <b>(Default pull High)</b>                                                                                                       |                                 |       |                      |       |       |
|                      |    | MODE                                                                                                                                                              | Function                        |       |                      |       |       |
|                      |    | 1                                                                                                                                                                 | DE mode                         |       |                      |       |       |
|                      |    | 0                                                                                                                                                                 | Sync mode                       |       |                      |       |       |
|                      |    | If MIPI mode, please fix this pin at VDD level.                                                                                                                   |                                 |       |                      |       |       |
| ZIGZAG_ENB<br>(VDD)  | In | Dual gata or Zig-zag panel setting. <b>(Default pull High)</b>                                                                                                    |                                 |       |                      |       |       |
|                      |    | ZIGZAG_ENB                                                                                                                                                        | Function                        |       |                      |       |       |
|                      |    | 1                                                                                                                                                                 | Dual gate panel                 |       |                      |       |       |
|                      |    | 0                                                                                                                                                                 | Dual gate+Zig-zag panel         |       |                      |       |       |
| PTYPE[1:0]<br>(VDD)  | In | Panel layout type selection. <b>(Default PTYPE[1:0]=00b)</b>                                                                                                      |                                 |       |                      |       |       |
|                      |    | PTYPE[1:0]                                                                                                                                                        | Dual gate panel                 |       | Zig-zag panel        |       |       |
|                      |    | 00                                                                                                                                                                | Dual gate layout type1          |       | Zig-zag layout type1 |       |       |
|                      |    | 01                                                                                                                                                                |                                 |       | Zig-zag layout type2 |       |       |
|                      |    | 10                                                                                                                                                                | Dual gate layout type2          |       | Zig-zag layout type3 |       |       |
|                      |    | 11                                                                                                                                                                |                                 |       | Zig-zag layout type4 |       |       |
| DCLKPOL<br>(VDD)     | In | TTL input clock edge selection. <b>(Default pull Low)</b>                                                                                                         |                                 |       |                      |       |       |
|                      |    | DCLKPOL                                                                                                                                                           | Function                        |       |                      |       |       |
|                      |    | 1                                                                                                                                                                 | Latch data at DCLK rising edge  |       |                      |       |       |
|                      |    | 0                                                                                                                                                                 | Latch data at DCLK falling edge |       |                      |       |       |
|                      |    | If MIPI/LVDS mode, please fix this pin at VSS level.                                                                                                              |                                 |       |                      |       |       |
| INVSEL<br>(VDD)      | In | Inversion type selection. <b>(Default pull High)</b><br>(only for dual gate panel, single gate and dual gate+zig-zag panel<br>please fix this pin at VDD level.). |                                 |       |                      |       |       |
|                      |    | INVSEL                                                                                                                                                            | Function                        |       |                      |       |       |
|                      |    | 1                                                                                                                                                                 | 1+2dot                          |       |                      |       |       |
|                      |    | 0                                                                                                                                                                 | 2 dot                           |       |                      |       |       |

|                       |     |                                                                                                                                                |                      |               |
|-----------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------|
| NBW<br>(VDD)          | In  | Normally black or normally white setting. <b>(Default pull Low)</b>                                                                            |                      |               |
|                       |     | NBW                                                                                                                                            | Function             |               |
|                       |     | 1                                                                                                                                              | Normally white panel |               |
|                       |     | 0                                                                                                                                              | Normally black panel |               |
| LVFMT<br>(VDD)        | In  | 8-bit input format select for LVDS mode. <b>(Default pull High)</b><br>(only for LVDS, TTL/MIPI mode please fix the pin at VDD level)          |                      |               |
|                       |     | LVFMT                                                                                                                                          | Function             |               |
|                       |     | 1                                                                                                                                              | VESA format          |               |
|                       |     | 0                                                                                                                                              | JEIDA format         |               |
| LVBIT<br>(VDD)        | In  | 6-bit / 8-bit input select for LVDS mode. <b>(Default pull High)</b><br>(only for LVDS, TTL/MIPI mode please fix the pin at VDD level)         |                      |               |
|                       |     | LVBIT                                                                                                                                          | Function             |               |
|                       |     | 1                                                                                                                                              | 8-bit                |               |
|                       |     | 0                                                                                                                                              | 6-bit                |               |
| CMD_SEL[1:0]<br>(VDD) | In  | Command interface selection. <b>(Default CMD_SEL[1:0]=00b).</b>                                                                                |                      |               |
|                       |     | CMD_SEL1                                                                                                                                       | CMD_SEL0             | Function      |
|                       |     | 0                                                                                                                                              | 0                    | 3-wire SPI I  |
|                       |     | 0                                                                                                                                              | 1                    | 3-wire SPI II |
|                       |     | 1                                                                                                                                              | X                    | I2C           |
|                       |     | MIPI and SPI/I2C command can't receive at the same time.                                                                                       |                      |               |
| VCOMIN_SEL<br>(VDD)   | In  | Internal/External VCOM selection. <b>(Default pull Low)</b>                                                                                    |                      |               |
|                       |     | VCOMIN_SEL                                                                                                                                     | Function             |               |
|                       |     | 1                                                                                                                                              | External VCOM        |               |
|                       |     | 0                                                                                                                                              | Internal VCOM        |               |
| VCOM_EN<br>(VDD)      | In  | VCOM buffer selection. <b>(Default pull High)</b>                                                                                              |                      |               |
|                       |     | VCOM_EN                                                                                                                                        | Function             |               |
|                       |     | 1                                                                                                                                              | VCOM buffer enable   |               |
|                       |     | 0                                                                                                                                              | VCOM buffer disable  |               |
| VMIDOP_EN<br>(VDD)    | In  | Internal/external VMIDOP selection. <b>(Default pull Low)</b>                                                                                  |                      |               |
|                       |     | VMIDOP_EN                                                                                                                                      | Function             |               |
|                       |     | 1                                                                                                                                              | VMIDOP Enable        |               |
|                       |     | 0                                                                                                                                              | VMIDOP Disable       |               |
| VDD18V_BPS            | In  | Dummy pin.<br>Please fix the pin at VSS or VDD level.                                                                                          |                      |               |
| LEDPWM                | Out | This pin is connected to the external LED driver.<br>PWM type control signal for brightness of the LED backlight.<br>If not used, let it open. |                      |               |
| TP_SYNC               | Out | Internal test pin.<br>Please keep test point in FPC.                                                                                           |                      |               |
| Panel control pins    |     |                                                                                                                                                |                      |               |
| S[1537:0]             | Out | Source Driver output signals.<br>If not used, let it open.                                                                                     |                      |               |
| GOUT_L[22:1]          | Out | GOA control signal at left side.<br>If not used, let it open.                                                                                  |                      |               |

|                            |     |                                                                                                                                                                                                  |
|----------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GOUT_R[22:1]               | Out | GOA control signal at right side.<br>If not used, let it open.                                                                                                                                   |
| Power pins                 |     |                                                                                                                                                                                                  |
| VDD_R/VDD_L                | In  | Power supply for digital circuits. VDD=1.71V to 1.89V / 2.8V to 3.6V                                                                                                                             |
| VDDIO                      | In  | Power supply for I/O circuits. VDDIO=1.71V to 1.89V / 2.8V to 3.6V<br>Connect to VDD_R/VDD_L in FPC.                                                                                             |
| AVDD                       | In  | Power supply for analog circuits. AVDD=9V to 12V.                                                                                                                                                |
| VGH_R/VGH_L                | In  | Power supply for GOA circuits. VGH=15V to 20V.<br>Connect these pins to VGH in FPC.                                                                                                              |
| VGL                        | In  | Power supply for GOA circuits. VGL= -6V to -12V.                                                                                                                                                 |
| VDD_15V                    | Out | Internal power supply for logic circuits.<br>Connect to a stabilizing capacitor.                                                                                                                 |
| VDD_15V_IF                 | Out | Internal power supply for interface circuits.<br>Connect to a stabilizing capacitor.                                                                                                             |
| VDD_12V_IF                 | Out | (Only for MIPI, TTL/LVDS mode please let it open).<br>VDDL P LDO output for MIPI LP mode TX use.<br>VDDL P LDO enable on MIPI interface. (IFSEL[1:0]=01b)<br>Connect to a stabilizing capacitor. |
| VMID_R[1:0]<br>VMID_L[1:0] | I/O | Power supply for analog circuits.<br>Connect these pins to VMID_OP in FPC.                                                                                                                       |
| VMID_OP                    | Out | VMID OP output ( $VMID\_OP=1/2*AVDD$ ).<br>Connect to a stabilizing capacitor.                                                                                                                   |
| VQHI_R/VQHI_L              | Out | Internal source reference voltage (7V to 7.5V).<br>Please let it open.                                                                                                                           |
| VQLI_R/VQLI_L              | Out | Internal source reference voltage (2.8V to 3V).<br>Please let it open.                                                                                                                           |
| VQHO                       | Out | Internal source reference voltage ( $0.75AVDD$ ).<br>Connect to a stabilizing capacitor.                                                                                                         |
| VQLO                       | Out | Internal source reference voltage ( $0.25AVDD$ ).<br>Connect to a stabilizing capacitor.                                                                                                         |
| VCOMIN                     | In  | VCOM buffer in.<br>If not used, let it open.                                                                                                                                                     |
| VCOM_OP                    | Out | VCOM buffer out (2.7V to 5.4V).<br>Connect to a stabilizing capacitor.<br>If not used, let it open.                                                                                              |
| VOTP                       | Out | Internal OTP voltage and operates at 8.25V.<br>Connect to a stabilizing capacitor.                                                                                                               |
| VPP_REN                    | Out | Internal test pin.<br>Please let it open.                                                                                                                                                        |
| VSS                        | In  | GND for the internal logic. VSS=0V. When using the COG method, connect to VSSA in FPC to prevent noise.                                                                                          |
| VSSA                       | In  | Analog ground. VSS=0V. When using the COG method, connect to VSS in FPC to prevent noise.                                                                                                        |

|                       |     |                                                                                                                                        |
|-----------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------|
| VSS_IF                | In  | Ground for interface, connect to VSS.                                                                                                  |
| Shielding_VSSA        | In  | Internal shielding ground pin.<br>Please let it open.                                                                                  |
| Serial interface pins |     |                                                                                                                                        |
| CSB<br>(VDD)          | In  | Serial Interface chip enable signal for SPI interface.<br>SPI_CSB=0: Selected (Accessible).<br>SPI_CSB=1: Not selected (Inaccessible). |
| SCL<br>(VDD)          | In  | Serial interface clock input for SPI interface.                                                                                        |
| SDI<br>(VDD)          | I/O | Serial interface address and data input/output for SPI interface.                                                                      |
| SDO<br>(VDD)          | I/O | Serial interface data output for SPI interface.                                                                                        |
| SDA_I2C<br>(VDD)      | I/O | Serial interface address and data input/output for I2C interface.<br>(I2C interface need external pull high resistance.)               |
| SCL_I2C<br>(VDD)      | In  | Serial interface clock input for I2C interface.<br>(I2C interface need external pull high resistance.)                                 |
| Pass line pins        |     |                                                                                                                                        |
| PASS2_R               | In  | VCOM pass line.                                                                                                                        |
| PASS1_R               | In  | VCOM pass line.                                                                                                                        |
| PASS2_L               | In  | VCOM pass line.                                                                                                                        |
| PASS1_L               | In  | VCOM pass line.                                                                                                                        |
| Test pins             |     |                                                                                                                                        |
| TEST_I[25:0]          | In  | Internal test pins.<br>Please let it open.                                                                                             |
| TEST_O[19:0]          | Out | Internal test pins.<br>Please let it open.                                                                                             |
| TEST_VDD_I[0]         | In  | Internal test pins.<br>Please let it open.                                                                                             |
| TEST_VDD_O[3:1]       | Out | Internal test pins.<br>Please let it open.                                                                                             |
| TEST_HV_I[1:0]        | In  | Internal test pins.<br>Please let it open.                                                                                             |
| TEST_MV_I[1:0]        | In  | Internal test pins.<br>Please let it open.                                                                                             |
| TEST_HV_O[1:0]        | Out | Internal test pins.<br>Please let it open.                                                                                             |
| TEST_MV_O[1:0]        | Out | Internal test pins.<br>Please let it open.                                                                                             |
| T_IRX                 | Out | Internal test pins.<br>Please let it open.                                                                                             |
| Dummy pins            |     |                                                                                                                                        |
| DUMMY                 | In  | Dummy pins.<br>Please let it open.                                                                                                     |
| DUMMY[1]/DUMMY36]     | In  | Dummy pins.<br>Please let it open.                                                                                                     |

# 5. FUNCTION DESCRIPTION

## 5.1 Power On Sequence

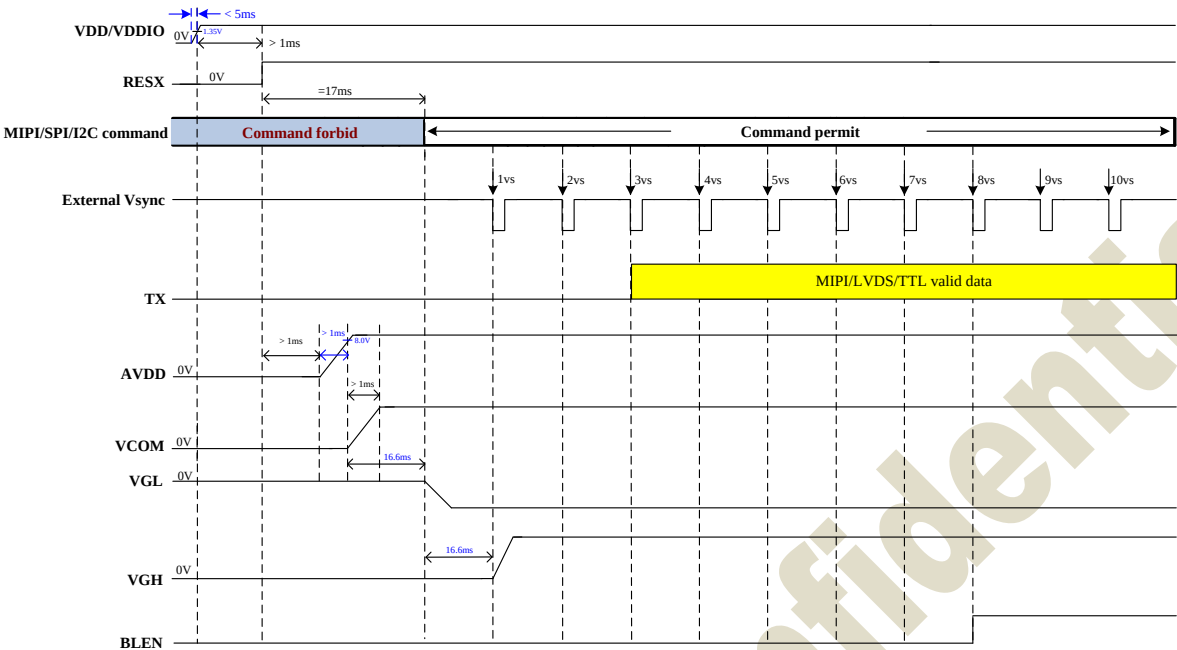


Figure 5.1: Power On timing chart

## 5.2 Power Off Sequence

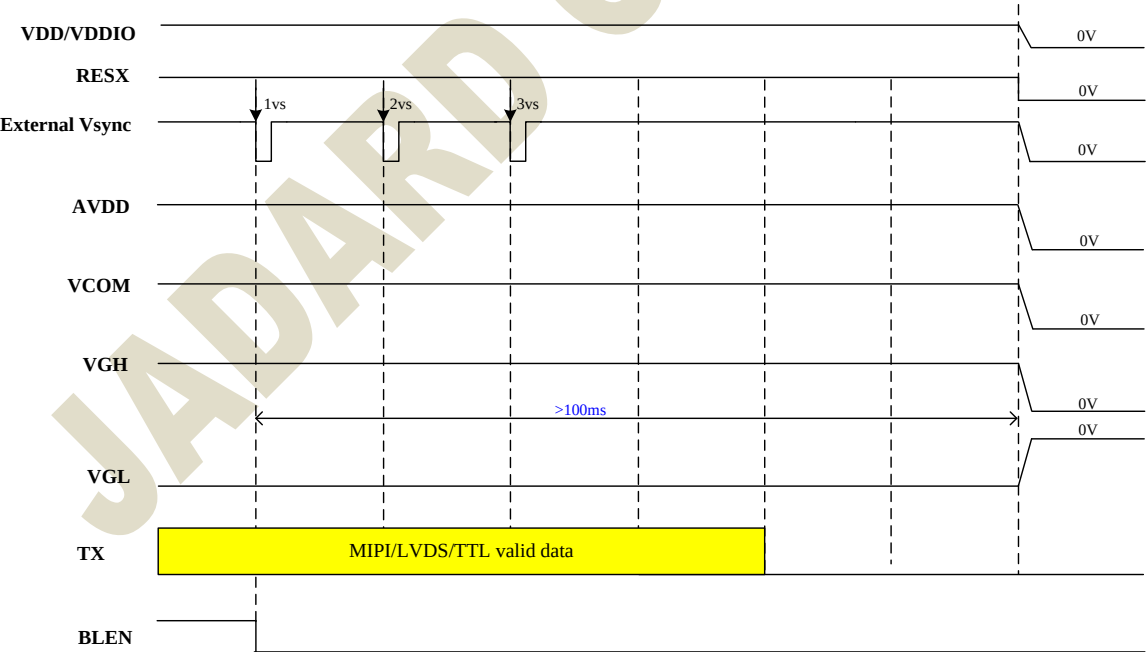


Figure 5.2: Power Off timing chart

### 5.3 Panel structure

JD9165 can support 3 types of driving method – Dual gate panel, Dual gate+Zig-Zag, and Single gate driving.

#### 5.3.1 Dual Gate Panel Configuration

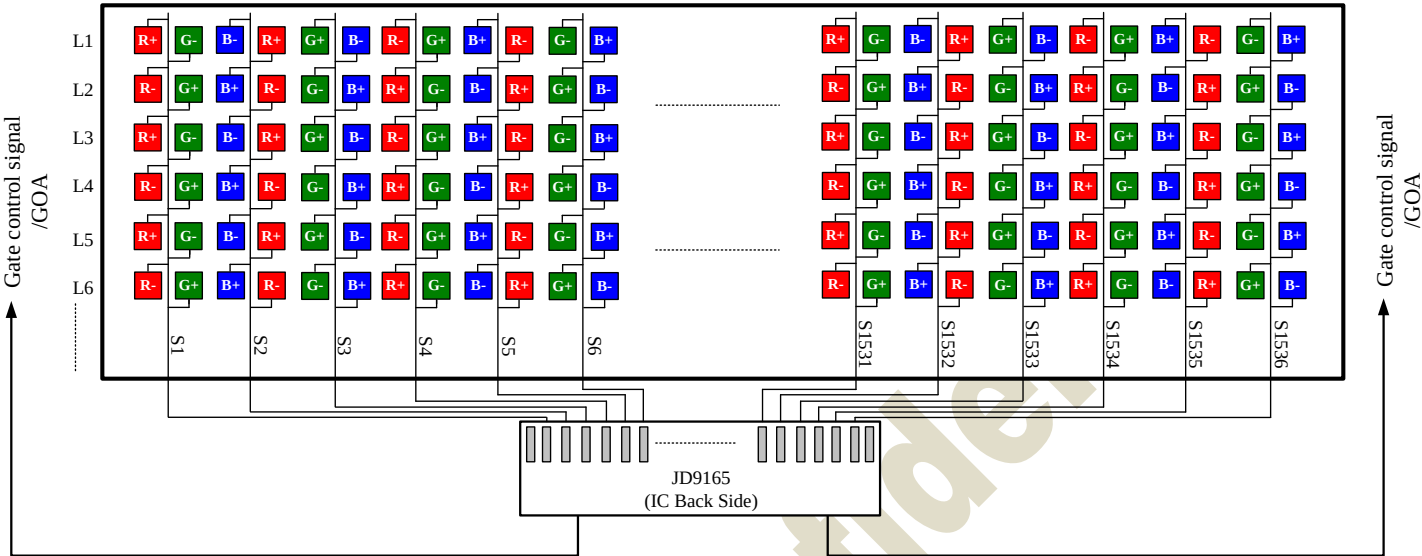


Figure 5.3: Dual gate type1 panel structure

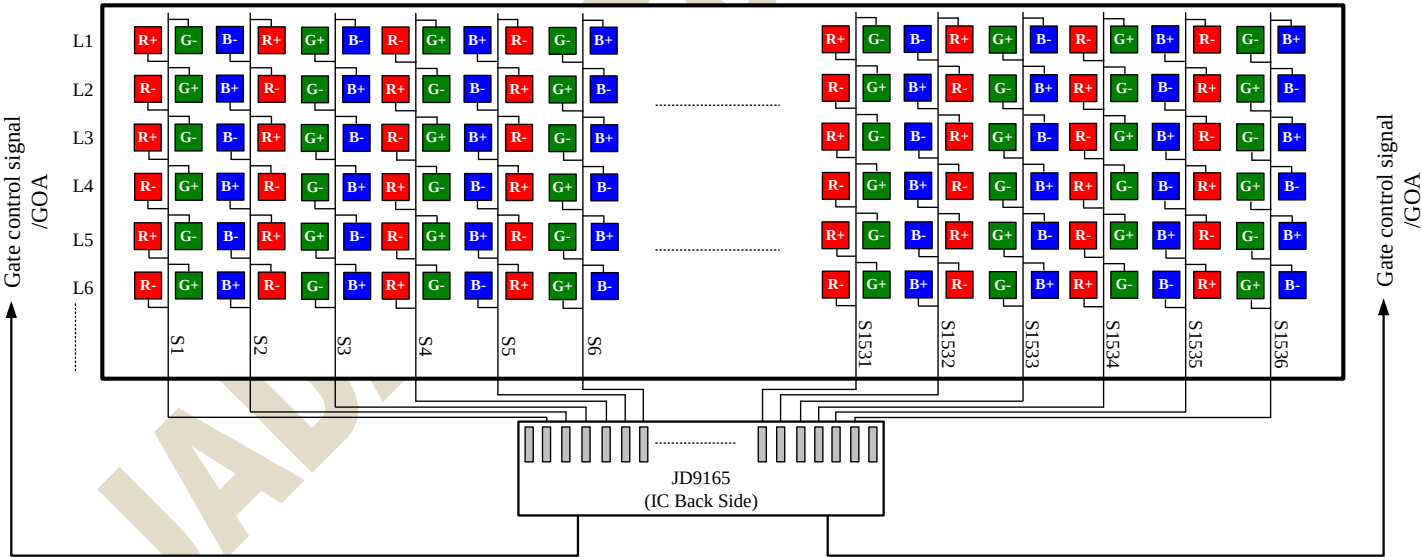


Figure 5.4: Dual gate type2 panel structure



### 5.3.2 Dual Gate + Zig-zag Panel Configuration

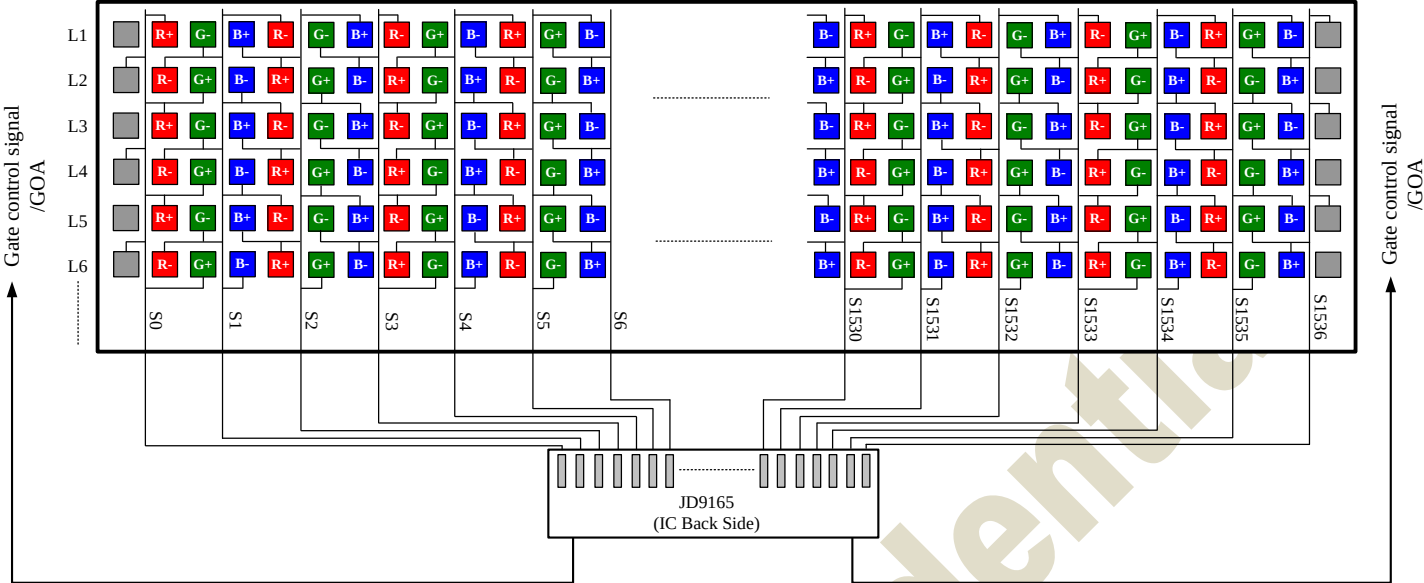


Figure 5.5: Dual gate+Zig-zag type1 panel structure

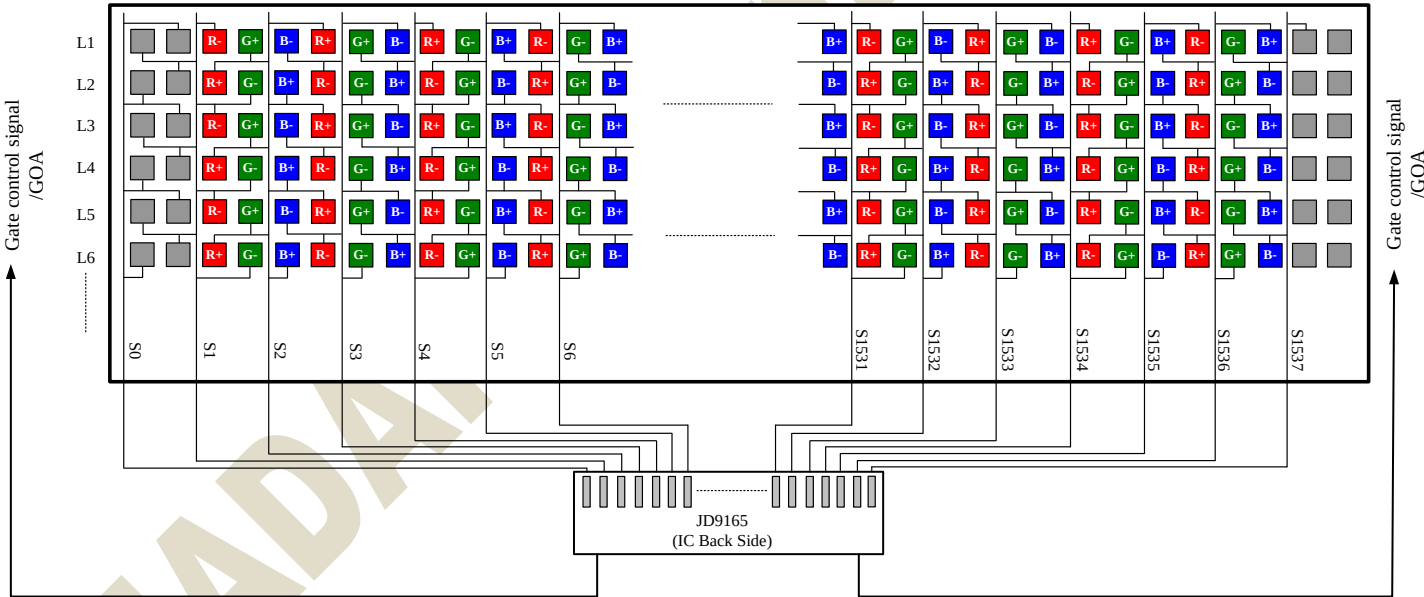


Figure 5.6: Dual gate+Zig-zag type2 panel structure

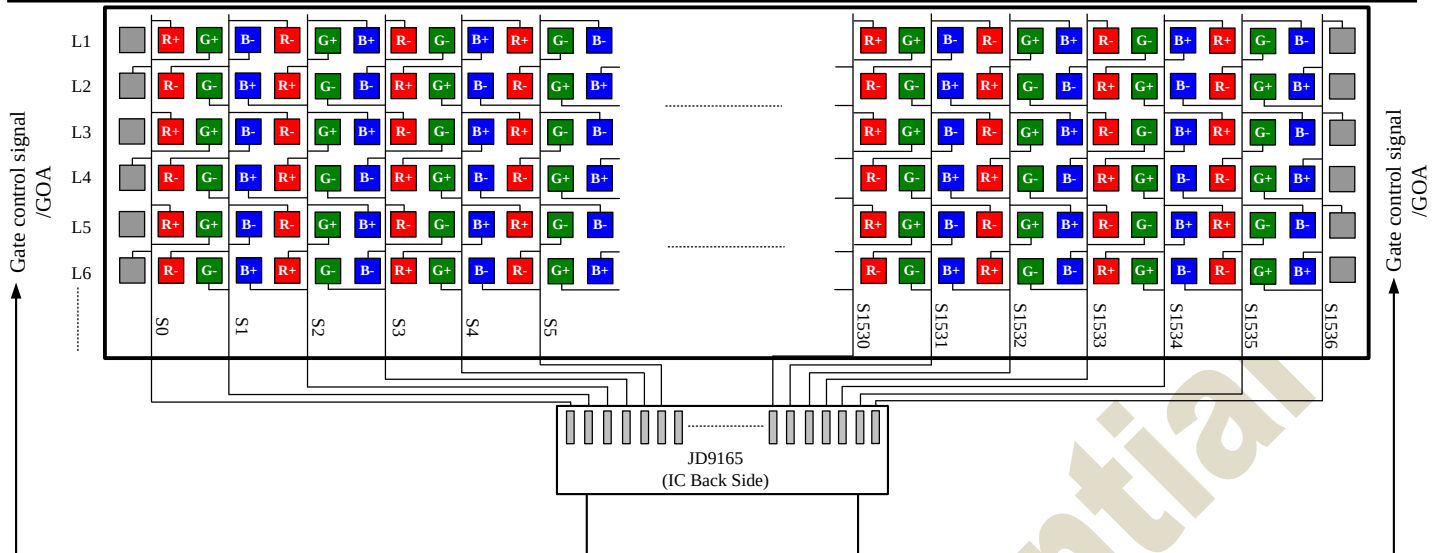


Figure 5.7: Dual gate+Zig-zag type3 panel structure

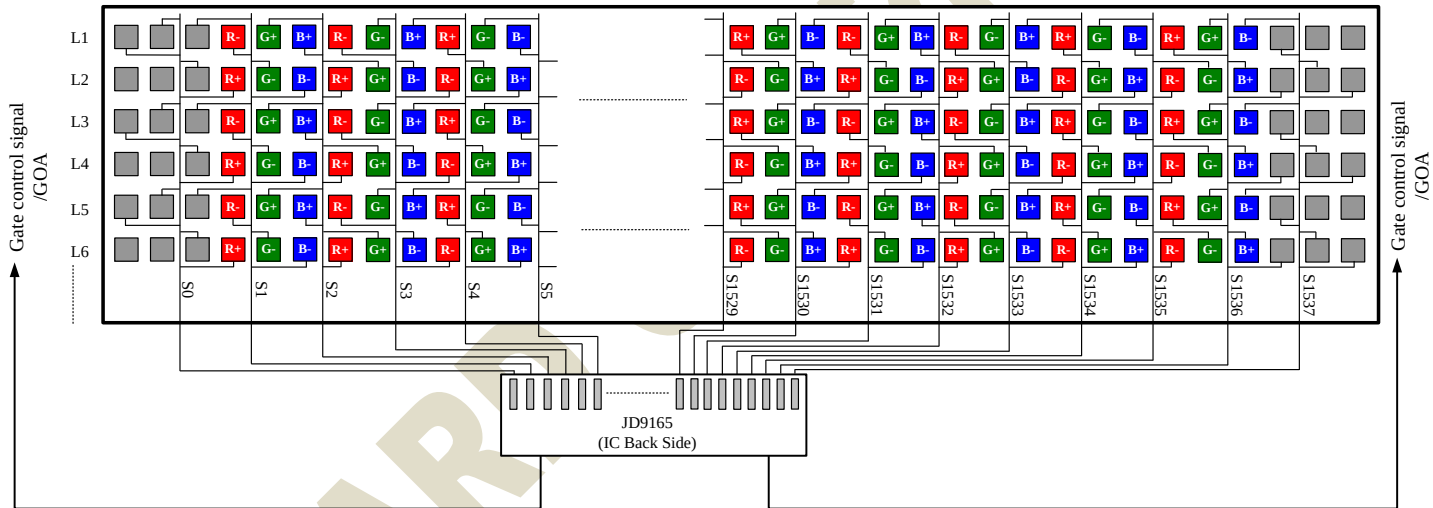


Figure 5.8: Dual gate+Zig-zag type4 panel structure

5.3.3 Single Gate Panel Configuration

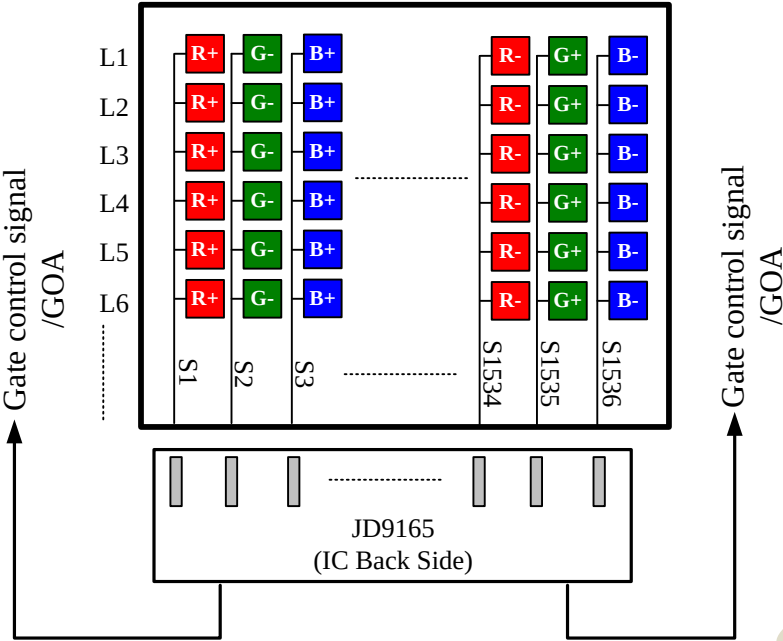
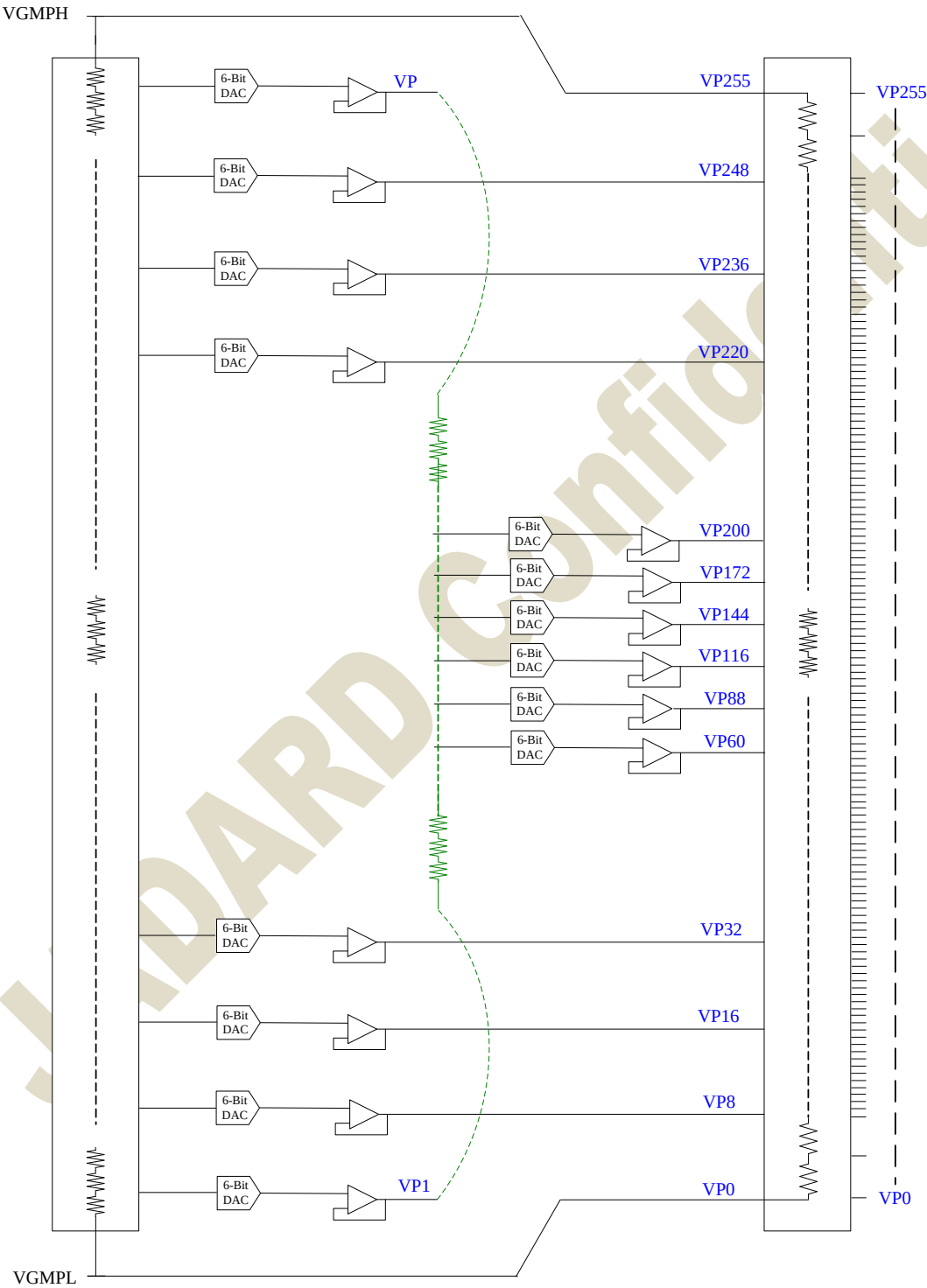


Figure 5.9: Single gate panel structure

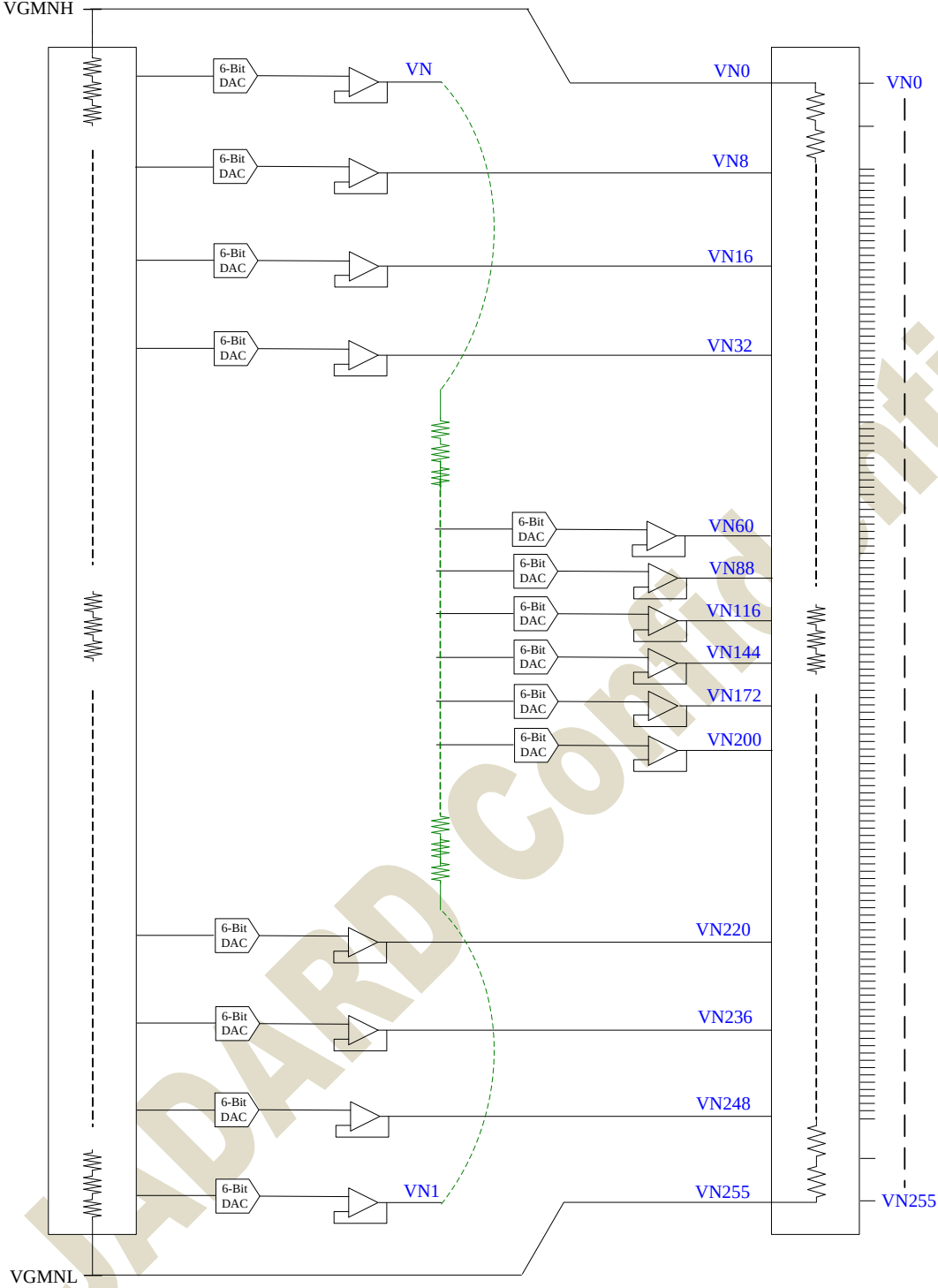
## 5.4 Gamma correction

JD9165 gamma correction are done by 13-segment piecewise linear interpolation. The 13 segments are defined with 14 register values for level 0, 8, 16, 32, 60, 88, 116, 144, 172, 200, 220, 236, 248, 255. These gamma register stream has 64-1 selector to adjust voltage.

### 5.4.1 Gamma structure



**Figure 5.10: Positive analog gamma structure**



**Figure 5.11: Negative analog gamma structure**

### 5.4.2 Gamma voltage reference table

The reference voltages can be set by registers, as described in the following table.

#### Setting for positive gamma voltage

| Gamma code | Reference voltage                                            |
|------------|--------------------------------------------------------------|
| VP1        | $VGMPL + (VGMPH - VGMPL) / 200 * (12 + REG\_VP0\_S[5:0])$    |
| VP8        | $VGMPL + (VGMPH - VGMPL) / 200 * (1 + REG\_VP8\_S[5:0])$     |
| VP16       | $VGMPL + (VGMPH - VGMPL) / 200 * (6 + REG\_VP16\_S[5:0])$    |
| VP32       | $VGMPL + (VGMPH - VGMPL) / 200 * (12 + REG\_VP32\_S[5:0])$   |
| VP60       | $VP32 + (VP220 - VP32) / 150 * (6 + REG\_VP60\_S[5:0])$      |
| VP88       | $VP32 + (VP220 - VP32) / 150 * (10 + REG\_VP88\_S[5:0])$     |
| VP116      | $VP32 + (VP220 - VP32) / 150 * (24 + REG\_VP116\_S[5:0])$    |
| VP144      | $VP32 + (VP220 - VP32) / 150 * (39 + REG\_VP144\_S[5:0])$    |
| VP172      | $VP32 + (VP220 - VP32) / 150 * (58 + REG\_VP172\_S[5:0])$    |
| VP200      | $VP32 + (VP220 - VP32) / 150 * (80 + REG\_VP200\_S[5:0])$    |
| VP220      | $VGMPL + (VGMPH - VGMPL) / 200 * (86 + REG\_VP220\_S[5:0])$  |
| VP236      | $VGMPL + (VGMPH - VGMPL) / 200 * (116 + REG\_VP236\_S[5:0])$ |
| VP248      | $VGMPL + (VGMPH - VGMPL) / 200 * (137 + REG\_VP248\_S[5:0])$ |
| VP         | $VGMPL + (VGMPH - VGMPL) / 200 * (86 + REG\_VP255\_S[5:0])$  |

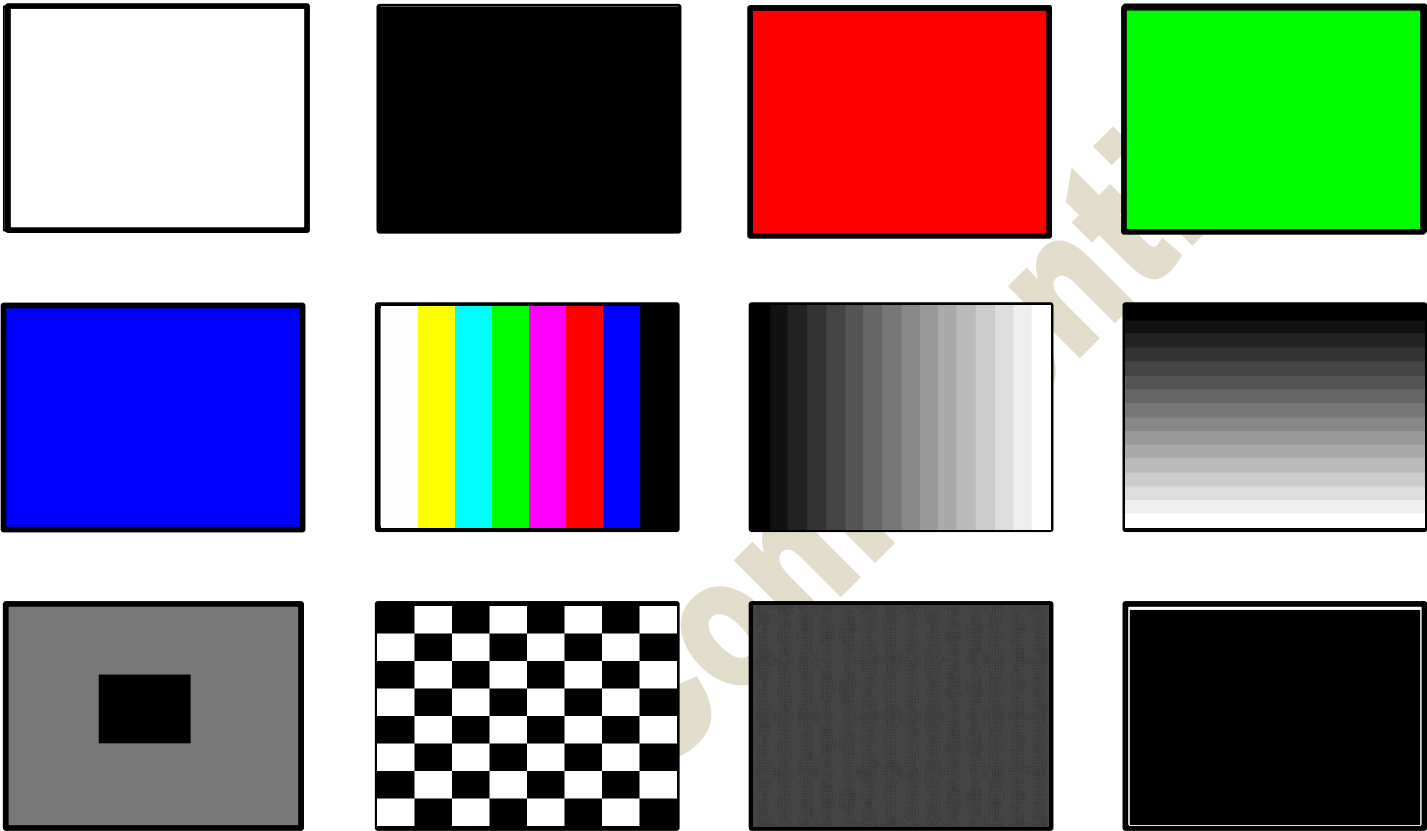
#### Setting for negative voltage

| Gamma code | Reference voltage                                            |
|------------|--------------------------------------------------------------|
| VN         | $VGMNL + (VGMNH - VGMNL) / 200 * (12 + REG\_VN0\_S[5:0])$    |
| VN8        | $VGMNL + (VGMNH - VGMNL) / 200 * (1 + REG\_VN8\_S[5:0])$     |
| VN16       | $VGMNL + (VGMNH - VGMNL) / 200 * (6 + REG\_VN16\_S[5:0])$    |
| VN32       | $VGMNL + (VGMNH - VGMNL) / 200 * (12 + REG\_VN32\_S[5:0])$   |
| VN60       | $VN32 + (VN220 - VN32) / 150 * (6 + REG\_VN60\_S[5:0])$      |
| VN88       | $VN32 + (VN220 - VN32) / 150 * (10 + REG\_VN88\_S[5:0])$     |
| VN116      | $VN32 + (VN220 - VN32) / 150 * (24 + REG\_VN116\_S[5:0])$    |
| VN144      | $VN32 + (VN220 - VN32) / 150 * (39 + REG\_VN144\_S[5:0])$    |
| VN172      | $VN32 + (VN220 - VN32) / 150 * (58 + REG\_VN172\_S[5:0])$    |
| VN200      | $VN32 + (VN220 - VN32) / 150 * (80 + REG\_VN200\_S[5:0])$    |
| VN220      | $VGMNL + (VGMNH - VGMNL) / 200 * (86 + REG\_VN220\_S[5:0])$  |
| VN236      | $VGMNL + (VGMNH - VGMNL) / 200 * (116 + REG\_VN236\_S[5:0])$ |
| VN248      | $VGMNL + (VGMNH - VGMNL) / 200 * (137 + REG\_VN248\_S[5:0])$ |
| VN1        | $VGMNL + (VGMNH - VGMNL) / 200 * (86 + REG\_VN255\_S[5:0])$  |

5.5 BIST function

When BIST is trigger to high, then JD9165 will leave normal operation mode and starts to generate the BIST pattern to LCD panel without TX input signals.

5.5.1 BIST pattern



White→Black→R→G→B→Color Bar→Horizontal 256 gray scale→Vertical 256 gray scale→Crosstalk pattern→Chess board (L255/L0)→Gray128→Black background with white out frame

**5.6 GAS function**

When battery is removed from an electronic device, the image still keeps on the LCD panel for a long time. GAS function can speed the process that image disappears.

The GAS function is a voltage detector. By GAS circuit, JD9165 can detect voltage of VDD/AVDD/VGH and sent ideal GAS signal to discharge residual potential in LCD panel and removes image. The detect voltage could be selected by register setting.

When VDD recovers, power on procedure must be done to return normal mode.

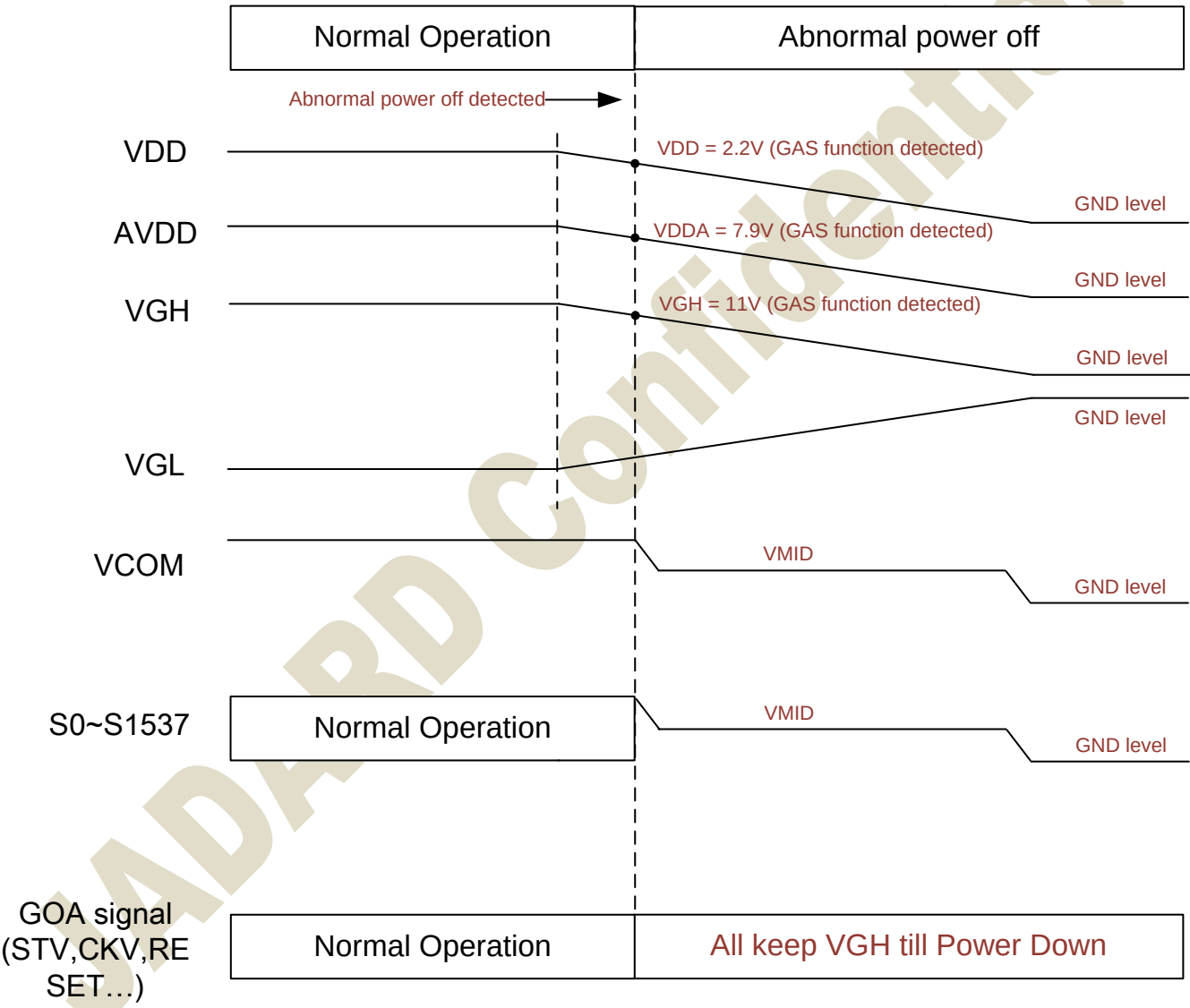


Figure 5.12: GAS discharge timing



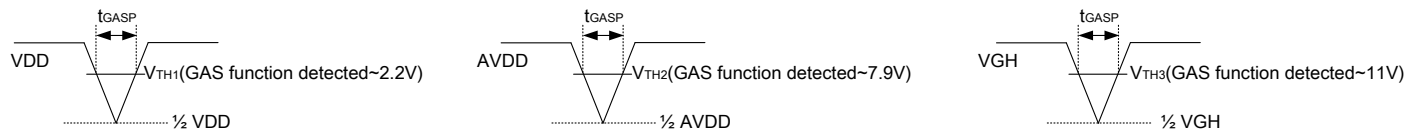


Figure 5.13: GAS V.S. VDD/AVDD/VGH

| Parameter                                                                                          | Symbol     | Step | Spec |     |     | Unit |
|----------------------------------------------------------------------------------------------------|------------|------|------|-----|-----|------|
|                                                                                                    |            |      | Min  | Typ | Max |      |
| VDD/AVDD power drop noise filter period.                                                           | $T_{GASP}$ | -    | 100  | -   | -   | ns   |
| GAS function detection threshold voltage. VDD lower than $V_{TH1}$ , IC will execute GAS function  | $V_{TH1}$  | 0.1  | 1.5  | 2.2 | 3.0 | V    |
| GAS function detection threshold voltage. AVDD lower than $V_{TH2}$ , IC will execute GAS function | $V_{TH2}$  | 0.5  | 7.11 | 7.9 | 8.3 | V    |
| GAS function detection threshold voltage. VGH lower than $V_{TH3}$ , IC will execute GAS function  | $V_{TH3}$  | 0.5  | 11   | 11  | 15  | V    |

6. TTL INPUT TIMING

6.1 24-bit parallel RGB data input format

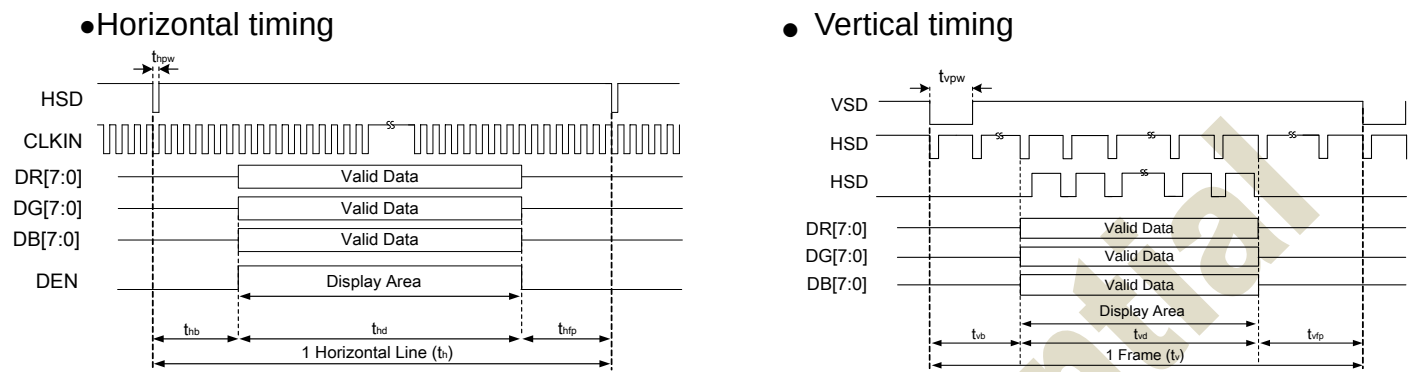


Figure 6.1: 24-bit parallel RGB mode horizontal/vertical input timing

## 6.2 TTL timing characteristic

| 24-bit parallel RGB Input Timing | Symbol | 1024RGBx768 |      |       | 1024RGBx600 |      |       | 800RGBx600 |      |       | Unit |
|----------------------------------|--------|-------------|------|-------|-------------|------|-------|------------|------|-------|------|
|                                  |        | Min         | Typ  | Max   | Min         | Typ  | Max   | Min        | Typ  | Max   |      |
| DCLK Frequency                   | -      | 52.7        | 65   | 71    | 41.4        | 51.2 | 67.2  | 33.1       | 39.6 | 62.4  | MHZ  |
| Horizontal Total                 | tht    | 1114        | 1344 | 1400  | 1114        | 1344 | 1400  | 890        | 1000 | 1300  | DCLK |
| Hsync Pulse width                | ths    | 1           | 24   | HBP-1 | 1           | 24   | HBP-1 | 1          | 24   | HBP-1 | DCLK |
| Horizontal Back Porch            | thb    | 60          | 160  | 160   | 60          | 160  | 160   | 60         | 160  | 250   | DCLK |
| Horizontal Valid Data            | thd    | 1024        |      |       | 1024        |      |       | 800        |      |       | DCLK |
| Horizontal Front Porch           | thfp   | 30          | 160  | 216   | 30          | 160  | 216   | 30         | 40   | 250   | DCLK |
| Vertical Total                   | vtv    | 788         | 806  | 845   | 620         | 635  | 800   | 620        | 660  | 800   | THT  |
| Vsync Pulse Width                | tvv    | 1           | 2    | VBP-1 | 1           | 2    | VBP-1 | 1          | 2    | VBP-1 | THT  |
| Vertical Back Porch              | tvb    | 8           | 23   | 33    | 8           | 23   | 100   | 8          | 23   | 100   | THT  |
| Vertical Valid Data              | tvv    | 768         |      |       | 600         |      |       | 600        |      |       | THT  |
| Vertical Front Porch             | tvfp   | 12          | 15   | 44    | 12          | 12   | 100   | 12         | 37   | 100   | THT  |

| 24-bit parallel RGB Input Timing | Symbol | 800RGBx480 |      |       | 640RGBx480 |      |       | 480RGBx272 |      |       | Unit |
|----------------------------------|--------|------------|------|-------|------------|------|-------|------------|------|-------|------|
|                                  |        | Min        | Typ  | Max   | Min        | Typ  | Max   | Min        | Typ  | Max   |      |
| DCLK Frequency                   | -      | 29         | 32.4 | 62    | 32.6       | 39.6 | 53.4  | 24.8       | 32.6 | 37.6  | MHZ  |
| Horizontal Total                 | tht    | 900        | 1000 | 1300  | 890        | 1000 | 1114  | 830        | 890  | 950   | DCLK |
| Hsync Pulse width                | ths    | 4          | 24   | HBP-1 | 1          | 24   | HBP-1 | 1          | 24   | HBP-1 | DCLK |
| Horizontal Back Porch            | thb    | 62         | 160  | 250   | 140        | 88   | 220   | 180        | 210  | 240   | DCLK |
| Horizontal Valid Data            | thd    | 800        |      |       | 640        |      |       | 480        |      |       | DCLK |
| Horizontal Front Porch           | thfp   | 38         | 40   | 250   | 110        | 272  | 254   | 170        | 200  | 230   | DCLK |
| Vertical Total                   | vtv    | 500        | 540  | 680   | 610        | 660  | 800   | 498        | 610  | 660   | THT  |
| Vsync Pulse Width                | tvv    | 1          | 2    | VBP-1 | 1          | 2    | VBP-1 | 1          | 2    | VBP-1 | THT  |
| Vertical Back Porch              | tvb    | 8          | 23   | 100   | 28         | 39   | 160   | 126        | 180  | 210   | THT  |
| Vertical Valid Data              | tvv    | 480        |      |       | 480        |      |       | 272        |      |       | THT  |
| Vertical Front Porch             | tvfp   | 12         | 37   | 100   | 102        | 141  | 160   | 100        | 158  | 178   | THT  |

### 6.3 TTL AC electrical characteristic

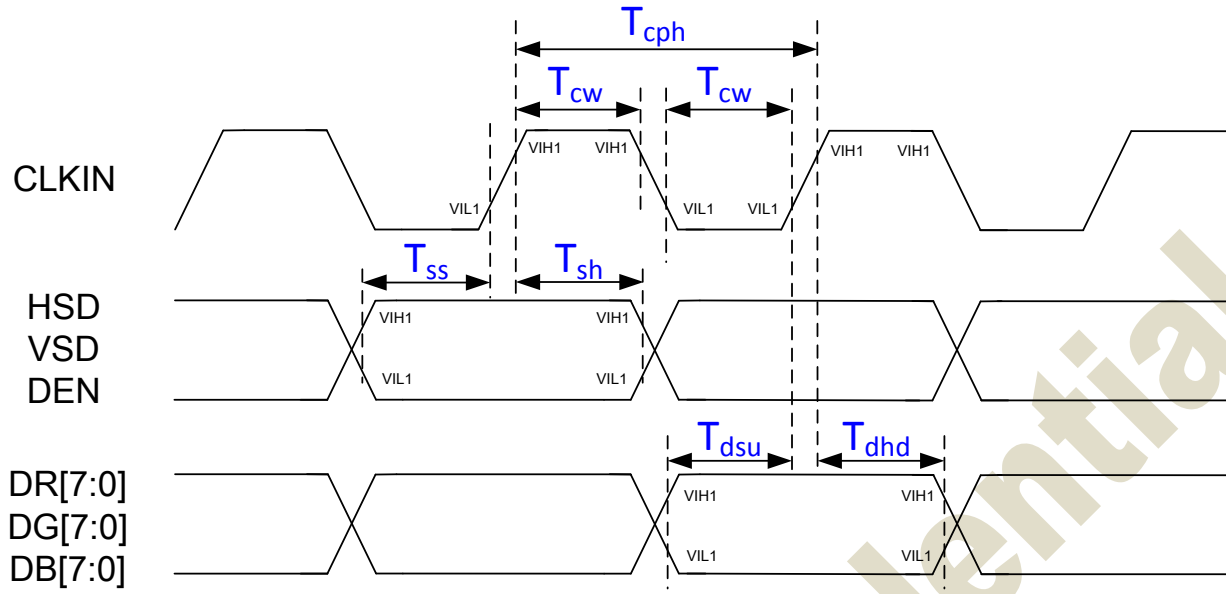


Figure 6.2: Input signal timing

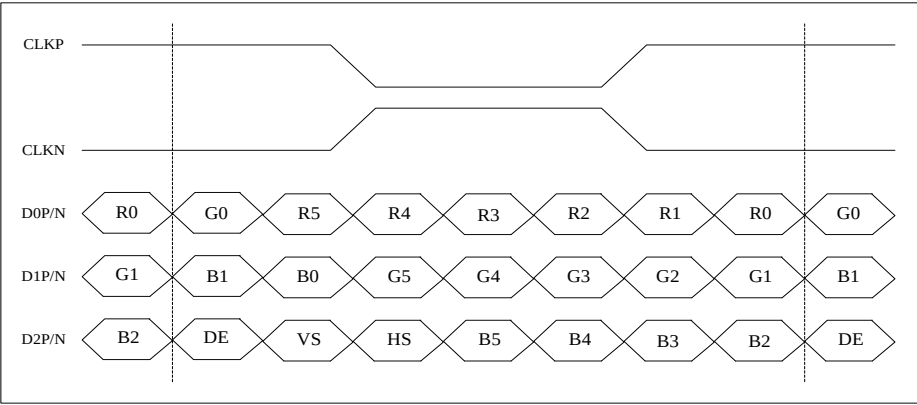
#### Input data /sync parameters

| Parameter        | Symbol    | Spec. |      |      | Unit |
|------------------|-----------|-------|------|------|------|
|                  |           | Min.  | Typ. | Max. |      |
| CLKIN period     | $T_{cph}$ | 14    | -    | -    | ns   |
| CLKIN duty ratio | $T_{cw}$  | 40    | 50   | 60   | %    |
| Data setup time  | $T_{dsu}$ | 5     | -    | -    | ns   |
| Data hold time   | $T_{dhd}$ | 5     | -    | -    | ns   |
| VSD setup time   | $T_{ss}$  | 5     | -    | -    | ns   |
| VSD hold time    | $T_{sh}$  | 5     | -    | -    | ns   |
| HSD setup time   | $T_{ss}$  | 5     | -    | -    | ns   |
| HSD hold time    | $T_{sh}$  | 5     | -    | -    | ns   |
| DEN setup time   | $T_{ss}$  | 5     | -    | -    | ns   |
| DEN hold time    | $T_{sh}$  | 5     | -    | -    | ns   |

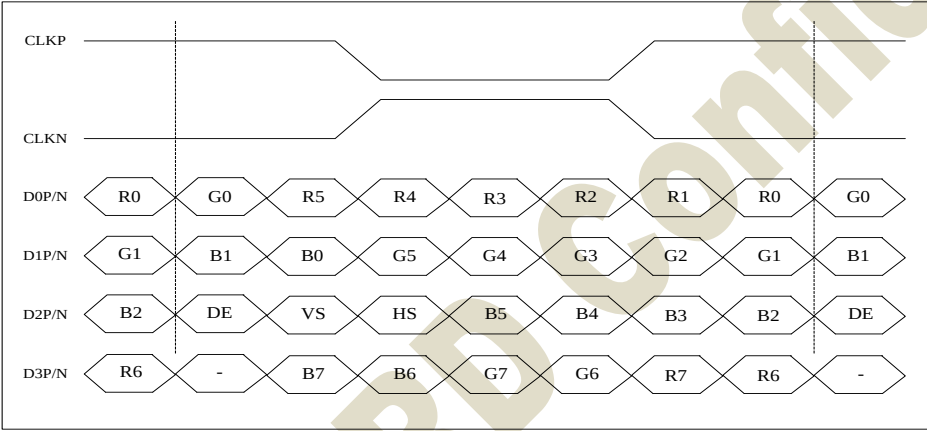
Table 6.1: TTL mode AC electrical characteristics

# 7. LVDS INPUT TIMING

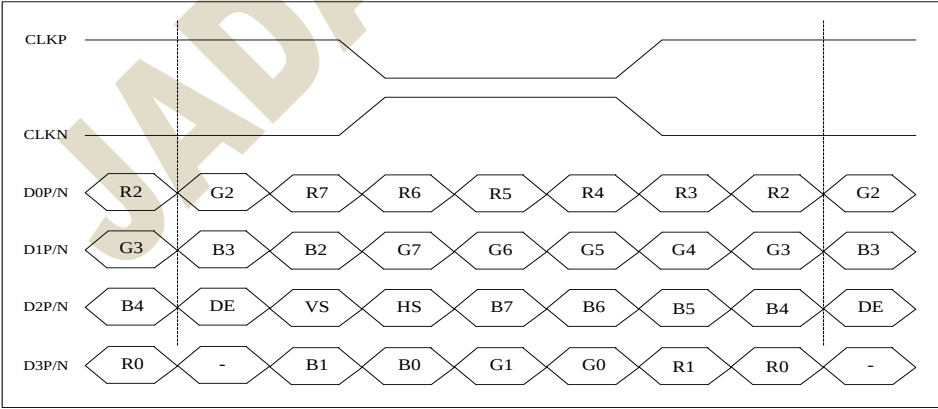
## 7.1 LVDS data input format



**Figure 7.1: 6-bit RGB LVDS input timing**



**Figure 7.2: 8-bit RGB LVDS VESA input timing**



**Figure 7.3: 8-bit RGB LVDS JEIDA input timing**

## 7.2 LVDS timing characteristic

| LVDS Input Timing      | Symbol | 1024RGBx768 |      |       | 1024RGBx600 |      |       | 800RGBx600 |      |       | Unit |
|------------------------|--------|-------------|------|-------|-------------|------|-------|------------|------|-------|------|
|                        |        | Min         | Typ  | Max   | Min         | Typ  | Max   | Min        | Typ  | Max   |      |
| DCLK Frequency         | -      | 52.7        | 65   | 71    | 41.4        | 51.2 | 67.2  | 33.1       | 39.6 | 62.4  | MHZ  |
| Horizontal Total       | tht    | 1114        | 1344 | 1400  | 1114        | 1344 | 1400  | 890        | 1000 | 1300  | DCLK |
| Hsync Pulse width      | ths    | 1           | 24   | HBP-1 | 1           | 24   | HBP-1 | 1          | 24   | HBP-1 | DCLK |
| Horizontal Back Porch  | thb    | 60          | 160  | 160   | 60          | 160  | 160   | 60         | 160  | 250   | DCLK |
| Horizontal Valid Data  | thd    | 1024        |      |       | 1024        |      |       | 800        |      |       | DCLK |
| Horizontal Front Porch | thfp   | 30          | 160  | 216   | 30          | 160  | 216   | 30         | 40   | 250   | DCLK |
| Vertical Total         | tv     | 788         | 806  | 845   | 620         | 635  | 800   | 620        | 660  | 800   | THT  |
| Vsync Pulse Width      | tv     | 1           | 2    | VBP-1 | 1           | 2    | VBP-1 | 1          | 2    | VBP-1 | THT  |
| Vertical Back Porch    | tvb    | 8           | 23   | 33    | 8           | 23   | 100   | 8          | 23   | 100   | THT  |
| Vertical Valid Data    | tvd    | 768         |      |       | 600         |      |       | 600        |      |       | THT  |
| Vertical Front Porch   | tvfp   | 12          | 15   | 44    | 12          | 12   | 100   | 12         | 37   | 100   | THT  |

| LVDS Input Timing      | Symbol | 800RGBx480 |      |       | 640RGBx480 |      |       | 480RGBx272 |      |       | Unit |
|------------------------|--------|------------|------|-------|------------|------|-------|------------|------|-------|------|
|                        |        | Min        | Typ  | Max   | Min        | Typ  | Max   | Min        | Typ  | Max   |      |
| DCLK Frequency         | -      | 29         | 32.4 | 62    | 32.6       | 39.6 | 53.4  | 24.8       | 32.6 | 37.6  | MHZ  |
| Horizontal Total       | tht    | 900        | 1000 | 1300  | 890        | 1000 | 1114  | 830        | 890  | 950   | DCLK |
| Hsync Pulse width      | ths    | 4          | 24   | HBP-1 | 1          | 24   | HBP-1 | 1          | 24   | HBP-1 | DCLK |
| Horizontal Back Porch  | thb    | 62         | 160  | 250   | 140        | 88   | 220   | 180        | 210  | 240   | DCLK |
| Horizontal Valid Data  | thd    | 800        |      |       | 640        |      |       | 480        |      |       | DCLK |
| Horizontal Front Porch | thfp   | 38         | 40   | 250   | 110        | 272  | 254   | 170        | 200  | 230   | DCLK |
| Vertical Total         | tv     | 500        | 540  | 680   | 610        | 660  | 800   | 498        | 610  | 660   | THT  |
| Vsync Pulse Width      | tv     | 1          | 2    | VBP-1 | 1          | 2    | VBP-1 | 1          | 2    | VBP-1 | THT  |
| Vertical Back Porch    | tvb    | 8          | 23   | 100   | 28         | 39   | 160   | 126        | 180  | 210   | THT  |
| Vertical Valid Data    | tvd    | 480        |      |       | 480        |      |       | 272        |      |       | THT  |
| Vertical Front Porch   | tvfp   | 12         | 37   | 100   | 102        | 141  | 160   | 100        | 158  | 178   | THT  |

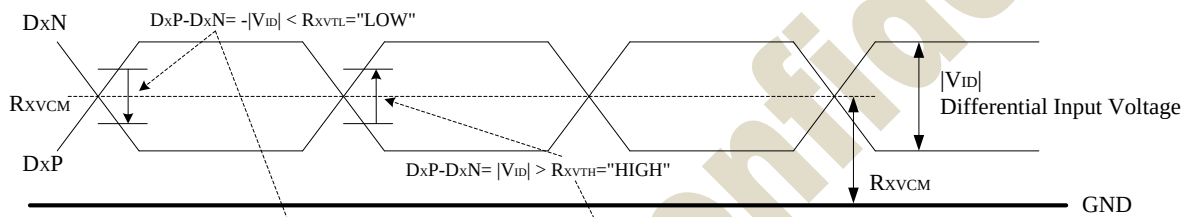
## 7.3 LVDS DC electrical characteristic

(Test condition: VDD=2.8V~3.6V, VSS=0V )

| Parameter                                 | Symbol       | Min. | Typ. | Max. | Unit | Condition                                                                                           |
|-------------------------------------------|--------------|------|------|------|------|-----------------------------------------------------------------------------------------------------|
| Differential input high threshold voltage | $R_{XVTH}$   |      |      | +37  | mV   | 1. $R_{XVCM} + 1/2 V_{ID}  \leq 1650\text{mV}$ .<br>2. $R_{XVCM} - 1/2 V_{ID}  \geq 400\text{mV}$ . |
| Differential input low threshold voltage  | $R_{XVTL}$   | -37  |      |      | mV   |                                                                                                     |
| Input voltage range (singled-end)         | $R_{XVIN}$   | 400  | -    | 1650 | mV   |                                                                                                     |
| Differential input common mode voltage    | $R_{XVCM}$   | 600  | 1200 | 1375 | mV   |                                                                                                     |
| Differential input voltage                | $ V_{ID} $   | 100  | -    | 400  | mV   |                                                                                                     |
| Differential input impedance              | $Z_{ID}$     | 80   | 100  | 120  | ohm  |                                                                                                     |
| Differential input leakage current        | $I_{LCLVDS}$ | -10  | -    | +10  | uA   |                                                                                                     |

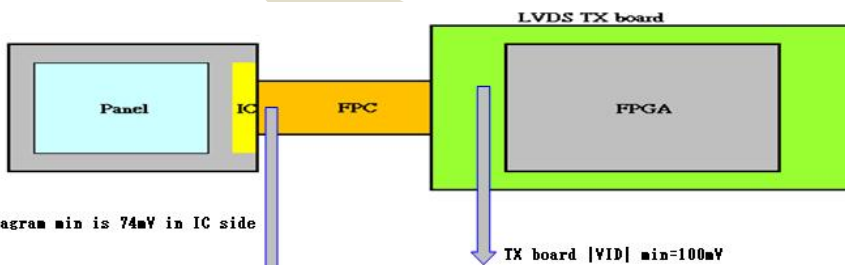
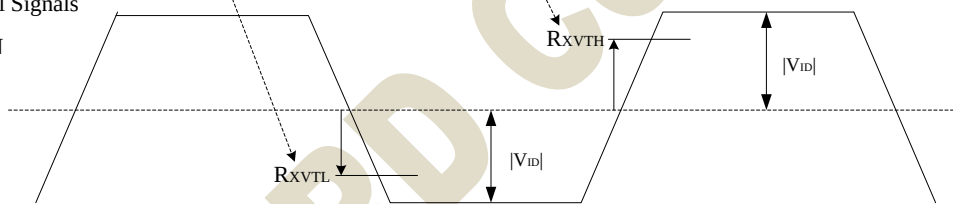
Table 7.1: LVDS mode DC electrical characteristics

Single-end Signals

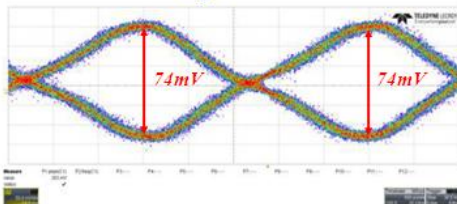


Differential Signals

$DxP - DxN$



Differential Signals  
 $DxP/N, CLKP/N$



## 7.4 LVDS AC electrical characteristic

| Parameter              | Symbol      | Spec. |                            |      | Unit | Condition                                                                                   |
|------------------------|-------------|-------|----------------------------|------|------|---------------------------------------------------------------------------------------------|
|                        |             | Min.  | Typ.                       | Max. |      |                                                                                             |
| Clock frequency        | $R_{xFCLK}$ | 20    | -                          | 71   | MHz  | Refer to input timing table for each display resolution                                     |
| Input data skew margin | $T_{RSKM}$  | -0.2  | -                          | 0.2  | UI   | $ VID  = 200\text{mV}$<br>$R_{xVCM} = 1.2\text{V}$<br>$1\text{UI} = 1/(R_{xFCLK} \times 7)$ |
| Clock high time        | $T_{LVCH}$  | -     | $3.5/(7 \times R_{xFCLK})$ | -    | ns   |                                                                                             |
| Clock low time         | $T_{LVCL}$  | -     | $3.5/(7 \times R_{xFCLK})$ | -    | ns   |                                                                                             |
| PLL wake-up time       | $T_{enPLL}$ | -     | -                          | 150  | us   |                                                                                             |

Table 7.2: LVDS mode AC electrical characteristics

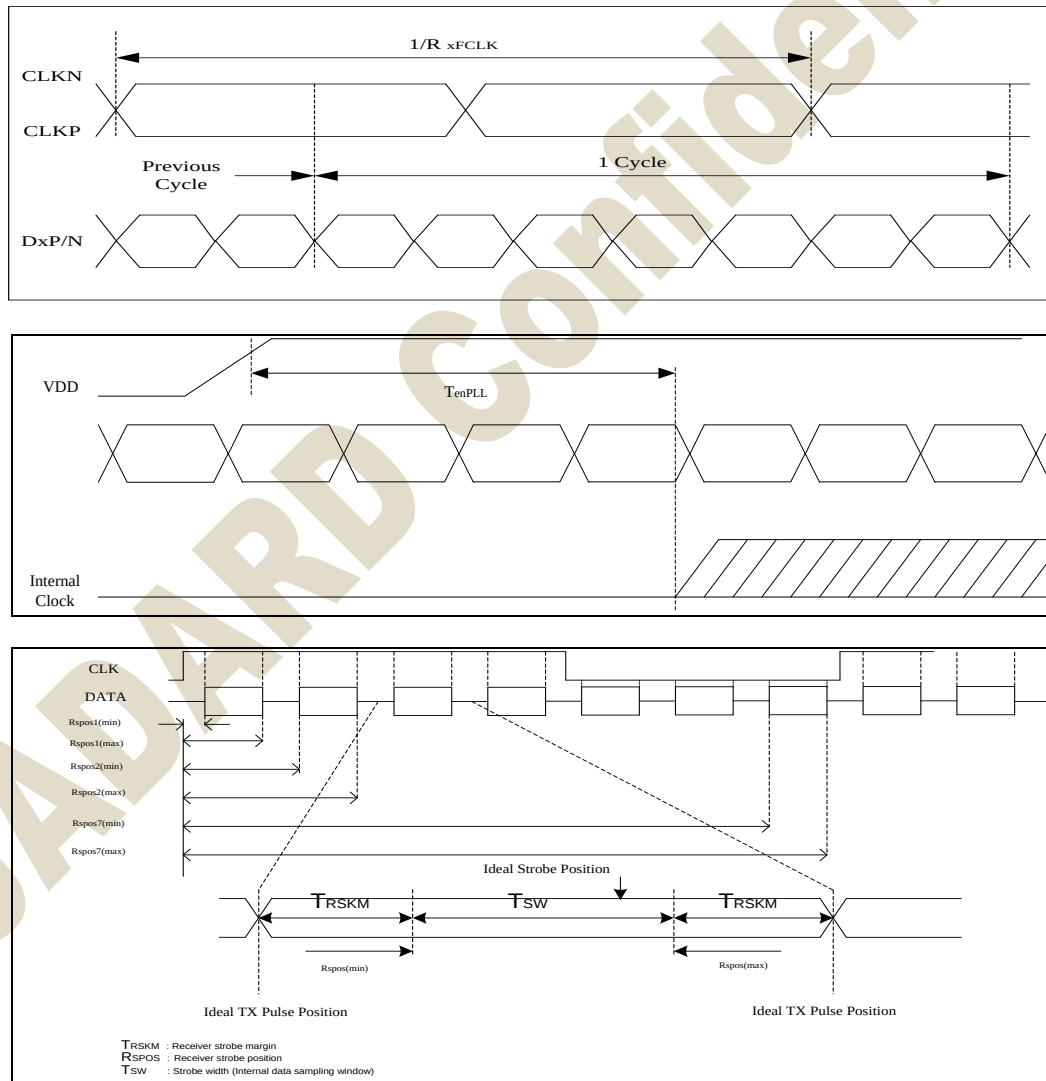


Figure 7.4: LVDS figure

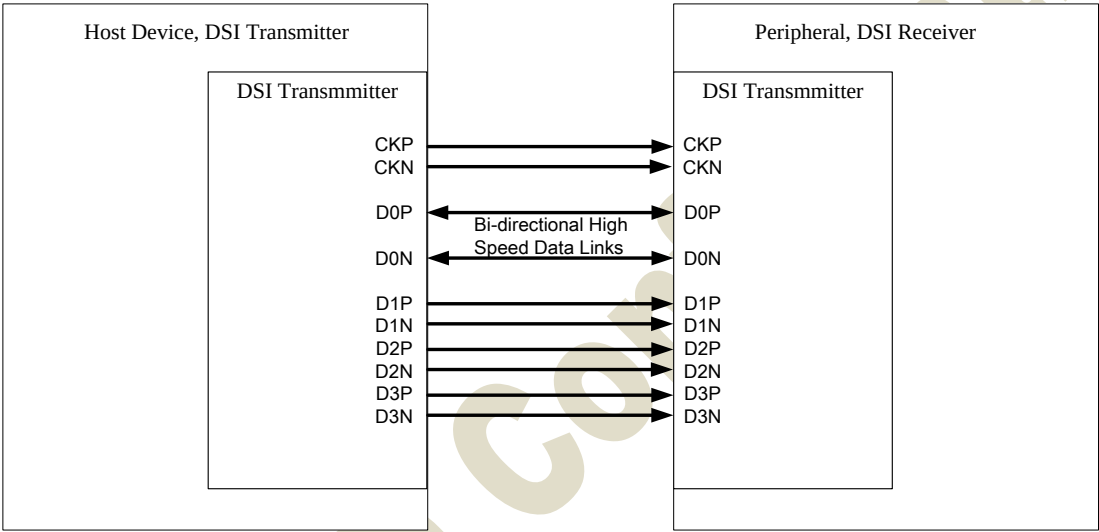


## 8. MIPI INTERFACE

### 8.1 DSI system interface

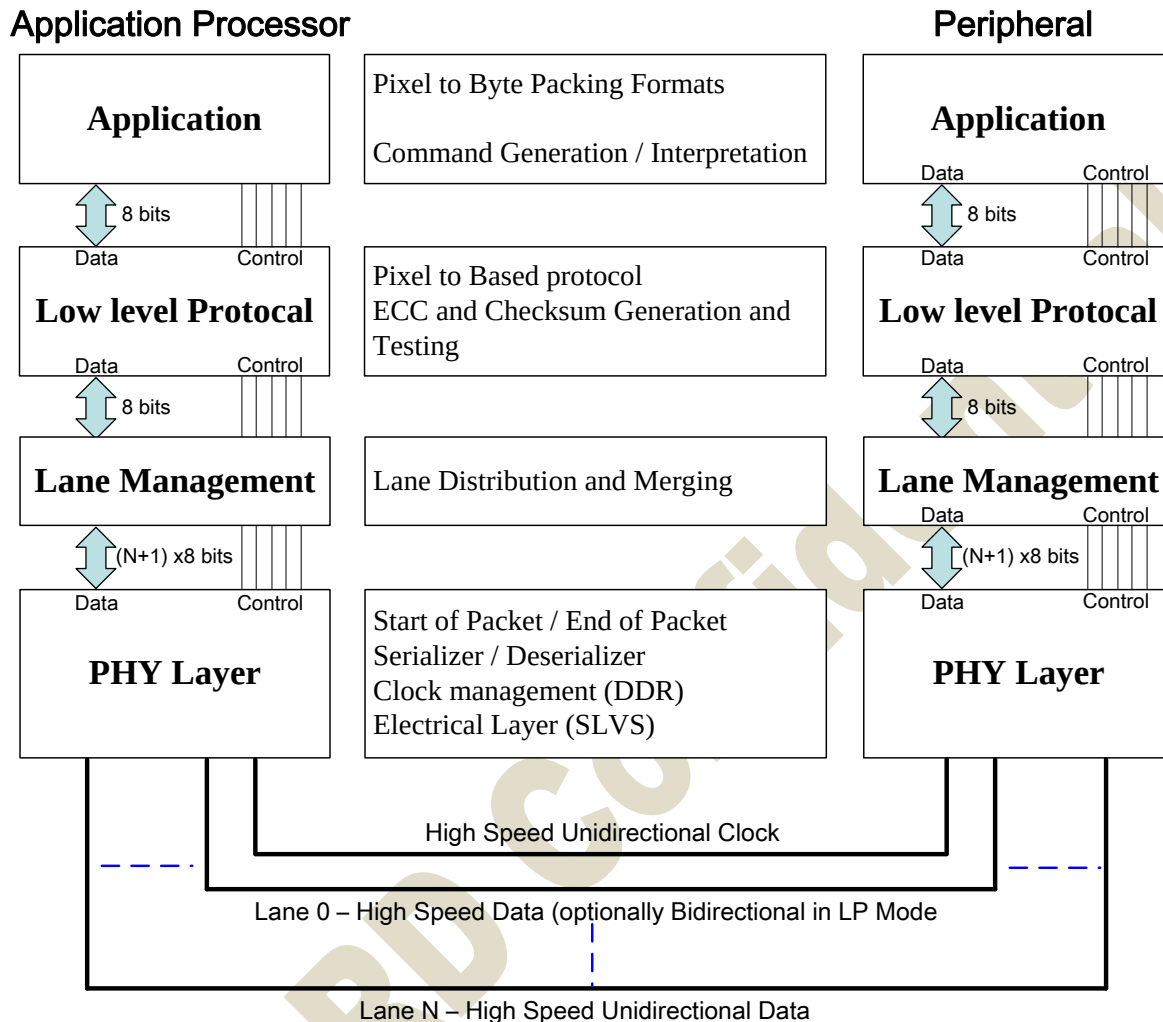
The Display Serial Interface (DSI) specifies the interface between a host processor and a peripheral. DSI builds on existing MIPI Alliance specification by adopting pixel formats and command set specified in DP1-2, DBI-2 and DCS standards.

Figure 8.1 shows a simplified DSI interface. DSI sends display data or commands to the peripheral, and can read back status or pixel information from the peripheral. The main difference is that DSI serializes all pixel data, commands, and events that, in traditional or legacy interfaces, are normally conveyed to and from the peripheral on a parallel data bus with additional control signals.



**Figure 8.1: DSI transmitter and receiver interface**

A conceptual view of DSI organizes the interface into several functional layers. A description of the layers follows and is also shown in Figure 8.2



**Figure 8.2: DSI Layer**

**PHY Layer:** The PHY Layer specifies transmission medium (electrical conductors), the input/output circuitry and the clocking mechanism that captures “ones” and “zeroes” from the serial bit stream. Bit-level and byte-level synchronization mechanisms are included as part of the PHY.

**Lane Management Layer:** DSI is Lane-scalable for increased performance. The number of data signals may be 1,2,3, or 4 depending on the bandwidth requirements of the application. The transmitter side of the interface distributes the outgoing data stream to one or more Lanes (“distributor” function). On the receiving end, the interface collects bytes from the Lanes and merges them together into recombined data stream that restores the original stream sequence (“merger” function).

**Protocol Layer:** At the lowest level, DSI protocol specifies the sequence and value of bits and bytes traversing the interface. It specifies how bytes are organized into defined groups called packets. The protocol defines required headers for each packet, and how header information is generated and interpreted. The transmitting side of the interface appends header and error-checking information to data being transmitted. On the receiving side, the header is stripped off and interpreted by corresponding logic in the receiver. Error-checking information may be used to test the integrity of incoming data. DSI protocol also documents how packets may be tagged for interleaving multiple command or data stream to

separate destinations using a single DSI.

**Application Layer:** This layer describes higher-level encoding and interpretation of data contained in the data stream. Depending on the display subsystem architecture, it may consist of pixels having a prescribed format, or of commands that are interpreted by the display controller inside a display module. The DSI specification describes the mapping of pixel values, commands and command parameters to bytes in the packet assembly.

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## 8.2 Video mode and Virtual Channel

### Video Mode

Video Mode refers to operation in which transfers from the host processor to the peripheral take the form of a real-time pixel stream. In normal operation, the display module relies on the host processor to provide image data at sufficient bandwidth to avoid flicker or other visible artifacts in the display image. Video information should only be transmitted using High Speed Mode.

Some Video Mode architectures may include a simple timing controller and partial line buffer, used to maintain a partial-screen or lower-resolution image in standby or Low Power Mode. This permits the interface to be shut down to reduce power consumption. To reduce complexity and cost, systems that only operate in Video Mode may use a unidirectional data path.

### Virtual Channel Capability

While this specification only addresses the connection of a host processor to a single peripheral, DSI incorporates a virtual channel capability for communication between a host processor and multiple, physical display modules. Since interface bandwidth is shared between peripherals, there are constraints that limit the physical extent and performance of multiple-peripheral systems. The DSI protocol permits up to four virtual channels, enabling traffic for multiple peripherals to share a common DSI Link. The DSI specification makes no requirements on the specific value assigned to each virtual channel used to designate interlaced fields. For clarity, the first interlaced video field may be assigned as  $DI[7:6]=2'b00$  and the second interlaced video field may be assigned  $DI[7:6]=2'b01$ .

### 8.3 DSI Format

Information is transferred between host processor and peripheral using one or more serial data signals and accompanying serial clock. The action of sending high-speed serial data across the bus is called a HS transmission or burst. Between transmissions, the differential data signal or Lane goes to a low-power state (LPS). Interfaces should be in LPS when they are not actively transmitting or receiving high-speed data. Figure 8.3 shows the basic structure of a HS transmission. N is the total number of bytes sent in the transmission.

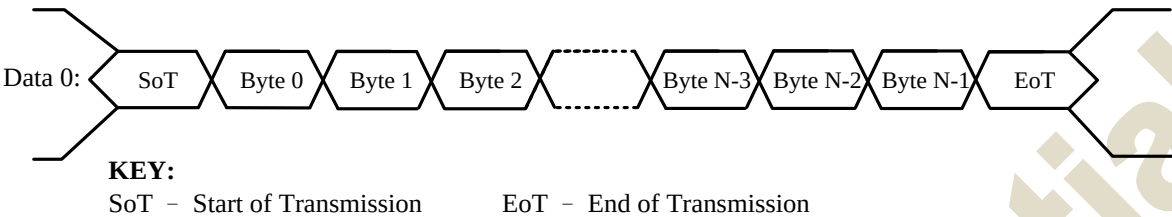


Figure 8.3: Basic HS Transmission Structure

#### Multi Lane Distribution and Merging

DSI is a Lane-scalable interface. Applications requiring more bandwidth than that provided by one Data Lane may expand the data path to two, three, or four Lanes wide and obtain approximately linear increases in peak bus bandwidth.

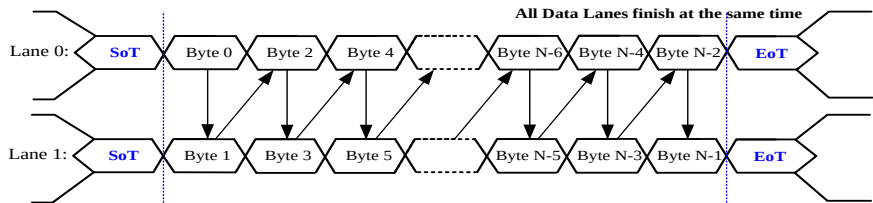
Multi-Lane implementations shall use a single common clock signal, shared by all Data Lanes. Conceptually, between the PHY and higher functional blocks is a layer that enables multi-Lane operation.

Since a HS transmission is composed of an arbitrary number of bytes that may not be an integer multiple of the number of Lanes, some Lanes may run out of data before others. Therefore, the Lane Management layer, as it buffers up the final set of less-than-N bytes, de-asserts its “valid data” signal into all Lanes for which there is no further data.

Although all Lanes start simultaneously with parallel Sots, each Lane operates independently and may complete the HS transmission before the other Lanes, sending an EoT one cycle (byte) earlier.

The N PHYs on the receiving end of the Link collect bytes in parallel and feed them into the Lane Management layer. The Lane Management layer reconstructs the original sequence of bytes in the transmission. Figure 8.4 & 8.5 illustrate a variety of ways a HS transmission can terminate for different number of Lanes and packet lengths.

Number of Bytes, N transmitted is an integer multiple of the number of lanes



Number of Bytes, N transmitted is NOT an integer multiple of the number of lanes

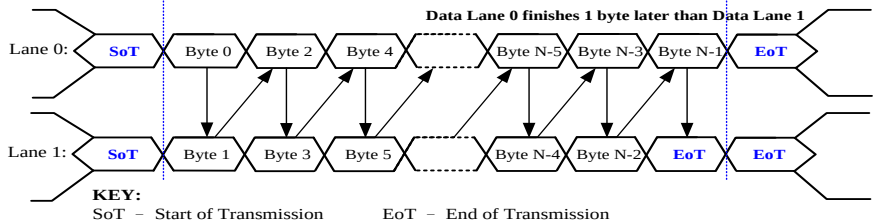
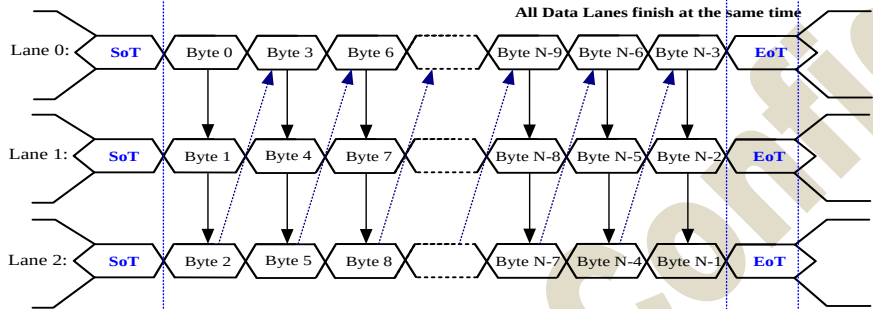
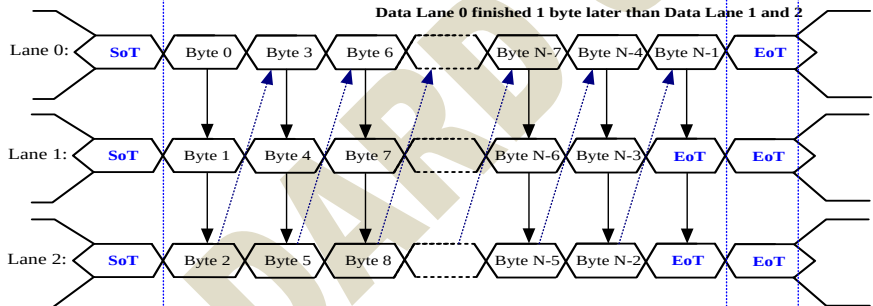


Figure 8.4: Two Lane HS Transmission Example

Number of Bytes, N transmitted is an integer multiple of the number of lanes



Number of Bytes, N transmitted is NOT an integer multiple of the number of lanes (Example 1):



Number of Bytes, N transmitted is NOT an integer multiple of the number of lanes (Example 2):

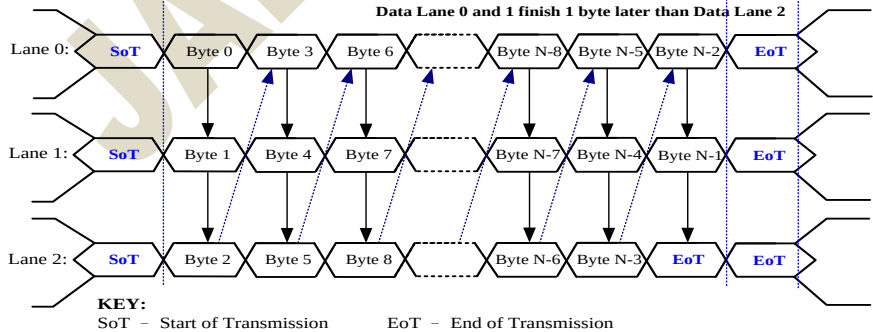


Figure 8.5: Three Lane HS Transmission Example

### 8.4 DSI Protocol

On the transmitter side of a DSI Link, parallel data, signal events, and commands are converted in the Protocol layer to packets, following the packet organization documented in this section. The Protocol layer appends packet-protocol information and headers, and then sends complete bytes through the Lane Management layer to the PHY.

### 8.5 Multiple Packets per Transmission

In its simplest form, a transmission may contain one packet. If many packets are to be transmitted, the overhead of frequent switching between LPS and High-Speed-Mode will severely limit bandwidth if packets are sent separately, e.g. one packet per transmission.

The DSI protocol permits multiple packets to be concatenated, which substantially boots effective bandwidth. This is useful for events such as peripheral initialization, where many registers may be loaded with separate write commands at system startup.

There are two modes of data transmission, HS and LP transmission modes, at the PHY layer. Before a HS transmission can be started, the transmitter PHY issues a SoT sequence to the receiver. After that, data or command packets can be transmitted in HS mode. Multiple packets may exist within a single HS transmission and the end of transmission is always signaled at the PHY layer using a dedicated EoT sequence. In order to enhance the overall robustness of the system, DSI defines a dedicated EoT packet (EoTp) at the protocol layer for signaling the end of HS transmission. For backwards compatibility with earlier DSI systems, the capability of generating and interpreting this EoTp can be enabled or disabled. The method of enabling or disabling this capability is out of scope for this document.

The top diagram in Figure 8.6 illustrates a case where multiple packets are being sent separately with EoTp support disabled. In HS mode, time gaps between packets shall result in separate HS transmissions for each packet, with a SoT, LPS, and EoT issued by the PHY layer between packets. This constraint does not apply to LP transmissions. The bottom diagram in Figure 8.6 demonstrates a case where multiple packets are concatenated within a single HS transmission.

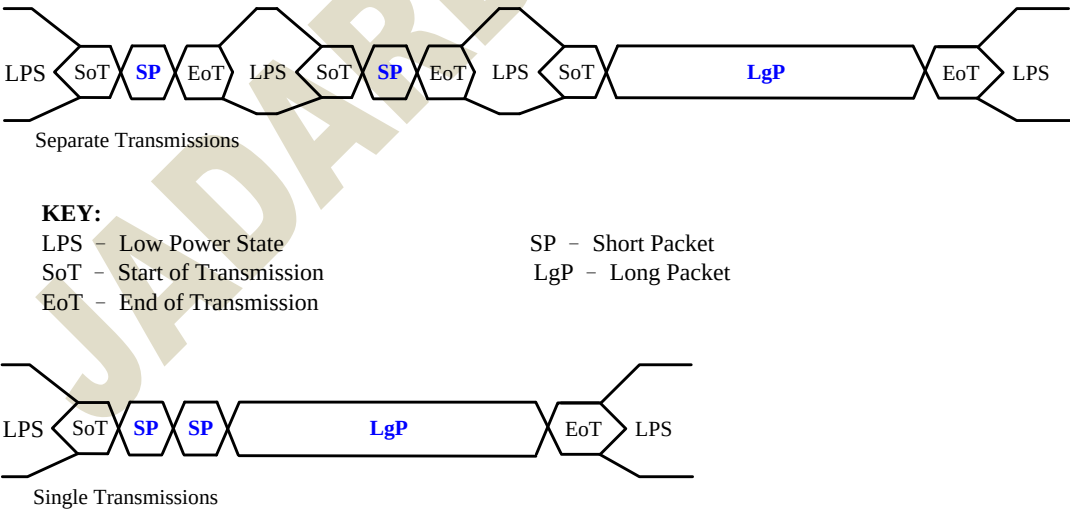


Figure 8.6: HS Transmission Examples with EoTp disabled

Figure 8.7 depicts HS transmission cases where EoTp generation is enabled. In the figure, EoT short packets are highlighted in red. The top diagram illustrates a case where a host is intending to send a short packet followed by a long packet using two separate transmissions. In this case, an additional EoT short packet is generated before each transmission ends. This mechanism provides a more robust environment, at the expense of increased overhead (four extra bytes per transmission) compared to cases where EoTp generation is disabled, i.e. the system only relies on the PHY layer EoT sequence for signaling the end of HS transmission. The overhead imposed by enabling EoTp can be minimized by sending multiple long and short packets within a single transmission as illustrated by the bottom diagram in Figure 8.7.

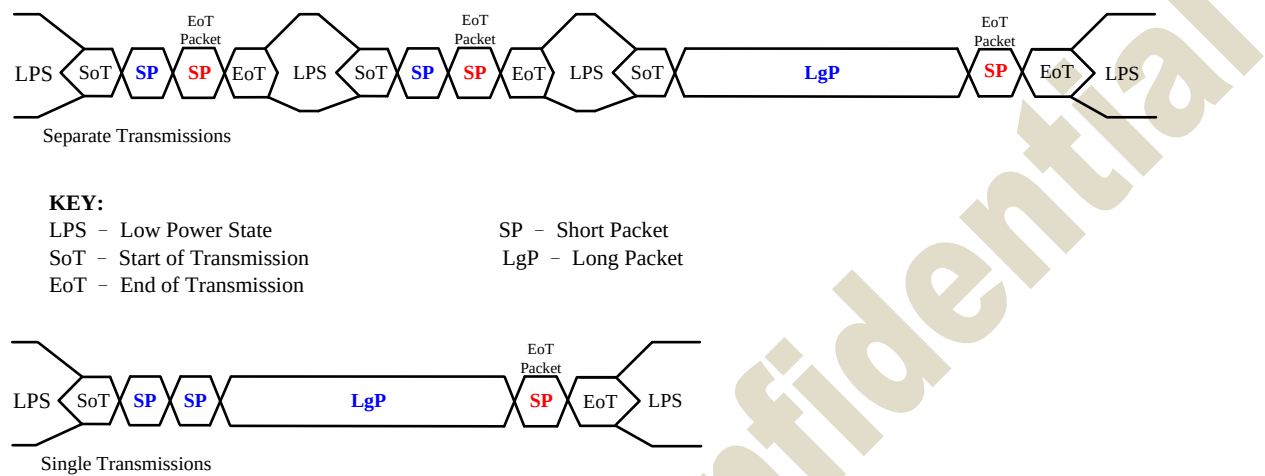
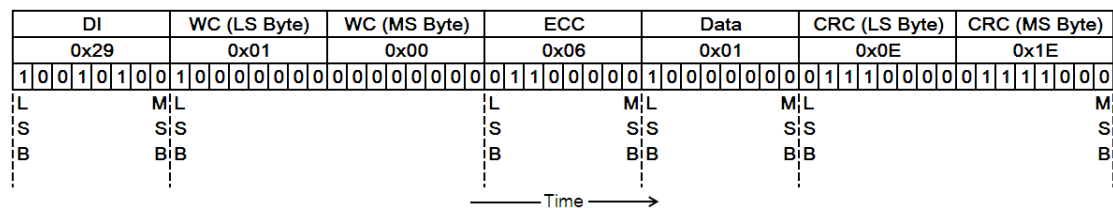


Figure 8.7: HS Transmission Examples with EoTp enabled



**8.6 Endian Policy**

All packet data traverses the interface as bytes. Sequentially, a transmitter shall send data LSB first, MSB last. For packets with multi-byte fields, the least significant byte shall be transmitted first unless otherwise specified. Figure 8.8 shows a complete Long packet data transmission. Note, the figure shows the byte values in standard positional notation, i.e. MSB on the left and LSB on the right, while the bits are shown in chronological order with the LSB on the left, the MSB on the right and time increasing left to right.



**Figure 8.8: Endian Example (Long Packet)**

## 8.7 Packet Structure

The first byte of the packet, the Data Identifier (DI), includes information specifying the type of the packet. Packet sizes fall into two categories:

- **Long packets** specify the payload length using a two-byte Word Count field. Payloads may be from 0 to 216- 1 bytes long. Therefore, a Long packet may be up to 65,541 bytes in length. Long packets permit transmission of large blocks of pixel or other data.
- **Short packets** are four bytes in length including the ECC. Short packets are used for most Command Mode commands and associated parameters. Other Short packets convey events like H Sync and V Sync edges. Because they are Short packets they can convey accurate timing information to logic at the peripheral.

The Set Maximum Return Packet Size command allows the host processor to limit the size of response packets coming from a peripheral.

8.8 Long Packet

Figure 8.9 shows the structure of the Long packet. A Long packet shall consist of three elements: a 32-bit Packet Header (PH), an application-specific Data Payload with a variable number of bytes, and a 16-bit Packet Footer (PF). The Packet Header is further composed of three elements: an 8-bit Data Identifier, a 16-bit Word Count, and 8-bit ECC. The Packet Footer has one element, a 16-bit checksum. Long packets can be from 6 to 65,541 bytes in length.

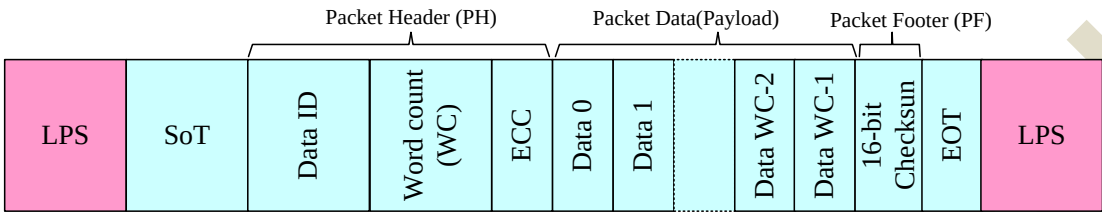


Figure 8.9: Long Packet Structure

The Data Identifier defines the Virtual Channel for the data and the Data Type for application specific payload data.

The Word Count defines the number of bytes in the Data Payload between the end of the Packet Header and the start of the Packet Footer. Neither the Packet Header nor the Packet Footer shall be included in the Word Count.

The Error Correction Code (ECC) bytes allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header. This includes both the Data Identifier and Word Count field.

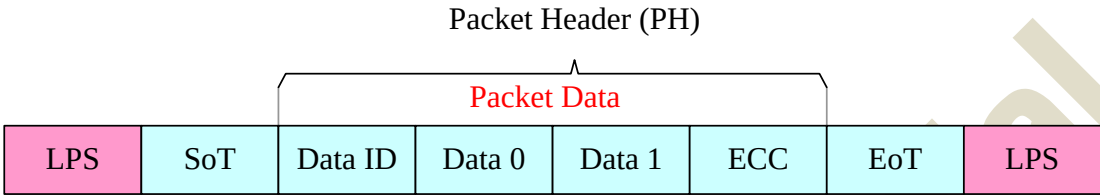
After the end of the Packet Header, the receiver reads the next Word Count \* bytes of the Data Payload. Within the Data Payload block, there are no limitations on the value of a data word, i.e. no embedded codes are used.

Once the receiver has read the Data Payload it reads the Checksum in the Packet Footer. The host processor shall always calculate and transmit a Checksum in the Packet Footer. Peripherals are not required to calculate a Checksum. Also note the special case of zero-byte Data Payload: if the payload has length 0, then the Checksum calculation results in (0xFFFF). If the Checksum is not calculated, the Packet Footer shall consist of two bytes of all zeros (0x0000). In the generic case, the length of the Data Payload shall be a multiple of bytes.

Each byte shall be transmitted least significant bit first. Payload data may be transmitted in any byte order restricted only by data format requirements. Multi-byte elements such as Word Count and Checksum shall transmitted least significant byte first.

**8.9 Short Packet**

Figure 8.10 shows the structure of the Short packet. A Short packet shall contain an 8-bit Data ID followed by two command or data bytes and an 8-bit ECC; a Packet Footer shall not be present. Short packets shall be four bytes in length. The Error Correction Code (ECC) byte allows single-bit errors to be corrected and 2-bit errors to be detected in the Short packet.



**Figure 8.10: Short Packet Structure**

## 8.10 Command Packet Elements

Long and Short packets have several common elements that are described in this section.

### 8.10.1 Data Identifier Byte

The first byte of any packet is the DI (Data Identifier) byte. Figure 8.11 shows the composition of the Data Identifier (DI) byte. DI[7:6]: These two bits identify the data as directed to one of four virtual channels. DI[5:4]: These two bits specify the Data Type.

| DI7                     | DI6 | DI5               | DI4 | DI3 | DI2 | DI1 | DI0 |
|-------------------------|-----|-------------------|-----|-----|-----|-----|-----|
| VC<br>(Virtual Channel) |     | DT<br>(Data Type) |     |     |     |     |     |

Figure 8.11: Data Identifier Byte

### 8.10.2 Virtual Channel Identifier – VC field, DI[7:6]

A processor may service up to four peripherals with tagged commands or blocks of data, using the Virtual Channel ID field of the header for packets targeted at different peripherals. The Virtual Channel ID enables one serial stream to service two or more virtual peripherals by multiplexing packets onto a common transmission channel.

### 8.10.3 Data Type Field DT[5:0]

The Data Type field specifies if the packet is a Long or Short packet type and the packet format. The Data Type field, along with the Word Count field for Long packets, informs the receiver of how many bytes to expect in the remainder of the packet. This is necessary because there are no special packet start / end sync codes to indicate the beginning and end of a packet. This permits packets to convey arbitrary data, but it also requires the packet header to explicitly specify the size of the packet. When the receiving logic has counted down to the end of a packet, it shall assume the next data is either the header of a new packet or the EoT (End of Transmission) sequence.

### 8.10.4 ECC

The Error Correction Code allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header. The host processor shall always calculate and transmit an ECC byte. Peripherals shall support ECC in both forward- and reverse-direction communication.

## 8.11 DSI packet

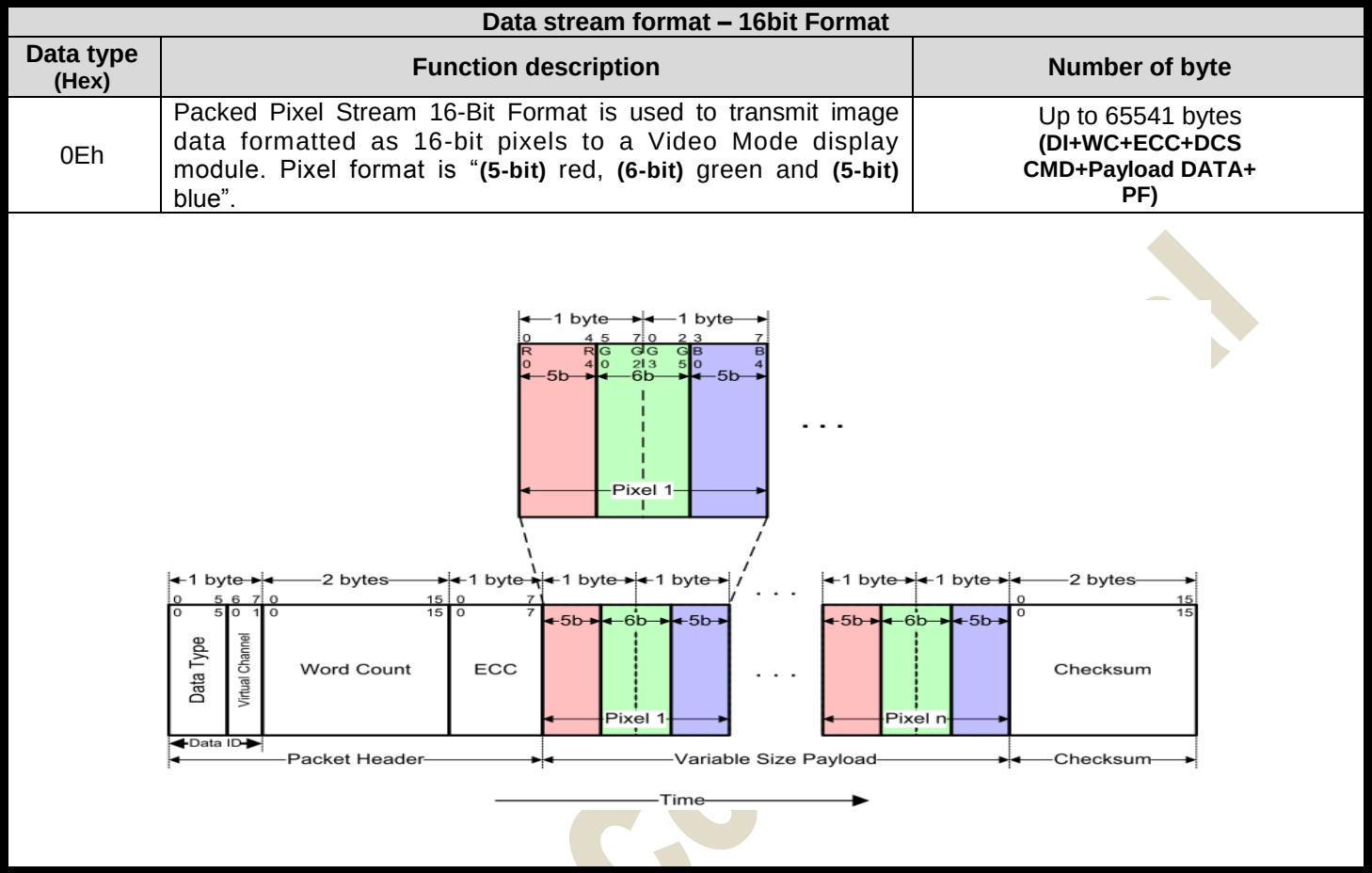
### 8.11.1 Processor-sourced Packets

The set of transaction types sent from the host processor to a peripheral, such as a display module, are shown in Table 8.1.

| Data Type (Hex) | Data Type (Binary) | Description                                           | Packet Size |
|-----------------|--------------------|-------------------------------------------------------|-------------|
| 0x01            | 00 0001            | Sync Event, V Sync Start                              | Short       |
| 0x11            | 01 0001            | Sync Event, V Sync End                                | Short       |
| 0x21            | 10 0001            | Sync Event, H Sync Start                              | Short       |
| 0x31            | 11 0001            | Sync Event, H Sync End                                | Short       |
| 0x08            | 00 1000            | End of Transmission Packet (EoTP) <sup>Note1</sup>    | Short       |
| 0x02            | 00 0010            | Color Mode (CM) Off Command                           | Short       |
| 0x12            | 01 0010            | Color Mode (CM) On Command                            | Short       |
| 0x22            | 10 0010            | Shut Down Peripheral Command                          | Short       |
| 0x32            | 11 0010            | Turn On Peripheral Command                            | Short       |
| 0x03            | 00 0011            | Generic Short WRITE, no parameters                    | Short       |
| 0x13            | 01 0011            | Generic Short WRITE, 1 parameters                     | Short       |
| 0x23            | 10 0011            | Generic Short WRITE, 2 parameters                     | Short       |
| 0x04            | 00 0100            | Generic Short READ, no parameters                     | Short       |
| 0x14            | 01 0100            | Generic Short READ, 1 parameters                      | Short       |
| 0x24            | 10 0100            | Generic Short READ, 2 parameters                      | Short       |
| 0x05            | 00 0101            | DCS Write, No Parameter                               | Short       |
| 0x15            | 01 0101            | DCS Write, 1 Parameter                                | Short       |
| 0x06            | 00 0110            | DCS Read, No Parameter                                | Short       |
| 0x37            | 11 0111            | Set Maximum Return Packet Size                        | Short       |
| 0x09            | 00 1001            | Null Packet, No Data, <sup>Note2</sup>                | Long        |
| 0x19            | 01 1001            | Blanking Packet, no data                              | Long        |
| 0x29            | 10 1001            | Generic Long Write                                    | Long        |
| 0x39            | 11 1001            | DCS Write Long                                        | Long        |
| 0x0E            | 00 1110            | Packed Pixel Stream, 16-bit RGB, 5-6-5 Format         | Long        |
| 0x1E            | 01 1110            | Packed Pixel Stream, 18-bit RGB, 6-6-6 Format         | Long        |
| 0x2E            | 10 1110            | Loosely Packed Pixel Stream, 18-bit RGB, 6-6-6 Format | Long        |
| 0x3E            | 11 1110            | Packed Pixel Stream, 24-bit RGB, 8-8-8 Format         | Long        |
| 0xX0            | xx 0000            | DO NOT USE                                            | --          |
| 0xFF            | xx 1111            | All unspecified codes are reserved                    | --          |

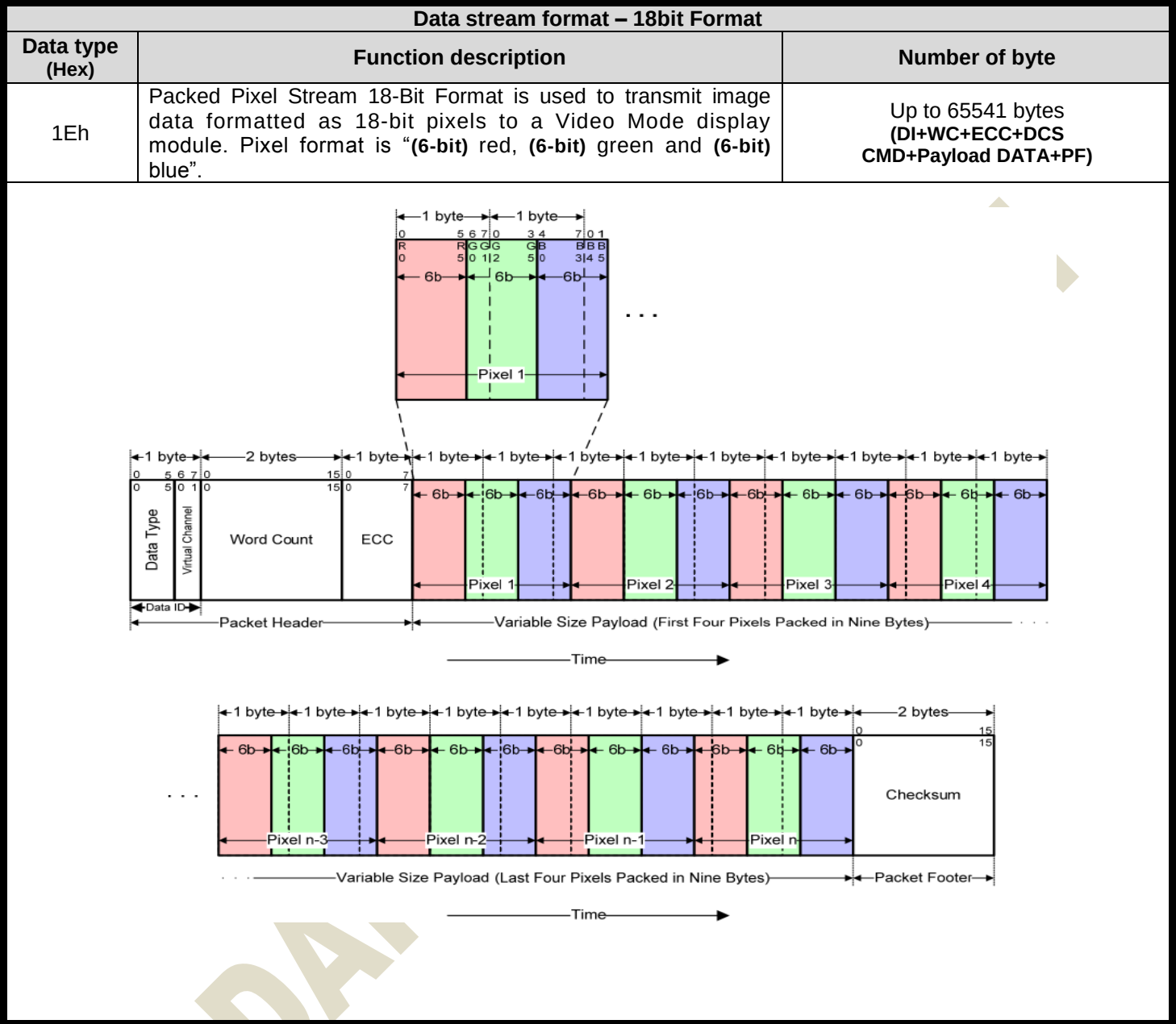
Table 8.1: Data Types for supported Processor-sourced Packets

8.11.2 Packed Pixel Stream, 16-bit Format, Long Packet



Note: (1) Within a color component, the “LSB is sent first, the MSB last “.

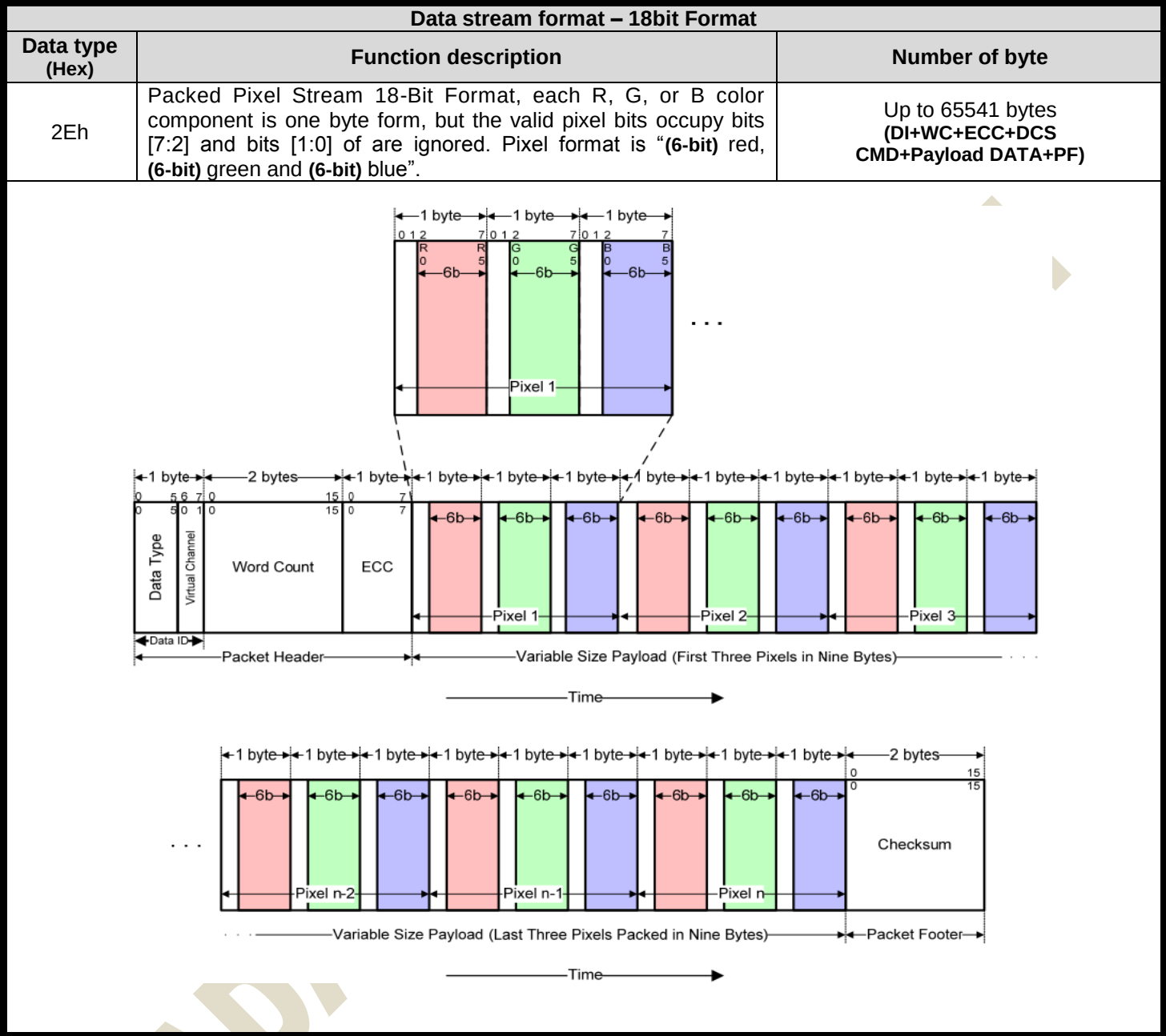
8.11.3 Packed Pixel Stream, 18-bit Format, Long Packet



**Note:** (1) Within a color component, the LSB is sent first and the MSB last and pixel boundaries only line up with byte boundaries every four pixels (**nine bytes**). Preferably, display modules employing this format have a horizontal extent (**width in pixels**) evenly divisible by four, so no partial bytes remain at the end of the display line data. It is possible to send pixel data that represent a line width that is not a multiple of four pixels, but display logic on the receiver end shall dispose of the extra bits of the partial byte at the end of active display and ensure a “clean start” for the next line.

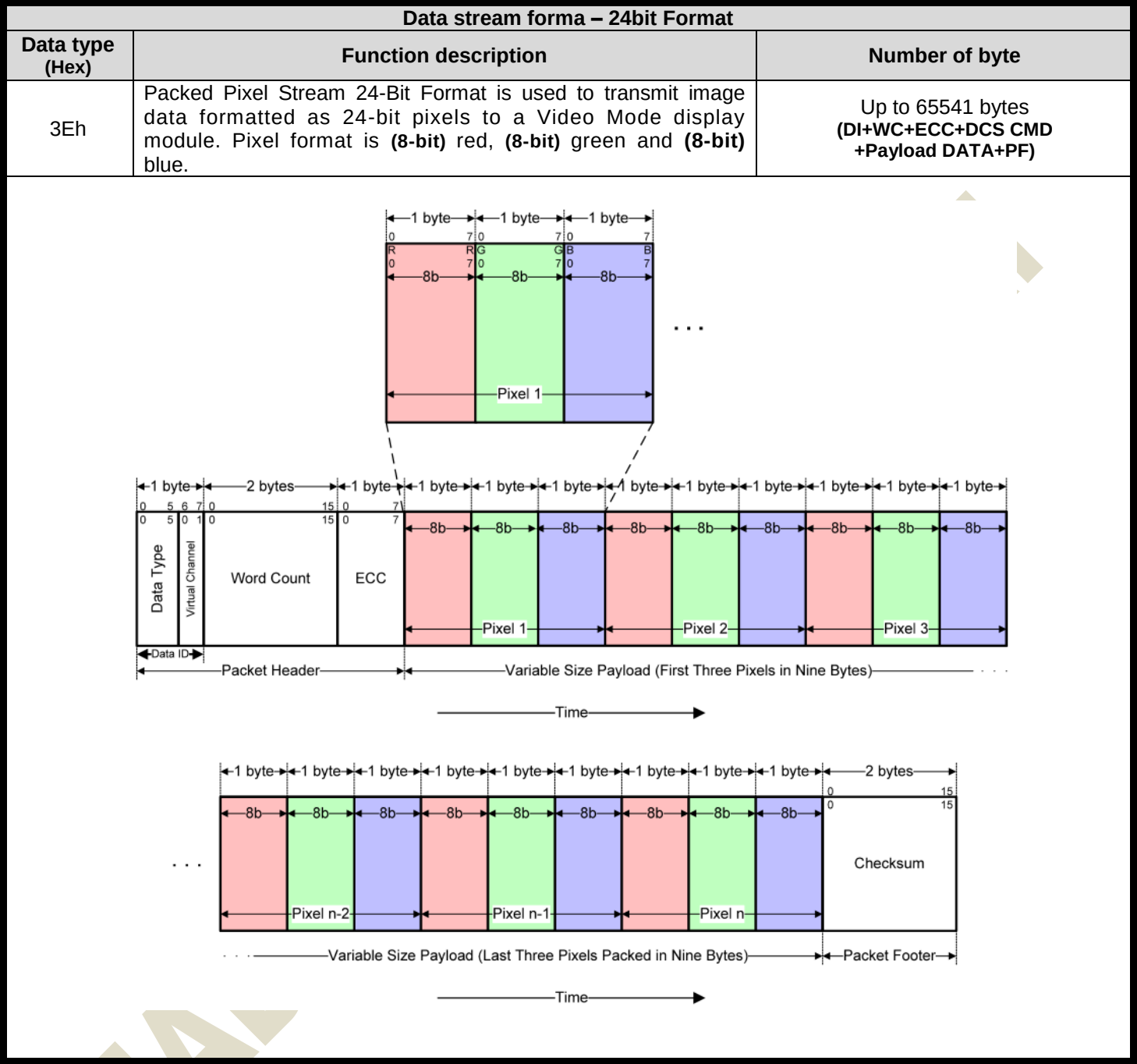


8.11.4 Packed Pixel Stream, 18-bit Loosely Format, Long Packet



**Note:** (1) Within a color component, the LSB is sent first, the MSB last and with this format, pixel boundaries line up with byte boundaries every three bytes.

8.11.5 Packed Pixel Stream, 24-bit Format, Long Packet



**Note:** (1) Within a color component, the LSB is sent first, the MSB last and with this format, pixel boundaries line up with byte boundaries every three bytes.

## 8.12 Peripheral to Processor Transmission

JD9165A-A3 has bidirectional capability for returning READ data, acknowledge, or error information to the host processor. BTA shall take place after every peripheral-to-processor transaction. This returns bus control to the host processor following the completion of the LP transmission from the peripheral.

Peripheral-to-processor transactions are of four basic types:

- **Acknowledge** is a Trigger message sent when the current transmission, as well as all preceding transmissions since the last peripheral to host communication, i.e. either trigger or packets, is received by the peripheral with no errors.
- **Acknowledge and Error Report** is a Short packet sent if any errors were detected in preceding transmissions from the host processor. Once reported, accumulated errors in the error register are cleared.
- **Response to Read Request** may be a Short or Long packet that returns data requested by the preceding READ command from the processor.

### 8.12.1 Appropriate Responses to Commands and ACK Requests

In general, if the host processor completes a transmission to the peripheral with BTA asserted, the peripheral shall respond with one or more appropriate packet(s), and then return bus ownership to the host processor. If BTA is not asserted following a transmission from the host processor, the peripheral shall not communicate an Acknowledge or error information back to the host processor.

Interpretation of processor-to-peripheral transactions with BTA asserted, and the expected response, are as follows:

- Following a non-Read command, the peripheral shall respond with Acknowledge if no errors were detected and stored since the last peripheral to host communication, i.e. either triggers or packets.
- Following a Read request, the peripheral shall send the requested READ data if no errors were detected and stored since the last peripheral to host communication, i.e. either triggers or packets.
- Following a Read request if only a single-bit ECC error was detected and corrected, the peripheral shall send the requested READ data in a Long or Short packet, followed by a 4-byte Acknowledge and Error Report packet in the same LP transmission. The Error Report shall have the ECC Error – Single Bit flag set, as well as any error bits from any preceding transmissions stored since the last peripheral to host communication.
- Following a non-Read command if only a single-bit ECC error was detected and corrected, the peripheral shall proceed to execute the command, and shall respond to BTA by sending a 4-byte Acknowledge and Error Report packet. The Error Report shall have the ECC Error – Single Bit flag set, as well as any error bits from any preceding transmissions stored since the last peripheral to host communication.
- Following a non-Read request, if multi-bit ECC errors were detected and not corrected, the peripheral shall send a 4-byte Acknowledge and Error Report packet without sending Read data. The Error Report shall have the ECC Error – Multi-Bit flag set, as well as any error bits from any preceding transmissions stored since the last peripheral to host communication.
- Following a non-Read command, if multi-bit ECC errors were detected and not corrected, the peripheral shall not execute the command, and shall send a 4-byte Acknowledge and Error Report packet. The Error Report shall have the ECC Error – Multi-Bit flag set, as well as any error bits from any preceding transmissions stored since the last peripheral to host communication.

- Following any command, if SoT Error, SoT Sync Error or DSI VC ID Invalid or DSI protocol violation was detected, or the DSI command was not recognized, the peripheral shall send a 4-byte Acknowledge and Error Report response, with the appropriate error flags set, as well as any error bits from any preceding transmissions stored since the last peripheral to host communication, in the two-byte error field. Only the Acknowledge and Error Report packet shall be transmitted; no read or write accesses shall take place on the peripheral in response.

- Following any command, if EoT Sync Error or LP Transmit Sync Error is detected, or a checksum error is detected in the payload, the peripheral shall send a 4-byte Acknowledge and Error Report packet with the appropriate error flag set, as well as any error bits from any preceding transmissions stored since the last peripheral to host communication. For a read command, only the Acknowledge and Error Report packet shall be transmitted; no read data shall be sent by the peripheral in response.

Once reported to the host processor, all errors documented in this section are cleared from the Error Register.

### 8.12.2 Peripheral-to-Processor Packet Description

Table 8.2 presents the complete set of peripheral-to-processor Data Types.

| Data Type (Hex) | Data Type (Binary) | Description                                  | Packet Size |
|-----------------|--------------------|----------------------------------------------|-------------|
| 0x02            | 00 0010            | Acknowledge and Error Report                 | Short       |
| 0x11            | 01 0001            | Generic Short READ Response, 1 byte returned | Short       |
| 0x12            | 01 0010            | Generic Short READ Response, 2 byte returned | Short       |
| 0x1A            | 01 1010            | Generic Long READ Response                   | Long        |
| 0x1C            | 01 1100            | DCS Long READ Response                       | Long        |
| 0x21            | 10 0001            | DCS Short READ Response, 1 byte returned     | Short       |
| 0x22            | 10 0010            | DCS Short READ Response, 2 byte returned     | Short       |

**Table 8.2: Data Types for Peripheral-sourced Packets**

## 8.13 Video Mode Interface Timing

Video Mode peripherals require pixel data delivered in real time. This section specifies the format and timing of DSI traffic for this type of display module.

### 8.13.1 Transmission Packet Sequences

DSI supports several formats, or packet sequences, for Video Mode data transmission. In the following sections, Burst Mode refers to time-compression of the RGB pixel (active video) portion of the transmission. In addition, these terms are used throughout the following sections:

- **Non-Burst Mode with Sync Pulse** – enables the peripheral to accurately reconstruct original video timing, including sync pulse widths.
- **Non-Burst Mode with Sync Events** – similar to above, but accurate reconstruction of sync pulse widths is not required, so a single Sync Event is substituted.
- **Burst Mode** – RGB pixel packets are time-compressed, leaving more time during a scan line for LP mode (saving power) or for multiplexing other transmissions onto the DSI link.

In the following figures the Blanking or Low-Power Interval (BLLP) is defined as a period during which video packets such as pixel-stream and sync event packets are not actively transmitted to the peripheral.

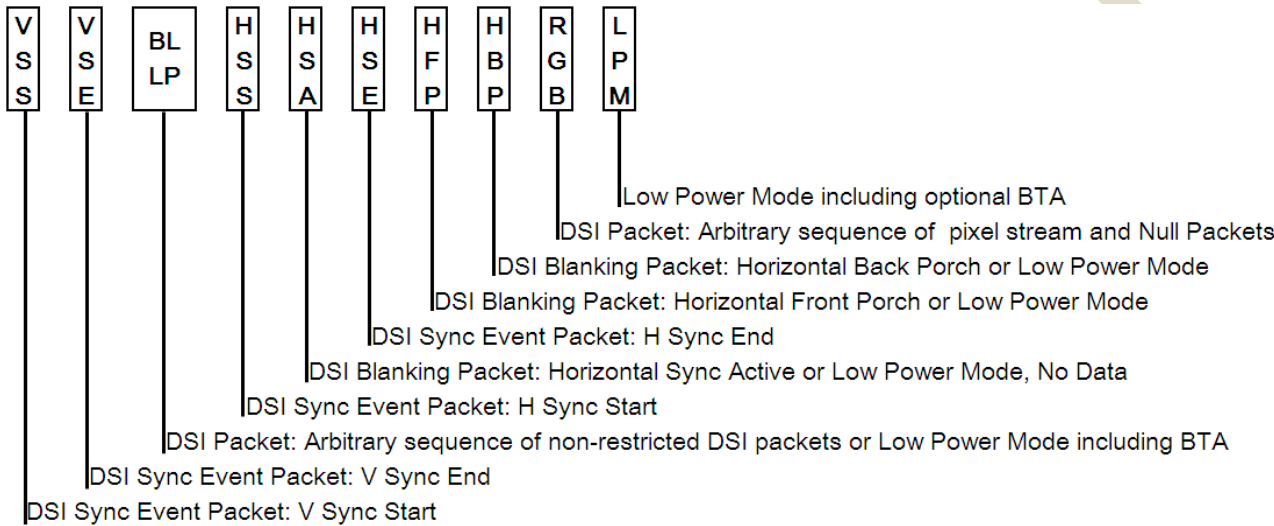
To enable PHY synchronization the host processor should periodically end HS transmission and drive the Data Lanes to the LP state. This transition should take place at least once per frame; shown as LPM in the figures in this section. The host processor should return to LP state once per scanline during the horizontal blanking time.

During the BLLP the DSI Link may do any of the following:

- Remain in Idle Mode with the host processor in LP-11 state and the peripheral in LP-RX
- Transmit one or more non-video packets from the host processor to the peripheral using Escape Mode
- Transmit one or more non-video packets from the host processor to the peripheral using HS Mode
- If the previous processor-to-peripheral transmission ended with BTA, transmit one or more packets from the peripheral to the host processor using Escape Mode
- Transmit one or more packets from the host processor to a different peripheral using a different Virtual Channel ID

The sequence of packets within the BLLP or RGB portion of a HS transmission is arbitrary. The host processor may compose any sequence of packets, including iterations, within the limits of the packet format definitions. For all timing cases, the first line of a frame shall start with VSS; all other lines shall start with VSE or HSS. Note that the position of synchronization packets, such as VSS and HSS, in time is of utmost importance since this has a direct impact on the visual performance of the display panel.

Normally, RGB pixel data is sent with one full scan line of pixels in a single packet. Transmission packet components used in the figures in this section are defined in Figure 8.12 unless otherwise specified.

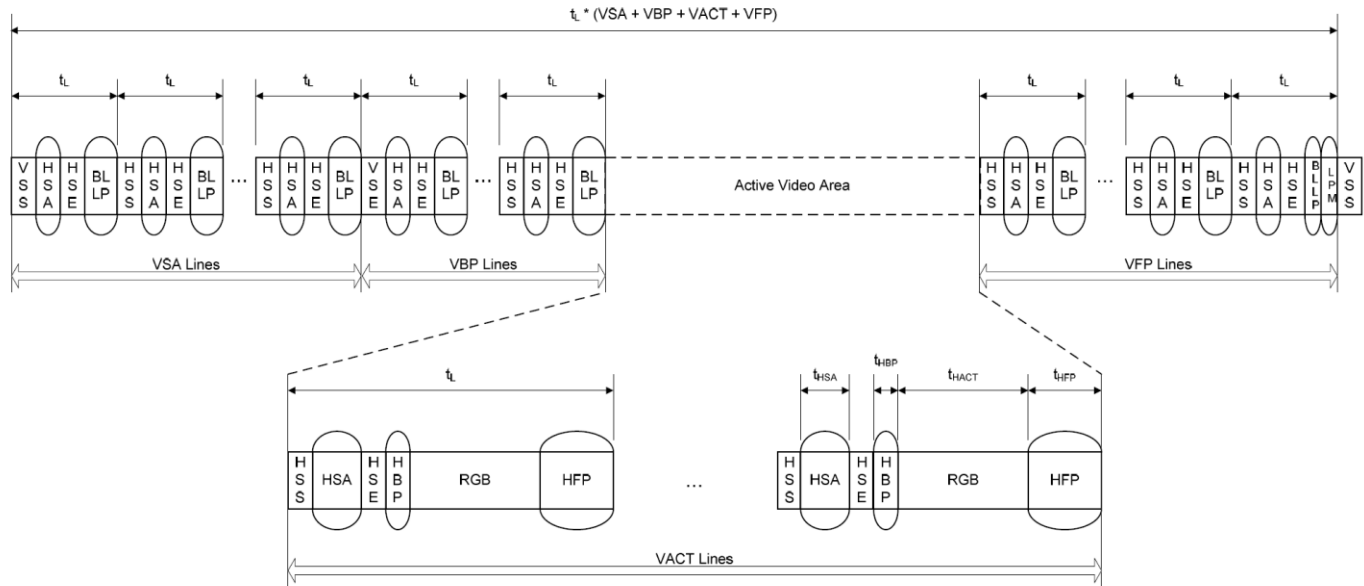


**Figure 8.12: Video Mode Interface Timing Legend**

If a peripheral timing specification for HBP or HFP minimum period is zero, the corresponding Blanking Packet may be omitted. If the HBP or HFP maximum period is zero, the corresponding blanking packet shall be omitted.

### 8.13.2 Non-Burst sync pulse mode

With this format, the goal is to accurately convey DPI-type timing over the DSI serial Link. This includes matching DPI pixel-transmission rates, and width of timing events like sync pulses. Accordingly, synchronization periods are defined using packets transmitting both start and end of sync pulses. An example of this mode is shown in Figure 8.13



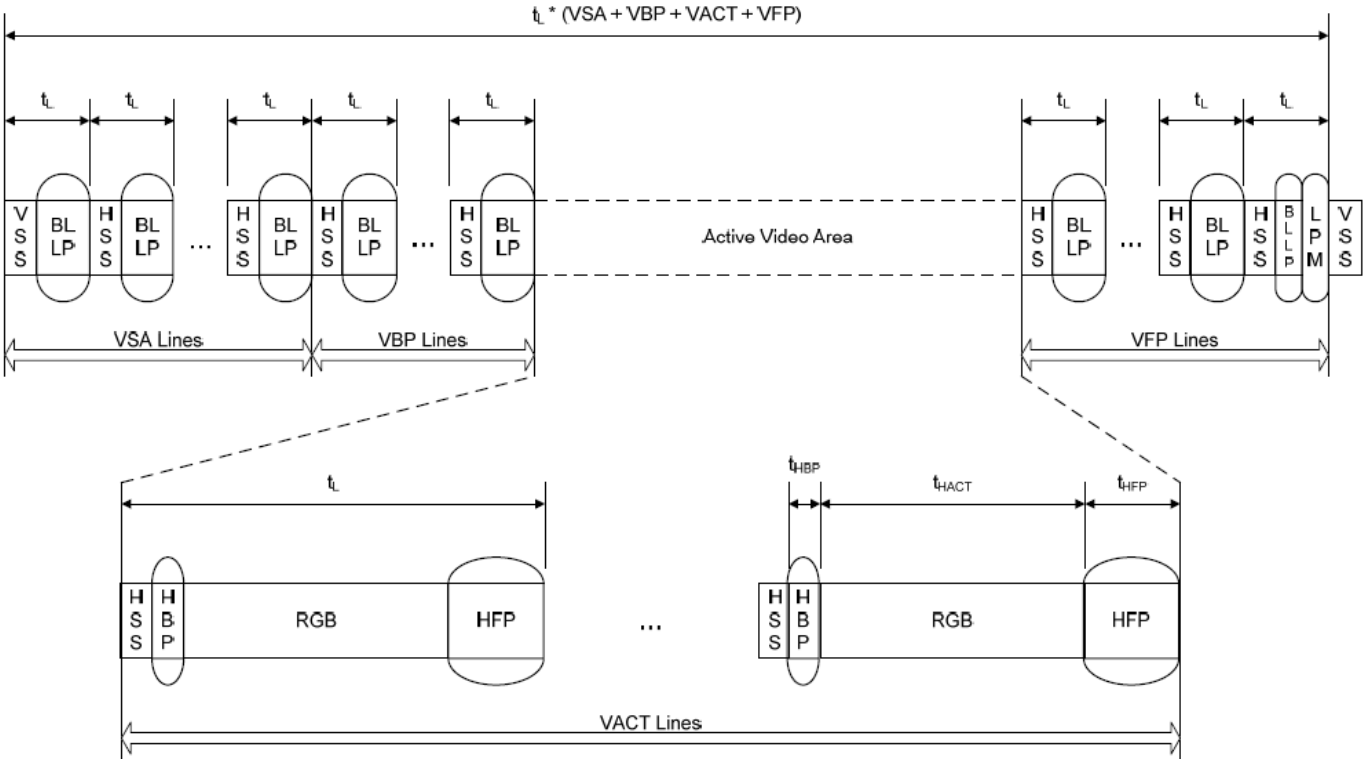
**Figure 8.13: Video Mode Interface Timing: Non-Burst Transmission with Sync Start and End**

Normally, periods shown as HAS (Horizontal Sync Active), HBP (Horizontal Back Porch) and HFP (Horizontal Front Porch) are filled by Blanking Packets, with lengths (including packet overhead) calculated to match the period specified by the peripheral's data sheet. Alternatively, if there is sufficient time to transition from HS to LP mode and back again, a timed interval in LP mode may substitute for a Blanking Packet, thus saving power. During HAS, HBP and HFP periods, the bus should stay in the LP-11 state.



### 8.13.3 Non-Burst sync event mode

This mode is a simplification of the “Non-Burst Mode with Sync Pulses” format. Only the start of each synchronization pulse is transmitted. The peripheral may regenerate sync pulses as needed from each Sync Event packet received. An example of this mode is shown in Figure 8.14.



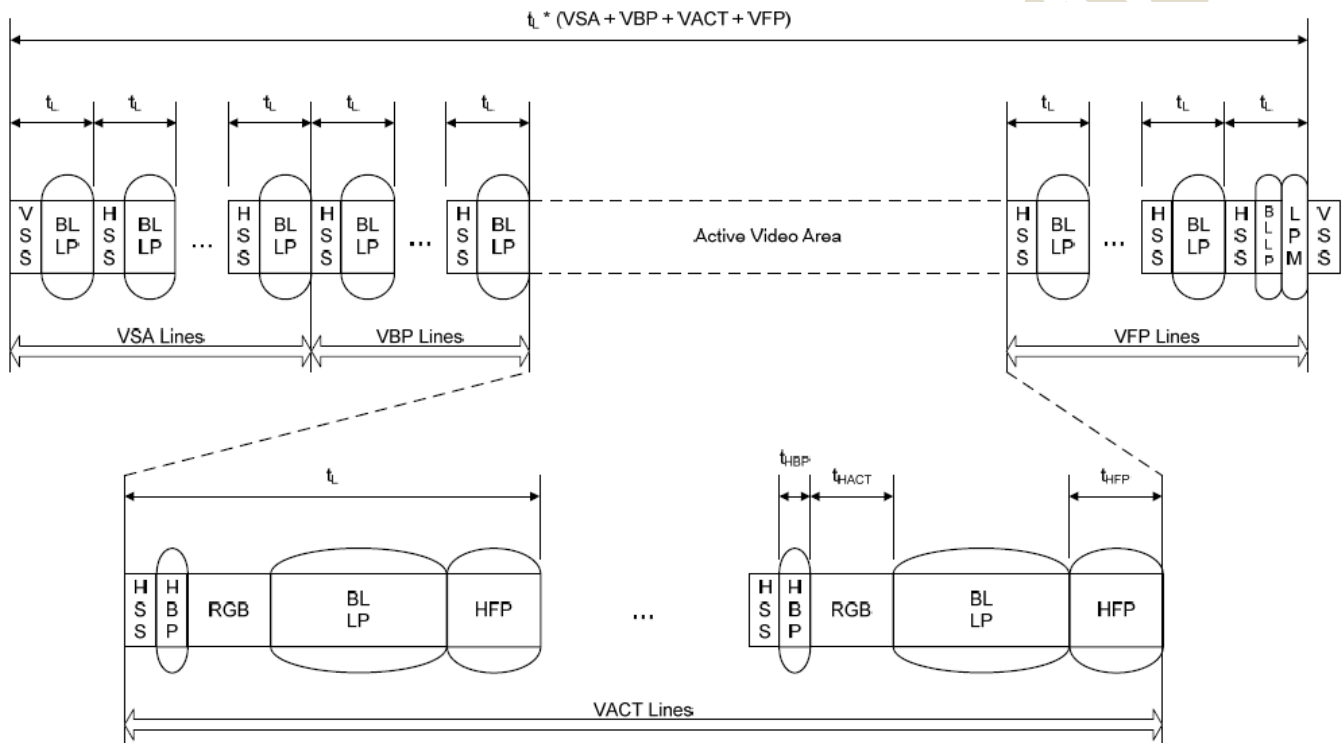
**Figure 8.14: Video Mode Interface Timing: Non-Burst Transmission with Sync Events**

As with the previous Non-Burst Mode, if there is sufficient time to transition from HS to LP mode and back again, a timed interval in LP mode may substitute for a Blanking Packet, thus saving power.

### 8.13.4 Burst mode

In this mode, blocks of pixel data can be transferred in a shorter time using a time-compressed burst format. This is a good strategy to reduce overall DSI power consumption, as well as enabling larger blocks of time for other data transmissions over the Link in either direction.

Following HS pixel data transmission, the bus may stay in HS Mode for sending blanking packets or go to Low Power Mode, during which it may remain idle, i.e. the host processor remains in LP-11 state, or LP transmission may take place in either direction. If the peripheral takes control of the bus for sending data to the host processor, its transmission time shall be limited to ensure data underflow does not occur from its interval buffer memory to the display device. An example of this mode is shown in Figure 8.15



**Figure 8.15: Video Mode Interface Timing: Burst Transmission**

Similar to the Non-Burst Mode scenario, if there is sufficient time to transition from HS to LP mode and back again, a timed interval in LP mode may substitute for a Blanking Packet, thus saving power.

## 8.14 Error-Correcting Code and Checksum

### 8.14.1 Error-Correcting Code(ECC)

MIPI DSI uses Hamming Code Theory as ECC generate rule. The parity of each bits in ECC are showed as below.

$$P7=0$$

$$P6=0$$

$$P5=D10 \wedge D11 \wedge D12 \wedge D13 \wedge D14 \wedge D15 \wedge D16 \wedge D17 \wedge D18 \wedge D19 \wedge D20 \wedge D21 \wedge D22 \wedge D23$$

$$P4=D4 \wedge D5 \wedge D6 \wedge D7 \wedge D8 \wedge D9 \wedge D16 \wedge D17 \wedge D18 \wedge D19 \wedge D20 \wedge D21 \wedge D22 \wedge D23$$

$$P3=D1 \wedge D2 \wedge D3 \wedge D7 \wedge D8 \wedge D9 \wedge D13 \wedge D14 \wedge D15 \wedge D19 \wedge D20 \wedge D21 \wedge D23$$

$$P2=D0 \wedge D2 \wedge D3 \wedge D5 \wedge D6 \wedge D9 \wedge D11 \wedge D12 \wedge D15 \wedge D18 \wedge D20 \wedge D21 \wedge D22$$

$$P1=D0 \wedge D1 \wedge D3 \wedge D4 \wedge D6 \wedge D8 \wedge D10 \wedge D12 \wedge D14 \wedge D17 \wedge D20 \wedge D21 \wedge D22 \wedge D23$$

$$P0=D0 \wedge D1 \wedge D2 \wedge D4 \wedge D5 \wedge D7 \wedge D10 \wedge D11 \wedge D13 \wedge D16 \wedge D20 \wedge D21 \wedge D22 \wedge D23$$

ECC is generated from the twenty-four bits with in the Packet Header as illustrated in Figure 8.16 which also serves as an ECC calculation example.

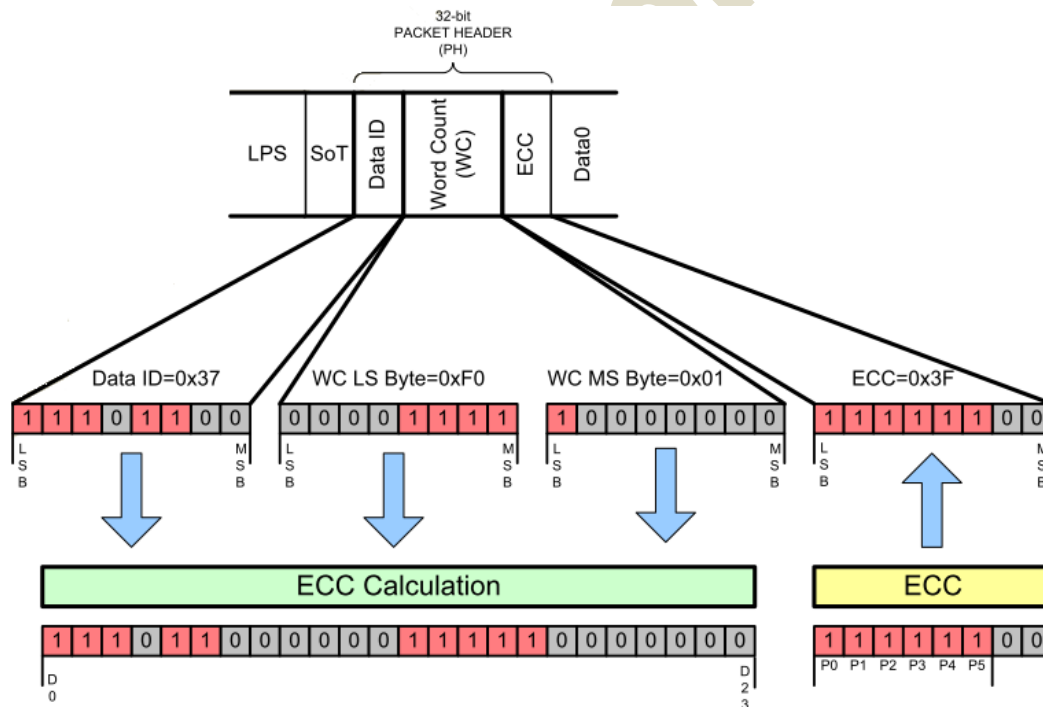
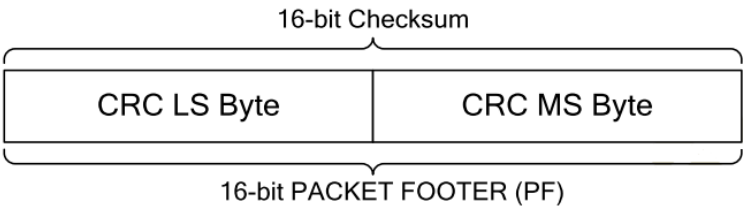


Figure 8.16: 24-bit ECC generation Example

### 8.14.2 Checksum Generation for Long Packet Payloads

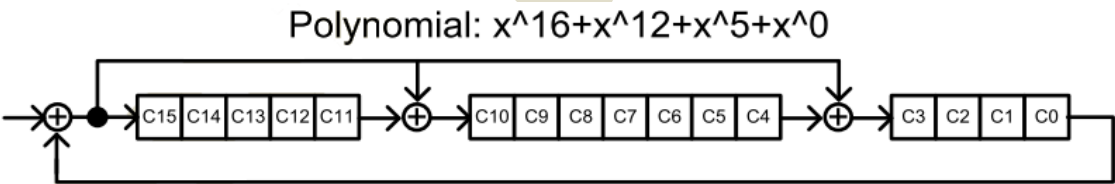
To detect errors in transmission of Long packets, a checksum is calculated over the payload portion of the data packet. Note that, for the special case of a zero-length payload, the 2-byte checksum is set to 0Xffff. The checksum shall be realized as a 16-bit CRC with a generator polynomial of  $x^{16}+x^{12}+x^5+x^0$ .

The transmission of the checksum is illustrated in Figure 8.17. The LS byte is sent first, followed by the MS byte. Note that within the byte, the LS bit is sent first.



**Figure 8.17: Checksum Transmission**

The CRC implementation is presented in Figure 8.18. The CRC shift register shall be initialized to 0Xffff before packet data enters. Packet data not including the Packet Header then enters as a bitwise data stream from the left, LS bit first. Each bit is fed through the CRC shift register before it is passed to the output for transmission to the peripheral. After all bytes in the packet payload have passed through the CRC shift register, the shift register contains the checksum. C15 contains the checksum’s MSB and C0 the LSB of the 16-bit checksum. The checksum is then appended to the data stream and sent to the register. The receiver uses its own generated CRC to verify that no errors have occurred in transmission.



**Figure 8.18: 16-bit CRC Generation Using a Shift Register**

## 8.15 DPHY

### 8.15.1 Lane Module

A PHY configuration contains a Check Lane Module and one or more Data Lane Modules. Each of these PHY Lane Modules communicates via two Lines to a complementary part at the other side of the Lane Interconnect. Each Lane Module consists of one or more differential High-Speed functions utilizing both interconnect wires simultaneously, one or more single-ended Low-Power functions operating on each of the interconnect wires individually, and control & interface logic. For proper operation, the set of functions in the Lane Modules on both sides of the Lane Interconnect has to be matched.

### 8.15.2 Lane Module Type of Clock Lane, Data0, Data1 and Data2

The required functions in a Lane Module depend on the Lane type and which side (master or slave) of the Lane Interconnect the Lane Module is located. There are three main Lane types: Clock Lane, Unidirectional Data Lane and Bi-directional Data Lane. Several PHY configurations can be constructed with these Lane types. Below show the lane module architecture of each lane.

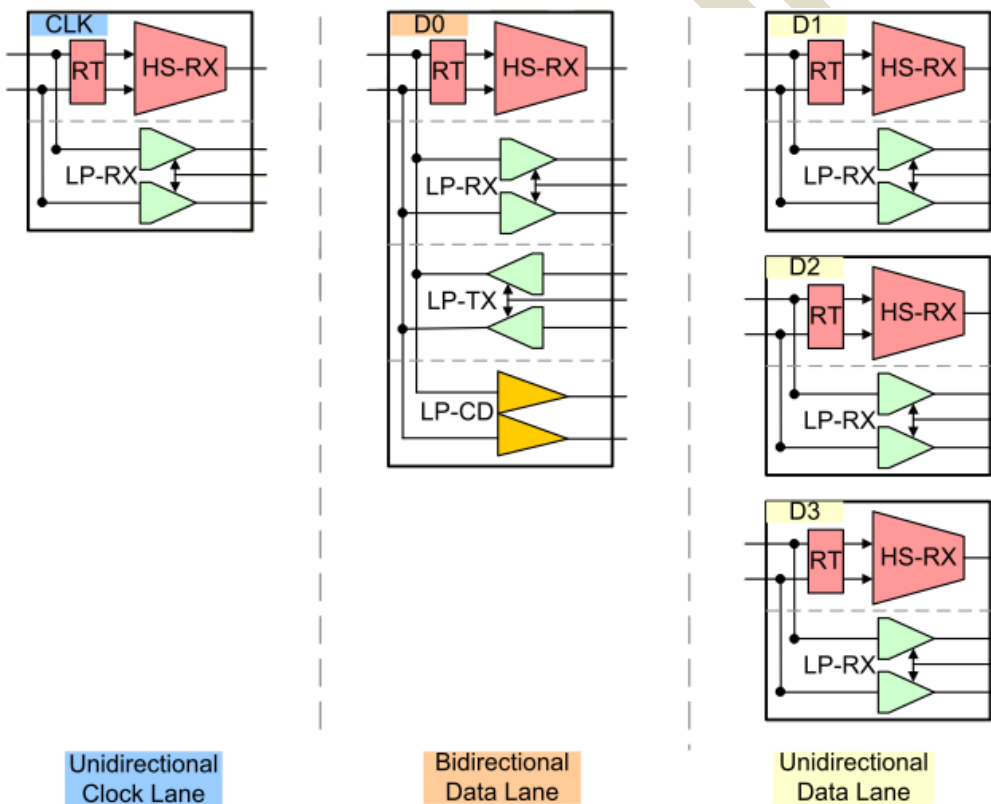


Figure 8.19: Lane Module Type

### 8.15.3 Master and Slave

Each Link has a Master and a Slave side. The Master provides the High-Speed DDR Clock signal to the Clock Lane and is the main data source. The Slave receives the clock signal at the Clock Lane and the main data sink. The main direction of data communication, from source to sink, is denoted as the Forward direction. Data communication in the opposite direction is called Reverse transmission. Only bi-directional Data Lanes can transmit in the Reverse direction. In all cases, the Clock Lane remains in the Forward direction, but bi-directional Data Lane(s) can be turned around, sourcing data from the Slave side.

JD9165A-A3 serves as Slave side.

### 8.15.4 Lane States and Line Levels

Transmitter functions determine the Lane state by driving certain Line levels. During normal operation either a HS-TX or a LP-TX is driving a Lane. A HS-TX always drives the Lane differentially. The two LP-TX's drive the two Lines of a Lane independently and single-ended. This results in two possible High-Speed Lane states and four possible Low-Power Lane states. The High-Speed Lane states are Differential-0 and Differential-1. The interpretation of Low-Power Lane states depends on the mode of operation. The LP-Receivers shall always interpret both High-Speed differential states as LP-00.

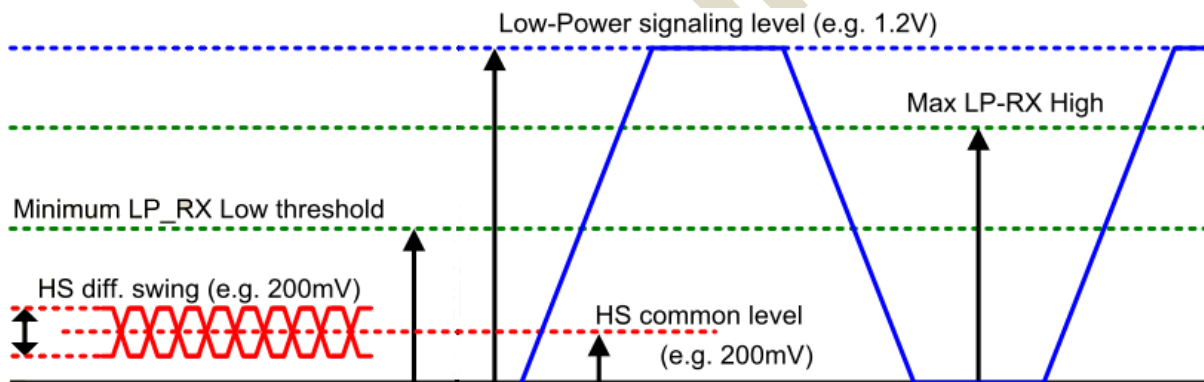


Figure 8.20: Line Levels

The Stop state has a very exclusive and central function. If the Line levels show a Stop state for the minimum required time, the PHY state machine shall return to the Stop state regardless of the previous state. This can be in RX or TX mode depending on the most recent operating direction. Table 8.3 lists all the states that can appear on a Lane during normal operation. All LP state periods shall be at least TLPX in duration. State transitions shall be smooth and exclude glitch effects. A clock signal can be reconstructed by exclusive-ORing the Dp and Dn Lines. Ideally, the reconstructed clock has a duration of at least 2\*TLPX, but may have a duty cycle other than 50% due to signal slope and trip levels effects.

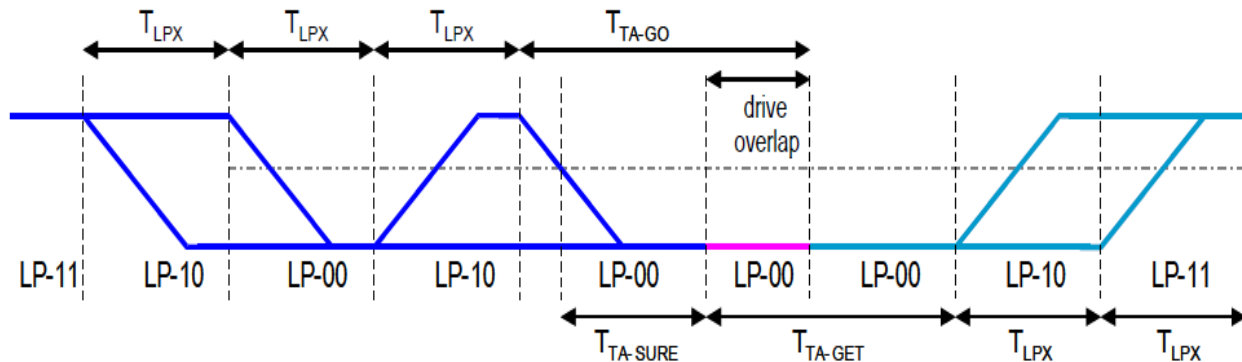
| Start Code | Line Voltage Levels |         | High-Speed     | Low-Power    |             |
|------------|---------------------|---------|----------------|--------------|-------------|
|            | Dp-Line             | Dn-Line | Burst Mode     | Control Mode | Escape Mode |
| HS-0       | HS Low              | HS High | Differential-0 | N/A          | N/A         |
| HS-1       | HS High             | HS Low  | Differential-1 | N/A          | N/A         |
| LP-00      | LP Low              | LP Low  | N/A            | Bridge       | Space       |
| LP-01      | LP Low              | LP High | N/A            | HS-Rqst      | Mark-0      |
| LP-10      | LP High             | LP Low  | N/A            | LP-Rqst      | Mark-1      |
| LP-11      | LP High             | LP High | N/A            | Stop         | N/A         |

**Table 8.3: Lane State Description**

### 8.15.5 Bi-directional Data Lane Turnaround

The transmission direction of a bi-directional Data Lane can be swapped by means of a Link Turnaround procedure. This procedure enables information transfer in the opposite direction of the current direction. The procedure is the same for either a change from Forward-to-Reverse direction or Reverse-to-Forward direction. Notice that Master and Slave side shall not be changed by Turnaround.

Figure 8.21 shows the Turnaround procedure graphically.



**Figure 8.21: Turnaround Procedure**

### 8.15.6 Escape Mode

Escape mode is a special mode of operation for Data Lanes using Low-Power states. With this mode some additional functionality becomes available. A Data Lane shall enter Escape mode via an Escape mode Entry procedure (LP-11, LP-10, LP-00, LP-01, LP-00). As soon as the final Bridge state (LP-00) is observed on the Lines the Lane shall enter Escape mode in Space state (LP-00). If an LP-11 is detected at any time before the final Bridge state (LP-00), the Escape mode Entry procedure shall be aborted and the receive side shall wait for, or return to, the Stop state.

For Data Lanes, once Escape mode is entered, the transmitter shall send an 8-bit entry command, by Spaced-One-Hot coding, to indicate the requested action. Table 8.4 lists all supported Escape mode commands and actions.

Spaced-One-Hot coding means that each Mark state is interleaved with a Space state. Each symbol consists therefore of two parts: a One-Hot phase (Mark-0 or Mark-1) and a Space phase. The TX shall send Mark-0 followed by a Space to transmit a 'zero-bit' and it shall send a Mark-1 followed by a Space to transmit a 'one-bit'. A Mark that is not followed by a Space does not represent a bit. The last phase before exiting Escape mode with a Stop state shall be a Mark-1 state that is not part of the communicated bits, as it is not followed by a Space state.

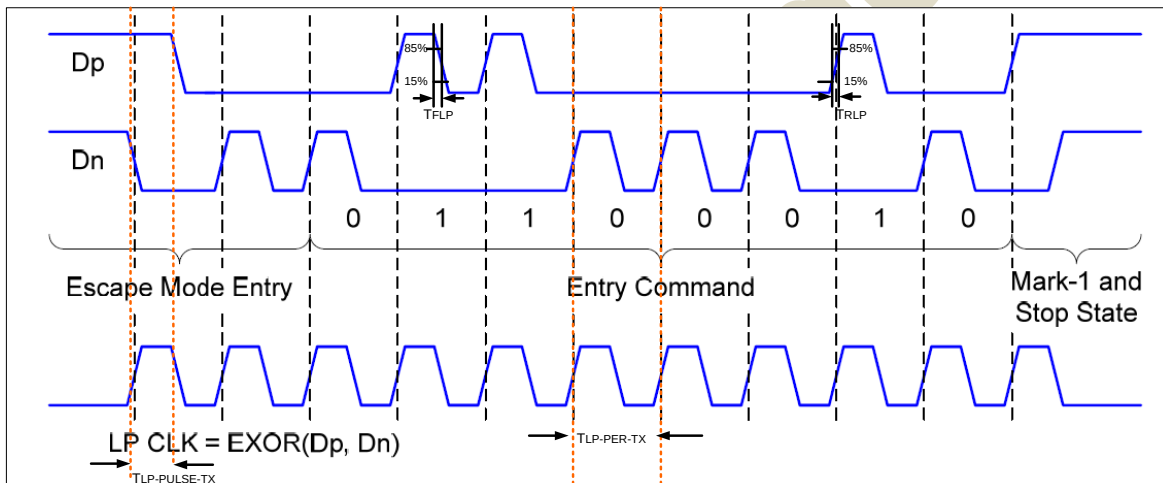


Figure 8.22: Trigger-Reset Command in Escape Mode

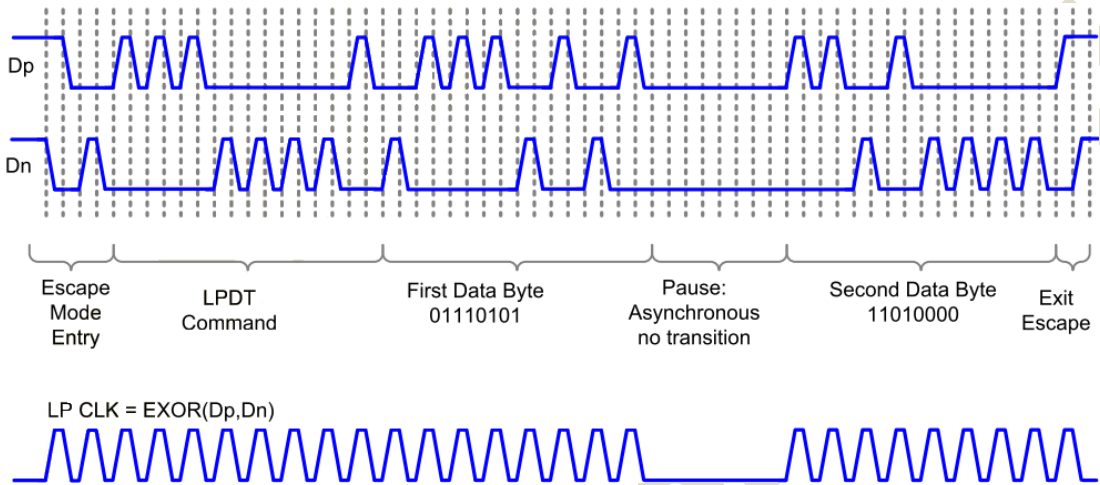
| Escape Mode Action          | Command Type | Entry Command Pattern<br>(first bit transmitted to<br>last bit transmitted) |
|-----------------------------|--------------|-----------------------------------------------------------------------------|
| Low-Power Data Transmission | mode         | 11100001                                                                    |
| Reset-Trigger               | Trigger      | 01100010                                                                    |
| Acknowledge                 | Trigger      | 00100001                                                                    |

Table 8.4: Escape Entry Codes



### 8.15.7 Low-Power Data Transmission(LPDT)

If the Escape mode Entry procedure is followed-up by the Entry Command for Low-Power Data Transmission (LPDT), Data can be communicated by the protocol at low speed, while the Lane remains in Low-Power mode. Data shall be encoded on the lines with the same Spaced-One-Hot code as used for the Entry Commands. The data is self-clocked by the applied bit encoding and does not rely on the Clock Lane. The Lane can pause while using LPDT by maintaining a Space state on the Lines. A Stop state on the Lines stops LPDT, exits Escape mode, and switches the Lane to Control mode. The last phase before Stop state shall be a Mark-1 state, which does not represent a data-bit. At the end of LPDT the Lane shall return to the Stop state.



**Figure 8.23: Two Data Byte Low-Power Data Transmission Example**

## 8.16 High Speed Transmission

### 8.16.1 Burst Payload Data

The payload data of a burst shall always represent an integer number of payload data bytes with a minimum length of one byte. Note that for short bursts the Start and End overhead consumes much more time than the actual transfer of the payload data. There is no maximum number of bytes implied by the PHY. However, in the PHY there is no autonomous way of error recovery during a HS data burst and the practical BER will not be zero. Therefore, it is important to consider for every individual protocol what the best choice is for maximum burst length.

### 8.16.2 Start-of-Transmission

After a Transmit request, a Data Lane leaves the Stop state and prepares for High-Speed mode by means of a Start-of-Transmission (SoT) procedure. Table 8.5 describes the sequence of events on TX and RX side.

| TX side                                                                 | RX side                                                                                                  |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Drives Stop state (LP-11)                                               | Observes Stop state                                                                                      |
| Drives HS-Rqst state (LP-01) for time $T_{LPX}$                         | Observes transition from LP-11 to LP-01 on the Line                                                      |
| Drives Bridge state (LP-00) for time $T_{HS-PREPARE}$                   | Observes transition from LP-01 to LP-00 on the Line, enables Line Termination after time $T_{D-TERM-EN}$ |
| Enables High-Speed driver and disables Low-Powerdrivers simultaneously  |                                                                                                          |
| Drives HS-0 for a time $T_{HS-ZERO}$                                    | Enables HS-RX and waits for timer $T_{HS-SETTLE}$ to expire in order to neglect transition effects       |
|                                                                         | Starts looking for Leader-Sequence                                                                       |
| Insert the HS Sync-Sequence '00011101' beginning on a rising Clock edge |                                                                                                          |
|                                                                         | Synchronizes upon recognition of Leader Sequence '011101'                                                |
| Continues to Transmit High-Speed payload data                           |                                                                                                          |
|                                                                         | Receives payload data                                                                                    |

Table 8.5: Start-of-Transmission Sequence

**8.16.3 End-of-Transmission**

At the end of a Data Burst, a Data Lane Levels High-Speed Transmission mode and enters the Stop state by means of an End-of-Transmission (EoT) procedure. Table 8.6 shows a possible sequence of events during the EoT procedure. Note, EoT processing may be handled by the protocol or by the D-PHY.

| TX side                                                                                                           | RX side                                                                                        |
|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Completes Transmission of payload data                                                                            | Receives payload data                                                                          |
| Toggles differential state immediately after last payload data bit and keeps that state for a time $T_{HS-TRAIL}$ |                                                                                                |
| Disables the HS-TX, enables the LP-TX, and drives Stop state (LP-11) for a time $T_{HS-EXIT}$                     | Detects the Lines leaving LP-00 state and entering Stop state (LP-11) and disables Termination |
|                                                                                                                   | Neglect bits of last period $T_{HS-SKIP}$ to hide transition effects                           |
|                                                                                                                   | Detect last transition in valid Data, determine last valid Data byte and skip trailer sequence |

**Table 8.6: End-of-Transmission Sequence**

### Figure 8.24: High-Speed Data Transmission in Bursts

**Figure 8.25: Switching the Clock Lane between Clock Transmission and Low-Power Mode**

# 8.17 System Power state

Each Lane within a PHY configuration, that is powered and enabled, has potentially three different power consumption levels: High-Speed Transmission mode, Low-Power mode and Ultra-Low-Power State.

## 8.17.1 Initialization

After power-up, the Slave side PHY shall be initialized when the Master PHY drives a Stop State (LP-11) for a period longer than  $T_{INIT}$ . The first Stop state longer than the specified  $T_{INIT}$  is called the Initialization period. The Master side shall ensure that a Stop State longer than  $T_{INIT}$  does not occur on the Lines before the Master is initialized.

## 8.17.2 Global Operation Flow Diagram

Figure 8.26 shows the operational flow diagram for a Data Lane Module. Within both TX and RX four main processes can be distinguished: High-Speed Transmission, Escape mode, Turnaround, and Initialization.

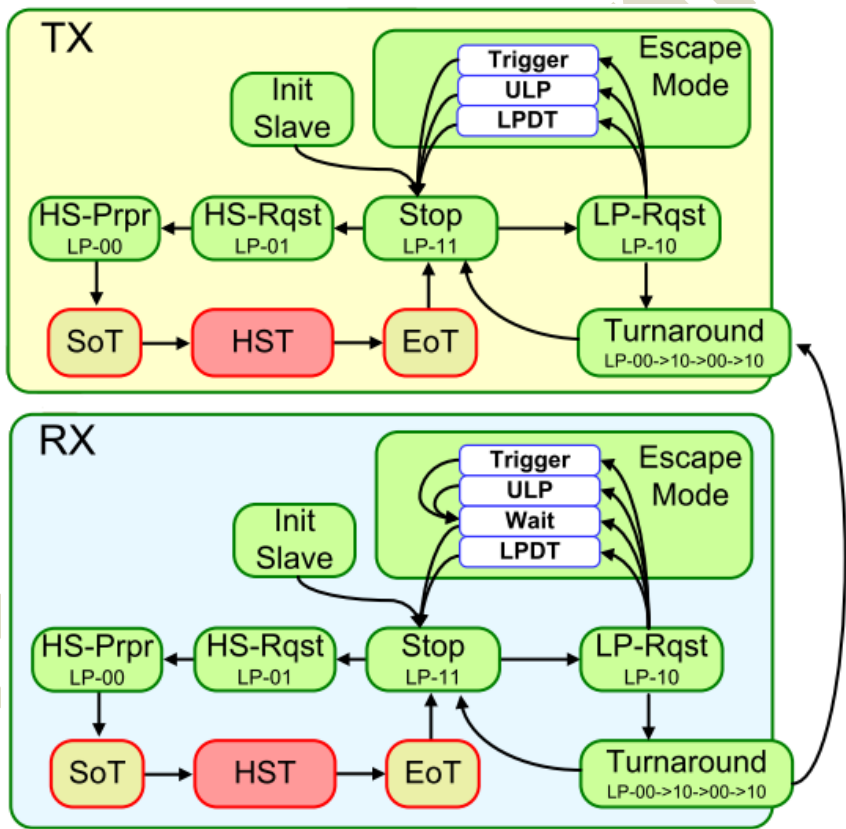


Figure 8.26: Data Lane Module State Diagram

Figure 8.27 shows the state diagram for a Clock Lane Module. The Clock Lane Module has four major operational states: Init (of unspecified duration), Low-Power Stop state, Ultra-Low Power state, and High-Speed clock transmission.

SLAVE

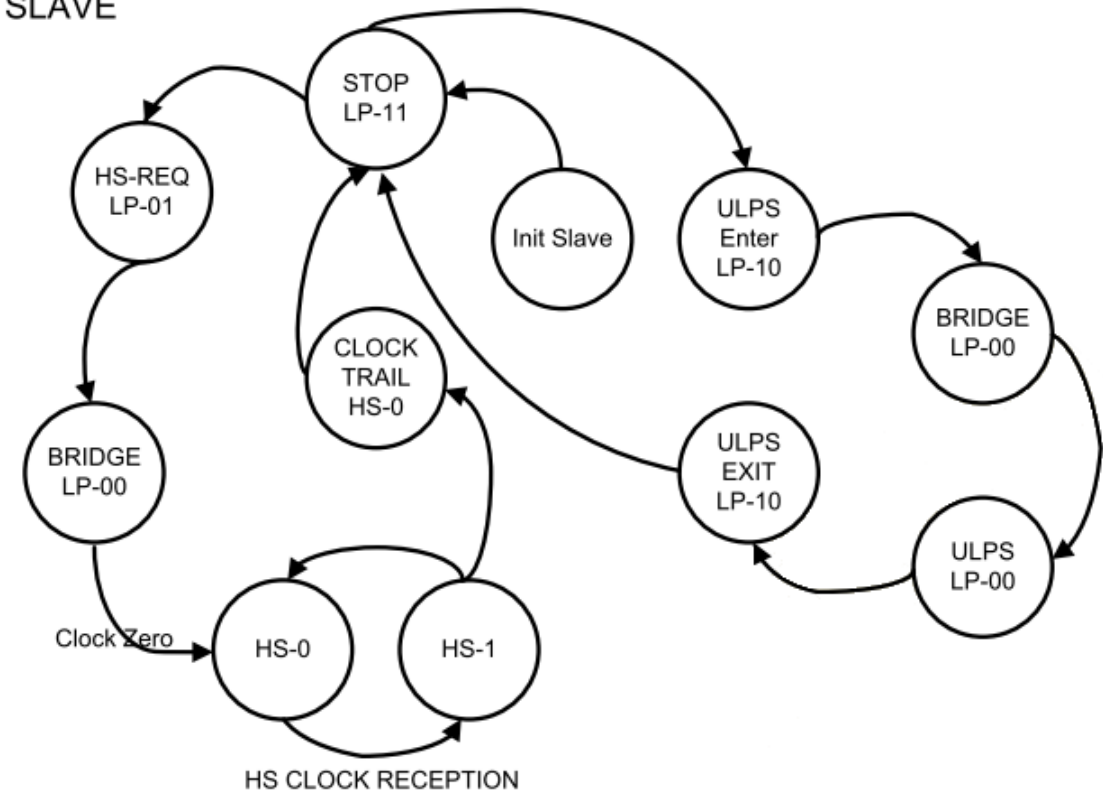


Figure 8.27: Clock Lane Module State Diagram

## 8.18 Command Description

| Command   | R/W | Hex | Description                       | Number of Parameters |
|-----------|-----|-----|-----------------------------------|----------------------|
| NOP       | W   | 00h | No operation.                     | 0                    |
| SWRESET   | W   | 01h | Software reset.                   | 0                    |
| RDNUMPE   | R   | 05h | Read number of the parity errors. | 1                    |
| REDRD     | R   | 06h | Read red color.                   | 1                    |
| REDGREEN  | R   | 07h | Read green color.                 | 1                    |
| REDBLUE   | R   | 08h | Read blue color.                  | 1                    |
| RDDST     | R   | 09h | Read display status.              | 1                    |
| RDDPM     | R   | 0Ah | Read display power mode.          | 1                    |
| RDDMATCDL | R   | 0Bh | Read display MADCTL.              | 1                    |
| RDDIM     | R   | 0Dh | Read display image mode.          | 1                    |
| SLPIN     | W   | 10h | Sleep In.                         | 0                    |
| SLPOUT    | W   | 11h | Sleep out.                        | 0                    |
| INVOFF    | W   | 20h | Display inversion off.            | 0                    |
| INVON     | W   | 21h | Display inversion on.             | 0                    |
| DISPOFF   | W   | 28h | Display off.                      | 0                    |
| DISPON    | W   | 29h | Display on.                       | 0                    |
| MADCTL    | W   | 36h | Memory access control.            | 1                    |
| IDMOFF    | W   | 38h | Idle mode off.                    | 0                    |
| IDMON     | W   | 39h | Idle mode on.                     | 0                    |
| WRIMC     | W   | 80h | Color selection.                  | 1                    |
| RDRIMC    | R   | 81h | Read color selection.             | 1                    |

**8.18.1 NOP (00h)**

| 00H         | NOP (No Operation)                                                         |    |    |    |    |    |    |    |    |     |
|-------------|----------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                        | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                          | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 00  |
| Parameter   | No Parameter                                                               |    |    |    |    |    |    |    |    |     |
| Description | - This command is empty command. It does have effect on the display module |    |    |    |    |    |    |    |    |     |
| Restriction | -                                                                          |    |    |    |    |    |    |    |    |     |
| Flow Chart  | -                                                                          |    |    |    |    |    |    |    |    |     |



8.18.2 SWRESET (01h)

| 01H         | SWRESET                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                          | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                                                                                                                                                                                                                                                                                                            | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 01  |
| Parameter   | No Parameter                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |    |    |    |    |    |     |
| Description | - When the Software Reset command is written, it causes a software reset. It resets the commands and parameters to their S/W Reset default values.                                                                                                                                                                                                                                                                           |    |    |    |    |    |    |    |    |     |
| Restriction | - It will be necessary to wait 5msec before sending new command following software reset.<br>- The display module loads all display supplier's factory default values to the registers during 5msec.<br>- If Software Reset is applied during Sleep Out mode, it will be necessary to wait 120msec before sending Sleep Out command.<br>- Software Reset should not be sent when the display module is not in Sleep In mode. |    |    |    |    |    |    |    |    |     |
| Flow Chart  | <div><div>soft_reset</div><div></div><div>Blank Display</div><div></div><div>Load S/W Defaults</div><div></div><div>SLPIN mode</div></div>                                                                                                                                                                                                                                                                                   |    |    |    |    |    |    |    |    |     |

## 8.18.3 RDNUMPE (05h)

| 05H         | RDNUMPE                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |    |    |    |     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                           | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 1  | 05  |
| Parameter 1 | R                                                                                                                                                                                                                                                                                                                                                                                                                                             | P7 | P6 | P5 | P4 | P3 | P2 | P1 | P0 |     |
| Description | <p>- The first parameter is telling a number of the errors on DSI. The more detailed description of the bits is below.</p> <p>- P[6:0] bits are telling a number of the errors.</p> <p>- P[7] is set to '1' if there is overflow with P[6..0] bits.</p> <p>- P[7:0] bits are set to '0's (as well as RDDSM(0Eh)'s D0 is set '0' at the same time) after there is sent the second parameter information (=The read function is completed).</p> |    |    |    |    |    |    |    |    |     |
| Restriction | -                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |    |    |    |    |    |    |    |     |
| Flow Chart  | <p style="text-align: center;">DSI I/F Mode</p> <pre> graph TD     A[RDNUMPE 05h] --&gt; B[/Send 1st parameter/]     B --&gt; C([RDDSM0Eh's D0 is set '0'<br/>P7:0='00'h])   </pre> <p style="text-align: right;">Host<br/>-----<br/>Driver</p>                                                                                                                                                                                               |    |    |    |    |    |    |    |    |     |

8.18.4 REDRD (06h)

| 06H         | REDRD                                                                                                                                                                                                                                                                                                                                                             |        |    |    |    |    |    |    |    |    |     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                                                                                                                                                                                                                                                                               | D15-D8 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                                                                                                                                                                                                                                                 | -      | 0  | 0  | 0  | 0  | 0  | 1  | 1  | 0  | 06  |
| Parameter 1 | R                                                                                                                                                                                                                                                                                                                                                                 | -      | R7 | R6 | R5 | R4 | R3 | R2 | R1 | R0 |     |
| Description | <div>- The first parameter is telling red color value of the first pixel of the frame when there is used DPI I/F.</div> <div>- 16 bit format: R5 is MSB and R1 is LSB. R7, R6 and R0 are set to '0'.</div> <div>- 18 bit format: R5 is MSB and R0 is LSB. R7 and R6 are set to '0'.</div> <div>- 24 bit format: R7 is MSB and R0 is LSB. All bits are used.</div> |        |    |    |    |    |    |    |    |    |     |
| Restriction | -                                                                                                                                                                                                                                                                                                                                                                 |        |    |    |    |    |    |    |    |    |     |
| Flow Chart  | <div><div>get_red_channel (06h)</div><div><div></div><div>Send D[7:0]</div></div><div>Host<br/>Driver</div></div>                                                                                                                                                                                                                                                 |        |    |    |    |    |    |    |    |    |     |

8.18.5 REDGREEN (07h)

| 07H         | REDGREEN                                                                                                                                                                                                                                                                                                                                                        |    |    |    |    |    |    |    |    |     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                                                                                                                                                                                                                                                                             | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                                                                                                                                                                                                                                               | 0  | 0  | 0  | 0  | 0  | 1  | 1  | 1  | 07  |
| Parameter 1 | R                                                                                                                                                                                                                                                                                                                                                               | G7 | G6 | G5 | G4 | G3 | G2 | G1 | G0 |     |
| Description | <div>- The first parameter is telling green color value of the first pixel of the frame when there is used DPI I/F.</div> <div>- 16 bit format: G5 is MSB and G0 is LSB. G7 and G6 are set to '0'.</div> <div>- 18 bit format: G5 is MSB and G0 is LSB. G7 and G6 are set to '0'.</div> <div>- 24 bit format: G7 is MSB and G0 is LSB. All bits are used.</div> |    |    |    |    |    |    |    |    |     |
| Restriction | -                                                                                                                                                                                                                                                                                                                                                               |    |    |    |    |    |    |    |    |     |
| Flow Chart  | <div><div>get_green_channel (07h)</div><div><div></div><div>Send D[7:0]</div></div><div><div>Host</div><div>Driver</div></div></div>                                                                                                                                                                                                                            |    |    |    |    |    |    |    |    |     |

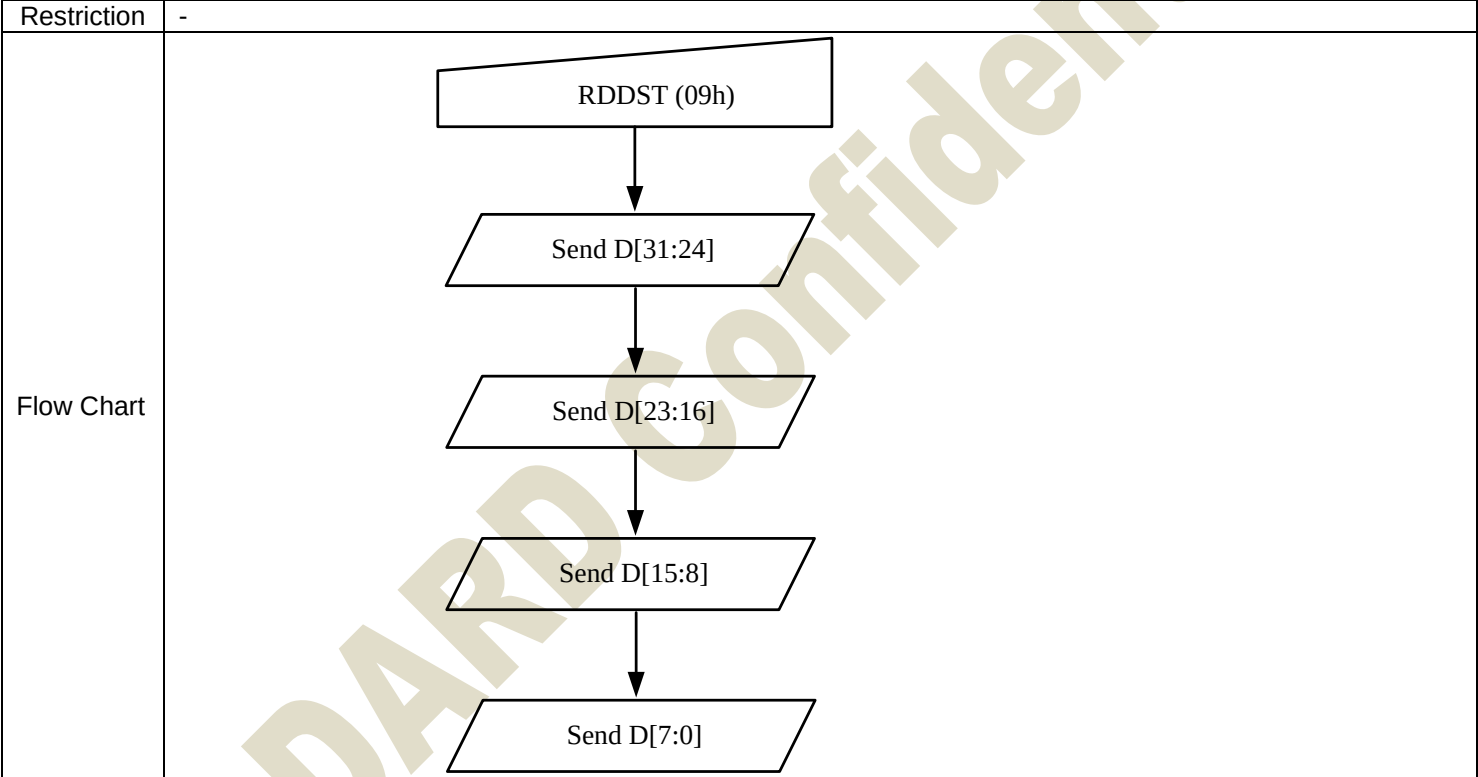
8.18.6 REDBLUE (08h)

| 08H         | REDBLUE                                                                                                                                                                                                                                                                                                                                                            |    |    |    |    |    |    |    |    |     |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                                                                                                                                                                                                                                                                                | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                                                                                                                                                                                                                                                  | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 08  |
| Parameter 1 | R                                                                                                                                                                                                                                                                                                                                                                  | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |     |
| Description | <div>- The first parameter is telling blue color value of the first pixel of the frame when there is used DPI I/F.</div> <div>- 16 bit format: B5 is MSB and B1 is LSB. B7, B6 and B0 are set to '0'.</div> <div>- 18 bit format: B5 is MSB and B0 is LSB. B7 and B6 are set to '0'.</div> <div>- 24 bit format: B7 is MSB and B0 is LSB. All bits are used.</div> |    |    |    |    |    |    |    |    |     |
| Restriction | -                                                                                                                                                                                                                                                                                                                                                                  |    |    |    |    |    |    |    |    |     |
| Flow Chart  | <div><div>get_blue_channel (08h)</div><div><div></div><div>Send D[7:0]</div></div><div>Host<br/>Driver</div></div>                                                                                                                                                                                                                                                 |    |    |    |    |    |    |    |    |     |

## 8.18.7 RDDST (09h)

| 09H         | RDDST                                                                                      |                        |    |    |    |                                                                                                        |    |    |    |     |
|-------------|--------------------------------------------------------------------------------------------|------------------------|----|----|----|--------------------------------------------------------------------------------------------------------|----|----|----|-----|
|             | R/W                                                                                        | D7                     | D6 | D5 | D4 | D3                                                                                                     | D2 | D1 | D0 | HEX |
| Command     | W                                                                                          | 0                      | 0  | 0  | 0  | 1                                                                                                      | 0  | 0  | 1  | 09  |
| Parameter 1 | R                                                                                          | D[31:24]               |    |    |    |                                                                                                        |    |    |    |     |
| Parameter 2 | R                                                                                          | D[23:16]               |    |    |    |                                                                                                        |    |    |    |     |
| Parameter 3 | R                                                                                          | D[15:8]                |    |    |    |                                                                                                        |    |    |    |     |
| Parameter 4 | R                                                                                          | D[7:0]                 |    |    |    |                                                                                                        |    |    |    |     |
| Description | -This command indicates the current status of the display as described in the table below: |                        |    |    |    |                                                                                                        |    |    |    |     |
|             | Bit                                                                                        | Description            |    |    |    | Value                                                                                                  |    |    |    |     |
|             | D31                                                                                        | Booster voltage status |    |    |    | '0' = booster off<br>'1' = booster on                                                                  |    |    |    |     |
|             | D30                                                                                        | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D29                                                                                        | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D28                                                                                        | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D27                                                                                        | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D26                                                                                        | RGB/BGR order          |    |    |    | '0' = RGB (MADCTL B4='0')<br>'1' = BGR (MADCTL B4='1')                                                 |    |    |    |     |
|             | D25                                                                                        | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D24                                                                                        | Source scan sequence   |    |    |    | '0' = Source output Left to Right (MADCTL B1='0')<br>'1' = Source output Right to Left (MADCTL B1='1') |    |    |    |     |
|             | D23                                                                                        | Gate scan sequence     |    |    |    | '0' = Gate output Top to Bottom (MADCTL B0='0')<br>'1' = Gate output Bottom to Top (MADCTL B0='1')     |    |    |    |     |
|             | D22                                                                                        | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D21                                                                                        | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D20                                                                                        | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D19                                                                                        | Idle mode on/off       |    |    |    | '0' = Idle mode off.<br>'1' = Idle mode on.                                                            |    |    |    |     |
|             | D18                                                                                        | Partial mode on/off    |    |    |    | '0' = Partial mode off.<br>'1' = Partial mode on.                                                      |    |    |    |     |
|             | D17                                                                                        | Sleep in/out           |    |    |    | '0' = Sleep in mode.<br>'1' = Sleep out mode.                                                          |    |    |    |     |
|             | D16                                                                                        | Not defined            |    |    |    | Set to '1'                                                                                             |    |    |    |     |
|             | D13                                                                                        | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D12                                                                                        | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D11                                                                                        | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D10                                                                                        | Display on/off         |    |    |    | '0' = Display is off.<br>'1' = Display is on.                                                          |    |    |    |     |
|             | D9                                                                                         | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D8                                                                                         | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D7                                                                                         | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |
|             | D6                                                                                         | Not defined            |    |    |    | Set to '0'                                                                                             |    |    |    |     |

|             |    |             |            |
|-------------|----|-------------|------------|
|             | D5 | Not defined | Set to '0' |
|             | D4 | Not defined | Set to '0' |
|             | D3 | Not defined | Set to '0' |
|             | D2 | Not defined | Set to '0' |
|             | D1 | Not defined | Set to '0' |
|             | D0 | Not defined | Set to '0' |
|             |    |             |            |
|             |    |             |            |
|             |    |             |            |
|             |    |             |            |
|             |    |             |            |
|             |    |             |            |
| Restriction | -  |             |            |



8.18.8 RDDPM (0Ah)

| 0AH         | RDDPM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                |                                             |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------|----|----|----|----|----|----|-----|-----|-------------|-------|----|-------------|-------------------|----|-------------|------------|----|-------------|------------|----|--------------|---------------------------------------------|----|-------------|------------|----|----------------|---------------------------------------------|----|-------------|------------|----|------------|---------------------------------------------|
|             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | D7             | D6                                          | D5 | D4 | D3 | D2 | D1 | D0 | HEX |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
| Command     | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0              | 0                                           | 0  | 0  | 1  | 0  | 1  | 0  | 0A  |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
| Parameter 1 | R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | D7             | D6                                          | D5 | D4 | D3 | D2 | 0  | 0  |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
| Description | <p>-This command indicates the current status of the display as described in the table below:</p> <table><tr><th>Bit</th><th>Description</th><th>Value</th></tr><tr><td>D7</td><td>Not Defined</td><td>Set to '0' or '1'</td></tr><tr><td>D6</td><td>Not Defined</td><td>Set to '0'</td></tr><tr><td>D5</td><td>Not Defined</td><td>Set to '0'</td></tr><tr><td>D4</td><td>Sleep in/out</td><td>'0' = sleep in mode<br/>'1' = sleep out mode</td></tr><tr><td>D3</td><td>Not Defined</td><td>Set to '1'</td></tr><tr><td>D2</td><td>Display on/off</td><td>'0' = display is off<br/>'1' = display is on</td></tr><tr><td>D1</td><td>Not defined</td><td>Set to '0'</td></tr><tr><td>D0</td><td>ESD detect</td><td>'0' = ESD detect off<br/>'1' = ESD detect on</td></tr></table> |                |                                             |    |    |    |    |    |    |     | Bit | Description | Value | D7 | Not Defined | Set to '0' or '1' | D6 | Not Defined | Set to '0' | D5 | Not Defined | Set to '0' | D4 | Sleep in/out | '0' = sleep in mode<br>'1' = sleep out mode | D3 | Not Defined | Set to '1' | D2 | Display on/off | '0' = display is off<br>'1' = display is on | D1 | Not defined | Set to '0' | D0 | ESD detect | '0' = ESD detect off<br>'1' = ESD detect on |
|             | Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Description    | Value                                       |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
|             | D7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Not Defined    | Set to '0' or '1'                           |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
|             | D6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Not Defined    | Set to '0'                                  |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
|             | D5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Not Defined    | Set to '0'                                  |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
|             | D4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Sleep in/out   | '0' = sleep in mode<br>'1' = sleep out mode |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
|             | D3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Not Defined    | Set to '1'                                  |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
|             | D2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Display on/off | '0' = display is off<br>'1' = display is on |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
|             | D1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Not defined    | Set to '0'                                  |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
|             | D0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ESD detect     | '0' = ESD detect off<br>'1' = ESD detect on |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
| Restriction | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |                                             |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |
| Flow Chart  | <div><div>get_power_mode (0Ah)</div><div>↓</div><div>Send D[7:0]</div></div> <div><div>Host</div><div>-----</div><div>Driver</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |                                             |    |    |    |    |    |    |     |     |             |       |    |             |                   |    |             |            |    |             |            |    |              |                                             |    |             |            |    |                |                                             |    |             |            |    |            |                                             |



8.18.9 RDDMATCDL (0Bh)

| 0BH         | RDDMATCDL                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                      |    |    |    |    |    |    |    |     |     |             |       |    |                      |                                                                                      |    |                    |                                                                                      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----|----|----|----|----|----|----|-----|-----|-------------|-------|----|----------------------|--------------------------------------------------------------------------------------|----|--------------------|--------------------------------------------------------------------------------------|
|             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | D7                                                                                   | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |     |             |       |    |                      |                                                                                      |    |                    |                                                                                      |
| Command     | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0                                                                                    | 0  | 0  | 0  | 1  | 0  | 1  | 1  | 0B  |     |             |       |    |                      |                                                                                      |    |                    |                                                                                      |
| Parameter 1 | R                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0                                                                                    | 0  | 0  | 0  | 0  | 0  | D1 | D0 |     |     |             |       |    |                      |                                                                                      |    |                    |                                                                                      |
| Description | <div>-This command indicates the current status of the display as described in the table below:</div> <table><tr><th>Bit</th><th>Description</th><th>Value</th></tr><tr><td>D1</td><td>Source scan sequence</td><td>'0' = left to right (when MADCTL B1='0')<br/>'1' = right to left (when MADCTL B1='1')</td></tr><tr><td>D0</td><td>Gate scan sequence</td><td>'0' = top to bottom (when MADCTL B0='0')<br/>'1' = bottom to top (when MADCTL B0='1')</td></tr></table> |                                                                                      |    |    |    |    |    |    |    |     | Bit | Description | Value | D1 | Source scan sequence | '0' = left to right (when MADCTL B1='0')<br>'1' = right to left (when MADCTL B1='1') | D0 | Gate scan sequence | '0' = top to bottom (when MADCTL B0='0')<br>'1' = bottom to top (when MADCTL B0='1') |
| Bit         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Value                                                                                |    |    |    |    |    |    |    |     |     |             |       |    |                      |                                                                                      |    |                    |                                                                                      |
| D1          | Source scan sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                     | '0' = left to right (when MADCTL B1='0')<br>'1' = right to left (when MADCTL B1='1') |    |    |    |    |    |    |    |     |     |             |       |    |                      |                                                                                      |    |                    |                                                                                      |
| D0          | Gate scan sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                       | '0' = top to bottom (when MADCTL B0='0')<br>'1' = bottom to top (when MADCTL B0='1') |    |    |    |    |    |    |    |     |     |             |       |    |                      |                                                                                      |    |                    |                                                                                      |
| Restriction | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                      |    |    |    |    |    |    |    |     |     |             |       |    |                      |                                                                                      |    |                    |                                                                                      |
| Flow Chart  | <div><div>get_address_mode (0Bh)</div><div><div></div><div></div></div><div>Send D[7:0]</div><div>Host<br/>Driver</div></div>                                                                                                                                                                                                                                                                                                                                            |                                                                                      |    |    |    |    |    |    |    |     |     |             |       |    |                      |                                                                                      |    |                    |                                                                                      |

8.18.10 RDDIM (0Dh)

| 0DH         | RDDIM                                                                                                                                           |                             |    |    |                                                            |    |    |    |    |     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----|----|------------------------------------------------------------|----|----|----|----|-----|
|             | R/W                                                                                                                                             | D7                          | D6 | D5 | D4                                                         | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                               | 0                           | 0  | 0  | 0                                                          | 1  | 1  | 0  | 1  | 0D  |
| Parameter 1 | R                                                                                                                                               | 0                           | D6 | D5 | 0                                                          | 0  | 0  | 0  | 0  |     |
| Description | -This is command indicates the current status of the display as described in the table below.                                                   |                             |    |    |                                                            |    |    |    |    |     |
|             | Bit                                                                                                                                             | Description                 |    |    | Value                                                      |    |    |    |    |     |
|             | D6                                                                                                                                              | Horizontal Scrolling Status |    |    | This bit is not applicable for this project, set it to '0' |    |    |    |    |     |
|             | D5                                                                                                                                              | Inversion On/Off            |    |    | '0' = Inversion is Off.<br>'1' = Inversion is On.          |    |    |    |    |     |
| Restriction | -                                                                                                                                               |                             |    |    |                                                            |    |    |    |    |     |
| Flow Chart  | <div><div>get_display_mode (0Dh)</div><div><div></div><div></div></div><div>Send D[7:0]</div></div> <div><div>Host</div><div>Driver</div></div> |                             |    |    |                                                            |    |    |    |    |     |

8.18.11 SLPIN (10h)

| 10H         | SLPIN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |    |    |    |    |    |    |    |     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 10  |
| Description | <p>- This command causes the LCD module to enter the minimum power consumption mode. In this mode, all unnecessary blocks inside the display module are disabled except interface communication. This is the lowest power mode the display module supports.</p> <p>DBI or DSI Command Mode remains operational and the line buffer maintains its contents. The host processor continues to send PCLK, HS and VS information to DPI IF for two frames after this command is sent when the display module is in Normal mode.</p> <p>In this mode the DC/DC converter is stopped, Internal oscillator is stopped, and panel scanning is stopped.</p> |    |    |    |    |    |    |    |    |     |
| Restriction | <p>- This command has no effect when module is already in sleep in mode. Sleep In Mode can only be left by the Sleep Out Command (11h). It will be necessary to wait 5msec before sending next command; this is to allow time for the supply voltages and clock circuits to stabilize. It will be necessary to wait 120msec after sending Sleep Out command (when in Sleep In Mode) before Sleep In command can be sent.</p>                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |
| Flow Chart  | <p>It takes 120msec to get into Sleep In mode after SLPIN command issued.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |    |    |    |    |    |    |    |     |

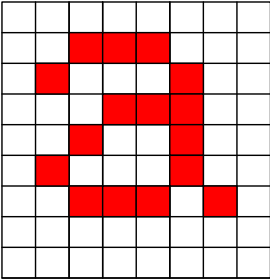
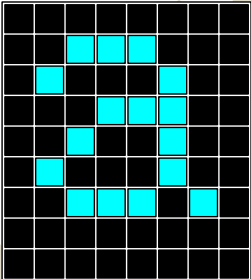
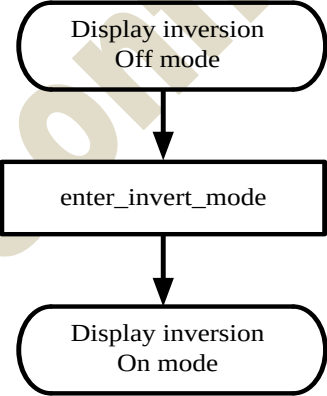
8.18.12 SLPOUT (11h)

| 11H         | SLPOUT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |    |    |    |    |    |    |     |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 11  |
| Description | <p>This command turns off sleep mode. In this mode the DC/DC converter is enabled, Internal oscillator is started, and panel scanning is started</p> <p>User can start to send PCLK, HS and VS information on DPI IF before Sleep Out command and this information is valid at least 1 frames before Sleep Out command, if there is left Sleep In -mode to Sleep Out-mode in Normal Mode On. There is used an internal oscillator for blank display.</p>                                                                                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |    |    |    |     |
| Restriction | <p>This command has no effect when module is already in sleep out mode. Sleep Out Mode can only be left by the Sleep In Command (10h).<br/>It will be necessary to wait 5msec before sending next command, this is to allow time for the supply voltages and clock circuits to stabilize.<br/>The display module loads all display supplier's factory default values to the registers during this 5msec and there cannot be any abnormal visual effect on the display image if factory default and register values are same when this load is done and when the display module is already Sleep Out –mode.<br/>The display module is doing self-diagnostic functions during this 5msec. It will be necessary to wait 120msec after sending Sleep In command (when in Sleep Out mode) before Sleep Out command can be sent.</p> |    |    |    |    |    |    |    |    |     |
| Flow Chart  | <p>It takes 120msec to become Sleep Out mode after SLPOUT command issued.</p> <pre> graph TD     A[exit_sleep_mode] --&gt; B{{Start internal Oscillator}}     B --&gt; C{{Start up DC/DC converter}}     C --&gt; D{{Charge Offset voltage For LCD panel}}     D --&gt; E{{Display whole blank screen for 2 frames<br/>(Automatic No effect to DISP ON/OFF Commands)}}     E --&gt; F{{Display Memory contents<br/>In accordance with the current command table setting}}     F --&gt; G([Sleep out mode])     </pre>                                                                                                                                                                                                                                                                                                          |    |    |    |    |    |    |    |    |     |

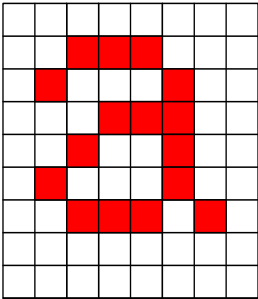
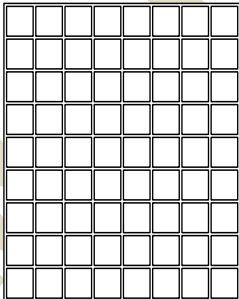
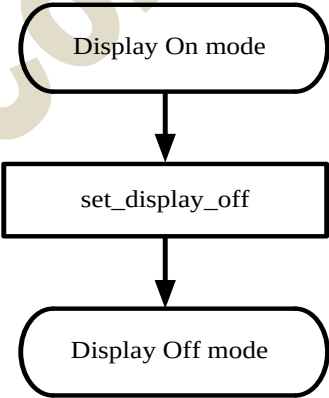
8.18.13 INVOFF (20h)

| 20H         | INVOFF                                                                                                                                                                              |    |    |    |    |    |    |    |    |     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                                                                                                 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                                                                   | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 0  | 20  |
| Description | <p>This command is used to recover from display inversion mode.</p> <p>This command makes no change of contents of image.</p> <p>This command does not change any other status.</p> |    |    |    |    |    |    |    |    |     |
|             | <p>(Example)</p> <div><div>Image</div></div> <div><div>Display Panel</div></div>                                                                                                    |    |    |    |    |    |    |    |    |     |
| Restriction | This command has no effect when module is already in inversion off mode.                                                                                                            |    |    |    |    |    |    |    |    |     |
| Flow Chart  | <pre>graph TD; A([Display inversion On mode]) --&gt; B[exit_invert_mode]; B --&gt; C([Display inversion Off mode]);</pre>                                                           |    |    |    |    |    |    |    |    |     |

8.18.14 INVON (21h)

| 21H         | INVON                                                                                                                                                                                                                                    |    |    |    |    |    |    |    |    |     |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                                                                                                                                                      | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                                                                                                                        | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 1  | 21  |
| Description | <p>This command is used to enter into display inversion mode.</p> <p>This command makes change of contents of image. Every bit is inverted from the image to the display.</p> <p>This command does not change any other status.</p>      |    |    |    |    |    |    |    |    |     |
|             | <p>(Example)</p> <div><div>Image</div><div>Display Panel</div></div> |    |    |    |    |    |    |    |    |     |
| Restriction | This command has no effect when module is already in inversion on mode.                                                                                                                                                                  |    |    |    |    |    |    |    |    |     |
| Flow Chart  |                                                                                                                                                       |    |    |    |    |    |    |    |    |     |

8.18.15 DISPOFF (28h)

| 28H         | DISPOFF                                                                                                                                                                                                                                                                                                                          |    |    |    |    |    |    |    |    |     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                                                                                                                                                                                                                                              | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                                                                                                                                                                                                                | 0  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | 28  |
| Description | <p>This command is used to enter into DISPLAY OFF mode. In this mode, the output from Line Buffer is disabled and blank page inserted.</p> <p>This command makes no change of contents of line buffer.</p> <p>This command does not change any other status.</p> <p>There will be no abnormal visible effect on the display.</p> |    |    |    |    |    |    |    |    |     |
|             | <p>(Example)</p> <div><div>Memory</div><div>Display Panel</div></div>                                                                                        |    |    |    |    |    |    |    |    |     |
| Restriction | It will be necessary to wait 40msec and send Sleep In command for power off sequence.                                                                                                                                                                                                                                            |    |    |    |    |    |    |    |    |     |
| Flow Chart  |                                                                                                                                                                                                                                              |    |    |    |    |    |    |    |    |     |

8.18.16 DISPON (29h)

| 29H         | DISPON                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                                                                                                                                         | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                                                                                                                                           | 0  | 0  | 1  | 0  | 1  | 0  | 0  | 1  | 29  |
| Description | <p>This command is used to recover from DISPLAY OFF mode. Output from the Line Buffer is enabled.</p> <p>This command makes no change of contents of line buffer.</p> <p>This command does not change any other status.</p> |    |    |    |    |    |    |    |    |     |
|             | <p>(Example)</p> <div><div>Memory</div><div>Display Panel</div></div>                                                                                                                                                       |    |    |    |    |    |    |    |    |     |
| Restriction | This command has no effect when module is already in display on mode.                                                                                                                                                       |    |    |    |    |    |    |    |    |     |
| Flow Chart  | <pre>graph TD; A([Display Off mode]) --&gt; B[set_display_on]; B --&gt; C([Display On mode]);</pre>                                                                                                                         |    |    |    |    |    |    |    |    |     |



8.18.17 MADCTL (36h)

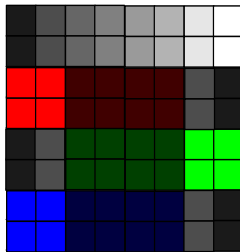
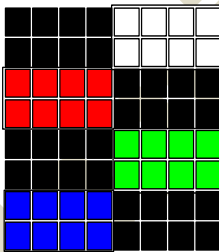
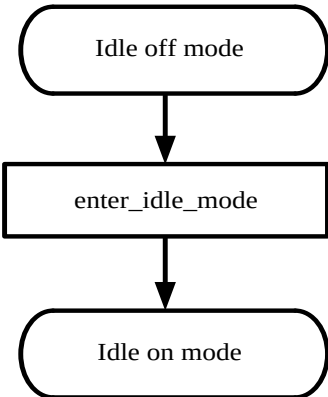
| 36H                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                | MADCTL               |                                                                                       |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------|----|----|----|----|----|----|-----|-----|-------------|-------|----|---------------|---------------------------------------------------------------------------------------|----|----------------------|-------------------------------------------------------|----|--------------------|-----------------------------------------------------|
|                                                                                                                                                                                                                                                                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                            | D7                   | D6                                                                                    | D5 | D4 | D3 | D2 | D1 | D0 | HEX |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
| Command                                                                                                                                                                                                                                                                       | W                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0                    | 0                                                                                     | 1  | 1  | 0  | 1  | 1  | 0  | 36  |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
| Parameter 1                                                                                                                                                                                                                                                                   | W                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0                    | 0                                                                                     | 0  | 0  | D3 | 0  | D1 | D0 |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
| Description                                                                                                                                                                                                                                                                   | This command defines read/write scanning direction of line buffer.                                                                                                                                                                                                                                                                                                                                                                             |                      |                                                                                       |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
|                                                                                                                                                                                                                                                                               | This command makes no change on the other driver status.                                                                                                                                                                                                                                                                                                                                                                                       |                      |                                                                                       |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
|                                                                                                                                                                                                                                                                               | <table><tr><th>Bit</th><th>Description</th><th>Value</th></tr><tr><td>D3</td><td>RGB-BGR ORDER</td><td>Color selector switch control<br/>0=RGB color filter panel<br/>1=BGR color filter panel</td></tr><tr><td>D1</td><td>Source scan sequence</td><td>Horizontal scan<br/>0=Left to right<br/>1=Right to left</td></tr><tr><td>D0</td><td>Gate scan sequence</td><td>Vertical scan<br/>0=Top to bottom<br/>1=Bottom to top</td></tr></table> |                      |                                                                                       |    |    |    |    |    |    |     | Bit | Description | Value | D3 | RGB-BGR ORDER | Color selector switch control<br>0=RGB color filter panel<br>1=BGR color filter panel | D1 | Source scan sequence | Horizontal scan<br>0=Left to right<br>1=Right to left | D0 | Gate scan sequence | Vertical scan<br>0=Top to bottom<br>1=Bottom to top |
|                                                                                                                                                                                                                                                                               | Bit                                                                                                                                                                                                                                                                                                                                                                                                                                            | Description          | Value                                                                                 |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
|                                                                                                                                                                                                                                                                               | D3                                                                                                                                                                                                                                                                                                                                                                                                                                             | RGB-BGR ORDER        | Color selector switch control<br>0=RGB color filter panel<br>1=BGR color filter panel |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
|                                                                                                                                                                                                                                                                               | D1                                                                                                                                                                                                                                                                                                                                                                                                                                             | Source scan sequence | Horizontal scan<br>0=Left to right<br>1=Right to left                                 |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
|                                                                                                                                                                                                                                                                               | D0                                                                                                                                                                                                                                                                                                                                                                                                                                             | Gate scan sequence   | Vertical scan<br>0=Top to bottom<br>1=Bottom to top                                   |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
|                                                                                                                                                                                                                                                                               | <div><div><div>D0=0</div><div><div>Host Processor</div><div><div>Top Left</div><div>SP</div><div>EP</div><div>S</div><div>E</div></div></div><div><div>Display Device</div><div><div>Top Left</div><div>SP</div><div>EP</div><div>S</div><div>E</div></div></div></div></div>                                                                                                                                                                  |                      |                                                                                       |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
|                                                                                                                                                                                                                                                                               | <div><div><div>D0=1</div><div><div>Host Processor</div><div><div>Top Left</div><div>SP</div><div>EP</div><div>S</div><div>E</div></div></div><div><div>Display Device</div><div><div>Top Left</div><div>SP</div><div>EP</div><div>S</div><div>E</div></div></div></div></div>                                                                                                                                                                  |                      |                                                                                       |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
|                                                                                                                                                                                                                                                                               | <div><div><div>D1=0</div><div><div>Host Processor</div><div><div>Top Left</div><div>SP</div><div>EP</div><div>S</div><div>E</div></div></div><div><div>Display Device</div><div><div>Top Left</div><div>SP</div><div>EP</div><div>S</div><div>E</div></div></div></div></div>                                                                                                                                                                  |                      |                                                                                       |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
| <div><div><div>D1=1</div><div><div>Host Processor</div><div><div>Top Left</div><div>SP</div><div>EP</div><div>S</div><div>E</div></div></div><div><div>Display Device</div><div><div>Top Left</div><div>SP</div><div>EP</div><div>E</div><div>S</div></div></div></div></div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                       |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
|                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                       |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |
|                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                       |    |    |    |    |    |    |     |     |             |       |    |               |                                                                                       |    |                      |                                                       |    |                    |                                                     |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <div><div><div>D3 = 0</div><div><div><div><div><div><div>R</div><div>G</div><div>B</div></div></div><div>S1G1</div></div><div><div><div>R</div><div>G</div><div>B</div></div><div>S1G2</div></div><div>.....</div><div><div><div>R</div><div>G</div><div>B</div></div><div>S1G480</div></div></div><div>Driver IC</div></div><div><div><div><div><div>R</div><div>G</div><div>B</div></div><div>S1G1</div></div><div><div><div>R</div><div>G</div><div>B</div></div><div>S1G2</div></div><div>.....</div><div><div><div>R</div><div>G</div><div>B</div></div><div>S1G480</div></div></div><div>LCD panel</div></div></div><div><div>RGB-BGR Order</div><div><div><div>D3 = 1</div><div><div><div><div><div>R</div><div>G</div><div>B</div></div></div><div>S1G1</div></div><div><div><div>R</div><div>G</div><div>B</div></div><div>S1G2</div></div><div>.....</div><div><div><div>R</div><div>G</div><div>B</div></div><div>S1G480</div></div></div><div>Driver IC</div></div><div><div><div><div><div>B</div><div>G</div><div>R</div></div><div>S1G1</div></div><div><div><div>B</div><div>G</div><div>R</div></div><div>S1G2</div></div><div>.....</div><div><div><div>B</div><div>G</div><div>R</div></div><div>S1G480</div></div></div><div>LCD panel</div></div></div></div></div> |
| Restriction |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Flow Chart  | <div><div>set_address_mode</div><div>↓</div><div>1st parameter<br/>B[7:0]</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

8.18.18 IDMOFF (38h)

| 38H         | IDMOFF                                                                                                         |    |    |    |    |    |    |    |    |     |
|-------------|----------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | R/W                                                                                                            | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| Command     | W                                                                                                              | 0  | 0  | 1  | 1  | 1  | 0  | 0  | 0  | 38  |
| Description | This command is used to recover from Idle mode on. In the idle off mode, LCD can display maximum 16.7M colors. |    |    |    |    |    |    |    |    |     |
| Restriction | This command has no effect when module is already in idle off mode.                                            |    |    |    |    |    |    |    |    |     |
| Flow Chart  | <div><div>Idle on mode</div><div>↓</div><div>exit_idle_mode</div><div>↓</div><div>Idle off mode</div></div>    |    |    |    |    |    |    |    |    |     |

8.18.19 IDMON (39h)

| 39H                 | IDMON                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |         |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|---------|----|----|----|----|----|-----|--|---------|---------|---------|-------|--------|---------|---------|------|---------|---------|---------|-----|---------|---------|---------|--------|---------|---------|---------|-------|---------|---------|---------|------|---------|---------|---------|--------|---------|---------|---------|-------|---------|---------|---------|
|                     | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | D7      | D6      | D5      | D4 | D3 | D2 | D1 | D0 | HEX |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| Command             | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0       | 0       | 1       | 1  | 1  | 0  | 0  | 1  | 39  |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| Description         | <p>This command is used to enter into Idle mode on. In the idle on mode, color expression is reduced. The primary and the secondary colors using MSB of each R, G and B in the line buffer, 8 color depth data is displayed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |         |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
|                     | <p>(Example)</p> <div><div>Memory</div><div>Display Panel</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |         |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
|                     | <p>Memory contents vs. Display Color</p> <table><tr><th></th><th>R7 – R0</th><th>G7 – G0</th><th>B7 – B0</th></tr><tr><td>Black</td><td>0XXXXX</td><td>0 XXXXX</td><td>0 XXXXX</td></tr><tr><td>Blue</td><td>0 XXXXX</td><td>0 XXXXX</td><td>1 XXXXX</td></tr><tr><td>Red</td><td>1 XXXXX</td><td>0 XXXXX</td><td>0 XXXXX</td></tr><tr><td>Magent</td><td>1 XXXXX</td><td>0 XXXXX</td><td>1 XXXXX</td></tr><tr><td>Green</td><td>0 XXXXX</td><td>1 XXXXX</td><td>0 XXXXX</td></tr><tr><td>Cyan</td><td>0 XXXXX</td><td>1 XXXXX</td><td>1 XXXXX</td></tr><tr><td>Yellow</td><td>1 XXXXX</td><td>1 XXXXX</td><td>0 XXXXX</td></tr><tr><td>White</td><td>1 XXXXX</td><td>1 XXXXX</td><td>1 XXXXX</td></tr></table> |         |         |         |    |    |    |    |    |     |  | R7 – R0 | G7 – G0 | B7 – B0 | Black | 0XXXXX | 0 XXXXX | 0 XXXXX | Blue | 0 XXXXX | 0 XXXXX | 1 XXXXX | Red | 1 XXXXX | 0 XXXXX | 0 XXXXX | Magent | 1 XXXXX | 0 XXXXX | 1 XXXXX | Green | 0 XXXXX | 1 XXXXX | 0 XXXXX | Cyan | 0 XXXXX | 1 XXXXX | 1 XXXXX | Yellow | 1 XXXXX | 1 XXXXX | 0 XXXXX | White | 1 XXXXX | 1 XXXXX | 1 XXXXX |
|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | R7 – R0 | G7 – G0 | B7 – B0 |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| Black               | 0XXXXX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0 XXXXX | 0 XXXXX |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| Blue                | 0 XXXXX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0 XXXXX | 1 XXXXX |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| Red                 | 1 XXXXX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0 XXXXX | 0 XXXXX |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| Magent              | 1 XXXXX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0 XXXXX | 1 XXXXX |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| Green               | 0 XXXXX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 XXXXX | 0 XXXXX |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| Cyan                | 0 XXXXX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 XXXXX | 1 XXXXX |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| Yellow              | 1 XXXXX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 XXXXX | 0 XXXXX |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| White               | 1 XXXXX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 XXXXX | 1 XXXXX |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| <p>X=don't care</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |         |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| Restriction         | <p>This command has no effect when module is already in idle on mode.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |         |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |
| Flow Chart          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |         |         |    |    |    |    |    |     |  |         |         |         |       |        |         |         |      |         |         |         |     |         |         |         |        |         |         |         |       |         |         |         |      |         |         |         |        |         |         |         |       |         |         |         |

# 8.18.20 WRIMC (80h)

| 80H             | WRIMC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|--------------------------------|---|---|---|-------|---|---|---|------|---|---|---|-------|---|---|---|------|---|---|---|-----|---|---|---|---------|---|---|---|--------|---|---|---|-------|---|---|---|
|                 | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| Command         | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 80  |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| Parameter 1     | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | X  | X  | X  | X  | X  | R  | G  | B  |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| Description     | This command is used to select color for 1bpp Idle Mode. Color selection is configured with bits [D2:D0].<br>D2: R Component<br>D1: G Component<br>D0: B Component<br>Color selection is defined in following table:                                                                                                                                                                                                                                                                                                                 |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | <table><tr><td>1bpp Idle Mode Color Selection</td><td>R</td><td>G</td><td>B</td></tr><tr><td>Black</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Blue</td><td>0</td><td>0</td><td>1</td></tr><tr><td>Green</td><td>0</td><td>1</td><td>0</td></tr><tr><td>Cyan</td><td>0</td><td>1</td><td>1</td></tr><tr><td>Red</td><td>1</td><td>0</td><td>0</td></tr><tr><td>Magenta</td><td>1</td><td>0</td><td>1</td></tr><tr><td>Yellow</td><td>1</td><td>1</td><td>0</td></tr><tr><td>White</td><td>1</td><td>1</td><td>1</td></tr></table> |    |    |    |    |    |    |    |    |     | 1bpp Idle Mode Color Selection | R | G | B | Black | 0 | 0 | 0 | Blue | 0 | 0 | 1 | Green | 0 | 1 | 0 | Cyan | 0 | 1 | 1 | Red | 1 | 0 | 0 | Magenta | 1 | 0 | 1 | Yellow | 1 | 1 | 0 | White | 1 | 1 | 1 |
|                 | 1bpp Idle Mode Color Selection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R  | G  | B  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Black                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0  | 0  | 0  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Blue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0  | 0  | 1  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Green                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0  | 1  | 0  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Cyan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0  | 1  | 1  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Red                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1  | 0  | 0  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Magenta                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1  | 0  | 1  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Yellow                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1  | 1  | 0  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| White           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1  | 1  |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| X = Don't care. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| Restriction     | -This command is only applicable to 1bpp Compression in Idle Mode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| Flow Chart      | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |

**8.18.21 RDRIMC (81h)**

| 81H             | RDRIMC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|--------------------------------|---|---|---|-------|---|---|---|------|---|---|---|-------|---|---|---|------|---|---|---|-----|---|---|---|---------|---|---|---|--------|---|---|---|-------|---|---|---|
|                 | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| Command         | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 81  |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| Parameter 1     | R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | X  | X  | X  | X  | X  | R  | G  | B  |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| Description     | This command returns the current color selection of 1bpp Idle Mode which color selection is configured with bits [D2:D0].<br>D2: R Component<br>D1: G Component<br>D0: B Component<br>Color selection is defined in following table:                                                                                                                                                                                                                                                                                                 |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | <table><tr><td>1bpp Idle Mode Color Selection</td><td>R</td><td>G</td><td>B</td></tr><tr><td>Black</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Blue</td><td>0</td><td>0</td><td>1</td></tr><tr><td>Green</td><td>0</td><td>1</td><td>0</td></tr><tr><td>Cyan</td><td>0</td><td>1</td><td>1</td></tr><tr><td>Red</td><td>1</td><td>0</td><td>0</td></tr><tr><td>Magenta</td><td>1</td><td>0</td><td>1</td></tr><tr><td>Yellow</td><td>1</td><td>1</td><td>0</td></tr><tr><td>White</td><td>1</td><td>1</td><td>1</td></tr></table> |    |    |    |    |    |    |    |    |     | 1bpp Idle Mode Color Selection | R | G | B | Black | 0 | 0 | 0 | Blue | 0 | 0 | 1 | Green | 0 | 1 | 0 | Cyan | 0 | 1 | 1 | Red | 1 | 0 | 0 | Magenta | 1 | 0 | 1 | Yellow | 1 | 1 | 0 | White | 1 | 1 | 1 |
|                 | 1bpp Idle Mode Color Selection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R  | G  | B  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Black                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0  | 0  | 0  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Blue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0  | 0  | 1  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Green                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0  | 1  | 0  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Cyan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0  | 1  | 1  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Red                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1  | 0  | 0  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Magenta                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1  | 0  | 1  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 | Yellow                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1  | 1  | 0  |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| White           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1  | 1  |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| X = Don't care. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| Restriction     | -This command is only applicable to 1bpp Compression in Idle Mode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |
| Flow Chart      | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |    |    |    |    |    |    |     |                                |   |   |   |       |   |   |   |      |   |   |   |       |   |   |   |      |   |   |   |     |   |   |   |         |   |   |   |        |   |   |   |       |   |   |   |

## 8.19 MIPI DC characteristic

| Parameter                                                  | Symbol         | Min. | Typ. | Max. | Unit |
|------------------------------------------------------------|----------------|------|------|------|------|
| MIPI Characteristics for High Speed Receiver               |                |      |      |      |      |
| Single-ended input low voltage                             | $V_{ILHS}$     | -40  | -    | -    | mV   |
| Single-ended input high voltage                            | $V_{IHHS}$     | -    | -    | 460  | mV   |
| Common-mode voltage                                        | $V_{CMRXDC}$   | 70   | -    | 330  | mV   |
| Differential input impedance                               | $Z_{ID}$       | 80   | 100  | 120  | ohm  |
| HS transmit differential voltage( $V_{OD}=V_{DP}-V_{DN}$ ) | $ V_{OD} $     | 100  | 200  | 250  | mV   |
| MIPI Characteristics for Low Power Mode                    |                |      |      |      |      |
| Pad signal voltage range                                   | $V_I$          | -50  | -    | 1350 | mV   |
| Ground shift                                               | $V_{GNDSH}$    | -50  | -    | 50   | mV   |
| Logic 0 input threshold                                    | $V_{IL}$       | 0    | -    | 550  | mV   |
| Logic 1 input threshold                                    | $V_{IH}$       | 1000 | -    | 1350 | mV   |
| Input hysteresis                                           | $V_{HYST}$     | 25   | -    | -    | mV   |
| Output low level                                           | $V_{OL}$       | -50  | -    | 50   | mV   |
| Output high level                                          | $V_{OH}$       | 1.1  | 1.2  | 1.3  | V    |
| Output impedance of Low Power Transmitter                  | $Z_{OLP}$      | 110  |      |      | ohm  |
| Logic 0 contention threshold                               | $V_{ILCD,MAX}$ | -    | -    | 200  | mV   |
| Logic 1 contention threshold                               | $V_{IHCD,MIN}$ | 450  | -    | -    | mV   |
| MIPI Digital Operating Current                             | $I_{VDDMIPI}$  | -    | 15   | 20   | mA   |
| MIPI Digital Stand-by Current                              | $I_{STMPI}$    | -    | -    | 250  | uA   |

Note: MIPI Digital Operating and Stand-by Current is at RT 25°C condition.

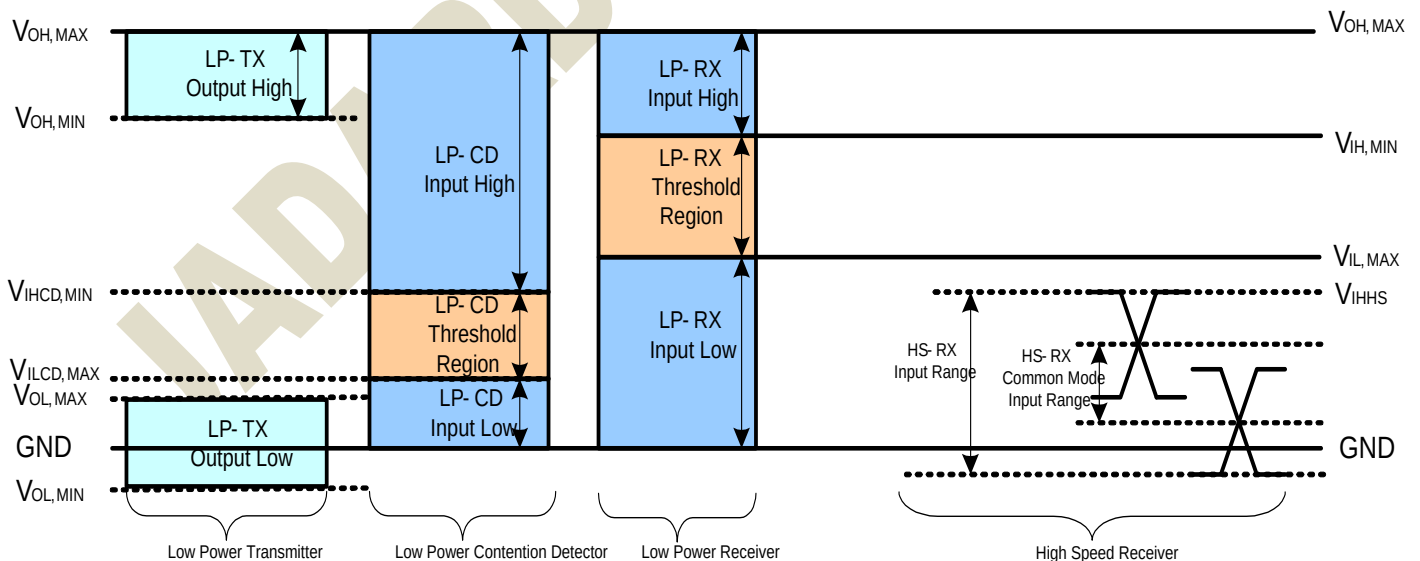


Figure 8.28: MIPI signaling and contention voltage levels

## 8.20 MIPI AC characteristic

## 8.20.1 MIPI Low Power Transmitter AC Specification

| Parameter                             | Symbol                                                                     | Min | Typ | Max | Units | Notes |
|---------------------------------------|----------------------------------------------------------------------------|-----|-----|-----|-------|-------|
| 15%~85% rising time and falling time  | $T_{RLP} / T_{FLP}$                                                        | -   | -   | 25  | ns    | -     |
| 30%~85% rising time and falling time  | $T_{REOT}$                                                                 | -   | -   | 35  | ns    | -     |
| Pulse width of LP exclusive-OR clock  | First LP EXOR clock pulse after STOP state or Last pulse before stop state | 100 | -   | -   | ns    | -     |
|                                       | All other pulses                                                           |     |     |     |       |       |
| Period of the LP EXOR clock(LP Speed) | $T_{LP-PER-TX}$                                                            | 200 | -   | -   | ns    | -     |
| Slew Rate @CLOAD =0pF                 | $\delta V / \delta t_{SR}$                                                 | 20  | -   | 500 | mV/ns | -     |
| Slew Rate @CLOAD =5pF                 |                                                                            | 20  | -   | 200 | mV/ns | -     |
| Slew Rate @CLOAD =20pF                |                                                                            | 20  | -   | 150 | mV/ns | -     |
| Slew Rate @CLOAD =70pF                |                                                                            | 20  | -   | 100 | mV/ns | -     |
| Load Capacitance                      | $T_{RLP}$                                                                  | -   | -   | 70  | pF    | -     |

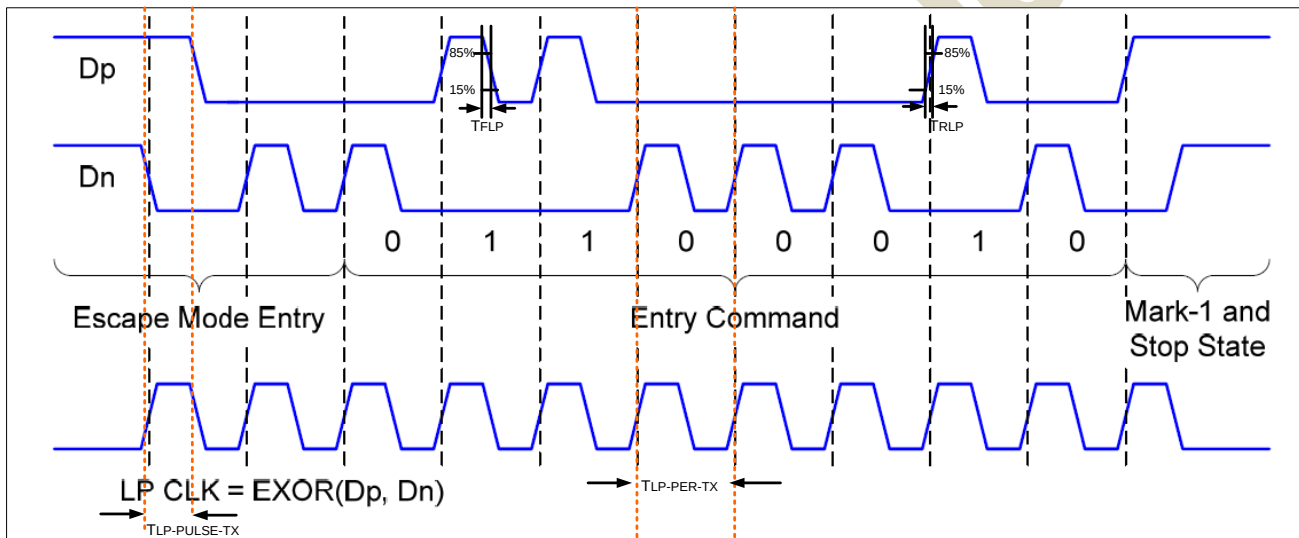


Figure 8.29: MIPI LP AC timing



8.20.2 MIPI Low Power Turnaround Procedure

Turnaround Procedure Operation Timing Parameters

| Parameter                                    | Symbol        | Min       | Typ        | Max        | Units |
|----------------------------------------------|---------------|-----------|------------|------------|-------|
| Length of any Low-Power state period         | $T_{LPX}$     | 100       | -          | -          | ns    |
| Time-out before new TX side start driving    | $T_{TA-Sure}$ | $T_{LPX}$ | -          | $2T_{LPX}$ | ns    |
| Time to drive LP-00 by new TX                | $T_{TA-GET}$  | -         | $5T_{LPX}$ | -          | ns    |
| Time to drive LP-00 after Turnaround Request | $T_{TA-GO}$   | -         | $4T_{LPX}$ | -          | ns    |

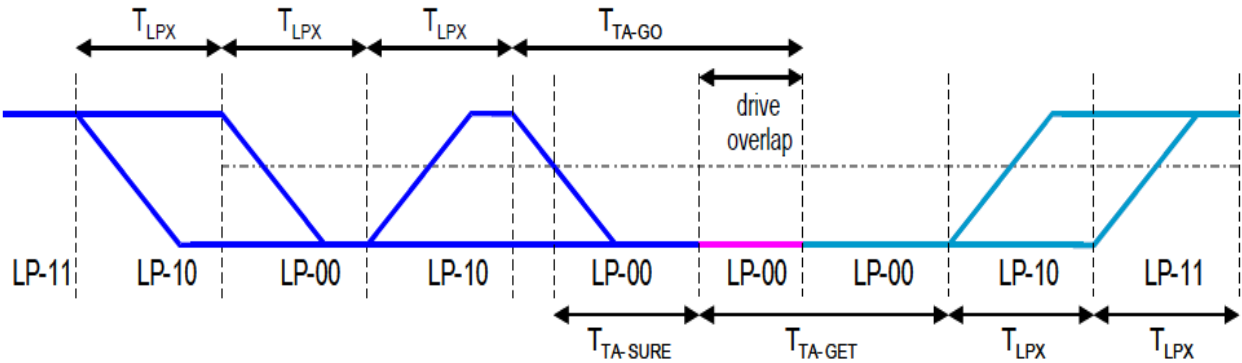


Figure 8.30: Turnaround Procedure

## 8.20.3 MIPI High Speed AC characteristics

| Parameter                     | Descript                                                                                                                                                                                                                     | Spec.                              |      |                          | Unit |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------|--------------------------|------|
|                               |                                                                                                                                                                                                                              | Min.                               | Typ. | Max.                     |      |
| $T_{REOT}$                    | 30%-85% rise time and fall time                                                                                                                                                                                              | -                                  | -    | 35                       | ns   |
| $T_{CLK-MISS}$                | Timeout for receiver to detect absence of Clock transitions and disable the Clock Lane HS-RX.                                                                                                                                | -                                  | -    | 60                       | ns   |
| $T_{CLK-POST}^{*1}$           | Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. Interval is defined as the period from the end of $T_{HS-TRAIL}$ to the beginning of $T_{CLK-TRAIL}$ . | $60\text{ ns} + 52*UI$             | -    |                          | ns   |
| $T_{CLK-PRE}$                 | Time that the HS clock shall be driven by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.                                                                                     | 8                                  | -    |                          | UI   |
| $T_{CLK-SETTLE}$              | Time interval during which the HS receiver shall ignore any Clock Lane HS transitions, starting from the beginning of $T_{CLK-PRE}$ .                                                                                        | 95                                 | -    | 300                      | ns   |
| $T_{CLK-TERM-EN}$             | Time for the Clock Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$ .                                                                                              | Time for Dn to reach $V_{TERM-EN}$ | -    | 38                       | ns   |
| $T_{HS-SETTLE}$               | Time interval during which the HS receiver shall ignore any Data Lane HS transitions, starting from the beginning of $T_{HSPREPRE}$ .                                                                                        | $85\text{ ns} + 6*UI$              | -    | $145\text{ ns} + 10*UI$  | ns   |
| $T_{EOT}$                     | Time from start of $T_{HS-TRAIL}$ or $T_{CLK-TRAIL}$ period to start of LP-11 state                                                                                                                                          | -                                  | -    | $105\text{ns} + n*12*UI$ | -    |
| $T_{HS-EXIT}^{(1)}$           | time to drive LP-11 after HS burst                                                                                                                                                                                           | 100                                | -    | -                        | ns   |
| $T_{HS-PREPRE}$               | Time to drive LP-00 to prepare for HS transmission                                                                                                                                                                           | $40\text{ns} + 4*UI$               | -    | $85\text{ns} + 6*UI$     | ns   |
| $T_{HS-PREPRE} + T_{HS-ZERO}$ | $T_{HS-PREPRE}$ + Time to drive HS-0 before the Sync sequence                                                                                                                                                                | $145\text{ns} + 10*UI$             | -    | -                        | ns   |
| $T_{HS-SKIP}$                 | Time-out at RX to ignore transition period of EoT                                                                                                                                                                            | 40                                 | -    | $55\text{ns} + 4*UI$     | ns   |
| $T_{HS-TRAIL}$                | Time to drive flipped differential state after last payload data bit of a HS transmission burst                                                                                                                              | $60 + 4*UI$                        | -    | -                        | ns   |
| $T_{LPX}$                     | Length of any Low-Power state period                                                                                                                                                                                         | 100                                | -    | -                        | ns   |
| Ratio $T_{LPX}$               | Ratio of $T_{LPX(MASTER)}/T_{LPX(SLAVE)}$ between Master and Slave side                                                                                                                                                      | 2/3                                | -    | 3/2                      | -    |
| $T_{TA-GET}$                  | Time to drive LP-00 by new TX                                                                                                                                                                                                | $5*T_{LPX}$                        |      |                          | ns   |
| $T_{TA-GO}$                   | Time to drive LP-00 after Turnaround Request                                                                                                                                                                                 | $4*T_{LPX}$                        |      |                          | ns   |
| $T_{TA-SURE}$                 | Time-out before new TX side starts driving                                                                                                                                                                                   | $T_{LPX}$                          | -    | $2*T_{LPX}$              | ns   |

**Note:** (1) For  $T_{CLK-POST}$  example:

$T_{CLK-POST}$  min value =  $164UI$  when MIPI max frequency per lane = 0.5Gbps.

$T_{CLK-POST}$  min value =  $112UI$  when MIPI max frequency per lane = 1Gbps

(2) For  $T_{EOT}$ :

When  $n = 1$  for Forward-direction HS mode and  $n=4$  for Reverse-direction HS mode.

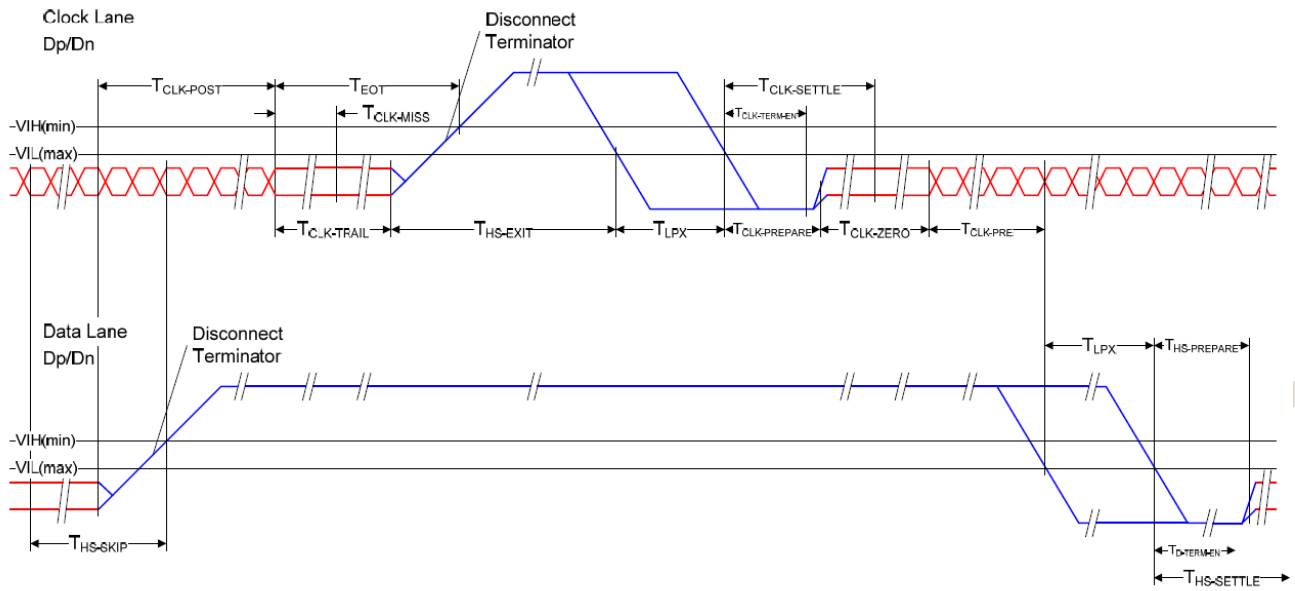


Figure 8.31: Switching the clock lane between clock transmission and low-power mode

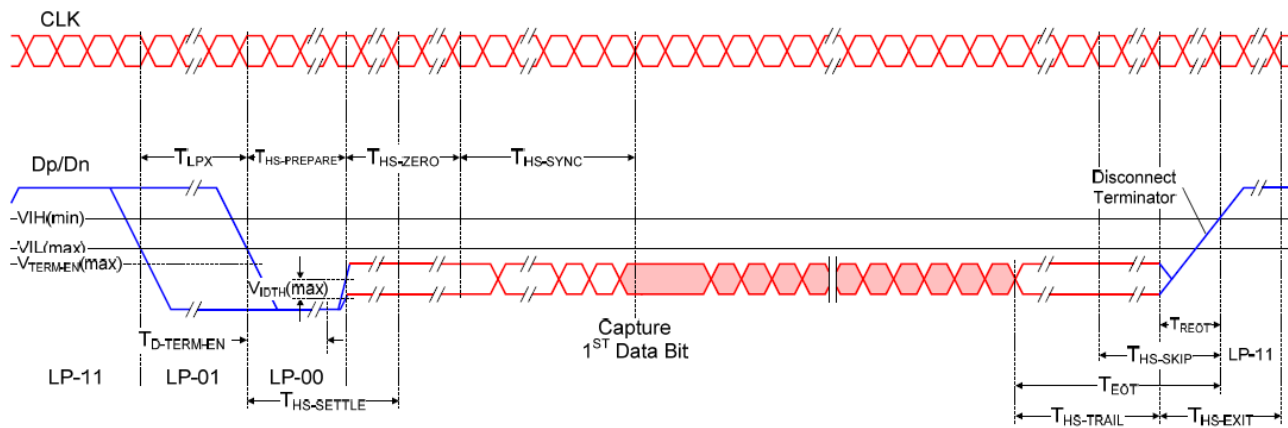


Figure 8.32: Timing of high-speed data transmission in bursts

8.20.4 MIPI data-clock timing specification

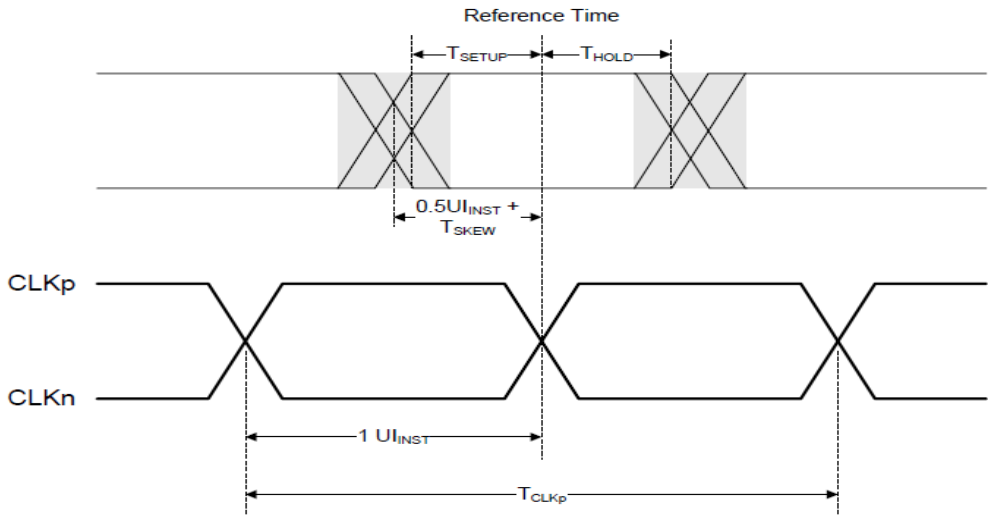


Figure 8.33: Data to clock timing

| Parameter                | Symbol                 | Min          | Typ | Max | Units                |
|--------------------------|------------------------|--------------|-----|-----|----------------------|
| Data to clock setup time | $T_{\text{SETUP[RX]}}$ | $0.15^{(1)}$ | -   | -   | $U_{\text{I\_INST}}$ |
| Data to clock hold time  | $T_{\text{HOLD[RX]}}$  | $0.15^{(1)}$ | -   | -   | $U_{\text{I\_INST}}$ |

**Note:** (1) Total setup and hold window for receiver of  $0.3 \cdot U_{\text{I\_INST}}$ .

Table 8.8: Data to Clock Timing Specifications

8.21 MIPI video input timing

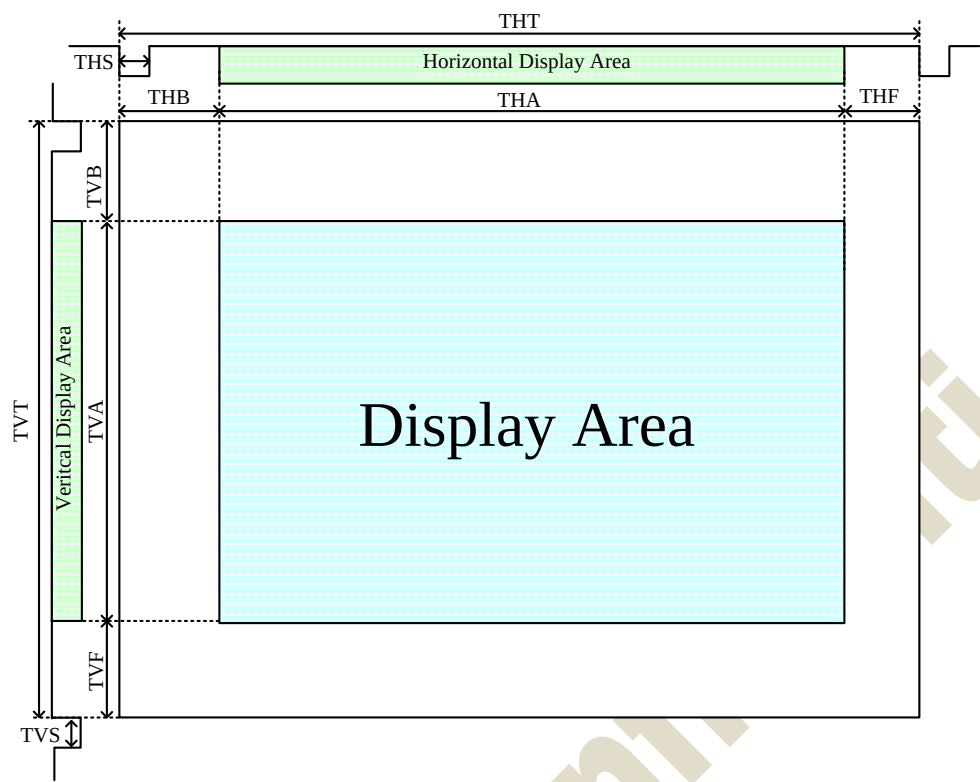


Figure 8.34: MIPI video input timing

## 8.21.1 MIPI timing characteristic

| MIPI Input Timing                           | Symbol | 1024RGBx768 |      |       | 1024RGBx600 |      |       | 800RGBx600 |      |       | Unit |
|---------------------------------------------|--------|-------------|------|-------|-------------|------|-------|------------|------|-------|------|
|                                             |        | Min         | Typ  | Max   | Min         | Typ  | Max   | Min        | Typ  | Max   |      |
| MIPI 24-bit RGB@ 2 lane Operating Frequency | -      | 500         | -    | 750   | 400         | 616  | 750   | 330        | 476  | 750   | Mbps |
| MIPI 24-bit RGB@ 4 lane Operating Frequency | -      | 250         | 390  | 500   | 200         | 308  | 500   | 165        | 238  | 500   | Mbps |
| Frame Rate@ 2 lane                          | -      | 47          | -    | -     | 48          | 60   | -     | 49         | 60   | -     | Hz   |
| Frame Rate@ 4 lane                          | -      | 47          | 60   | -     | 48          | 60   | -     | 49         | 60   | -     | Hz   |
| Horizontal Total                            | tht    | 1114        | 1344 | 1400  | 1114        | 1344 | 1400  | 890        | 1000 | 1300  | DCLK |
| Hsync Pulse width                           | ths    | 1           | 24   | HBP-1 | 1           | 24   | HBP-1 | 1          | 24   | HBP-1 | DCLK |
| Horizontal Back Porch                       | thb    | 60          | 160  | 160   | 60          | 160  | 160   | 60         | 160  | 250   | DCLK |
| Horizontal Valid Data                       | thd    | 1024        |      |       | 1024        |      |       | 800        |      |       | DCLK |
| Horizontal Front Porch                      | thfp   | 30          | 160  | 216   | 30          | 160  | 216   | 30         | 40   | 250   | DCLK |
| Vertical Total                              | vt     | 788         | 806  | 845   | 620         | 635  | 800   | 620        | 660  | 800   | THT  |
| Vsync Pulse Width                           | tv     | 1           | 2    | VBP-1 | 1           | 2    | VBP-1 | 1          | 2    | VBP-1 | THT  |
| Vertical Back Porch                         | tvb    | 8           | 23   | 33    | 8           | 23   | 100   | 8          | 39   | 100   | THT  |
| Vertical Valid Data                         | tv     | 768         |      |       | 600         |      |       | 600        |      |       | THT  |
| Vertical Front Porch                        | tvfp   | 12          | 15   | 44    | 12          | 12   | 100   | 12         | 21   | 100   | THT  |

| MIPI Input Timing                           | Symbol | 800RGBx480 |      |       | 640RGBx480 |      |       | 480RGBx272 |     |       | Unit |
|---------------------------------------------|--------|------------|------|-------|------------|------|-------|------------|-----|-------|------|
|                                             |        | Min        | Typ  | Max   | Min        | Typ  | Max   | Min        | Typ | Max   |      |
| MIPI 24-bit RGB@ 2 lane Operating Frequency | -      | 270        | 390  | 650   | 330        | 476  | 650   | 260        | 392 | 500   | Mbps |
| MIPI 24-bit RGB@ 4 lane Operating Frequency | -      | 135        | 195  | 400   | 165        | 238  | 400   | 130        | 196 | 300   | Mbps |
| Frame Rate@ 2 lane                          | -      | 50         | 60   | -     | 50         | 60   | -     | 52         | 60  | -     | Hz   |
| Frame Rate@ 4 lane                          | -      | 50         | 60   | -     | 50         | 60   | -     | 52         | 60  | -     | Hz   |
| Horizontal Total                            | tht    | 900        | 1000 | 1300  | 890        | 1000 | 1114  | 830        | 890 | 950   | DCLK |
| Hsync Pulse width                           | ths    | 4          | 24   | HBP-1 | 1          | 24   | HBP-1 | 1          | 24  | HBP-1 | DCLK |
| Horizontal Back Porch                       | thb    | 62         | 160  | 250   | 140        | 88   | 220   | 180        | 210 | 240   | DCLK |
| Horizontal Valid Data                       | thd    | 800        |      |       | 640        |      |       | 480        |     |       | DCLK |
| Horizontal Front Porch                      | thfp   | 38         | 40   | 250   | 110        | 272  | 254   | 170        | 200 | 230   | DCLK |
| Vertical Total                              | vt     | 500        | 540  | 680   | 610        | 660  | 800   | 498        | 610 | 660   | THT  |
| Vsync Pulse Width                           | tv     | 1          | 2    | VBP-1 | 1          | 2    | VBP-1 | 1          | 2   | VBP-1 | THT  |
| Vertical Back Porch                         | tvb    | 8          | 23   | 100   | 28         | 39   | 160   | 126        | 180 | 210   | THT  |
| Vertical Valid Data                         | tv     | 480        |      |       | 480        |      |       | 272        |     |       | THT  |
| Vertical Front Porch                        | tvfp   | 12         | 37   | 100   | 102        | 141  | 160   | 100        | 158 | 178   | THT  |

# 9. SPI FORMAT

## 9.1 3-wire SPI format

The JD9165 supports the 3-wire serial peripheral interface (**SPI**) to set internal register. The basic operation of SPI interface is shown below. The Host asserts the CSB when it wants to initiate a read or write transaction. This is followed by the Host sending 9 pulses on the SPI clock (**SCL**). The 1th bit of SCL pulses is defined to write command or parameter. (**0=command** , **1=parameter**). The Host de-asserts the CSB to indicate end of transfer.

### 9.1.1 3-wire SPI interface I

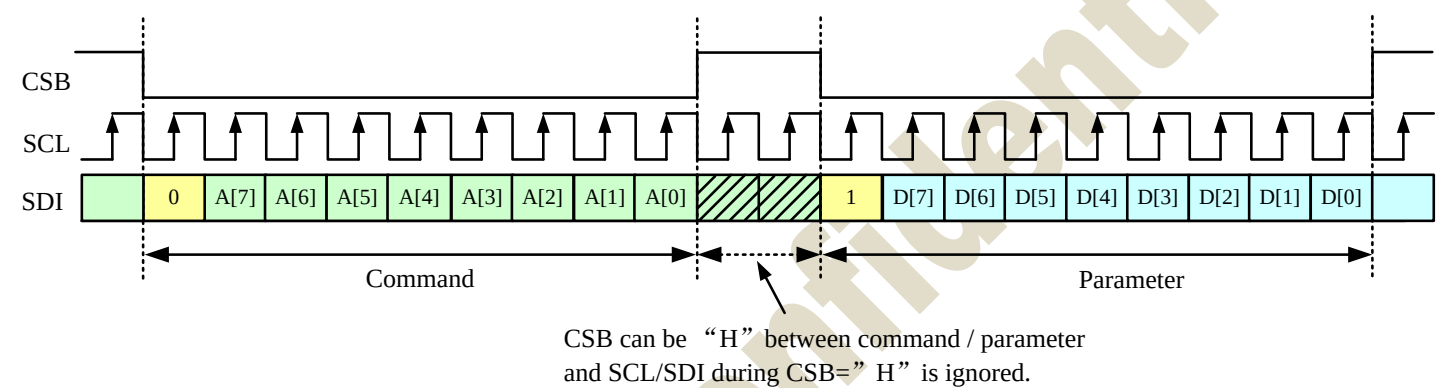


Figure 9.1: 3-wire SPI I single write operation

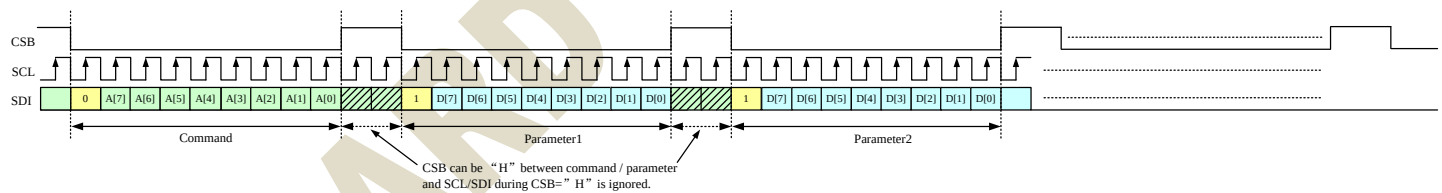
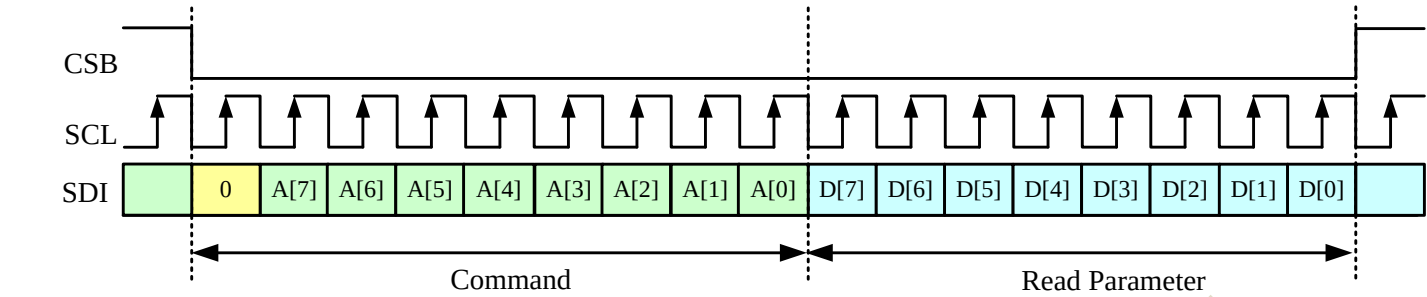
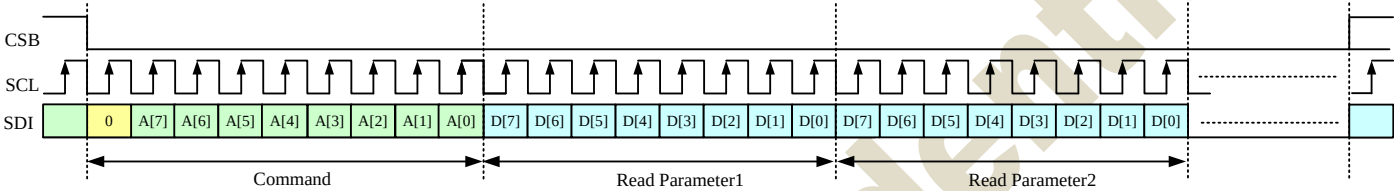


Figure 9.2: 3-wire SPI I multi write operation



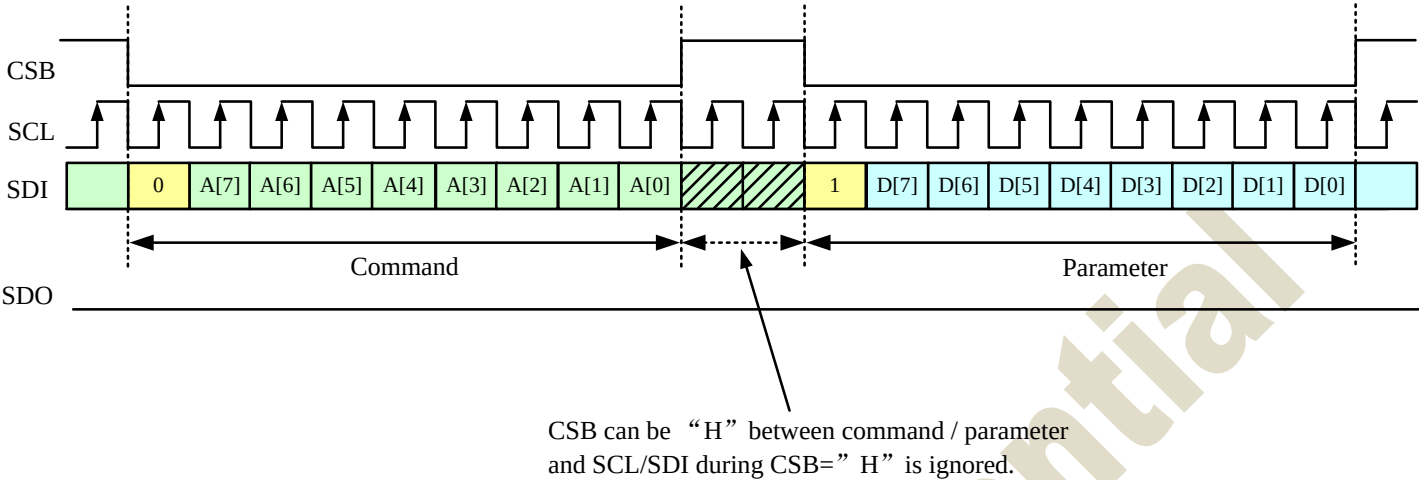
**Figure 9.3: 3-wire SPI I single read operation**



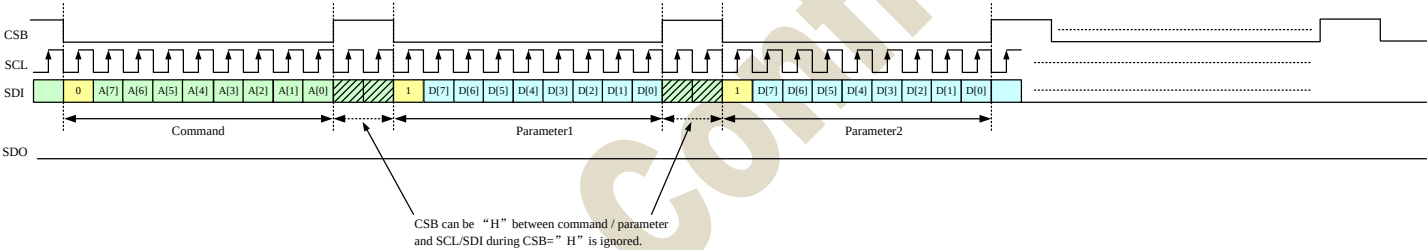
**Figure 9.4: 3-wire SPI I multi read operation**



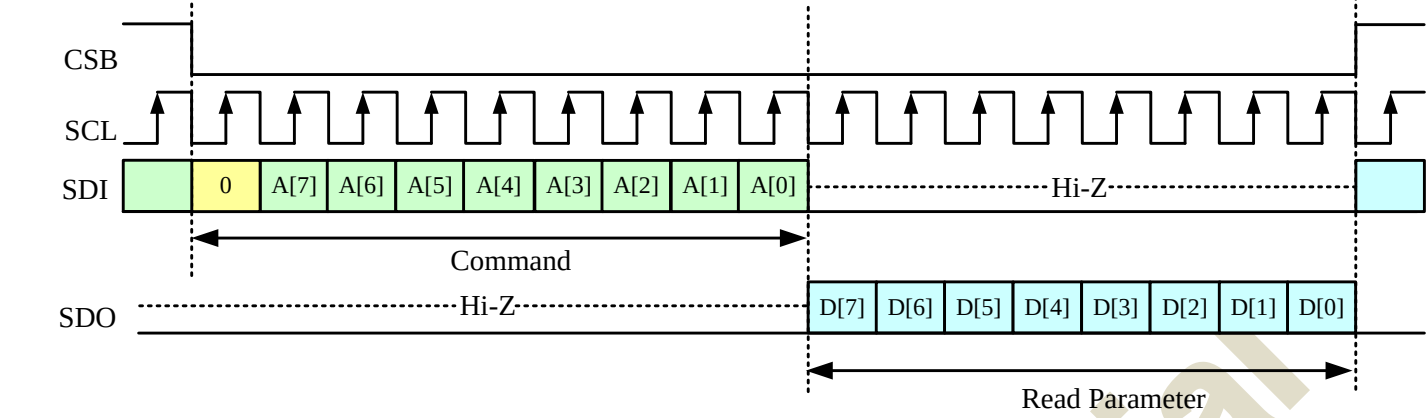
9.1.2 3-wire SPI interface II



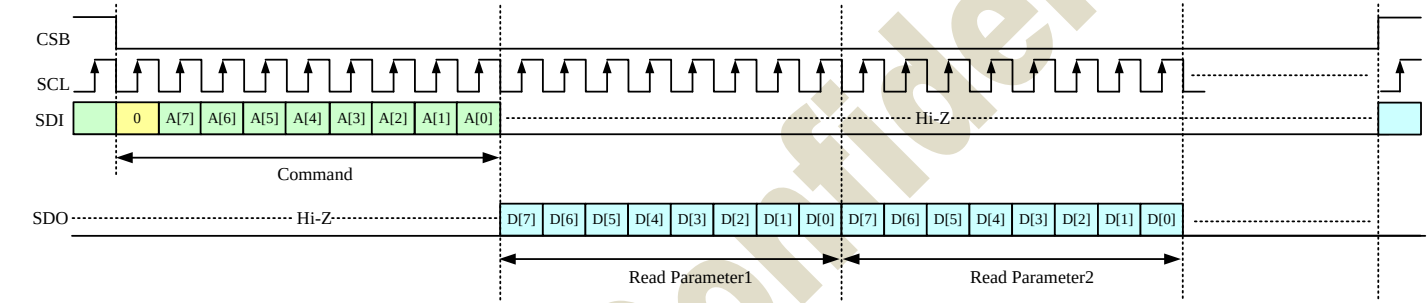
**Figure 9.5: 3-wire SPI II single write operation**



**Figure 9.6: 3-wire SPI II multi write operation**



**Figure 9.7: 3-wire SPI II single read operation**



**Figure 9.8: 3-wire SPI II multi read operation**

## 9.2 3-wire interface characteristics

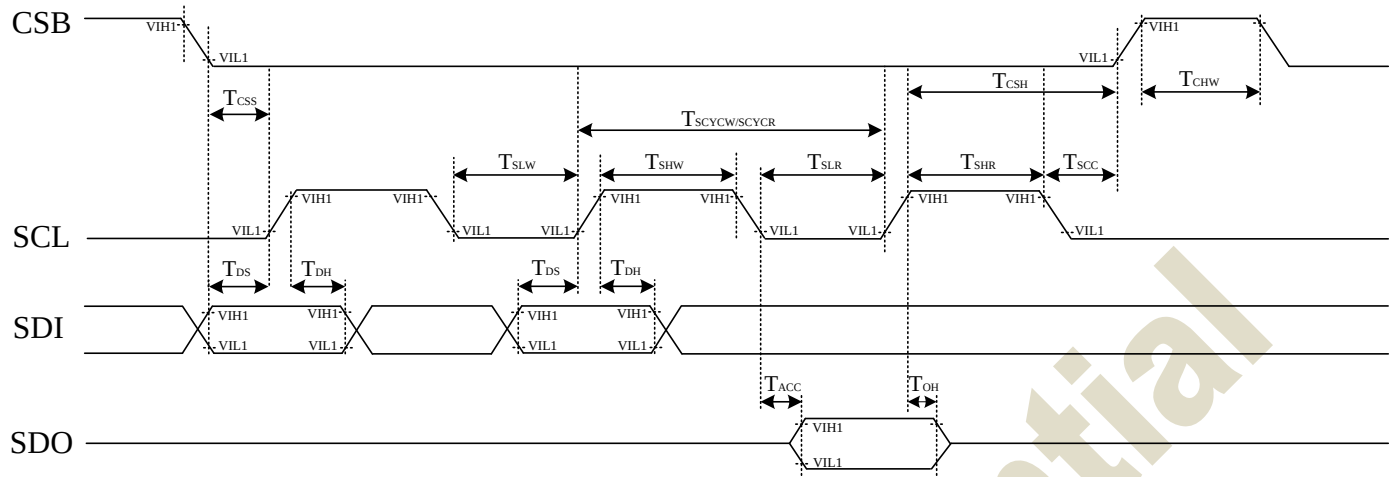


Figure 9.9: 3-wire SPI interface characteristics

| Parameter                   | Symbol      | Spec. |      |     | Unit |
|-----------------------------|-------------|-------|------|-----|------|
|                             |             | Min.  | Typ. | Max |      |
| Chip select setup time      | $T_{CSS}$   | 120   | -    | -   | ns   |
| Chip select hold time       | $T_{CSH}$   | 220   | -    | -   | ns   |
| Chip select setup time      | $T_{SCC}$   | 120   | -    | -   | ns   |
| Chip select high width      | $T_{CHW}$   | 100   | -    | -   | ns   |
| Serial clock cycle (Write)  | $T_{SCYCW}$ | 74    | -    | -   | ns   |
| SCL "H" pulse width (Write) | $T_{SHW}$   | 37    | -    | -   | ns   |
| SCL "L" pulse width (Write) | $T_{SLW}$   | 37    | -    | -   | ns   |
| Serial clock cycle (Read)   | $T_{SCYCR}$ | 200   | -    | -   | ns   |
| SCL "H" pulse width (Read)  | $T_{SHR}$   | 100   | -    | -   | ns   |
| SCL "L" pulse width (Read)  | $T_{SHL}$   | 100   | -    | -   | ns   |
| Data setup time             | $T_{SDS}$   | 18    | -    | -   | ns   |
| Data hold time              | $T_{SDH}$   | 18    | -    | -   | ns   |
| Access time                 | $T_{ACC}$   | 10    | -    | 75  | ns   |
| Output disable time         | $T_{OH}$    | 10    | -    | 75  | ns   |

# 10. I2C FORMAT

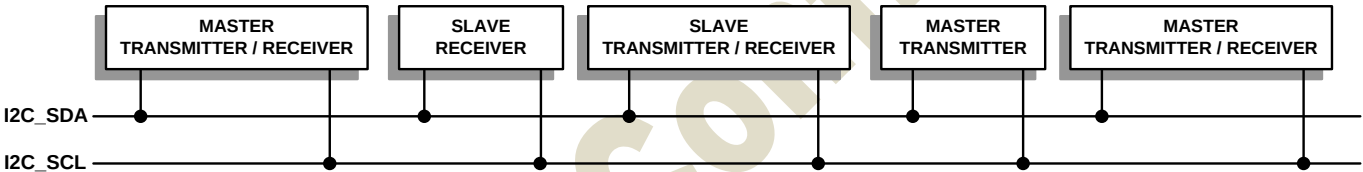
The I<sup>2</sup>C-bus is for bi-directional, two-line communication between different ICs or modules. The two lines are the Serial Data line (I<sup>2</sup>C\_SDA) and the Serial Clock Line (I<sup>2</sup>C\_SCL). Both lines must be connected to a positive supply via pull-up resistors. Data transfer can be initiated only when the bus is not busy. Each byte of eight bits is followed by an acknowledge bit. The acknowledge bit is a HIGH level signal put on the bus by the transmitter during which time the master generates an extra acknowledgement related clock pulse. A slave receiver which is addressed must generate an acknowledgement after the reception of each byte. Also a master receiver must generate an acknowledgement after the reception of each byte that has been clocked out of the slave transmitter.

(a) I<sup>2</sup>C-Bus Protocol:

Before any data is transmitted on the I<sup>2</sup>C-bus, the device which should respond is addressed first. There are four slave address can be selected by MCU. The slave addressing is always carried out with the first byte transmitted after the START procedure.

(b) Definitions:

- Transmitter: The device which sends the data to the bus.
- Receiver: The device which receives the data from the bus.
- Master: The device which initiates a transfer, generates clock signals and terminates a transfer.
- Slave: The device addressed by a master.
- Multi-master: More than one master can attempt to control the bus at the same time without corrupting the message.
- Arbitration: Procedure to ensure that, if more than one master simultaneously tries to control the bus, only one is allowed to do so and the message is not corrupted.
- Synchronization: Procedure to synchronize the clock signals of two or more devices.



## 10.1 Register write sequence of I<sup>2</sup>C interface

JD9165 supports register write sequence via I<sup>2</sup>C-bus transfer. The register writing support single register write mode and multi-register write mode. The detail transference sequences are illustrated and described as below.

- (1) Data transfers for register writing follow the format is shown in below.
- (2) After the START condition (S), a slave address is sent. R/W bit is setting to "0" for WRITE.
- (3) The slave issues an ACK to master.
- (4) 8 bits register address transfer first then transfer the register data parameter.
- (5) A data transfer is always terminated by a STOP condition.
- (6) JD9165 DA[6:0]=110\_1000

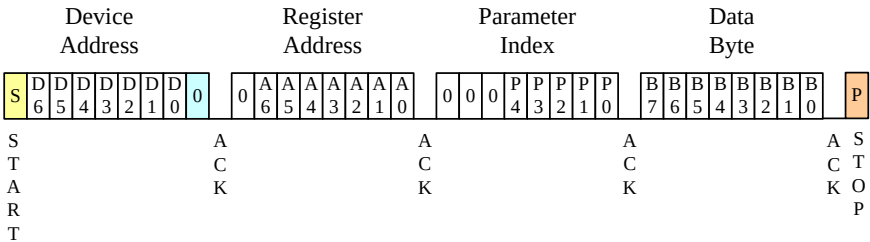


Figure 10.1: Single Register Writing Timing

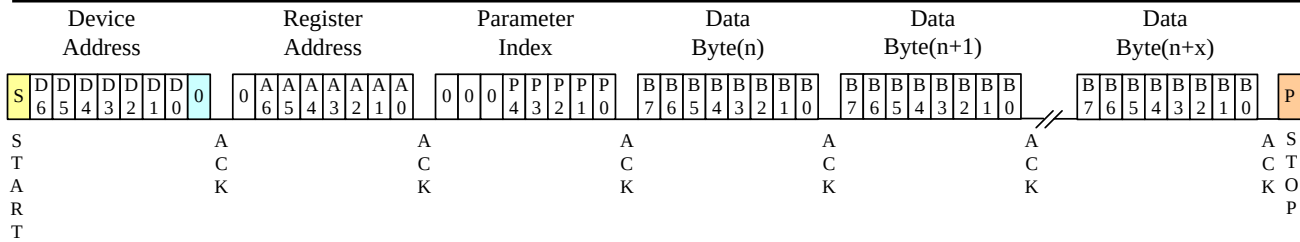


Figure 10.2: Multi Register Writing Timing

10.2 Register read sequence of I<sup>2</sup>C interface

JD9165 supports register read sequence via I<sup>2</sup>C-bus transfer. The register reading only support single register read mode.  
Register data reading transfers follow the format and is shown in below.

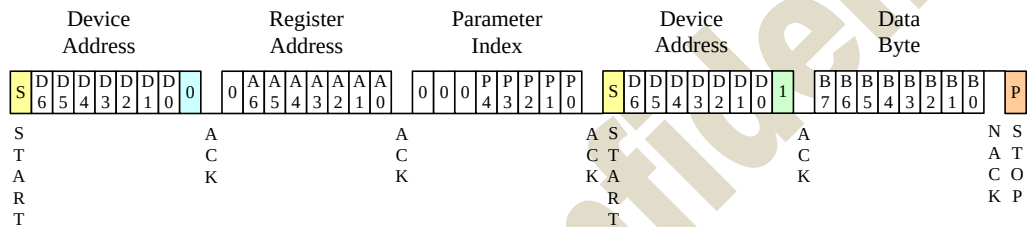


Figure 10.3: Single Register Reading Timing

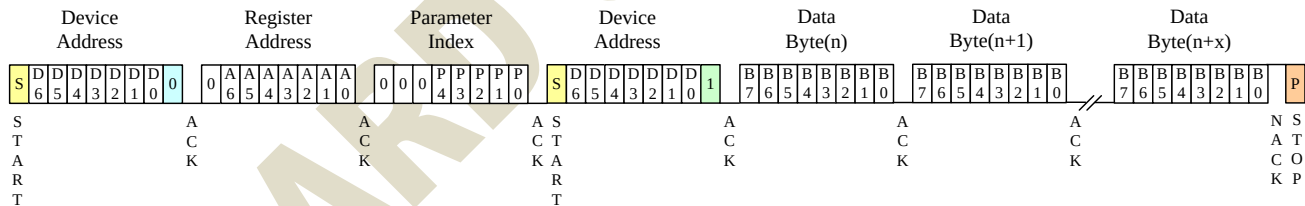


Figure 10.4: Multi Register Reading Timing

## 11. REGISTER FUNCTION

### 11.1 Register page selection

| X Address | Y Address | W/R | D7                      | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Default |
|-----------|-----------|-----|-------------------------|----|----|----|----|----|----|----|---------|
| 30        | 00        | W/R | register page selection |    |    |    |    |    |    |    | 00      |

R30h\_00

| Address | Bit                     |    |    |    |    |    |    |    |
|---------|-------------------------|----|----|----|----|----|----|----|
| R30h    | D7                      | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | register page selection |    |    |    |    |    |    |    |
| Default | 0                       | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

R30h[7:0]: Register page selection.

| R30h[7:0] | Function        | Note    |
|-----------|-----------------|---------|
| 00h       | Register page 0 | Default |
| 01h       | Register page 1 | -       |
| 02h       | Register page 2 | -       |
| 06h       | Register page 6 | -       |
| 07h       | Register page 7 | -       |
| 08h       | Register page 8 | -       |
| 09h       | Register page 9 | -       |
| 0Ah       | Register page A | -       |
| 0Dh       | Register page D | -       |

## 11.2 Page 0 command

| X Address | Y Address | W/R | D7      | D6 | D5 | D4 | D3  | D2 | D1 | D0    | Default |
|-----------|-----------|-----|---------|----|----|----|-----|----|----|-------|---------|
| 00        | 00        | W   | NOP     |    |    |    |     |    |    |       | --      |
| 01        | 00        | W   | SWRESET |    |    |    |     |    |    |       | --      |
| 10        | 00        | W   | SLPIN   |    |    |    |     |    |    |       | --      |
| 11        | 00        | W   | SLPOUT  |    |    |    |     |    |    |       | --      |
| 20        | 00        | W   | INVOFF  |    |    |    |     |    |    |       | --      |
| 21        | 00        | W   | INVON   |    |    |    |     |    |    |       | --      |
| 28        | 00        | W   | DISPOFF |    |    |    |     |    |    |       | --      |
| 29        | 00        | W   | DISPON  |    |    |    |     |    |    |       | --      |
| 34        | 00        | W   | tp_off  |    |    |    |     |    |    |       | --      |
| 35        | 00        | W   | -       | -  | -  | -  | -   | -  | -  | tp_on | 00      |
| 36        | 00        | W   | -       | -  | -  | -  | bgr | -  | ss | gs    | 00      |
| 7E        | 00        | W/R | -       | -  | -  | -  | -   | -  | -  | stbyb | 01      |
| F7        | 00        | W/R | key1    |    |    |    |     |    |    |       | 00      |
| F7        | 01        | W/R | key2    |    |    |    |     |    |    |       | 00      |
| F7        | 02        | W/R | key3    |    |    |    |     |    |    |       | 00      |
| F7        | 03        | W/R | key4    |    |    |    |     |    |    |       | 00      |

R00h: MIPI empty command (refer to 8.18.1 NOP section)

| Address | Bit |    |    |    |    |    |    |    |  |
|---------|-----|----|----|----|----|----|----|----|--|
| R00h    | D7  | D6 | D5 | D4 | D3 | D2 | D1 | D0 |  |
| Name    | NOP |    |    |    |    |    |    |    |  |

R01h: MIPI software reset command (refer to 8.18.2 SWRESET section)

| Address | Bit     |    |    |    |    |    |    |    |  |
|---------|---------|----|----|----|----|----|----|----|--|
| R01h    | D7      | D6 | D5 | D4 | D3 | D2 | D1 | D0 |  |
| Name    | SWRESET |    |    |    |    |    |    |    |  |

R10h: MIPI sleep in command (refer to 8.18.11 SLPIN section)

| Address | Bit   |    |    |    |    |    |    |    |  |
|---------|-------|----|----|----|----|----|----|----|--|
| R10h    | D7    | D6 | D5 | D4 | D3 | D2 | D1 | D0 |  |
| Name    | SLPIN |    |    |    |    |    |    |    |  |

R11h: MIPI sleep out command (refer to 8.18.12 SLPOUT section)

| Address | Bit     |    |    |    |    |    |    |    |  |
|---------|---------|----|----|----|----|----|----|----|--|
| R11h    | D7      | D6 | D5 | D4 | D3 | D2 | D1 | D0 |  |
| Name    | SLPOUT- |    |    |    |    |    |    |    |  |

R20h: This command makes no change of contents of image (refer to 8.18.13 INVOFF section)

| Address | Bit    |    |    |    |    |    |    |    |
|---------|--------|----|----|----|----|----|----|----|
| R20h    | D7     | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | INVOFF |    |    |    |    |    |    |    |

R21h: Every bit is inverted from the image to the display (refer to 8.18.14 INVON section)

| Address | Bit   |    |    |    |    |    |    |    |
|---------|-------|----|----|----|----|----|----|----|
| R21h    | D7    | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | INVON |    |    |    |    |    |    |    |

R28h: MIPI display off command (refer to 8.18.15 DISPOFF section)

| Address | Bit     |    |    |    |    |    |    |    |
|---------|---------|----|----|----|----|----|----|----|
| R28h    | D7      | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | DISPOFF |    |    |    |    |    |    |    |

R29h: MIPI display on command (refer to 8.18.16 DISPON section)

| Address | Bit    |    |    |    |    |    |    |    |
|---------|--------|----|----|----|----|----|----|----|
| R29h    | D7     | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | DISPON |    |    |    |    |    |    |    |

R34h: TP\_SYNC off command

| Address | Bit    |    |    |    |    |    |    |    |
|---------|--------|----|----|----|----|----|----|----|
| R34h    | D7     | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | tp_off |    |    |    |    |    |    |    |

R35h: TP\_SYNC on command

| Address | Bit |    |    |    |    |    |    |       |
|---------|-----|----|----|----|----|----|----|-------|
| R35h    | D7  | D6 | D5 | D4 | D3 | D2 | D1 | D0    |
| Name    | -   | -  | -  | -  | -  | -  | -  | tp_on |
| Default | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0     |

R35h[0]: Vertical scan direction.

| R35h[0] | Function                 | Note    |
|---------|--------------------------|---------|
| 0       | TP_SYNC = no signal      | Default |
| 1       | TP_SYNC = Internal Vsync | -       |



## R36h\_00

| Address | Bit |    |    |    |     |    |    |    |
|---------|-----|----|----|----|-----|----|----|----|
| R36h    | D7  | D6 | D5 | D4 | D3  | D2 | D1 | D0 |
| Name    | -   | -  | -  | -  | bgr | -  | ss | gs |
| Default | 0   | 0  | 0  | 0  | 0   | 0  | 0  | 0  |

R36h[0]: Vertical scan direction.

| R36h[0] | Function      | Note    |
|---------|---------------|---------|
| 0       | Top -> Bottom | Default |
| 1       | Bottom -> Top | -       |

Note: R36h[0] is only support for MIPI.

R36h[1]: Horizontal scan direction.

| R36h[1] | Function                           | Note    |
|---------|------------------------------------|---------|
| 0       | S0 -> S1-> ..... -> S1536 -> S1537 | Default |
| 1       | S1537 -> S1536 -> ..... -> S1 ->S0 | -       |

Note: R36h[1] is only support for MIPI.

R36h[3]: RGB-BGR order.

| R36h[3] | Function         | Note    |
|---------|------------------|---------|
| 0       | RGB color filter | Default |
| 1       | BGR color filter | -       |

Note: R36h[3] is only support for MIPI.

## R7Eh\_00

| Address | Bit |    |    |    |    |    |    |       |
|---------|-----|----|----|----|----|----|----|-------|
| R7Eh    | D7  | D6 | D5 | D4 | D3 | D2 | D1 | D0    |
| Name    | -   | -  | -  | -  | -  | -  | -  | stbyb |
| Default | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 1     |

R7Eh[0]: IC Standby mode selection.

| R7Eh[0] | Function       | Note    |
|---------|----------------|---------|
| 0       | Standby mode   | Default |
| 1       | Normal display | -       |

Note: R7Eh[0] is only support for TTL and LVDS.

## RF7h\_00/01/02/03

| Address | Bit  |    |    |    |    |    |    |    |
|---------|------|----|----|----|----|----|----|----|
| RF7h    | D7   | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | key1 |    |    |    |    |    |    |    |
| Name    | key2 |    |    |    |    |    |    |    |
| Name    | key3 |    |    |    |    |    |    |    |
| Name    | key4 |    |    |    |    |    |    |    |
| Default | 0    | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

RF7h\_00/01/02/03: IC inhouse command entry.

| RF7h | Function | Note |
|------|----------|------|
| 00   | 49h      | -    |
| 01   | 61h      | -    |
| 02   | 02h      | -    |
| 03   | 00h      | -    |

## 11.3 Page 1 command

| X Address | Y Address | W/R | D7                     | D6              | D5                 | D4               | D3             | D2                 | D1                       | D0              | Default |
|-----------|-----------|-----|------------------------|-----------------|--------------------|------------------|----------------|--------------------|--------------------------|-----------------|---------|
| 04        | 00        | W/R | r_nbw_sel_reg          | -               | -                  | -                | r_updn_sel_reg | r_shlr_sel_reg     | -                        | -               | 00      |
| 05        | 00        | W/R | r_lvbit_sel_req        | r_lvfmt_sel_req | -                  | -                | -              | -                  | -                        | r_bist_sel_req  | 00      |
| 06        | 00        | W/R |                        | r_bist          | r_nbw              | -                | -              | -                  | r_updn                   | r_shlr          | 00      |
| 07        | 00        | W/R | r_prog_hres[7:0]       |                 |                    |                  |                |                    |                          |                 | 00      |
| 08        | 00        | W/R | r_lvbit                | r_lvfmt         | -                  | -                | -              | r_prog_hres[10:8]  |                          |                 | 00      |
| 09        | 00        | W/R | r_prog_vres[7:0]       |                 |                    |                  |                |                    |                          |                 | 00      |
| 0A        | 00        | W/R | r_force_single_gate    | r_prog_vres_en  | r_prog_hres_en     | -                | -              | r_prog_vres[10:8]  |                          |                 | 00      |
| 0B        | 00        | W/R | -                      | r_res[2:0]      |                    |                  | -              | -                  | r_lansel[1:0]            |                 | 00      |
| 18        | 00        | W/R | IC_Version_ID[7:0]     |                 |                    |                  |                |                    |                          |                 | 02      |
| 18        | 01        | W/R | Panel_Version_ID1[7:0] |                 |                    |                  |                |                    |                          |                 | 00      |
| 18        | 02        | W/R | Panel_Version_ID2[7:0] |                 |                    |                  |                |                    |                          |                 | 00      |
| 1B        | 00        | W/R | -                      | -               | -                  | -                | -              | -                  | -                        | otp_cmd_program | 00      |
| 1F        | 00        | W/R | -                      | -               | -                  | r_hs_settle[4:0] |                |                    |                          |                 | 00      |
| 20        | 00        | W/R | -                      | -               | -                  | -                | -              | lansel_sel_req     | -                        | -               | 00      |
| 23        | 00        | W/R | -                      | -               | -                  | GAS_VDD_EN       | GAS_EN         | PSD_GAS_VDDA_EN    | -                        | -               | 3C      |
| 25        | 00        | W/R | -                      | -               | VCOM_LPC[2:0]      |                  |                | -                  | VCOM_MILLER_EN           | -               | 01      |
| 27        | 00        | W/R |                        |                 |                    |                  |                |                    | VCOM_DA[1:0]             |                 | AB      |
| 28        | 00        | W/R | -                      | -               | -                  | -                | -              | -                  | VCOM_GND_HIZ_ON_STB[1:0] |                 | 1A      |
| 29        | 00        | W/R | VCOMR_SEL[7:0]         |                 |                    |                  |                |                    |                          |                 | 04      |
| 2A        | 00        | W/R | -                      | -               | -                  | -                | -              | -                  | -                        | VCOMR_SEL[8]    | 01      |
| 2B        | 00        | W/R | VCOM_SEL[7:0]          |                 |                    |                  |                |                    |                          |                 | 04      |
| 2C        | 00        | W/R | -                      | -               | -                  | -                | -              | -                  | -                        | VCOM_SEL[8]     | 01      |
| 39        | 00        | W/R | -                      | r_ledpwm_en     | r_ledpwm_duty[2:0] |                  |                | r_ledpwm_freq[2:0] |                          |                 | 1C      |
| 3D        | 00        | W/R | -                      | -               | -                  | -                | -              | -                  | -                        | vcom_s          | 02      |

## R04h\_00

| Address | Bit           |    |    |    |                |                |    |    |
|---------|---------------|----|----|----|----------------|----------------|----|----|
| R04h    | D7            | D6 | D5 | D4 | D3             | D2             | D1 | D0 |
| Name    | r_nbw_sel_reg | -  | -  | -  | r_updn_sel_reg | r_shlr_sel_reg | -  | -  |
| Default | 0             | 0  | 0  | 0  | 0              | 0              | 0  | 0  |

R04h[2]: Horizontal scan direction selection.

| R04h[2] | Function                 | Note    |
|---------|--------------------------|---------|
| 0       | SHLR setting by H/W pin  | Default |
| 1       | SHLR setting by register | -       |

R04h[3]: Vertical scan direction selection.

| R04h[3] | Function                 | Note    |
|---------|--------------------------|---------|
| 0       | UPDN setting by H/W pin  | Default |
| 1       | UPDN setting by register | -       |

R04h[7]: Panel type selection.

| R04h[7] | Function               | Note    |
|---------|------------------------|---------|
| 0       | NBW setting by H/W pin | Default |
| 1       | NBW by register        | -       |

## R05h\_00

| Address | Bit             |                 |    |    |    |    |    |                |
|---------|-----------------|-----------------|----|----|----|----|----|----------------|
| R05h    | D7              | D6              | D5 | D4 | D3 | D2 | D1 | D0             |
| Name    | r_lvbit_sel_reg | r_lvfmt_sel_reg | -  | -  | -  | -  | -  | r_bist_sel_reg |
| Default | 0               | 0               | 0  | 0  | 0  | 0  | 0  | 0              |

R05h[0]: BIST mode selection.

| R05h[0] | Function                    | Note    |
|---------|-----------------------------|---------|
| 0       | BIST_EN setting by H/W pin  | Default |
| 1       | BIST_EN setting by register | -       |

R05h[6]: LVDS 8-bit input format selection.

| R05h[6] | Function                  | Note    |
|---------|---------------------------|---------|
| 0       | LVFMT setting by H/W pin  | Default |
| 1       | LVFMT setting by register | -       |

R05h[7]: LVDS 6-bit/8-bit input selection.

| R05h[7] | Function                  | Note    |
|---------|---------------------------|---------|
| 0       | LVBIT setting by H/W pin  | Default |
| 1       | LVBIT setting by register | -       |

## R06h\_00

| Address | Bit |        |       |    |    |    |        |        |
|---------|-----|--------|-------|----|----|----|--------|--------|
| R06h    | D7  | D6     | D5    | D4 | D3 | D2 | D1     | D0     |
| Name    | -   | r_bist | r_nbw | -  | -  | -  | r_updn | r_shlr |
| Default | 0   | 0      | 0     | 0  | 0  | 0  | 0      | 0      |

R06h[0]: Horizontal scan direction.

| R06h[0] | Function                           | Note    |
|---------|------------------------------------|---------|
| 0       | S0 -> S1-> ..... -> S1536 -> S1537 | Default |
| 1       | S1537 -> S1536 -> ..... -> S1 ->S0 | -       |

Note: R06h[0] is only support for TTL and LVDS.

R06h[1]: Vertical scan direction.

| R06h[1] | Function      | Note    |
|---------|---------------|---------|
| 0       | Top -> Bottom | Default |
| 1       | Bottom -> Top | -       |

Note: R06h[1] is only support for TTL and LVDS.

R06h[5]: Panel type selection.

| R06h[5] | Function             | Note    |
|---------|----------------------|---------|
| 0       | Normally black panel | Default |
| 1       | Normally white panel | -       |

R06h[6]: BIST mode selection.

| R06h[6] | Function    | Note    |
|---------|-------------|---------|
| 0       | Normal mode | Default |
| 1       | BIST mode   | -       |

## R07h\_00

| Address | Bit              |    |    |    |    |    |    |    |
|---------|------------------|----|----|----|----|----|----|----|
| R07h    | D7               | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | r_prog_hres[7:0] |    |    |    |    |    |    |    |
| Default | 0                | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

R07h[7:0]/R08h[2:0]: Free horizontal resolution selection.

Ex: 600RGB => R07h=58h/R08h=02h.

**R08h\_00**

| Address | Bit     |         |    |    |    |                   |    |    |
|---------|---------|---------|----|----|----|-------------------|----|----|
| R08h    | D7      | D6      | D5 | D4 | D3 | D2                | D1 | D0 |
| Name    | r_lvbit | r_lvfmt | -  | -  | -  | r_prog_hres[10:8] |    |    |
| Default | 0       | 0       | 0  | 0  | 0  | 0                 | 0  | 0  |

R08h[6]: LVDS 8-bit input format selection.

| R08h[6] | Function     | Note    |
|---------|--------------|---------|
| 0       | JEIDA format | Default |
| 1       | VESA format  | -       |

R08h[7]: LVDS 6-bit/8-bit input selection.

| R08h[7] | Function | Note    |
|---------|----------|---------|
| 0       | 6-bit    | Default |
| 1       | 8-bit    | -       |

**R09h\_00**

| Address | Bit              |    |    |    |    |    |    |    |
|---------|------------------|----|----|----|----|----|----|----|
| R09h    | D7               | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | r_prog_vres[7:0] |    |    |    |    |    |    |    |
| Default | 0                | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

R09h[7:0]/R0Ah[2:0]: Free vertical resolution selection.

Ex: 480line => R09h=E0h/R0Ah=01h

| Address | Bit                 |                |                |    |    |                   |    |    |
|---------|---------------------|----------------|----------------|----|----|-------------------|----|----|
| R0Ah    | D7                  | D6             | D5             | D4 | D3 | D2                | D1 | D0 |
| Name    | r_force_single_gate | r_prog_vres_en | r_prog_hres_en | -  | -  | r_prog_vres[10:8] |    |    |
| Default | 0                   | 0              | 0              | 0  | 0  | 0                 | 0  | 0  |

R0Ah[5]: Free horizontal resolution enable.

| R0Ah[5] | Function | Note    |
|---------|----------|---------|
| 0       | disable  | Default |
| 1       | enable   | -       |

R0Ah[6]: Free vertical resolution enable.

| R0Ah[6] | Function | Note    |
|---------|----------|---------|
| 0       | disable  | Default |
| 1       | enable   | -       |

R0Ah[7]: Dual gate / Single gate selection.

| R0Ah[7] | Function    | Note    |
|---------|-------------|---------|
| 0       | Dual gate   | Default |
| 1       | Single gate | -       |

R0Bh\_00

| Address | Bit |            |    |    |    |    |               |    |
|---------|-----|------------|----|----|----|----|---------------|----|
| R0Bh    | D7  | D6         | D5 | D4 | D3 | D2 | D1            | D0 |
| Name    | -   | r_res[2:0] |    |    | -  | -  | r_lansel[1:0] |    |
| Default | 0   | 0          | 0  | 0  | 0  | 0  | 0             | 0  |

R0Bh[1:0]: MIPI lane number selection.

| R0Bh[1:0] | Function | Note    |
|-----------|----------|---------|
| 00        | 1 lane   | Default |
| 01        | 2 lane   | -       |
| 10        | 3 lane   | -       |
| 11        | 4 lane   | -       |

Note: R20h[2]=1 set MIPI lane number function by register setting.

R0Bh[6:4]: Panel resolution selection.

| R0Bh[6:4] | Function      | Note    |
|-----------|---------------|---------|
| 000       | 1024RGB x 768 | Default |
| 001       | 1024RGB x 600 | -       |
| 010       | 800RGB x 600  | -       |
| 011       | 640RGB x 480  | -       |
| 100       | 480RGB x 272  | -       |
| 101       | T.B.D         | -       |
| 110       | T.B.D         | -       |
| 111       | T.B.D         | -       |

## R18h\_00: IC\_Version\_ID

| Address | Bit                |    |    |    |    |    |    |    |
|---------|--------------------|----|----|----|----|----|----|----|
| R18h    | D7                 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | IC_Version_ID[7:0] |    |    |    |    |    |    |    |
| Default | 0                  | 0  | 0  | 0  | 0  | 0  | 1  | 0  |

## R18h\_01: Panel\_Version\_ID1

| Address | Bit                    |    |    |    |    |    |    |    |
|---------|------------------------|----|----|----|----|----|----|----|
| R18h    | D7                     | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | Panel_Version_ID1[7:0] |    |    |    |    |    |    |    |
| Default | 0                      | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

## R18h\_02: IC\_Version\_ID2

| Address | Bit                    |    |    |    |    |    |    |    |
|---------|------------------------|----|----|----|----|----|----|----|
| R18h    | D7                     | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | Panel_Version_ID2[7:0] |    |    |    |    |    |    |    |
| Default | 0                      | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

Note: R18h\_00/01/02 is already OTP.

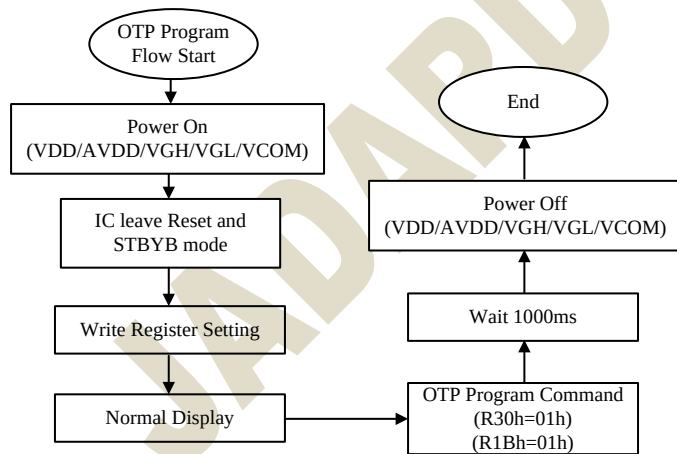
## R1Bh\_00

| Address | Bit |    |    |    |    |    |    |                 |
|---------|-----|----|----|----|----|----|----|-----------------|
| R1Bh    | D7  | D6 | D5 | D4 | D3 | D2 | D1 | D0              |
| Name    | -   | -  | -  | -  | -  | -  | -  | otp_cmd_program |
| Default | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0               |

R1Bh[0]: OTP program command.

| R1Bh[0] | Function | Note    |
|---------|----------|---------|
| 0       | disable  | Default |
| 1       | enable   | -       |

## OTP Flow:



## R1Fh\_00

| Address | Bit |    |    |                  |    |    |    |    |
|---------|-----|----|----|------------------|----|----|----|----|
| R1Fh    | D7  | D6 | D5 | D4               | D3 | D2 | D1 | D0 |
| Name    | -   | -  | -  | r_hs_settle[4:0] |    |    |    |    |
| Default | 0   | 0  | 0  | 0                | 0  | 0  | 0  | 0  |

R1Fh[4:0]: MIPI high speed settle time.

| R1Fh[4:0] | Function       | Note    |
|-----------|----------------|---------|
| 00h       | 100+8*UI(ns)   | Default |
| 01h       | 100+12*UI(ns)  | -       |
| 02h       | 100+16*UI(ns)  | -       |
| 03h       | 100+20*UI(ns)  | -       |
| 04h       | 100+24*UI(ns)  | -       |
| 05h       | 100+28*UI(ns)  | -       |
| :         | :              | -       |
| 10h       | 100+72*UI(ns)  | -       |
| 11h       | 100+76*UI(ns)  | -       |
| :         | :              | -       |
| 1Eh       | 100+128*UI(ns) | -       |
| 1Fh       | 100+132*UI(ns) | -       |

Ex: MIPI Speed=500Mbps, 1UI=2ns.

R20h\_00: MIPI lane number function.

| Address | Bit |    |    |    |    |                |    |    |
|---------|-----|----|----|----|----|----------------|----|----|
| R20h    | D7  | D6 | D5 | D4 | D3 | D2             | D1 | D0 |
| Name    | -   | -  | -  | -  | -  | lansel_sel_reg | -  | -  |
| Default | 0   | 0  | 0  | 0  | 0  | 0              | 0  | 0  |

R20h[2]: MIPI lane number function.

| R20h[2] | Function                             | Note    |
|---------|--------------------------------------|---------|
| 0       | MIPI lane number setting by H/W pin  | Default |
| 1       | MIPI lane number setting by register | -       |

## R23h\_00

| Address | Bit |    |    |            |        |                 |    |    |
|---------|-----|----|----|------------|--------|-----------------|----|----|
| R23h    | D7  | D6 | D5 | D4         | D3     | D2              | D1 | D0 |
| Name    | -   | -  | -  | GAS_VDD_EN | GAS_EN | PSD_GAS_VDDA_EN | -  | -  |
| Default | 0   | 0  | 1  | 1          | 1      | 1               | 0  | 0  |

R23h[2]: GAS detect by AVDD voltage

| R23h[2] | Function | Note    |
|---------|----------|---------|
| 0       | disable  | -       |
| 1       | enable   | Default |

R23h[3]: GAS function

| R23h[3] | Function | Note    |
|---------|----------|---------|
| 0       | disable  | -       |
| 1       | enable   | Default |

R23h[4]: GAS detect by VDD voltage

| R23h[4] | Function | Note    |
|---------|----------|---------|
| 0       | disable  | -       |
| 1       | enable   | Default |



## R25h\_00

| Address | Bit |    |               |    |    |    |                |    |
|---------|-----|----|---------------|----|----|----|----------------|----|
| R25h    | D7  | D6 | D5            | D4 | D3 | D2 | D1             | D0 |
| Name    | -   | -  | VCOM_LPC[2:0] |    |    | -  | VCOM_MILLER_EN | -  |
| Default | 0   | 0  | 0             | 0  | 0  | 0  | 0              | 1  |

R25h[1]: VCOM compensation method.

| R25h[1] | Function   | Note    |
|---------|------------|---------|
| 0       | millor off | Default |
| 1       | millor on  | -       |

R25h[5:3]: VCOM slew rate selection.

| R25h[5:3] | Function | Note    |
|-----------|----------|---------|
| 000       | 2us      | Default |
| 001       | 4us      | -       |
| 010       | 6us      | -       |
| 011       | 8us      | -       |
| 100       | 10us     | -       |
| 101       | 12us     | -       |
| 110       | 14us     | -       |
| 111       | 16us     | -       |

## R27h\_00

| Address | Bit |    |    |    |    |    |              |    |
|---------|-----|----|----|----|----|----|--------------|----|
| R27h    | D7  | D6 | D5 | D4 | D3 | D2 | D1           | D0 |
| Name    | -   | -  | -  | -  | -  | -  | VCOM_DA[1:0] |    |
| Default | 1   | 0  | 1  | 0  | 1  | 0  | 1            | 1  |

R27h[1:0]: VCOM output driving ability

| R27h[1:0] | Function        | Note    |
|-----------|-----------------|---------|
| 00        | Only Soft Start | -       |
| 01        | x1              | -       |
| 10        | x2              | -       |
| 11        | x3              | Default |

## R28h\_00

| Address | Bit |    |    |    |    |    |                          |    |
|---------|-----|----|----|----|----|----|--------------------------|----|
| R28h    | D7  | D6 | D5 | D4 | D3 | D2 | D1                       | D0 |
| Name    | -   | -  | -  | -  | -  | -  | VCOM_GND_HIZ_ON_STB[1:0] |    |
| Default | 0   | 0  | 0  | 1  | 1  | 0  | 1                        | 0  |

R28h[1:0]: VCOM voltage behavior in the Standby mode.

| R28h[1:0] | Function          | Note    |
|-----------|-------------------|---------|
| 00        | Discharge to VMID | -       |
| 01        | reserved          | -       |
| 10        | Hi-Z              | Default |
| 11        | reserved          | -       |

## R29h\_00

| Address | Bit            |    |    |    |    |    |    |    |
|---------|----------------|----|----|----|----|----|----|----|
| R29h    | D7             | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | VCOMR_SEL[7:0] |    |    |    |    |    |    |    |
| Default | 0              | 0  | 0  | 0  | 0  | 1  | 0  | 0  |

## R2Ah\_00

| Address | Bit |    |    |    |    |    |    |              |
|---------|-----|----|----|----|----|----|----|--------------|
| R2Ah    | D7  | D6 | D5 | D4 | D3 | D2 | D1 | D0           |
| Name    | -   | -  | -  | -  | -  | -  | -  | VCOMR_SEL[8] |
| Default | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 1            |

| R29h[7:0]/R2Ah[0]: VCOM voltage. (Panel bottom to top scan) |           |       |           |       |           |       |           |       |            |       |            |       |            |       |           |       |           |       |                    |  |  |  |  |  |
|-------------------------------------------------------------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|------------|-------|------------|-------|------------|-------|-----------|-------|-----------|-------|--------------------|--|--|--|--|--|
| 00000000 reserved                                           | 00010010  | 4.85  | 00101000  | 4.6   | 01001010  | 4.35  | 01001000  | 4.1   | 01111010   | 3.85  | 10010100   | 3.6   | 10101110   | 3.35  | 11001000  | 3.1   | 111000010 | 2.85  | 11111010 reserved  |  |  |  |  |  |
| 00000001 reserved                                           | 00010011  | 4.845 | 00100101  | 4.595 | 01001011  | 4.345 | 01001001  | 4.095 | 01111011   | 3.845 | 10010101   | 3.595 | 10101111   | 3.345 | 11001001  | 3.095 | 111000011 | 2.845 | 111110101 reserved |  |  |  |  |  |
| 00000010 reserved                                           | 00010100  | 4.84  | 00100110  | 4.59  | 01001100  | 4.34  | 01001010  | 4.09  | 01111100   | 3.84  | 10010110   | 3.59  | 10101000   | 3.34  | 11001010  | 3.09  | 111000100 | 2.84  | 111110110 reserved |  |  |  |  |  |
| 00000011 reserved                                           | 00010101  | 4.835 | 00100111  | 4.585 | 01001101  | 4.335 | 01001011  | 4.085 | 01111101   | 3.835 | 10010111   | 3.585 | 10101001   | 3.335 | 11001011  | 3.085 | 111000101 | 2.835 | 111110111 reserved |  |  |  |  |  |
| 000000100 reserved                                          | 00010110  | 4.83  | 00101000  | 4.58  | 010011010 | 4.33  | 010010100 | 4.08  | 01111110   | 3.83  | 100101000  | 3.58  | 10100100   | 3.33  | 110010100 | 3.08  | 111000110 | 2.83  | 111110100 reserved |  |  |  |  |  |
| 000000101 reserved                                          | 00010111  | 4.825 | 00101001  | 4.575 | 010011011 | 4.325 | 010010101 | 4.075 | 01111111   | 3.825 | 100101001  | 3.575 | 10100101   | 3.325 | 110010101 | 3.075 | 111000111 | 2.825 | 111110101 reserved |  |  |  |  |  |
| 000000110 reserved                                          | 00011000  | 4.82  | 00101010  | 4.57  | 010011100 | 4.32  | 01001110  | 4.07  | 100000000  | 3.82  | 100101010  | 3.57  | 101001000  | 3.32  | 110010110 | 3.07  | 111000100 | 2.82  | 111110110 reserved |  |  |  |  |  |
| 000000111 reserved                                          | 00011001  | 4.815 | 00101011  | 4.565 | 010011101 | 4.315 | 010011101 | 4.065 | 100000001  | 3.815 | 100101011  | 3.565 | 10100101   | 3.315 | 110010111 | 3.065 | 111000101 | 2.815 | 111110111 reserved |  |  |  |  |  |
| 00000100 reserved                                           | 00011010  | 4.81  | 00101100  | 4.56  | 010011110 | 4.31  | 01010000  | 4.06  | 100000010  | 3.81  | 100101010  | 3.56  | 10100110   | 3.31  | 110010100 | 3.06  | 111001010 | 2.81  | 111111000 reserved |  |  |  |  |  |
| 000001001 reserved                                          | 00011011  | 4.805 | 00101101  | 4.555 | 010011111 | 4.305 | 01010001  | 4.055 | 100000011  | 3.805 | 100101011  | 3.555 | 10100111   | 3.305 | 110010101 | 3.055 | 111001011 | 2.805 | 111111010 reserved |  |  |  |  |  |
| 000001010 reserved                                          | 00011100  | 4.8   | 00101110  | 4.55  | 010100000 | 4.3   | 01010010  | 4.05  | 100000100  | 3.8   | 100101010  | 3.55  | 10101000   | 3.3   | 110010100 | 3.05  | 111001100 | 2.8   | 111111010 reserved |  |  |  |  |  |
| 000001011 reserved                                          | 00011101  | 4.795 | 00101111  | 4.545 | 010100001 | 4.295 | 01010011  | 4.045 | 100000101  | 3.795 | 100101011  | 3.545 | 10101001   | 3.295 | 110010101 | 3.045 | 111001101 | 2.795 | 111111011 reserved |  |  |  |  |  |
| 00001100 reserved                                           | 00011110  | 4.79  | 00110000  | 4.54  | 010100100 | 4.29  | 010100100 | 4.04  | 100000110  | 3.79  | 100110000  | 3.54  | 10101010   | 3.29  | 11001100  | 3.04  | 111001100 | 2.79  |                    |  |  |  |  |  |
| 000011001 reserved                                          | 00011111  | 4.785 | 00110001  | 4.535 | 010100101 | 4.285 | 01010011  | 4.035 | 100000111  | 3.785 | 100110001  | 3.535 | 10101011   | 3.285 | 11001101  | 3.035 | 111001110 | 2.785 |                    |  |  |  |  |  |
| 00001110 reserved                                           | 00100000  | 4.78  | 00110010  | 4.53  | 010100100 | 4.28  | 010100110 | 4.03  | 100000100  | 3.78  | 100110100  | 3.53  | 10101100   | 3.28  | 11001110  | 3.03  | 111010000 | 2.78  |                    |  |  |  |  |  |
| 00001111 reserved                                           | 00100001  | 4.775 | 00110011  | 4.525 | 010100101 | 4.275 | 01010011  | 4.025 | 100000101  | 3.775 | 100110101  | 3.525 | 10101101   | 3.275 | 11001111  | 3.025 | 111010001 | 2.775 |                    |  |  |  |  |  |
| 00010000 reserved                                           | 00100010  | 4.77  | 00110100  | 4.52  | 010100110 | 4.27  | 010100100 | 4.02  | 1000001010 | 3.77  | 10011100   | 3.52  | 10101110   | 3.27  | 110100000 | 3.02  | 111010010 | 2.77  |                    |  |  |  |  |  |
| 00010001 reserved                                           | 00100011  | 4.765 | 00110101  | 4.515 | 010100111 | 4.265 | 01010011  | 4.015 | 100000111  | 3.765 | 10011101   | 3.515 | 10101111   | 3.265 | 110100001 | 3.015 | 111010011 | 2.765 |                    |  |  |  |  |  |
| 00010010 reserved                                           | 00100100  | 4.76  | 00110110  | 4.51  | 010100100 | 4.26  | 010101010 | 4.01  | 100001000  | 3.76  | 10011110   | 3.51  | 10110000   | 3.26  | 110100010 | 3.01  | 111010100 | 2.76  |                    |  |  |  |  |  |
| 00010011 reserved                                           | 00100101  | 4.755 | 00110111  | 4.505 | 010100101 | 4.255 | 01010101  | 4.005 | 100001001  | 3.755 | 10011111   | 3.505 | 10110001   | 3.255 | 110100011 | 3.005 | 111010101 | 2.755 |                    |  |  |  |  |  |
| 00010100 reserved                                           | 00100110  | 4.75  | 00110100  | 4.5   | 010101010 | 4.25  | 010101100 | 4     | 100001100  | 3.75  | 101000000  | 3.5   | 10110010   | 3.25  | 110100100 | 3     | 111010110 | 2.75  |                    |  |  |  |  |  |
| 00010101 reserved                                           | 00100111  | 4.745 | 00110101  | 4.495 | 010101011 | 4.245 | 010101101 | 3.995 | 100001111  | 3.745 | 101000001  | 3.495 | 10110011   | 3.245 | 110100101 | 2.995 | 111010111 | 2.745 |                    |  |  |  |  |  |
| 00010110 reserved                                           | 00101000  | 4.74  | 00110110  | 4.49  | 010101100 | 4.24  | 010101110 | 3.99  | 100010000  | 3.74  | 101000010  | 3.49  | 10110100   | 3.24  | 110100110 | 2.99  | 111010100 | 2.74  |                    |  |  |  |  |  |
| 00010111 reserved                                           | 00101001  | 4.735 | 00110111  | 4.485 | 010101011 | 4.235 | 01010111  | 3.985 | 100010001  | 3.735 | 101000011  | 3.485 | 10110101   | 3.235 | 110100111 | 2.985 | 11101001  | 2.735 |                    |  |  |  |  |  |
| 00011000 reserved                                           | 00101010  | 4.73  | 00111000  | 4.48  | 010101110 | 4.23  | 011100000 | 3.98  | 100010010  | 3.73  | 101000100  | 3.48  | 10110110   | 3.23  | 110101000 | 2.98  | 11101010  | 2.73  |                    |  |  |  |  |  |
| 00011001 reserved                                           | 00101011  | 4.725 | 00111001  | 4.475 | 010101111 | 4.225 | 011100001 | 3.975 | 100010011  | 3.725 | 101000101  | 3.475 | 10110111   | 3.225 | 110101001 | 2.975 | 11101011  | 2.725 |                    |  |  |  |  |  |
| 00011010 reserved                                           | 00101100  | 4.72  | 00111100  | 4.47  | 010100000 | 4.22  | 011100010 | 3.97  | 100010100  | 3.72  | 101000110  | 3.47  | 10111000   | 3.22  | 110101010 | 2.97  | 11101100  | 2.72  |                    |  |  |  |  |  |
| 00011011 reserved                                           | 00101101  | 4.715 | 00111101  | 4.465 | 010100001 | 4.215 | 011100011 | 3.965 | 100010101  | 3.715 | 101000111  | 3.465 | 10111001   | 3.215 | 110101011 | 2.965 | 11101101  | 2.715 |                    |  |  |  |  |  |
| 00011100 reserved                                           | 00100100  | 4.71  | 010000000 | 4.46  | 010100100 | 4.21  | 011100100 | 3.96  | 1000101010 | 3.71  | 1010001000 | 3.46  | 10111010   | 3.21  | 110101100 | 2.96  | 11101110  | 2.71  |                    |  |  |  |  |  |
| 00011101 reserved                                           | 00100101  | 4.705 | 010000001 | 4.455 | 010100101 | 4.205 | 011100101 | 3.955 | 1000101011 | 3.705 | 1010001001 | 3.455 | 10111011   | 3.205 | 110101101 | 2.955 | 11101111  | 2.705 |                    |  |  |  |  |  |
| 00011110 reserved                                           | 001001000 | 4.7   | 010000010 | 4.45  | 010100100 | 4.2   | 011100110 | 3.95  | 100010000  | 3.7   | 101000100  | 3.45  | 10111100   | 3.2   | 110101110 | 2.95  | 111100000 | 2.7   |                    |  |  |  |  |  |
| 00011111 reserved                                           | 001000001 | 4.695 | 010000011 | 4.445 | 010100101 | 4.195 | 01110011  | 3.945 | 100010001  | 3.695 | 101000101  | 3.445 | 10111101   | 3.195 | 11010111  | 2.945 | 111100001 |       | reserved           |  |  |  |  |  |
| 00100000 4.04                                               | 001001010 | 4.69  | 010000100 | 4.44  | 01010110  | 4.19  | 011100000 | 3.94  | 10001010   | 3.69  | 1010010    | 3.44  | 1011110    | 3.19  | 11010000  | 2.94  | 111100010 |       | reserved           |  |  |  |  |  |
| 00100001 4.935                                              | 001001011 | 4.685 | 010000101 | 4.435 | 01010111  | 4.185 | 01110001  | 3.935 | 10001011   | 3.685 | 10100101   | 3.435 | 10111111   | 3.185 | 11010001  | 2.935 | 111100011 |       | reserved           |  |  |  |  |  |
| 001000010 4.93                                              | 001010100 | 4.68  | 010000110 | 4.43  | 010110000 | 4.18  | 01110010  | 3.93  | 10001100   | 3.68  | 10100110   | 3.43  | 110000000  | 3.18  | 11010010  | 2.93  | 111100100 |       | reserved           |  |  |  |  |  |
| 001000011 4.925                                             | 001010101 | 4.675 | 010000111 | 4.425 | 010110001 | 4.175 | 01110011  | 3.925 | 100010101  | 3.675 | 10100111   | 3.425 | 110000001  | 3.175 | 110100101 | 2.925 | 111100101 |       | reserved           |  |  |  |  |  |
| 00100100 4.92                                               | 001010110 | 4.67  | 010001000 | 4.42  | 010110010 | 4.17  | 01110100  | 3.92  | 100011100  | 3.67  | 10101000   | 3.42  | 110000010  | 3.17  | 11010100  | 2.92  | 111100110 |       | reserved           |  |  |  |  |  |
| 00100101 4.915                                              | 001010111 | 4.665 | 010001001 | 4.415 | 010110011 | 4.165 | 01110101  | 3.915 | 100011101  | 3.665 | 10101001   | 3.415 | 110000011  | 3.165 | 11010101  | 2.915 | 111100111 |       | reserved           |  |  |  |  |  |
| 00100110 4.91                                               | 00101100  | 4.66  | 01010010  | 4.41  | 01011100  | 4.16  | 01110110  | 3.91  | 100100000  | 3.66  | 10100010   | 3.41  | 110000100  | 3.16  | 11010110  | 2.91  | 111101000 |       | reserved           |  |  |  |  |  |
| 001001011 4.905                                             | 001010101 | 4.655 | 010001011 | 4.405 | 01011101  | 4.155 | 01110111  | 3.905 | 100100001  | 3.655 | 101000101  | 3.405 | 1100000101 | 3.155 | 11010111  | 2.905 | 111101001 |       | reserved           |  |  |  |  |  |
| 00101000 4.9                                                | 00101010  | 4.65  | 010001100 | 4.4   | 01011110  | 4.15  | 011110000 | 3.9   | 100100010  | 3.65  | 10101000   | 3.4   | 110000110  | 3.15  | 110110000 | 2.9   | 111101010 |       | reserved           |  |  |  |  |  |
| 00101001 4.895                                              | 00101011  | 4.645 | 010001101 | 4.395 | 01011111  | 4.145 | 01111001  | 3.895 | 100100011  | 3.645 | 10101001   | 3.395 | 110000111  | 3.145 | 11011001  | 2.895 | 111101011 |       | reserved           |  |  |  |  |  |
| 00101010 4.89                                               | 00101100  | 4.64  | 010001100 | 4.39  | 011000000 | 4.14  | 011110010 | 3.89  | 100100100  | 3.64  | 10101010   | 3.39  | 110000100  | 3.14  | 11011010  | 2.89  | 111101100 |       | reserved           |  |  |  |  |  |
| 00101011 4.885                                              | 00101101  | 4.635 | 010001101 | 4.385 | 011000001 | 4.135 | 011110011 | 3.885 | 100100101  | 3.635 | 10101011   | 3.385 | 110000101  | 3.135 | 11011011  | 2.885 | 111101101 |       | reserved           |  |  |  |  |  |
| 00101100 4.88                                               | 00101100  | 4.63  | 010010000 | 4.38  | 011000010 | 4.13  | 011110010 | 3.88  | 100100110  | 3.63  | 10101000   | 3.38  | 110000100  | 3.13  | 11011100  | 2.88  | 111101100 |       | reserved           |  |  |  |  |  |
| 00101101 4.875                                              | 00101111  | 4.625 | 010010001 | 4.375 | 011000011 | 4.125 | 011110011 | 3.875 |            |       |            |       |            |       |           |       |           |       |                    |  |  |  |  |  |

R2Bh\_00

| Address | Bit           |    |    |    |    |    |    |    |
|---------|---------------|----|----|----|----|----|----|----|
| R2Bh    | D7            | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | VCOM_SEL[7:0] |    |    |    |    |    |    |    |
| Default | 0             | 0  | 0  | 0  | 0  | 1  | 0  | 0  |

R2Ch\_00

| Address | Bit |    |    |    |    |    |    |             |
|---------|-----|----|----|----|----|----|----|-------------|
| R2Ch    | D7  | D6 | D5 | D4 | D3 | D2 | D1 | D0          |
| Name    | -   | -  | -  | -  | -  | -  | -  | VCOM_SEL[8] |
| Default | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 1           |

| R2Bh[7:0]/R2Ch[0]: VCOM voltage. (Panel top to bottom scan) |           |           |            |           |           |           |           |           |            |            |            |            |           |            |           |           |           |           |                    |
|-------------------------------------------------------------|-----------|-----------|------------|-----------|-----------|-----------|-----------|-----------|------------|------------|------------|------------|-----------|------------|-----------|-----------|-----------|-----------|--------------------|
| 00000000 reserved                                           | 00011000  | 4.85      | 00100100   | 4.8       | 01001010  | 4.35      | 01001000  | 4.1       | 01111010   | 3.85       | 10010100   | 3.6        | 10101110  | 3.35       | 11001000  | 3.1       | 11100010  | 2.85      | 11111010 reserved  |
| 00000001 reserved                                           | 00011001  | 4.845     | 00100101   | 4.595     | 01001011  | 4.345     | 01001001  | 4.095     | 01111011   | 3.845      | 10010101   | 3.595      | 10101111  | 3.345      | 11001001  | 3.095     | 11100011  | 2.845     | 11111011 reserved  |
| 00000010 reserved                                           | 00011010  | 4.84      | 00100110   | 4.59      | 01001100  | 4.34      | 01001101  | 4.09      | 01111100   | 3.84       | 10010110   | 3.59       | 10110000  | 3.34       | 11001010  | 3.09      | 11100010  | 2.84      | 11111010 reserved  |
| 00000011 reserved                                           | 00011011  | 4.835     | 00100111   | 4.585     | 01001101  | 4.335     | 01001101  | 4.085     | 01111101   | 3.835      | 10010111   | 3.585      | 10110001  | 3.335      | 11001011  | 3.085     | 11100011  | 2.835     | 11111011 reserved  |
| 00000100 reserved                                           | 00011010  | 4.83      | 00101000   | 4.58      | 01001101  | 4.33      | 01001100  | 4.08      | 01111111   | 3.83       | 10011000   | 3.58       | 101100010 | 3.33       | 110010100 | 3.08      | 111000110 | 2.83      | 111110100 reserved |
| 00000101 reserved                                           | 00011011  | 4.825     | 00101001   | 4.575     | 01001101  | 4.325     | 01001101  | 4.075     | 01111111   | 3.825      | 10011001   | 3.575      | 101100011 | 3.325      | 110010101 | 3.075     | 111000111 | 2.825     | 111110101 reserved |
| 00000110 reserved                                           | 00011000  | 4.82      | 00101010   | 4.57      | 01001100  | 4.32      | 01001110  | 4.07      | 100000000  | 3.82       | 10010010   | 3.57       | 101000000 | 3.32       | 11001010  | 3.07      | 111000000 | 2.82      | 111110100 reserved |
| 00000111 reserved                                           | 00011001  | 4.815     | 00101011   | 4.565     | 01001101  | 4.315     | 01001111  | 4.065     | 100000001  | 3.815      | 100110011  | 3.565      | 10100001  | 3.315      | 110010111 | 3.065     | 11100001  | 2.815     | 111110101 reserved |
| 00001000 reserved                                           | 00011100  | 4.81      | 00101100   | 4.56      | 01001110  | 4.31      | 01010000  | 4.06      | 100000000  | 3.81       | 10010100   | 3.56       | 101001010 | 3.31       | 110011000 | 3.06      | 111001010 | 2.81      | 111111000 reserved |
| 00001001 reserved                                           | 00011101  | 4.805     | 00101101   | 4.555     | 01001111  | 4.305     | 01010001  | 4.055     | 100000001  | 3.805      | 10010101   | 3.555      | 101001011 | 3.305      | 110011001 | 3.055     | 111001011 | 2.805     | 111111001 reserved |
| 00001010 reserved                                           | 00011100  | 4.8       | 00101110   | 4.55      | 01010000  | 4.3       | 01010010  | 4.05      | 100000000  | 3.8        | 10010110   | 3.55       | 101010000 | 3.3        | 110011010 | 3.05      | 111001100 | 2.8       | 111111100 reserved |
| 00001011 reserved                                           | 00011101  | 4.795     | 00101111   | 4.545     | 01010001  | 4.295     | 01010011  | 4.045     | 100000001  | 3.795      | 10010111   | 3.545      | 10101001  | 3.295      | 110011011 | 3.045     | 111001101 | 2.795     | 111111101 reserved |
| 00001100 reserved                                           | 00011110  | 4.79      | 00101100   | 4.54      | 01010010  | 4.29      | 000000100 | 4.04      | 100000010  | 3.79       | 10011000   | 3.54       | 10101010  | 3.29       | 11001100  | 3.04      | 11100110  | 2.79      |                    |
| 00001101 reserved                                           | 00011111  | 4.785     | 00101101   | 4.535     | 01010011  | 4.285     | 01010101  | 4.035     | 100000011  | 3.785      | 10011001   | 3.535      | 10101011  | 3.285      | 11001101  | 3.035     | 11100111  | 2.785     |                    |
| 00001110 reserved                                           | 00100000  | 4.78      | 001011010  | 4.53      | 01010010  | 4.28      | 01010110  | 4.03      | 100000000  | 3.78       | 10011010   | 3.53       | 101010100 | 3.28       | 11001110  | 3.03      | 111000000 | 2.78      |                    |
| 00001111 reserved                                           | 00100001  | 4.775     | 001011011  | 4.525     | 01010011  | 4.275     | 01010111  | 4.025     | 100000001  | 3.775      | 10011011   | 3.525      | 101010111 | 3.275      | 11001111  | 3.025     | 11100001  | 2.775     |                    |
| 000010000 reserved                                          | 001000010 | 4.77      | 0010110100 | 4.52      | 010100110 | 4.27      | 010101000 | 4.02      | 100000010  | 3.77       | 10011100   | 3.52       | 10101010  | 3.27       | 110100000 | 3.02      | 111000010 | 2.77      |                    |
| 000010001 reserved                                          | 001000011 | 4.765     | 0010110101 | 4.515     | 010100111 | 4.265     | 01010101  | 4.015     | 100000011  | 3.765      | 10011101   | 3.515      | 101010111 | 3.265      | 110100001 | 3.015     | 111000011 | 2.765     |                    |
| 000010010 reserved                                          | 001000010 | 4.76      | 0010110110 | 4.51      | 010100100 | 4.26      | 010101010 | 4.01      | 100000100  | 3.76       | 10011110   | 3.51       | 101010000 | 3.26       | 110100010 | 3.01      | 111000100 | 2.76      |                    |
| 000010011 reserved                                          | 001000011 | 4.755     | 0010110111 | 4.505     | 010100101 | 4.255     | 010101011 | 4.005     | 100000101  | 3.755      | 10011111   | 3.505      | 101010001 | 3.255      | 110100011 | 3.005     | 111000101 | 2.755     |                    |
| 000010100 reserved                                          | 001000100 | 4.75      | 0010111000 | 4.5       | 010101010 | 4.25      | 001011100 | 4         | 100000110  | 3.75       | 101000000  | 3.5        | 101010010 | 3.25       | 110100100 | 3         | 111001010 | 2.75      |                    |
| 000010101 reserved                                          | 001000101 | 4.745     | 0010111001 | 4.495     | 010101011 | 4.245     | 010101101 | 3.995     | 100000111  | 3.745      | 101000001  | 3.495      | 101001011 | 3.245      | 110100101 | 2.995     | 111001011 | 2.745     |                    |
| 000010110 reserved                                          | 001000100 | 4.74      | 0010111010 | 4.49      | 010101100 | 4.24      | 010110110 | 3.99      | 100000000  | 3.74       | 101000010  | 3.49       | 101010100 | 3.24       | 110100110 | 2.99      | 111001000 | 2.74      |                    |
| 000010111 reserved                                          | 00100101  | 4.735     | 0010111011 | 4.485     | 010101101 | 4.235     | 001011111 | 3.985     | 100000001  | 3.735      | 101000011  | 3.485      | 101010101 | 3.235      | 110100111 | 2.985     | 111001001 | 2.735     |                    |
| 000011000 reserved                                          | 00101010  | 4.73      | 0010111100 | 4.48      | 010101110 | 4.23      | 011000000 | 3.98      | 100001010  | 3.73       | 101000100  | 3.48       | 101011010 | 3.23       | 110101000 | 2.98      | 111001010 | 2.73      |                    |
| 000011001 reserved                                          | 001010101 | 4.725     | 0010111101 | 4.475     | 010101111 | 4.225     | 001000001 | 3.975     | 100000001  | 3.725      | 101000101  | 3.475      | 101010111 | 3.225      | 110101001 | 2.975     | 111001011 | 2.725     |                    |
| 000011010 reserved                                          | 001010110 | 4.72      | 0010111110 | 4.47      | 010100000 | 4.22      | 011000010 | 3.97      | 100001010  | 3.72       | 1010001010 | 3.47       | 101011000 | 3.22       | 110101010 | 2.97      | 111001100 | 2.72      |                    |
| 000011011 reserved                                          | 001010111 | 4.715     | 0010111111 | 4.465     | 010100001 | 4.215     | 011000011 | 3.965     | 1000010101 | 3.715      | 1010001011 | 3.465      | 101011001 | 3.215      | 110101011 | 2.965     | 111001101 | 2.715     |                    |
| 000011100 reserved                                          | 001001100 | 4.71      | 010000000  | 4.46      | 010100000 | 4.21      | 011000000 | 3.96      | 1000010100 | 3.71       | 101000000  | 3.46       | 101011000 | 3.21       | 110010000 | 2.96      | 111001100 | 2.71      |                    |
| 000011101 reserved                                          | 001001101 | 4.705     | 010000001  | 4.455     | 010100001 | 4.205     | 011000001 | 3.955     | 1000010101 | 3.705      | 101000001  | 3.455      | 101011001 | 3.205      | 110010001 | 2.955     | 111001101 | 2.705     |                    |
| 000011110 reserved                                          | 001000000 | 4.7       | 010000010  | 4.45      | 010100010 | 4.2       | 011000010 | 3.95      | 1000010000 | 3.7        | 101000010  | 3.45       | 101011000 | 3.2        | 110010100 | 2.95      | 111000000 | 2.7       |                    |
| 000011111 reserved                                          | 001000001 | 4.695     | 010000011  | 4.445     | 010100011 | 4.195     | 011000011 | 3.945     | 1000010001 | 3.695      | 1010000101 | 3.445      | 101011001 | 3.195      | 110010111 | 2.945     | 111000001 | 2.695     | reserved           |
| 000100000                                                   | 4.64      | 001000010 | 4.49       | 010000000 | 4.44      | 010101010 | 4.19      | 011000000 | 3.94       | 1000010100 | 3.69       | 1010000100 | 3.44      | 101011000  | 3.19      | 110010000 | 2.94      | 111000000 | reserved           |
| 000100001                                                   | 4.635     | 001000011 | 4.485      | 010000001 | 4.435     | 010101011 | 4.185     | 011000001 | 3.935      | 1000010101 | 3.685      | 1010000101 | 3.435     | 101011001  | 3.185     | 110010001 | 2.935     | 111000001 | reserved           |
| 000100010                                                   | 4.63      | 001000010 | 4.48       | 010000010 | 4.43      | 010100000 | 4.18      | 011000010 | 3.93       | 1000010110 | 3.68       | 1010000110 | 3.43      | 1010000000 | 3.18      | 110010010 | 2.93      | 111000000 | reserved           |
| 000100011                                                   | 4.625     | 001000011 | 4.475      | 010000011 | 4.425     | 010100001 | 4.175     | 011000011 | 3.925      | 1000010111 | 3.675      | 1010000111 | 3.425     | 1010000001 | 3.175     | 110010011 | 2.925     | 111000001 | reserved           |
| 000100100                                                   | 4.62      | 001000010 | 4.42       | 010000000 | 4.42      | 010101010 | 4.17      | 011000000 | 3.92       | 100001110  | 3.67       | 101000000  | 3.42      | 1010000000 | 3.17      | 110010100 | 2.92      | 111000000 | reserved           |
| 000100101                                                   | 4.615     | 001000011 | 4.415      | 010000001 | 4.415     | 010101011 | 4.165     | 011000001 | 3.915      | 100001111  | 3.665      | 101000001  | 3.415     | 1010000001 | 3.165     | 110010101 | 2.915     | 111000001 | reserved           |
| 000100110                                                   | 4.61      | 001000010 | 4.41       | 010000000 | 4.41      | 010101100 | 4.16      | 011000000 | 3.91       | 100000000  | 3.66       | 101000010  | 3.41      | 1010000000 | 3.16      | 110010110 | 2.91      | 111000000 | reserved           |
| 000100111                                                   | 4.605     | 001000011 | 4.405      | 010000001 | 4.405     | 010101101 | 4.155     | 011000001 | 3.905      | 100000001  | 3.655      | 101000011  | 3.405     | 1010000001 | 3.155     | 110010111 | 2.905     | 111000001 | reserved           |
| 000101000                                                   | 4.6       | 001000010 | 4.4        | 010000000 | 4.4       | 010101110 | 4.15      | 011000000 | 3.9        | 100000000  | 3.65       | 101000010  | 3.4       | 1010000000 | 3.15      | 110010100 | 2.9       | 111000000 | reserved           |
| 000101001                                                   | 4.595     | 001000011 | 4.395      | 010000001 | 4.395     | 010101111 | 4.145     | 011000001 | 3.895      | 100000001  | 3.645      | 101000011  | 3.395     | 1010000001 | 3.145     | 110010101 | 2.895     | 111000001 | reserved           |
| 000101010                                                   | 4.59      | 001000010 | 4.39       | 010000000 | 4.39      | 010100000 | 4.14      | 011000000 | 3.89       | 100000000  | 3.64       | 101000010  | 3.39      | 1000000000 | 3.14      | 11001010  | 2.89      | 111000000 | reserved           |
| 000101011                                                   | 4.585     | 001000011 | 4.385      | 010000001 | 4.385     | 010100001 | 4.135     | 011000001 | 3.885      | 100000001  | 3.635      | 101000011  | 3.385     | 1000000001 | 3.135     | 110010101 | 2.885     | 111000001 | reserved           |
| 000101100                                                   | 4.58      | 001000010 | 4.38       | 010000000 | 4.38      | 010000000 | 4.13      | 011000000 | 3.88       | 100000000  | 3.63       | 101000000  | 3.38      | 1000000000 | 3.13      | 110010100 | 2.88      | 111000000 | reserved           |
| 000101101                                                   | 4.575     | 001000011 | 4.375      | 010000001 | 4.375     | 010000001 | 4.125     | 011000001 | 3.875      | 100000001  | 3.625      | 101000001  | 3.375     | 1000000011 | 3.125     | 110010101 | 2.875     | 111000001 | reserved           |
| 000101110                                                   | 4.57      | 001000000 | 4.37       | 010000000 | 4.37      | 010000000 | 4.12      | 011000000 | 3.87       | 100000000  | 3.62       | 101000000  | 3.37      | 1000000000 | 3.12      | 110010100 | 2.87      | 111000000 | reserved           |
| 000101111                                                   | 4.565     | 001000001 | 4.365      | 010000001 | 4.365     | 010000001 | 4.115     | 011000001 | 3.865      | 100000000  | 3.615      | 101000001  | 3.365     | 1000000001 | 3.115     | 110010101 | 2.865     | 111000001 | reserved           |
| 000110000                                                   | 4.56      | 001000010 | 4.36       | 010000000 | 4.36      | 010000000 | 4.11      | 011000000 | 3.86       | 100000000  | 3.61       | 101000000  | 3.        |            |           |           |           |           |                    |

R39h\_00

| Address | Bit |             |                    |    |    |                    |    |    |
|---------|-----|-------------|--------------------|----|----|--------------------|----|----|
| R39h    | D7  | D6          | D5                 | D4 | D3 | D2                 | D1 | D0 |
| Name    | -   | r_ledpwm_en | r_ledpwm_duty[2:0] |    |    | r_ledpwm_freq[2:0] |    |    |
| Default | 0   | 0           | 0                  | 1  | 1  | 1                  | 0  | 0  |

R39h[2:0]: LEDPWM frequency selection.

| R39h[2:0] | Function | Note    |
|-----------|----------|---------|
| 000       | 2kHz     | -       |
| 001       | 4kHz     | -       |
| 010       | 6kHz     | -       |
| 011       | 8kHz     | -       |
| 100       | 10kHz    | Default |
| 101       | 12kHz    | -       |
| 110       | 14kHz    | -       |
| 111       | 16kHz    | -       |

R39h[5:3]: LEDPWM duty selection.

| R39h[5:3] | Function | Note    |
|-----------|----------|---------|
| 000       | 15%      | -       |
| 001       | 25%      | -       |
| 010       | 35%      | -       |
| 011       | 50%      | Default |
| 100       | 65%      | -       |
| 101       | 75%      | -       |
| 110       | 85%      | -       |
| 111       | 100%     | -       |

R39h[6]: LEDPWM selection.

| R39h[6] | Function | Note    |
|---------|----------|---------|
| 0       | disable  | Default |
| 1       | enable   | -       |



## 11.4 Page 2 command

| X Address | Y Address | W/R | D7               | D6               | D5 | D4 | D3 | D2                   | D1               | D0     | Default |
|-----------|-----------|-----|------------------|------------------|----|----|----|----------------------|------------------|--------|---------|
| 00        | 00        | W/R | force_flg_clr_en | sub_flg_dis[1:0] |    |    |    | stv_ckv_dis[2:0]     | stv_ckv_s_tagger | GIP_DR | 05      |
| 0B        | 00        | W/R | -                | -                | -  |    |    | goutr1_sel_u2d[4:0]  |                  |        | 00      |
| 0B        | 01        | W/R | -                | -                | -  |    |    | goutr2_sel_u2d[4:0]  |                  |        | 00      |
| 0B        | 02        | W/R | -                | -                | -  |    |    | goutr3_sel_u2d[4:0]  |                  |        | 00      |
| 0B        | 03        | W/R | -                | -                | -  |    |    | goutr4_sel_u2d[4:0]  |                  |        | 00      |
| 0B        | 04        | W/R | -                | -                | -  |    |    | goutr5_sel_u2d[4:0]  |                  |        | 00      |
| 0B        | 05        | W/R | -                | -                | -  |    |    | goutr6_sel_u2d[4:0]  |                  |        | 00      |
| 0B        | 06        | W/R | -                | -                | -  |    |    | goutr7_sel_u2d[4:0]  |                  |        | 00      |
| 0B        | 07        | W/R | -                | -                | -  |    |    | goutr8_sel_u2d[4:0]  |                  |        | 00      |
| 0B        | 08        | W/R | -                | -                | -  |    |    | goutr9_sel_u2d[4:0]  |                  |        | 00      |
| 0B        | 09        | W/R | -                | -                | -  |    |    | goutr10_sel_u2d[4:0] |                  |        | 00      |
| 0B        | 0A        | W/R | -                | -                | -  |    |    | goutr11_sel_u2d[4:0] |                  |        | 00      |
| 0C        | 00        | W/R | -                | -                | -  |    |    | goutr12_sel_u2d[4:0] |                  |        | 00      |
| 0C        | 01        | W/R | -                | -                | -  |    |    | goutr13_sel_u2d[4:0] |                  |        | 00      |
| 0C        | 02        | W/R | -                | -                | -  |    |    | goutr14_sel_u2d[4:0] |                  |        | 00      |
| 0C        | 03        | W/R | -                | -                | -  |    |    | goutr15_sel_u2d[4:0] |                  |        | 00      |
| 0C        | 04        | W/R | -                | -                | -  |    |    | goutr16_sel_u2d[4:0] |                  |        | 00      |
| 0C        | 05        | W/R | -                | -                | -  |    |    | goutr17_sel_u2d[4:0] |                  |        | 00      |
| 0C        | 06        | W/R | -                | -                | -  |    |    | goutr18_sel_u2d[4:0] |                  |        | 00      |
| 0C        | 07        | W/R | -                | -                | -  |    |    | goutr19_sel_u2d[4:0] |                  |        | 00      |
| 0C        | 08        | W/R | -                | -                | -  |    |    | goutr20_sel_u2d[4:0] |                  |        | 00      |
| 0C        | 09        | W/R | -                | -                | -  |    |    | goutr21_sel_u2d[4:0] |                  |        | 00      |
| 0C        | 0A        | W/R | -                | -                | -  |    |    | goutr22_sel_u2d[4:0] |                  |        | 00      |
| 0D        | 00        | W/R | -                | -                | -  |    |    | goutl1_sel_u2d[4:0]  |                  |        | 00      |
| 0D        | 01        | W/R | -                | -                | -  |    |    | goutl2_sel_u2d[4:0]  |                  |        | 00      |
| 0D        | 02        | W/R | -                | -                | -  |    |    | goutl3_sel_u2d[4:0]  |                  |        | 00      |
| 0D        | 03        | W/R | -                | -                | -  |    |    | goutl4_sel_u2d[4:0]  |                  |        | 00      |
| 0D        | 04        | W/R | -                | -                | -  |    |    | goutl5_sel_u2d[4:0]  |                  |        | 00      |
| 0D        | 05        | W/R | -                | -                | -  |    |    | goutl6_sel_u2d[4:0]  |                  |        | 00      |
| 0D        | 06        | W/R | -                | -                | -  |    |    | goutl7_sel_u2d[4:0]  |                  |        | 00      |
| 0D        | 07        | W/R | -                | -                | -  |    |    | goutl8_sel_u2d[4:0]  |                  |        | 00      |
| 0D        | 08        | W/R | -                | -                | -  |    |    | goutl9_sel_u2d[4:0]  |                  |        | 00      |
| 0D        | 09        | W/R | -                | -                | -  |    |    | goutl10_sel_u2d[4:0] |                  |        | 00      |
| 0D        | 0A        | W/R | -                | -                | -  |    |    | goutl11_sel_u2d[4:0] |                  |        | 00      |
| 0E        | 00        | W/R | -                | -                | -  |    |    | goutl12_sel_u2d[4:0] |                  |        | 00      |
| 0E        | 01        | W/R | -                | -                | -  |    |    | goutl13_sel_u2d[4:0] |                  |        | 00      |
| 0E        | 02        | W/R | -                | -                | -  |    |    | goutl14_sel_u2d[4:0] |                  |        | 00      |
| 0E        | 03        | W/R | -                | -                | -  |    |    | goutl15_sel_u2d[4:0] |                  |        | 00      |
| 0E        | 04        | W/R | -                | -                | -  |    |    | goutl16_sel_u2d[4:0] |                  |        | 00      |

|    |    |     |   |   |   |                      |    |
|----|----|-----|---|---|---|----------------------|----|
| 0E | 05 | W/R | - | - | - | goutl17_sel_u2d[4:0] | 00 |
| 0E | 06 | W/R | - | - | - | goutl18_sel_u2d[4:0] | 00 |
| 0E | 07 | W/R | - | - | - | goutl19_sel_u2d[4:0] | 00 |
| 0E | 08 | W/R | - | - | - | goutl20_sel_u2d[4:0] | 00 |
| 0E | 09 | W/R | - | - | - | goutl21_sel_u2d[4:0] | 00 |
| 0E | 0A | W/R | - | - | - | goutl22_sel_u2d[4:0] | 00 |
| 0F | 00 | W/R | - | - | - | goutr1_sel_d2u[4:0]  | 00 |
| 0F | 01 | W/R | - | - | - | goutr2_sel_d2u[4:0]  | 00 |
| 0F | 02 | W/R | - | - | - | goutr3_sel_d2u[4:0]  | 00 |
| 0F | 03 | W/R | - | - | - | goutr4_sel_d2u[4:0]  | 00 |
| 0F | 04 | W/R | - | - | - | goutr5_sel_d2u[4:0]  | 00 |
| 0F | 05 | W/R | - | - | - | goutr6_sel_d2u[4:0]  | 00 |
| 0F | 06 | W/R | - | - | - | goutr7_sel_d2u[4:0]  | 00 |
| 0F | 07 | W/R | - | - | - | goutr8_sel_d2u[4:0]  | 00 |
| 0F | 08 | W/R | - | - | - | goutr9_sel_d2u[4:0]  | 00 |
| 0F | 09 | W/R | - | - | - | goutr10_sel_d2u[4:0] | 00 |
| 0F | 0A | W/R | - | - | - | goutr11_sel_d2u[4:0] | 00 |
| 10 | 00 | W/R | - | - | - | goutr12_sel_d2u[4:0] | 00 |
| 10 | 01 | W/R | - | - | - | goutr13_sel_d2u[4:0] | 00 |
| 10 | 02 | W/R | - | - | - | goutr14_sel_d2u[4:0] | 00 |
| 10 | 03 | W/R | - | - | - | goutr15_sel_d2u[4:0] | 00 |
| 10 | 04 | W/R | - | - | - | goutr16_sel_d2u[4:0] | 00 |
| 10 | 05 | W/R | - | - | - | goutr17_sel_d2u[4:0] | 00 |
| 10 | 06 | W/R | - | - | - | goutr18_sel_d2u[4:0] | 00 |
| 10 | 07 | W/R | - | - | - | goutr19_sel_d2u[4:0] | 00 |
| 10 | 08 | W/R | - | - | - | goutr20_sel_d2u[4:0] | 00 |
| 10 | 09 | W/R | - | - | - | goutr21_sel_d2u[4:0] | 00 |
| 10 | 0A | W/R | - | - | - | goutr22_sel_d2u[4:0] | 00 |
| 11 | 00 | W/R | - | - | - | goutl1_sel_d2u[4:0]  | 00 |
| 11 | 01 | W/R | - | - | - | goutl2_sel_d2u[4:0]  | 00 |
| 11 | 02 | W/R | - | - | - | goutl3_sel_d2u[4:0]  | 00 |
| 11 | 03 | W/R | - | - | - | goutl4_sel_d2u[4:0]  | 00 |
| 11 | 04 | W/R | - | - | - | goutl5_sel_d2u[4:0]  | 00 |
| 11 | 05 | W/R | - | - | - | goutl6_sel_d2u[4:0]  | 00 |
| 11 | 06 | W/R | - | - | - | goutl7_sel_d2u[4:0]  | 00 |
| 11 | 07 | W/R | - | - | - | goutl8_sel_d2u[4:0]  | 00 |
| 11 | 08 | W/R | - | - | - | goutl9_sel_d2u[4:0]  | 00 |
| 11 | 09 | W/R | - | - | - | goutl10_sel_d2u[4:0] | 00 |
| 11 | 0A | W/R | - | - | - | goutl11_sel_d2u[4:0] | 00 |
| 12 | 00 | W/R | - | - | - | goutl12_sel_d2u[4:0] | 00 |
| 12 | 01 | W/R | - | - | - | goutl13_sel_d2u[4:0] | 00 |

|    |    |     |                  |   |   |                      |    |
|----|----|-----|------------------|---|---|----------------------|----|
| 12 | 02 | W/R | -                | - | - | goutl14_sel_d2u[4:0] | 00 |
| 12 | 03 | W/R | -                | - | - | goutl15_sel_d2u[4:0] | 00 |
| 12 | 04 | W/R | -                | - | - | goutl16_sel_d2u[4:0] | 00 |
| 12 | 05 | W/R | -                | - | - | goutl17_sel_d2u[4:0] | 00 |
| 12 | 06 | W/R | -                | - | - | goutl18_sel_d2u[4:0] | 00 |
| 12 | 07 | W/R | -                | - | - | goutl19_sel_d2u[4:0] | 00 |
| 12 | 08 | W/R | -                | - | - | goutl20_sel_d2u[4:0] | 00 |
| 12 | 09 | W/R | -                | - | - | goutl21_sel_d2u[4:0] | 00 |
| 12 | 0A | W/R | -                | - | - | goutl22_sel_d2u[4:0] | 00 |
| 14 | 00 | W/R | gckv_rise_o[7:0] |   |   |                      | 00 |
| 14 | 01 | W/R | gckv_rise_e[7:0] |   |   |                      | 00 |
| 14 | 02 | W/R | gckv_fall_o[7:0] |   |   |                      | 41 |
| 14 | 03 | W/R | gckv_fall_e[7:0] |   |   |                      | 41 |



R00h\_00

| Address | Bit |    |    |                  |    |    |    |    |
|---------|-----|----|----|------------------|----|----|----|----|
| R00h    | D7  | D6 | D5 | D4               | D3 | D2 | D1 | D0 |
| Name    | -   | -  | -  | stv_ckv_dis[2:0] |    |    | -  | -  |
| Default | 0   | 0  | 0  | 0                | 0  | 1  | 0  | 1  |

R00h[4:2]: stv rising before ckv.

| R00h[4:2] | Function                      | Note    |
|-----------|-------------------------------|---------|
| 000       | stv rising before ckv 1 linne | -       |
| 001       | stv rising before ckv 2 linne | Default |
| 010       | stv rising before ckv 3 linne | -       |
| 011       | stv rising before ckv 4 linne | -       |
| 100       | stv rising before ckv 5 linne | -       |
| 101       | stv rising before ckv 6 linne | -       |
| 110       | stv rising before ckv 7 linne | -       |
| 111       | stv rising before ckv 8 linne | -       |

R0Bh\_00~0A: right side goa flexible select at u2d.

| Address | Bit |    |    |                      |    |    |    |    |
|---------|-----|----|----|----------------------|----|----|----|----|
| R0Bh    | D7  | D6 | D5 | D4                   | D3 | D2 | D1 | D0 |
| 00      | -   | -  | -  | goutr1_sel_u2d[4:0]  |    |    |    |    |
| 01      | -   | -  | -  | goutr2_sel_u2d[4:0]  |    |    |    |    |
| 02      | -   | -  | -  | goutr3_sel_u2d[4:0]  |    |    |    |    |
| 03      | -   | -  | -  | goutr4_sel_u2d[4:0]  |    |    |    |    |
| 04      | -   | -  | -  | goutr5_sel_u2d[4:0]  |    |    |    |    |
| 05      | -   | -  | -  | goutr6_sel_u2d[4:0]  |    |    |    |    |
| 06      | -   | -  | -  | goutr7_sel_u2d[4:0]  |    |    |    |    |
| 07      | -   | -  | -  | goutr8_sel_u2d[4:0]  |    |    |    |    |
| 08      | -   | -  | -  | goutr9_sel_u2d[4:0]  |    |    |    |    |
| 09      | -   | -  | -  | goutr10_sel_u2d[4:0] |    |    |    |    |
| 0A      | -   | -  | -  | goutr11_sel_u2d[4:0] |    |    |    |    |
| Default | 0   | 0  | 0  | 0                    | 0  | 0  | 0  | 0  |

R0Ch\_00~0A: right side goa flexible select at u2d.

| Address | Bit |    |    |                      |    |    |    |    |
|---------|-----|----|----|----------------------|----|----|----|----|
| R0Ch    | D7  | D6 | D5 | D4                   | D3 | D2 | D1 | D0 |
| 00      | -   | -  | -  | goutr12_sel_u2d[4:0] |    |    |    |    |
| 01      | -   | -  | -  | goutr13_sel_u2d[4:0] |    |    |    |    |
| 02      | -   | -  | -  | goutr14_sel_u2d[4:0] |    |    |    |    |
| 03      | -   | -  | -  | goutr15_sel_u2d[4:0] |    |    |    |    |
| 04      | -   | -  | -  | goutr16_sel_u2d[4:0] |    |    |    |    |
| 05      | -   | -  | -  | goutr17_sel_u2d[4:0] |    |    |    |    |
| 06      | -   | -  | -  | goutr18_sel_u2d[4:0] |    |    |    |    |
| 07      | -   | -  | -  | goutr19_sel_u2d[4:0] |    |    |    |    |
| 08      | -   | -  | -  | goutr20_sel_u2d[4:0] |    |    |    |    |
| 09      | -   | -  | -  | goutr21_sel_u2d[4:0] |    |    |    |    |
| 0A      | -   | -  | -  | goutr22_sel_u2d[4:0] |    |    |    |    |
| Default | 0   | 0  | 0  | 0                    | 0  | 0  | 0  | 0  |

R0Dh\_00~0A: left side goa flexible select at u2d.

| Address | Bit |    |    |                      |    |    |    |    |
|---------|-----|----|----|----------------------|----|----|----|----|
| R0Dh    | D7  | D6 | D5 | D4                   | D3 | D2 | D1 | D0 |
| 00      | -   | -  | -  | goutl1_sel_u2d[4:0]  |    |    |    |    |
| 01      | -   | -  | -  | goutl2_sel_u2d[4:0]  |    |    |    |    |
| 02      | -   | -  | -  | goutl3_sel_u2d[4:0]  |    |    |    |    |
| 03      | -   | -  | -  | goutl4_sel_u2d[4:0]  |    |    |    |    |
| 04      | -   | -  | -  | goutl5_sel_u2d[4:0]  |    |    |    |    |
| 05      | -   | -  | -  | goutl6_sel_u2d[4:0]  |    |    |    |    |
| 06      | -   | -  | -  | goutl7_sel_u2d[4:0]  |    |    |    |    |
| 07      | -   | -  | -  | goutl8_sel_u2d[4:0]  |    |    |    |    |
| 08      | -   | -  | -  | goutl9_sel_u2d[4:0]  |    |    |    |    |
| 09      | -   | -  | -  | goutl10_sel_u2d[4:0] |    |    |    |    |
| 0A      | -   | -  | -  | goutl11_sel_u2d[4:0] |    |    |    |    |
| Default | 0   | 0  | 0  | 0                    | 0  | 0  | 0  | 0  |

R0Eh\_00~0A: left side goa flexible select at u2d.

| Address | Bit |    |    |                      |    |    |    |    |
|---------|-----|----|----|----------------------|----|----|----|----|
| R0Eh    | D7  | D6 | D5 | D4                   | D3 | D2 | D1 | D0 |
| 00      | -   | -  | -  | goutl12_sel_u2d[4:0] |    |    |    |    |
| 01      | -   | -  | -  | goutl13_sel_u2d[4:0] |    |    |    |    |
| 02      | -   | -  | -  | goutl14_sel_u2d[4:0] |    |    |    |    |
| 03      | -   | -  | -  | goutl15_sel_u2d[4:0] |    |    |    |    |
| 04      | -   | -  | -  | goutl16_sel_u2d[4:0] |    |    |    |    |
| 05      | -   | -  | -  | goutl17_sel_u2d[4:0] |    |    |    |    |
| 06      | -   | -  | -  | goutl18_sel_u2d[4:0] |    |    |    |    |
| 07      | -   | -  | -  | goutl19_sel_u2d[4:0] |    |    |    |    |
| 08      | -   | -  | -  | goutl20_sel_u2d[4:0] |    |    |    |    |
| 09      | -   | -  | -  | goutl21_sel_u2d[4:0] |    |    |    |    |
| 0A      | -   | -  | -  | goutl22_sel_u2d[4:0] |    |    |    |    |
| Default | 0   | 0  | 0  | 0                    | 0  | 0  | 0  | 0  |

R0Fh\_00~0A: right side goa flexible select at d2u.

| Address | Bit |    |    |                      |    |    |    |    |
|---------|-----|----|----|----------------------|----|----|----|----|
| R0Fh    | D7  | D6 | D5 | D4                   | D3 | D2 | D1 | D0 |
| 00      | -   | -  | -  | goutr1_sel_du2[4:0]  |    |    |    |    |
| 01      | -   | -  | -  | goutr2_sel_du2[4:0]  |    |    |    |    |
| 02      | -   | -  | -  | goutr3_sel_du2[4:0]  |    |    |    |    |
| 03      | -   | -  | -  | goutr4_sel_du2[4:0]  |    |    |    |    |
| 04      | -   | -  | -  | goutr5_sel_d2u[4:0]  |    |    |    |    |
| 05      | -   | -  | -  | goutr6_sel_d2u[4:0]  |    |    |    |    |
| 06      | -   | -  | -  | goutr7_sel_d2u[4:0]  |    |    |    |    |
| 07      | -   | -  | -  | goutr8_sel_d2u[4:0]  |    |    |    |    |
| 08      | -   | -  | -  | goutr9_sel_d2u[4:0]  |    |    |    |    |
| 09      | -   | -  | -  | goutr10_sel_d2u[4:0] |    |    |    |    |
| 0A      | -   | -  | -  | goutr11_sel_d2u[4:0] |    |    |    |    |
| Default | 0   | 0  | 0  | 0                    | 0  | 0  | 0  | 0  |

R10h\_00~0A: right side goa flexible select at d2u.

| Address | Bit |    |    |                      |    |    |    |    |
|---------|-----|----|----|----------------------|----|----|----|----|
| R10h    | D7  | D6 | D5 | D4                   | D3 | D2 | D1 | D0 |
| 00      | -   | -  | -  | goutr12_sel_d2u[4:0] |    |    |    |    |
| 01      | -   | -  | -  | goutr13_sel_d2u[4:0] |    |    |    |    |
| 02      | -   | -  | -  | goutr14_sel_d2u[4:0] |    |    |    |    |
| 03      | -   | -  | -  | goutr15_sel_d2u[4:0] |    |    |    |    |
| 04      | -   | -  | -  | goutr16_sel_d2u[4:0] |    |    |    |    |
| 05      | -   | -  | -  | goutr17_sel_d2u[4:0] |    |    |    |    |
| 06      | -   | -  | -  | goutr18_sel_d2u[4:0] |    |    |    |    |
| 07      | -   | -  | -  | goutr19_sel_d2u[4:0] |    |    |    |    |
| 08      | -   | -  | -  | goutr20_sel_d2u[4:0] |    |    |    |    |
| 09      | -   | -  | -  | goutr21_sel_d2u[4:0] |    |    |    |    |
| 0A      | -   | -  | -  | goutr22_sel_d2u[4:0] |    |    |    |    |
| Default | 0   | 0  | 0  | 0                    | 0  | 0  | 0  | 0  |

R11h\_00~0A: left side goa flexible select at d2u.

| Address | Bit |    |    |                      |    |    |    |    |
|---------|-----|----|----|----------------------|----|----|----|----|
| R11h    | D7  | D6 | D5 | D4                   | D3 | D2 | D1 | D0 |
| 00      | -   | -  | -  | goutl1_sel_du2[4:0]  |    |    |    |    |
| 01      | -   | -  | -  | goutl2_sel_du2[4:0]  |    |    |    |    |
| 02      | -   | -  | -  | goutl3_sel_du2[4:0]  |    |    |    |    |
| 03      | -   | -  | -  | goutl4_sel_du2[4:0]  |    |    |    |    |
| 04      | -   | -  | -  | goutl5_sel_d2u[4:0]  |    |    |    |    |
| 05      | -   | -  | -  | goutl6_sel_d2u[4:0]  |    |    |    |    |
| 06      | -   | -  | -  | goutl7_sel_d2u[4:0]  |    |    |    |    |
| 07      | -   | -  | -  | goutl8_sel_d2u[4:0]  |    |    |    |    |
| 08      | -   | -  | -  | goutl9_sel_d2u[4:0]  |    |    |    |    |
| 09      | -   | -  | -  | goutl10_sel_d2u[4:0] |    |    |    |    |
| 0A      | -   | -  | -  | goutl11_sel_d2u[4:0] |    |    |    |    |
| Default | 0   | 0  | 0  | 0                    | 0  | 0  | 0  | 0  |

R12h\_00~0A: left side goa flexible select at d2u.

| Address | Bit |    |    |                      |    |    |    |    |
|---------|-----|----|----|----------------------|----|----|----|----|
| R12h    | D7  | D6 | D5 | D4                   | D3 | D2 | D1 | D0 |
| 00      | -   | -  | -  | goutl12_sel_d2u[4:0] |    |    |    |    |
| 01      | -   | -  | -  | goutl13_sel_d2u[4:0] |    |    |    |    |
| 02      | -   | -  | -  | goutl14_sel_d2u[4:0] |    |    |    |    |
| 03      | -   | -  | -  | goutl15_sel_d2u[4:0] |    |    |    |    |
| 04      | -   | -  | -  | goutl16_sel_d2u[4:0] |    |    |    |    |
| 05      | -   | -  | -  | goutl17_sel_d2u[4:0] |    |    |    |    |
| 06      | -   | -  | -  | goutl18_sel_d2u[4:0] |    |    |    |    |
| 07      | -   | -  | -  | goutl19_sel_d2u[4:0] |    |    |    |    |
| 08      | -   | -  | -  | goutl20_sel_d2u[4:0] |    |    |    |    |
| 09      | -   | -  | -  | goutl21_sel_d2u[4:0] |    |    |    |    |
| 0A      | -   | -  | -  | goutl22_sel_d2u[4:0] |    |    |    |    |
| Default | 0   | 0  | 0  | 0                    | 0  | 0  | 0  | 0  |

Flexible GIP output mapping table.

| goutr/ <a href="#">ln_sel_u2d[4:0]</a><br>goutr/ <a href="#">ln_sel_d2u[4:0]</a><br>n=1~22 | output | goutr/ <a href="#">ln_sel_u2d[4:0]</a><br>goutr/ <a href="#">ln_sel_d2u[4:0]</a><br>n=1~22 | output   | goutr/ <a href="#">ln_sel_u2d[4:0]</a><br>goutr/ <a href="#">ln_sel_d2u[4:0]</a><br>n=1~22 | output |
|--------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------|--------|
| 0                                                                                          | GND    | 10                                                                                         | FLCA     | 20                                                                                         | STV5   |
| 1                                                                                          | CSV    | 11                                                                                         | FLCB     | 21                                                                                         | STV6   |
| 2                                                                                          | CSV_N  | 12                                                                                         | Sub-FLCA | 22                                                                                         | STV7   |
| 3                                                                                          | GAS    | 13                                                                                         | Sub-FLCB | 23                                                                                         | STV8   |
| 4                                                                                          | GAS_N  | 14                                                                                         | GCH      | 24                                                                                         | CKV9   |
| 5                                                                                          | RST1   | 15                                                                                         | GCL      | 25                                                                                         | CKV10  |
| 6                                                                                          | CKV4   | 16                                                                                         | STV2     | 26                                                                                         | CKV11  |
| 7                                                                                          | CKV3   | 17                                                                                         | RST2     | 27                                                                                         | CKV12  |
| 8                                                                                          | CKV2   | 18                                                                                         | STV3     | 28                                                                                         | CKV13  |
| 9                                                                                          | CKV1   | 19                                                                                         | STV4     | 29                                                                                         | CKV14  |
| A                                                                                          | STV1   | 1A                                                                                         | RST3     | 2A                                                                                         | CKV15  |
| B                                                                                          | VGH    | 1B                                                                                         | RST4     | 2B                                                                                         | CKV16  |
| C                                                                                          | VGL    | 1C                                                                                         | CKV5     | 2C                                                                                         | RST5   |
| D                                                                                          | VGL    | 1D                                                                                         | CKV6     | 2D                                                                                         | RST6   |
| E                                                                                          | VGL    | 1E                                                                                         | CKV7     | 2E                                                                                         | RST7   |
| F                                                                                          | VRSET  | 1F                                                                                         | CKV8     | 2F                                                                                         | RST8   |

Note: GAS=VGL@Normal display ; GAS=VGH@GAS event.

14h\_00~01[7:0]: ckv rising edge adjustment.

| Address | Bit              |    |    |    |    |    |    |    |
|---------|------------------|----|----|----|----|----|----|----|
| R14h    | D7               | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 00      | gckv_rise_o[7:0] |    |    |    |    |    |    |    |
| 01      | gckv_rise_e[7:0] |    |    |    |    |    |    |    |
| Default | 0                | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

R14h\_00[7:0]: odd ckv rising edge fine tune.

| R14h_00 | Function                     | Note    |
|---------|------------------------------|---------|
| 00h     | odd stv rising shift 0 clk   | Default |
| 01h     | odd stv rising shift 1 clk   | -       |
| 02h     | odd stv rising shift 2 clk   | -       |
| :       | :                            | -       |
| 41h     | odd stv rising shift 65 clk  | -       |
| 42h     | odd stv rising shift 66 clk  | -       |
| :       | :                            | -       |
| FEh     | odd stv rising shift 254 clk | -       |
| FFh     | odd stv rising shift 255 clk | -       |

R14h\_01[7:0]: even ckv rising edge fine tune.

| R14h_01 | Function                     | Note    |
|---------|------------------------------|---------|
| 00h     | odd stv rising shift 0 clk   | Default |
| 01h     | odd stv rising shift 1 clk   | -       |
| 02h     | odd stv rising shift 2 clk   | -       |
| :       | :                            | -       |
| 41h     | odd stv rising shift 65 clk  | -       |
| 42h     | odd stv rising shift 66 clk  | -       |
| :       | :                            | -       |
| FEh     | odd stv rising shift 254 clk | -       |
| FFh     | odd stv rising shift 255 clk | -       |

14h\_02~03[7:0]: ckv falling edge adjustment.

| Address | Bit              |    |    |    |    |    |    |    |
|---------|------------------|----|----|----|----|----|----|----|
| R14h    | D7               | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 02      | gckv_fall_o[7:0] |    |    |    |    |    |    |    |
| 03      | gckv_fall_e[7:0] |    |    |    |    |    |    |    |
| Default | 0                | 1  | 0  | 0  | 0  | 0  | 0  | 1  |

R14h\_02[7:0]: odd ckv falling edge fine tune.

| R14h_02 | Function                     | Note    |
|---------|------------------------------|---------|
| 00h     | odd stv rising shift 0 clk   | -       |
| 01h     | odd stv rising shift 1 clk   | -       |
| 02h     | odd stv rising shift 2 clk   | -       |
| :       | :                            | -       |
| 41h     | odd stv rising shift 65 clk  | Default |
| 42h     | odd stv rising shift 66 clk  | -       |
| :       | :                            | -       |
| FEh     | odd stv rising shift 254 clk | -       |
| FFh     | odd stv rising shift 255 clk | -       |

R14h\_03[7:0]: even ckv falling edge fine tune.

| R14h_03 | Function                     | Note    |
|---------|------------------------------|---------|
| 00h     | odd stv rising shift 0 clk   | -       |
| 01h     | odd stv rising shift 1 clk   | -       |
| 02h     | odd stv rising shift 2 clk   | -       |
| :       | :                            | -       |
| 41h     | odd stv rising shift 65 clk  | Default |
| 42h     | odd stv rising shift 66 clk  | -       |
| :       | :                            | -       |
| FEh     | odd stv rising shift 254 clk | -       |
| FFh     | odd stv rising shift 255 clk | -       |



## 11.5 Page 6 command

| X Address | Y Address | W/R | D7 | D6 | D5                 | D4 | D3 | D2 | D1 | D0 | Default |
|-----------|-----------|-----|----|----|--------------------|----|----|----|----|----|---------|
| 12        | 00        | W/R | -  | -  | REG_GA_GSH255[5:0] |    |    |    |    |    | 3F      |
| 12        | 01        | W/R | -  | -  | REG_GA_GSH248[5:0] |    |    |    |    |    | 2E      |
| 12        | 02        | W/R | -  | -  | REG_GA_GSH236[5:0] |    |    |    |    |    | 28      |
| 12        | 03        | W/R | -  | -  | REG_GA_GSH220[5:0] |    |    |    |    |    | 20      |
| 12        | 04        | W/R | -  | -  | REG_GA_GSH200[5:0] |    |    |    |    |    | 12      |
| 12        | 05        | W/R | -  | -  | REG_GA_GSH172[5:0] |    |    |    |    |    | 02      |
| 12        | 06        | W/R | -  | -  | REG_GA_GSH144[5:0] |    |    |    |    |    | 1E      |
| 12        | 07        | W/R | -  | -  | REG_GA_GSH116[5:0] |    |    |    |    |    | 20      |
| 12        | 08        | W/R | -  | -  | REG_GA_GSH88[5:0]  |    |    |    |    |    | 32      |
| 12        | 09        | W/R | -  | -  | REG_GA_GSH60[5:0]  |    |    |    |    |    | 32      |
| 12        | 0A        | W/R | -  | -  | REG_GA_GSH32[5:0]  |    |    |    |    |    | 28      |
| 12        | 0B        | W/R | -  | -  | REG_GA_GSH16[5:0]  |    |    |    |    |    | 22      |
| 12        | 0C        | W/R | -  | -  | REG_GA_GSH8[5:0]   |    |    |    |    |    | 1C      |
| 12        | 0D        | W/R | -  | -  | REG_GA_GSH0[5:0]   |    |    |    |    |    | 00      |
| 13        | 00        | W/R | -  | -  | REG_GA_GSL255[5:0] |    |    |    |    |    | 3F      |
| 13        | 01        | W/R | -  | -  | REG_GA_GSL248[5:0] |    |    |    |    |    | 2E      |
| 13        | 02        | W/R | -  | -  | REG_GA_GSL236[5:0] |    |    |    |    |    | 28      |
| 13        | 03        | W/R | -  | -  | REG_GA_GSL220[5:0] |    |    |    |    |    | 20      |
| 13        | 04        | W/R | -  | -  | REG_GA_GSL200[5:0] |    |    |    |    |    | 12      |
| 13        | 05        | W/R | -  | -  | REG_GA_GSL172[5:0] |    |    |    |    |    | 02      |
| 13        | 06        | W/R | -  | -  | REG_GA_GSL144[5:0] |    |    |    |    |    | 1E      |
| 13        | 07        | W/R | -  | -  | REG_GA_GSL116[5:0] |    |    |    |    |    | 20      |
| 13        | 08        | W/R | -  | -  | REG_GA_GSL88[5:0]  |    |    |    |    |    | 32      |
| 13        | 09        | W/R | -  | -  | REG_GA_GSL60[5:0]  |    |    |    |    |    | 32      |
| 13        | 0A        | W/R | -  | -  | REG_GA_GSL32[5:0]  |    |    |    |    |    | 28      |
| 13        | 0B        | W/R | -  | -  | REG_GA_GSL16[5:0]  |    |    |    |    |    | 22      |
| 13        | 0C        | W/R | -  | -  | REG_GA_GSL8[5:0]   |    |    |    |    |    | 1C      |
| 13        | 0D        | W/R | -  | -  | REG_GA_GSL0[5:0]   |    |    |    |    |    | 00      |

## R12h\_00

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH255[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                  | 1  | 1  | 1  | 1  | 1  |

R12h[5:0]: VP positive voltage(Normally black panel).

| R12h[5:0] | Function  | Note    |
|-----------|-----------|---------|
| 3Eh       | VP - 23mV | -       |
| 3Fh       | VP        | Default |

## R12h\_01

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH248[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                  | 0  | 1  | 1  | 1  | 0  |

R12h[5:0]: Gray248 positive voltage(Normally black panel).

| R12h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 2Dh       | Gray248 - 23mV |         |
| 2Eh       | Gray248        | Default |
| 2Fh       | Gray248 + 23mV | -       |

## R12h\_02

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH236[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                  | 0  | 1  | 0  | 0  | 0  |

R12h[5:0]: Gray236 positive voltage(Normally black panel).

| R12h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 27h       | Gray236 - 23mV |         |
| 28h       | Gray236        | Default |
| 29h       | Gray236 + 23mV | -       |

## R12h\_03

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH220[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                  | 0  | 0  | 0  | 0  | 0  |

R12h[5:0]: Gray220 positive voltage(Normally black panel).

| R12h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 1Fh       | Gray220 - 23mV |         |
| 20h       | Gray220        | Default |
| 21h       | Gray220 + 23mV | -       |

## R12h\_04

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH200[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 0                  | 1  | 0  | 0  | 1  | 0  |

R12h[5:0]: Gray200 positive voltage(Normally black panel).

| R12h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 11h       | Gray200 - 23mV |         |
| 12h       | Gray200        | Default |
| 13h       | Gray200 + 23mV | -       |

## R12h\_05

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH172[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 0                  | 0  | 0  | 0  | 1  | 0  |

R12h[5:0]: Gray172 positive voltage(Normally black panel).

| R12h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 01h       | Gray172 - 23mV |         |
| 02h       | Gray172        | Default |
| 03h       | Gray172 + 23mV | -       |

## R12h\_06

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH144[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 0                  | 1  | 1  | 1  | 1  | 0  |

R12h[5:0]: Gray144 positive voltage(Normally black panel).

| R12h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 1Dh       | Gray144 - 23mV |         |
| 1Eh       | Gray144        | Default |
| 1Fh       | Gray144 + 23mV | -       |

## R12h\_07

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH116[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                  | 0  | 0  | 0  | 0  | 0  |

R12h[5:0]: Gray116 positive voltage(Normally black panel).

| R12h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 1Fh       | Gray116 - 23mV |         |
| 20h       | Gray116        | Default |
| 21h       | Gray116 + 23mV | -       |

## R12h\_08

| Address | Bit |    |                   |    |    |    |    |    |
|---------|-----|----|-------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH88[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                 | 1  | 0  | 0  | 1  | 0  |

R12h[5:0]: Gray88 positive voltage(Normally black panel).

| R12h[5:0] | Function      | Note    |
|-----------|---------------|---------|
| 31h       | Gray88 - 23mV |         |
| 32h       | Gray88        | Default |
| 33h       | Gray88 + 23mV | -       |

## R12h\_09

| Address | Bit |    |                   |    |    |    |    |    |
|---------|-----|----|-------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH60[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                 | 1  | 0  | 0  | 1  | 0  |

R12h[5:0]: Gray60 positive voltage(Normally black panel).

| R12h[5:0] | Function      | Note    |
|-----------|---------------|---------|
| 31h       | Gray60 - 23mV |         |
| 32h       | Gray60        | Default |
| 33h       | Gray60 + 23mV | -       |

## R12h\_0A

| Address | Bit |    |                   |    |    |    |    |    |
|---------|-----|----|-------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH32[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                 | 0  | 1  | 0  | 0  | 0  |

R12h[5:0]: Gray32 positive voltage(Normally black panel).

| R12h[5:0] | Function      | Note    |
|-----------|---------------|---------|
| 27h       | Gray32 - 23mV |         |
| 28h       | Gray32        | Default |
| 29h       | Gray32 + 23mV | -       |

## R12h\_0B

| Address | Bit |    |                   |    |    |    |    |    |
|---------|-----|----|-------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5                | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH16[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                 | 0  | 0  | 0  | 1  | 0  |

R12h[5:0]: Gray16 positive voltage(Normally black panel).

| R12h[5:0] | Function      | Note    |
|-----------|---------------|---------|
| 21h       | Gray16 - 23mV |         |
| 22h       | Gray16        | Default |
| 23h       | Gray16 + 23mV | -       |

## R12h\_0C

| Address | Bit |    |                  |    |    |    |    |    |
|---------|-----|----|------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5               | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH8[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 0                | 1  | 1  | 1  | 0  | 0  |

R12h[5:0]: Gray8 positive voltage(Normally black panel).

| R12h[5:0] | Function     | Note    |
|-----------|--------------|---------|
| 1Bh       | Gray8 - 23mV |         |
| 1Ch       | Gray8        | Default |
| 1Dh       | Gray8 + 23mV | -       |

## R12h\_0D

| Address | Bit |    |                  |    |    |    |    |    |
|---------|-----|----|------------------|----|----|----|----|----|
| R12h    | D7  | D6 | D5               | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSH0[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 0                | 0  | 0  | 0  | 0  | 0  |

R12h[5:0]: VP1 positive voltage(Normally black panel).

| R12h[5:0] | Function   | Note    |
|-----------|------------|---------|
| 00h       | VP1        | Default |
| 01h       | VP1 + 23mV | -       |

## R13h\_00

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL255[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                  | 1  | 1  | 1  | 1  | 1  |

R13h[5:0]: VN1 negative voltage(Normally black panel).

| R13h[5:0] | Function   | Note    |
|-----------|------------|---------|
| 3Eh       | VN1 + 23mV | -       |
| 3Fh       | VN1        | Default |

## R13h\_01

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL248[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                  | 0  | 1  | 1  | 1  | 0  |

R13h[5:0]: Gray248 negative voltage(Normally black panel).

| R13h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 2Dh       | Gray248 + 23mV |         |
| 2Eh       | Gray248        | Default |
| 2Fh       | Gray248 - 23mV | -       |

## R13h\_02

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL236[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                  | 0  | 1  | 0  | 0  | 0  |

R13h[5:0]: Gray236 negative voltage(Normally black panel).

| R13h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 27h       | Gray236 + 23mV |         |
| 28h       | Gray236        | Default |
| 29h       | Gray236 - 23mV | -       |

## R13h\_03

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL220[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                  | 0  | 0  | 0  | 0  | 0  |

R13h[5:0]: Gray220 negative voltage(Normally black panel).

| R13h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 1Fh       | Gray220 + 23mV |         |
| 20h       | Gray220        | Default |
| 21h       | Gray220 - 23mV | -       |

## R13h\_04

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL200[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 0                  | 1  | 0  | 0  | 1  | 0  |

R13h[5:0]: Gray200 negative voltage(Normally black panel).

| R13h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 11h       | Gray200 + 23mV |         |
| 12h       | Gray200        | Default |
| 13h       | Gray200 - 23mV | -       |

## R13h\_05

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL172[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 0                  | 0  | 0  | 0  | 1  | 0  |

R13h[5:0]: Gray172 negative voltage(Normally black panel).

| R13h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 01h       | Gray172 + 23mV |         |
| 02h       | Gray172        | Default |
| 03h       | Gray172 - 23mV | -       |

## R13h\_06

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL144[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 0                  | 1  | 1  | 1  | 1  | 0  |

R13h[5:0]: Gray144 negative voltage(Normally black panel).

| R13h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 1Dh       | Gray144 + 23mV |         |
| 1Eh       | Gray144        | Default |
| 1Fh       | Gray144 - 23mV | -       |

## R13h\_07

| Address | Bit |    |                    |    |    |    |    |    |
|---------|-----|----|--------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                 | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL116[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                  | 0  | 0  | 0  | 0  | 0  |

R13h[5:0]: Gray116 negative voltage(Normally black panel).

| R13h[5:0] | Function       | Note    |
|-----------|----------------|---------|
| 1Fh       | Gray116 + 23mV |         |
| 20h       | Gray116        | Default |
| 21h       | Gray116 - 23mV | -       |

## R13h\_08

| Address | Bit |    |                   |    |    |    |    |    |
|---------|-----|----|-------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL88[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                 | 1  | 0  | 0  | 1  | 0  |

R13h[5:0]: Gray88 negative voltage(Normally black panel).

| R13h[5:0] | Function      | Note    |
|-----------|---------------|---------|
| 31h       | Gray88 + 23mV |         |
| 32h       | Gray88        | Default |
| 33h       | Gray88 - 23mV | -       |

## R13h\_09

| Address | Bit |    |                   |    |    |    |    |    |
|---------|-----|----|-------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL60[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                 | 1  | 0  | 0  | 1  | 0  |

R13h[5:0]: Gray60 negative voltage(Normally black panel).

| R13h[5:0] | Function      | Note    |
|-----------|---------------|---------|
| 31h       | Gray60 + 23mV |         |
| 32h       | Gray60        | Default |
| 33h       | Gray60 - 23mV | -       |

## R13h\_0A

| Address | Bit |    |                   |    |    |    |    |    |
|---------|-----|----|-------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL32[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                 | 0  | 1  | 0  | 0  | 0  |

R13h[5:0]: Gray32 negative voltage(Normally black panel).

| R13h[5:0] | Function      | Note    |
|-----------|---------------|---------|
| 27h       | Gray32 + 23mV |         |
| 28h       | Gray32        | Default |
| 29h       | Gray32 - 23mV | -       |

## R13h\_0B

| Address | Bit |    |                   |    |    |    |    |    |
|---------|-----|----|-------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5                | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL16[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 1                 | 0  | 0  | 0  | 1  | 0  |

R13h[5:0]: Gray16 negative voltage(Normally black panel).

| R13h[5:0] | Function      | Note    |
|-----------|---------------|---------|
| 21h       | Gray16 + 23mV |         |
| 22h       | Gray16        | Default |
| 23h       | Gray16 - 23mV | -       |

## R13h\_0C

| Address | Bit |    |                  |    |    |    |    |    |
|---------|-----|----|------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5               | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL8[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 0                | 1  | 1  | 1  | 0  | 0  |

R13h[5:0]: Gray8 negative voltage(Normally black panel).

| R13h[5:0] | Function     | Note    |
|-----------|--------------|---------|
| 1Bh       | Gray8 + 23mV |         |
| 1Ch       | Gray8        | Default |
| 1Dh       | Gray8 - 23mV | -       |

## R13h\_0D

| Address | Bit |    |                  |    |    |    |    |    |
|---------|-----|----|------------------|----|----|----|----|----|
| R13h    | D7  | D6 | D5               | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | REG_GA_GSL0[5:0] |    |    |    |    |    |
| Default | 0   | 0  | 0                | 0  | 0  | 0  | 0  | 0  |

R13h[5:0]: VN negative voltage(Normally black panel).

| R13h[5:0] | Function  | Note    |
|-----------|-----------|---------|
| 00h       | VN        | Default |
| 01h       | VN - 23mV | -       |



# 11.6 Page 7 command

| X Address | Y Address | W/R | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0                 | Default |
|-----------|-----------|-----|----|----|----|----|----|----|----|--------------------|---------|
| 0D        | 00        | W/R | -  | -  | -  | -  | -  | -  | -  | ulpslp11_check_enb | 00      |

R0Dh\_00

| Address | Bit |    |    |    |    |    |    |                    |
|---------|-----|----|----|----|----|----|----|--------------------|
| R0Dh    | D7  | D6 | D5 | D4 | D3 | D2 | D1 | D0                 |
| Name    | -   | -  | -  | -  | -  | -  | -  | ulpslp11_check_enb |
| Default | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0                  |

R0Dh[0]: MIPI clk lane sequence check.

| R00h[0] | Function | Note    |
|---------|----------|---------|
| 0       | enable   | default |
| 1       | disable  | -       |

## 11.7 Page 8 command

| X Address | Y Address | W/R | D7 | D6 | D5 | D4 | D3 | D2 | D1                | D0         | Default |
|-----------|-----------|-----|----|----|----|----|----|----|-------------------|------------|---------|
| 05        | 00        | W/R |    | -  | -  | -  | -  | -  | -                 | esd_rst_en | 00      |
| 0C        | 00        | W/R | -  | -  | -  | -  | -  | -  | esd_det_en        | -          | 18      |
| 0D        | 00        | W/R | -  | -  | -  | -  | -  | -  | analog_esd_det_en | -          | 0C      |

## R05h\_00

| Address | Bit |    |    |    |    |    |    |            |
|---------|-----|----|----|----|----|----|----|------------|
| R05h    | D7  | D6 | D5 | D4 | D3 | D2 | D1 | D0         |
| Name    | -   | -  | -  | -  | -  | -  | -  | esd_rst_en |
| Default | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0          |

R05h[0]: ESD event trig IC reset function.

| R05h[0] | Function | Note    |
|---------|----------|---------|
| 0       | disable  | Default |
| 1       | enable   | -       |

## R0Ch\_00

| Address | Bit |    |    |    |    |    |            |    |
|---------|-----|----|----|----|----|----|------------|----|
| R0Ch    | D7  | D6 | D5 | D4 | D3 | D2 | D1         | D0 |
| Name    | -   | -  | -  | -  | -  | -  | esd_det_en | -  |
| Default | 0   | 0  | 0  | 1  | 1  | 0  | 0          | 0  |

R0Ch[1]: ESD event detection

| R0Ch[1] | Function | Note    |
|---------|----------|---------|
| 0       | disable  | Default |
| 1       | enable   | -       |

## R0Dh\_00

| Address | Bit |    |    |    |    |    |                   |    |
|---------|-----|----|----|----|----|----|-------------------|----|
| R0Dh    | D7  | D6 | D5 | D4 | D3 | D2 | D1                | D0 |
| Name    | -   | -  | -  | -  | -  | -  | analog_esd_det_en | -  |
| Default | 0   | 0  | 0  | 0  | 1  | 1  | 0                 | 0  |

R0Dh[1]: Analog ESD event detection

| R0Dh[1] | Function | Note    |
|---------|----------|---------|
| 0       | disable  | Default |
| 1       | enable   | -       |

## 11.8 Page 9 command

| X Address | Y Address | W/R | D7 | D6 | D5                      | D4                      | D3                      | D2                      | D1                      | D0                      | Default |
|-----------|-----------|-----|----|----|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|---------|
| 17        | 00        | R   | -  | -  | index_revers<br>e_vcom6 | index_revers<br>e_vcom5 | index_revers<br>e_vcom4 | index_revers<br>e_vcom3 | index_revers<br>e_vcom2 | index_revers<br>e_vcom1 | 00      |
| 1A        | 00        | R   | -  | -  | -                       | -                       | -                       | -                       | index_gamm<br>a2        | index_gamm<br>a1        | 00      |
| 20        | 00        | R   | -  | -  | index_norma<br>l_vcom6  | index_norma<br>l_vcom5  | index_norma<br>l_vcom4  | index_norma<br>l_vcom3  | index_norma<br>l_vcom2  | index_norma<br>l_vcom1  | 00      |

## R17h\_00

| Address | Bit |    |                         |                         |                         |                         |                         |                         |
|---------|-----|----|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| R17h    | D7  | D6 | D5                      | D4                      | D3                      | D2                      | D1                      | D0                      |
| Name    | -   | -  | index_reverse_vco<br>m6 | index_reverse_vco<br>m5 | index_reverse_vco<br>m4 | index_reverse_vco<br>m3 | index_reverse_vco<br>m2 | index_reverse_vco<br>m1 |
| Default | 0   | 0  | 0                       | 0                       | 0                       | 0                       | 0                       | 0                       |

R17h[5:0]: Reverse Scan VCOM OTP number of times.

| R17h[5:0] | Function         | Note    |
|-----------|------------------|---------|
| 00h       | VCOM OTP 0 times | Default |
| 01h       | VCOM OTP 1 times | -       |
| 03h       | VCOM OTP 2 times | -       |
| 07h       | VCOM OTP 3 times | -       |
| 0Fh       | VCOM OTP 4 times | -       |
| 1Fh       | VCOM OTP 5 times | -       |
| 3Fh       | VCOM OTP 6 times | -       |

## R1Ah\_00

| Address | Bit |    |    |    |    |    |              |              |
|---------|-----|----|----|----|----|----|--------------|--------------|
| R1Ah    | D7  | D6 | D5 | D4 | D3 | D2 | D1           | D0           |
| Name    | -   | -  | -  | -  | -  | -  | index_gamma2 | index_gamma1 |
| Default | 0   | 0  | 0  | 0  | 0  | 0  | 0            | 0            |

R1Ah[1:0]: Gamma OTP number of times.

| R1Ah[1:0] | Function          | Note    |
|-----------|-------------------|---------|
| 00h       | Gamma OTP 0 times | Default |
| 01h       | Gamma OTP 1 times | -       |
| 03h       | Gamma OTP 2 times | -       |

## R20h\_00

| Address | Bit |    |                        |                        |                        |                        |                        |                        |
|---------|-----|----|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| R20h    | D7  | D6 | D5                     | D4                     | D3                     | D2                     | D1                     | D0                     |
| Name    | -   | -  | index_normal_vco<br>m6 | index_normal_vco<br>m5 | index_normal_vco<br>m4 | index_normal_vco<br>m3 | index_normal_vco<br>m2 | index_normal_vco<br>m1 |
| Default | 0   | 0  | 0                      | 0                      | 0                      | 0                      | 0                      | 0                      |

R20h[5:0]: Normal Scan VCOM OTP number of times.

| R20h[5:0] | Function         | Note    |
|-----------|------------------|---------|
| 00h       | VCOM OTP 0 times | Default |
| 01h       | VCOM OTP 1 times | -       |
| 03h       | VCOM OTP 2 times | -       |
| 07h       | VCOM OTP 3 times | -       |
| 0Fh       | VCOM OTP 4 times | -       |
| 1Fh       | VCOM OTP 5 times | -       |
| 3Fh       | VCOM OTP 6 times | -       |

## 11.9 Page A command

| X Address | Y Address | W/R | D7               | D6 | D5                | D4 | D3                | D2                  | D1 | D0 | Default |
|-----------|-----------|-----|------------------|----|-------------------|----|-------------------|---------------------|----|----|---------|
| 01        | 00        | W/R | sd_hz_col_t[3:0] |    |                   |    | -                 | -                   | -  | -  | 27      |
| 02        | 00        | W/R | -                | -  | dummy             | -  |                   | -                   | -  | -  | 6F      |
| 03        | 00        | W/R | -                | -  | -                 | -  | -                 | sd_bias_sel_nd[2:0] |    |    | 12      |
| 09        | 00        | W/R | -                | -  | -                 | -  | sd_lr_osw_nd[1:0] |                     | -  | -  | 00      |
| 0B        | 00        | W/R | -                | -  | sd_slpin_opt[1:0] |    | -                 | -                   | -  | -  | 50      |

R01h\_00

| Address | Bit         |    |    |    |    |    |    |    |
|---------|-------------|----|----|----|----|----|----|----|
| R01h    | D7          | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | sd_hz_col_t |    |    |    | -  | -  | -  | -  |
| Default | 0           | 0  | 1  | 0  | 0  | 1  | 1  | 1  |

R01h[7:4]: Source Hi-Z area.

| R17h[5:0] | Function | Note    |
|-----------|----------|---------|
| 00h       | 200ns    | -       |
| 01h       | 400ns    | -       |
| 02h       | 600ns    | Default |
| 03h       | 800ns    | -       |
| 04h       | 1000ns   | -       |
| 05h       | 1200ns   | -       |
| 06h       | 1400ns   | -       |
| 07h       | 1600ns   | -       |
| 08h       | 1800ns   | -       |
| 09h       | 2000ns   | -       |
| 0Ah-0Fh   | reserved | -       |

R02h\_00

| Address | Bit |    |       |    |    |    |    |    |
|---------|-----|----|-------|----|----|----|----|----|
| R02h    | D7  | D6 | D5    | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | dummy | -  | -  | -  | -  | -  |
| Default | 0   | 1  | 1     | 0  | 1  | 1  | 1  | 1  |

R02h[5]: dummy function.

| R02h[5] | Function       | Note    |
|---------|----------------|---------|
| 0       | dummy function | -       |
| 1       | dummy function | Default |

## R03h\_00

| Address | Bit |    |    |    |    |                     |    |    |
|---------|-----|----|----|----|----|---------------------|----|----|
| R03h    | D7  | D6 | D5 | D4 | D3 | D2                  | D1 | D0 |
| Name    | -   | -  | -  | -  | -  | sd_bias_sel_nd[2:0] |    |    |
| Default | 0   | 0  | 0  | 1  | 0  | 0                   | 1  | 0  |

R03h[2:0]: Source bias current selection.

| R03h[2:0] | Function | Note    |
|-----------|----------|---------|
| 00h       | 0.8uA    | -       |
| 01h       | 1.6uA    | -       |
| 02h       | 2.4uA    | Default |
| 03h       | 3.2uA    | -       |
| 04h       | 4.0uA    | -       |
| 05h       | 4.8uA    | -       |
| 06h       | 5.6uA    | -       |
| 07h       | 7.2uA    | -       |

## R09h\_00

| Address | Bit |    |    |    |                   |    |    |    |
|---------|-----|----|----|----|-------------------|----|----|----|
| R09h    | D7  | D6 | D5 | D4 | D3                | D2 | D1 | D0 |
| Name    | -   | -  | -  | -  | sd_lr_osw_nd[1:0] |    | -  | -  |
| Default | 0   | 0  |    | 0  | 0                 |    | 0  | 0  |

R09h[3:2]: Source output equivalent resistance.

| R09h[3:2] | Function | Note    |
|-----------|----------|---------|
| 00h       | 5K ohm   | Default |
| 01h       | 1.3K ohm | -       |
| 10h       | 400 ohm  | -       |
| 11h       | 300 Ohm  | -       |

## R0Bh\_00

| Address | Bit |    |                   |    |    |    |    |    |
|---------|-----|----|-------------------|----|----|----|----|----|
| R0Bh    | D7  | D6 | D5                | D4 | D3 | D2 | D1 | D0 |
| Name    | -   | -  | sd_slpin_opt[1:0] |    | -  | -  | -  | -  |
| Default | 0   | 1  | 0                 | 1  | 0  | 0  | 0  | 0  |

R0Bh[5:4]: Source behavior in Standby mode.

| R0Bh[5:4] | Function     | Note    |
|-----------|--------------|---------|
| 00h       | VMID         | -       |
| 01h       | charge share | Default |
| 10h       | reserved     | -       |
| 11h       | reserved     | -       |

## 11.10 Page D command

| X Address | Y Address | W/R | D7         | D6 | D5 | D4 | D3      | D2 | D1 | D0 | Default |
|-----------|-----------|-----|------------|----|----|----|---------|----|----|----|---------|
| 0D        | 00        | W/R | -          | -  | -  | -  | GAS_VDD | -  | -  | -  | 04      |
| 10        | 00        | W/R | VGMPH[7:0] |    |    |    |         |    |    |    | 0C      |
| 11        | 00        | W/R | VGMPH[7:0] |    |    |    |         |    |    |    | 0C      |
| 12        | 00        | W/R | VGMNH[7:0] |    |    |    |         |    |    |    | 0C      |
| 13        | 00        | W/R | VGMNL[7:0] |    |    |    |         |    |    |    | 0C      |

R0Dh 00

| Address | Bit |    |    |    |         |    |    |    |
|---------|-----|----|----|----|---------|----|----|----|
| R0Dh    | D7  | D6 | D5 | D4 | D3      | D2 | D1 | D0 |
| Name    | -   | -  |    |    | GAS_VDD | -  | -  | -  |
| Default | 0   | 0  | 0  | 0  | 0       | 1  | 0  | 0  |

R0Dh[3]: GAS\_VDD detection voltage

| R0Dh[3] | Function | Note    |
|---------|----------|---------|
| 0       | 1.5V     | Default |
| 1       | 2.2V     | -       |

R10h 00

| Address | Bit         |    |    |    |    |    |    |    |
|---------|-------------|----|----|----|----|----|----|----|
| R10h    | D7          | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | VGMPH [7:0] |    |    |    |    |    |    |    |
| Default | 0           | 0  | 0  | 0  | 1  | 1  | 0  | 0  |

R10h[7:0]: Gamma positive high level reference voltage.

| R10h[7:0] | Function            | Note    |
|-----------|---------------------|---------|
| 0Bh       | AVDD - (15AVDD/800) | -       |
| 0Ch       | AVDD - (16AVDD/800) | Default |
| 0Dh       | AVDD - (17AVDD/800) | -       |
| 0Eh       | AVDD - (18AVDD/800) | -       |
| 0Fh       | AVDD - (19AVDD/800) | -       |
| :         | :                   | -       |
| 1Eh       | AVDD - (34AVDD/800) | -       |
| 1Fh       | AVDD - (35AVDD/800) | -       |

Operating voltage limits : AVDD – VGMPH ≥ 0.2V.

## R11h\_00

| Address | Bit         |    |    |    |    |    |    |    |
|---------|-------------|----|----|----|----|----|----|----|
| R11h    | D7          | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | VGMPL [7:0] |    |    |    |    |    |    |    |
| Default | 0           | 0  | 0  | 0  | 1  | 1  | 0  | 0  |

R11h[7:0]: Gamma positive low level reference voltage.

| R11h[7:0] | Function    | Note    |
|-----------|-------------|---------|
| 0Bh       | 415AVDD/800 | -       |
| 0Ch       | 416AVDD/800 | Default |
| 0Dh       | 417AVDD/800 | -       |
| 0Eh       | 418AVDD/800 | -       |
| 0Fh       | 419AVDD/800 | -       |
| :         | :           | -       |
| 1Eh       | 434AVDD/800 | -       |
| 1Fh       | 435AVDD/800 | -       |

Operating voltage limits : VGMPL – VMID  $\geq$  0.2V.

## R12h\_00

| Address | Bit         |    |    |    |    |    |    |    |
|---------|-------------|----|----|----|----|----|----|----|
| R12h    | D7          | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | VGMNH [7:0] |    |    |    |    |    |    |    |
| Default | 0           | 0  | 0  | 0  | 1  | 1  | 0  | 0  |

R12h[7:0]: Gamma negative high level reference voltage.

| R12h[7:0] | Function             | Note    |
|-----------|----------------------|---------|
| 0Bh       | AVDD – (415AVDD/800) | -       |
| 0Ch       | AVDD – (416AVDD/800) | Default |
| 0Dh       | AVDD – (417AVDD/800) | -       |
| 0Eh       | AVDD – (418AVDD/800) | -       |
| 0Fh       | AVDD – (419AVDD/800) | -       |
| :         | :                    | -       |
| 1Eh       | AVDD – (434AVDD/800) | -       |
| 1Fh       | AVDD – (435AVDD/800) | -       |

Operating voltage limits : VMID – VGMNH  $\geq$  0.2V.

## R13h\_00

| Address | Bit        |    |    |    |    |    |    |    |
|---------|------------|----|----|----|----|----|----|----|
| R13h    | D7         | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Name    | VGMNL[7:0] |    |    |    |    |    |    |    |
| Default | 0          | 0  | 0  | 0  | 1  | 1  | 0  | 0  |

R13h[7:0]: Gamma negative low level reference voltage.

| R13h[7:0] | Function   | Note    |
|-----------|------------|---------|
| 0Bh       | 15AVDD/800 | -       |
| 0Ch       | 16AVDD/800 | Default |
| 0Dh       | 17AVDD/800 | -       |
| 0Eh       | 18AVDD/800 | -       |
| 0Fh       | 19AVDD/800 | -       |
| :         | :          | -       |
| 1Eh       | 34AVDD/800 | -       |
| 1Fh       | 35AVDD/800 | -       |

Operating voltage limits : VGMNL – GND  $\geq$  0.2V.

## 12. DC CHARACTERISTICS

### 12.1 Absolute maximum ratings

| Parameter             | Symbol           | Spec.   |      |         | Unit | Note |
|-----------------------|------------------|---------|------|---------|------|------|
|                       |                  | Min.    | Typ. | Max.    |      |      |
| Supply power voltage  | VDD              | -0.30   | -    | 3.96    | V    |      |
| IO supply voltage     | VDDIO            | -0.30   |      | 3.96    | V    |      |
| AVDD voltage          | AVDD             | -0.30   | -    | 12      | V    |      |
| VGH voltage           | VGH              | -0.30   | -    | VGL+32V | V    |      |
| VGL voltage           | VGL              | VGH-32V | -    | 0.30    | V    |      |
| VMID voltage          | VMID             | -0.30   | -    | 6.6     | V    |      |
| VCOM voltage          | VCOM_OP          | -0.30   | -    | 5.43    | V    |      |
| VOTP (OTP power)      | VOTP             | -       | -    | 9       | V    |      |
| Operating Temperature | T <sub>OPR</sub> | -30     | -    | +85     | °C   |      |
| Storage temperature   | T <sub>STG</sub> | -55     | -    | 125     | °C   |      |

### 12.2 Typical operating condition

| Parameter                 | Symbol          | Spec.   |          |         | Unit | Note                                                              |
|---------------------------|-----------------|---------|----------|---------|------|-------------------------------------------------------------------|
|                           |                 | Min.    | Typ.     | Max.    |      |                                                                   |
| Supply power voltage      | VDD             | 1.71    | 1.8      | 1.89    | V    | MIPI mode condition1                                              |
| I/O power voltage         | VDDIO           | 1.71    | 1.8      | 1.89    | V    |                                                                   |
| Supply power voltage      | VDD             | 2.8     | 3.3      | 3.6     | V    | TTL mode condition<br>LVDS mode condition<br>MIPI mode condition2 |
| I/O power voltage         | VDDIO           | 2.8     | 3.3      | 3.6     | V    |                                                                   |
| AVDD voltage              | AVDD            | 9       | -        | 12      | V    |                                                                   |
| VGH voltage               | VGH             | 15      | 18       | 20      | V    |                                                                   |
| VGL voltage               | VGL             | -12     | -8       | -6      | V    |                                                                   |
| VMID voltage              | VMID            | 4.5     | 5        | 6.0     | V    |                                                                   |
| VMID_OP voltage           | VMID_OP         | 4.5     | 5        | 6.0     | V    |                                                                   |
| VQHO voltage              | VQHO            | -       | 0.75AVDD | -       | V    |                                                                   |
| VQLO voltage              | VQLO            | -       | 0.25AVDD | -       | V    |                                                                   |
| VCOM voltage              | VCOM_OP         | 2.7     | 3.8      | 5.4     | V    |                                                                   |
| VOTP                      | VOTP            | 8.0     | 8.25     | 8.5     | V    |                                                                   |
| Low level input voltage   | V <sub>IL</sub> | VSS     | -        | 0.2xVDD | V    |                                                                   |
| High level input voltage  | V <sub>IH</sub> | 0.8xVDD | -        | VDD     | V    |                                                                   |
| Low level output voltage  | V <sub>OL</sub> | VSS     | -        | 0.1xVDD | V    |                                                                   |
| High level output voltage | V <sub>OH</sub> | 0.9xVDD | -        | VDD     | V    |                                                                   |



## 12.3 VDD power on slew time

Be sure that VDD input can't result automatically in power on sequence.

| Parameter              | Symbol    | Conditions       | Spec. |      |     | Unit |
|------------------------|-----------|------------------|-------|------|-----|------|
|                        |           |                  | Min.  | Typ. | Max |      |
| VDD power on skew time | $T_{POR}$ | From 0V to 1.35V | 0.1   | -    | 5   | ms   |

## 12.4 Timing requirements for RESX

When RESX of the reset pin equals to Low, it will be in the condition of reset.

When it is in the condition of reset, it will make the device recover the initial set.

However, in order to avoid the reset noise cause reset, there is a mechanism to judge about whether the reset is needed or not.

The closed interval of Low can be shown as the following.

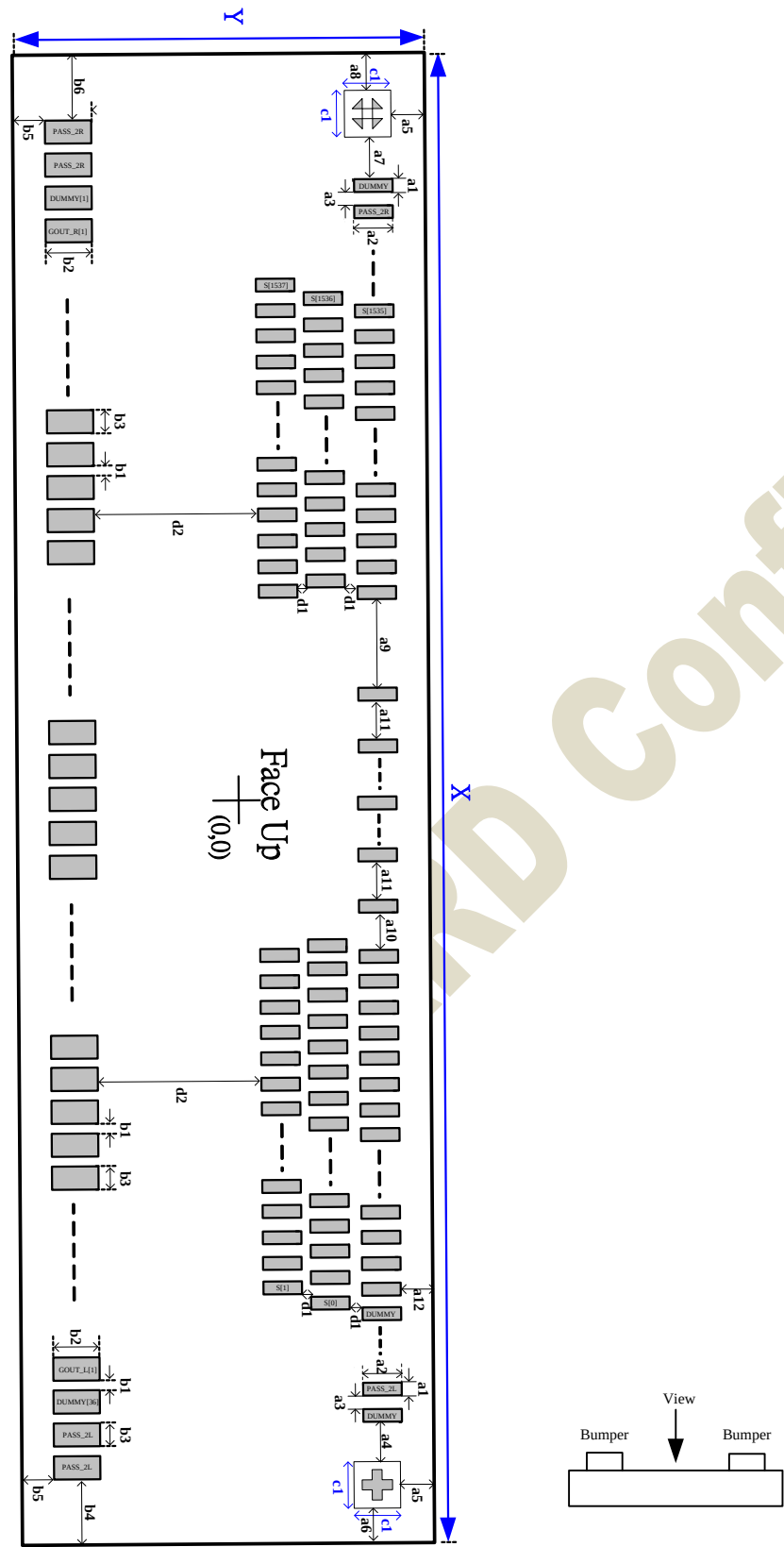
(Test condition: VDD=1.71V to 1.89V / 2.8V to 3.6V , VSS=0V,  $T_A$ =-30 ~+85 )

| Parameter             | Symbol | Conditions | Spec. |      |     | Unit    |
|-----------------------|--------|------------|-------|------|-----|---------|
|                       |        |            | Min.  | Typ. | Max |         |
| Reset low pulse width | $Trst$ |            | 6     | -    | -   | $\mu s$ |



13. CHIP INFORMATION

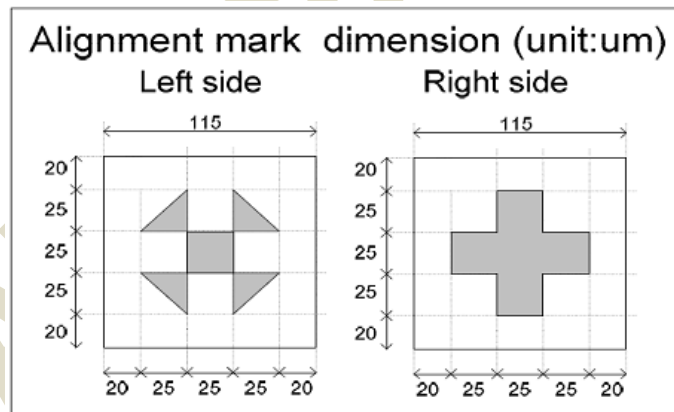
13.1 Chip outline dimensions



| symbol | Dimension (um) | symbol | Dimension (um) |
|--------|----------------|--------|----------------|
| a1     | 18             | b2     | 52             |
| a2     | 60             | b3     | 30             |
| a3     | 18             | b4     | 60             |
| a4     | 240            | b5     | 15             |
| a5     | 22             | b6     | 60             |
| a6     | 22             | c1     | 115            |
| a7     | 163            | d1     | 25             |
| a8     | 22             | d2     | 422.5          |
| a9     | 135            | X      | 28500          |
| a10    | 58             | Y      | 742            |
| a11    | 54             |        |                |
| a12    | 17.5           |        |                |
| b1     | 15             |        |                |

Table: Chip Outline Dimension (No Scribe Line)

### 13.2 Alignment mark



Alignment Mark

## 13.3 Pad Coordinate

| No. | Name       | X      | Y    | Bump size(um) | No. | Name           | X      | Y    | Bump size(um) |
|-----|------------|--------|------|---------------|-----|----------------|--------|------|---------------|
| 1   | PASS2_R    | -14175 | -335 | 30x52         | 57  | VGH_R          | -11655 | -335 | 30x52         |
| 2   | PASS2_R    | -14130 | -335 | 30x52         | 58  | VGH_R          | -11610 | -335 | 30x52         |
| 3   | DUMMY[1]   | -14085 | -335 | 30x52         | 59  | VGH_R          | -11565 | -335 | 30x52         |
| 4   | GOUT_R[1]  | -14040 | -335 | 30x52         | 60  | Shielding_VSSA | -11520 | -335 | 30x52         |
| 5   | GOUT_R[1]  | -13995 | -335 | 30x52         | 61  | PASS1_R        | -11475 | -335 | 30x52         |
| 6   | GOUT_R[2]  | -13950 | -335 | 30x52         | 62  | PASS1_R        | -11430 | -335 | 30x52         |
| 7   | GOUT_R[2]  | -13905 | -335 | 30x52         | 63  | Shielding_VSSA | -11385 | -335 | 30x52         |
| 8   | GOUT_R[3]  | -13860 | -335 | 30x52         | 64  | Shielding_VSSA | -11340 | -335 | 30x52         |
| 9   | GOUT_R[3]  | -13815 | -335 | 30x52         | 65  | TEST_I[0]      | -11295 | -335 | 30x52         |
| 10  | GOUT_R[4]  | -13770 | -335 | 30x52         | 66  | TEST_I[1]      | -11250 | -335 | 30x52         |
| 11  | GOUT_R[4]  | -13725 | -335 | 30x52         | 67  | TEST_I[2]      | -11205 | -335 | 30x52         |
| 12  | GOUT_R[5]  | -13680 | -335 | 30x52         | 68  | TEST_I[3]      | -11160 | -335 | 30x52         |
| 13  | GOUT_R[5]  | -13635 | -335 | 30x52         | 69  | TEST_I[4]      | -11115 | -335 | 30x52         |
| 14  | GOUT_R[6]  | -13590 | -335 | 30x52         | 70  | TEST_I[5]      | -11070 | -335 | 30x52         |
| 15  | GOUT_R[6]  | -13545 | -335 | 30x52         | 71  | TEST_I[6]      | -11025 | -335 | 30x52         |
| 16  | GOUT_R[7]  | -13500 | -335 | 30x52         | 72  | TEST_I[7]      | -10980 | -335 | 30x52         |
| 17  | GOUT_R[7]  | -13455 | -335 | 30x52         | 73  | TEST_I[8]      | -10935 | -335 | 30x52         |
| 18  | GOUT_R[8]  | -13410 | -335 | 30x52         | 74  | TEST_I[9]      | -10890 | -335 | 30x52         |
| 19  | GOUT_R[8]  | -13365 | -335 | 30x52         | 75  | TEST_I[10]     | -10845 | -335 | 30x52         |
| 20  | GOUT_R[9]  | -13320 | -335 | 30x52         | 76  | TEST_I[11]     | -10800 | -335 | 30x52         |
| 21  | GOUT_R[9]  | -13275 | -335 | 30x52         | 77  | TEST_I[12]     | -10755 | -335 | 30x52         |
| 22  | GOUT_R[10] | -13230 | -335 | 30x52         | 78  | TEST_I[13]     | -10710 | -335 | 30x52         |
| 23  | GOUT_R[10] | -13185 | -335 | 30x52         | 79  | TEST_I[14]     | -10665 | -335 | 30x52         |
| 24  | GOUT_R[11] | -13140 | -335 | 30x52         | 80  | TEST_I[15]     | -10620 | -335 | 30x52         |
| 25  | GOUT_R[11] | -13095 | -335 | 30x52         | 81  | TEST_I[16]     | -10575 | -335 | 30x52         |
| 26  | GOUT_R[12] | -13050 | -335 | 30x52         | 82  | TEST_I[17]     | -10530 | -335 | 30x52         |
| 27  | GOUT_R[12] | -13005 | -335 | 30x52         | 83  | TEST_I[18]     | -10485 | -335 | 30x52         |
| 28  | GOUT_R[13] | -12960 | -335 | 30x52         | 84  | TEST_I[19]     | -10440 | -335 | 30x52         |
| 29  | GOUT_R[13] | -12915 | -335 | 30x52         | 85  | TEST_I[20]     | -10395 | -335 | 30x52         |
| 30  | GOUT_R[14] | -12870 | -335 | 30x52         | 86  | TEST_I[21]     | -10350 | -335 | 30x52         |
| 31  | GOUT_R[14] | -12825 | -335 | 30x52         | 87  | TEST_I[22]     | -10305 | -335 | 30x52         |
| 32  | GOUT_R[15] | -12780 | -335 | 30x52         | 88  | VDD18V_BPS     | -10260 | -335 | 30x52         |
| 33  | GOUT_R[15] | -12735 | -335 | 30x52         | 89  | Shielding_VSSA | -10215 | -335 | 30x52         |
| 34  | GOUT_R[16] | -12690 | -335 | 30x52         | 90  | Shielding_VSSA | -10170 | -335 | 30x52         |
| 35  | GOUT_R[16] | -12645 | -335 | 30x52         | 91  | Shielding_VSSA | -10125 | -335 | 30x52         |
| 36  | GOUT_R[17] | -12600 | -335 | 30x52         | 92  | Shielding_VSSA | -10080 | -335 | 30x52         |
| 37  | GOUT_R[17] | -12555 | -335 | 30x52         | 93  | Shielding_VSSA | -10035 | -335 | 30x52         |
| 38  | GOUT_R[18] | -12510 | -335 | 30x52         | 94  | Shielding_VSSA | -9990  | -335 | 30x52         |
| 39  | GOUT_R[18] | -12465 | -335 | 30x52         | 95  | Shielding_VSSA | -9945  | -335 | 30x52         |
| 40  | GOUT_R[19] | -12420 | -335 | 30x52         | 96  | Shielding_VSSA | -9900  | -335 | 30x52         |
| 41  | GOUT_R[19] | -12375 | -335 | 30x52         | 97  | Shielding_VSSA | -9855  | -335 | 30x52         |
| 42  | GOUT_R[20] | -12330 | -335 | 30x52         | 98  | VQLI_R         | -9810  | -335 | 30x52         |
| 43  | GOUT_R[20] | -12285 | -335 | 30x52         | 99  | VQLI_R         | -9765  | -335 | 30x52         |
| 44  | GOUT_R[21] | -12240 | -335 | 30x52         | 100 | VQHI_R         | -9720  | -335 | 30x52         |
| 45  | GOUT_R[21] | -12195 | -335 | 30x52         | 101 | VQHI_R         | -9675  | -335 | 30x52         |
| 46  | GOUT_R[22] | -12150 | -335 | 30x52         | 102 | Shielding_VSSA | -9630  | -335 | 30x52         |
| 47  | GOUT_R[22] | -12105 | -335 | 30x52         | 103 | Shielding_VSSA | -9585  | -335 | 30x52         |
| 48  | VGL        | -12060 | -335 | 30x52         | 104 | Shielding_VSSA | -9540  | -335 | 30x52         |
| 49  | VGL        | -12015 | -335 | 30x52         | 105 | Shielding_VSSA | -9495  | -335 | 30x52         |
| 50  | VGL        | -11970 | -335 | 30x52         | 106 | Shielding_VSSA | -9450  | -335 | 30x52         |
| 51  | VGL        | -11925 | -335 | 30x52         | 107 | Shielding_VSSA | -9405  | -335 | 30x52         |
| 52  | VGL        | -11880 | -335 | 30x52         | 108 | Shielding_VSSA | -9360  | -335 | 30x52         |
| 53  | VGL        | -11835 | -335 | 30x52         | 109 | Shielding_VSSA | -9315  | -335 | 30x52         |
| 54  | VGH_R      | -11790 | -335 | 30x52         | 110 | Shielding_VSSA | -9270  | -335 | 30x52         |
| 55  | VGH_R      | -11745 | -335 | 30x52         | 111 | VSSA           | -9225  | -335 | 30x52         |
| 56  | VGH_R      | -11700 | -335 | 30x52         | 112 | VSSA           | -9180  | -335 | 30x52         |

| No. | Name           | X     | Y    | Bump size(um) | No. | Name           | X     | Y    | Bump size(um) |
|-----|----------------|-------|------|---------------|-----|----------------|-------|------|---------------|
| 113 | VSSA           | -9135 | -335 | 30x52         | 169 | Shielding_VSSA | -6615 | -335 | 30x52         |
| 114 | VSSA           | -9090 | -335 | 30x52         | 170 | Shielding_VSSA | -6570 | -335 | 30x52         |
| 115 | VSSA           | -9045 | -335 | 30x52         | 171 | Shielding_VSSA | -6525 | -335 | 30x52         |
| 116 | VSSA           | -9000 | -335 | 30x52         | 172 | Shielding_VSSA | -6480 | -335 | 30x52         |
| 117 | VMID_R[1]      | -8955 | -335 | 30x52         | 173 | Shielding_VSSA | -6435 | -335 | 30x52         |
| 118 | VMID_R[1]      | -8910 | -335 | 30x52         | 174 | Shielding_VSSA | -6390 | -335 | 30x52         |
| 119 | VMID_R[1]      | -8865 | -335 | 30x52         | 175 | Shielding_VSSA | -6345 | -335 | 30x52         |
| 120 | VMID_R[1]      | -8820 | -335 | 30x52         | 176 | Shielding_VSSA | -6300 | -335 | 30x52         |
| 121 | VMID_R[1]      | -8775 | -335 | 30x52         | 177 | TEST_VDD_O[1]  | -6255 | -335 | 30x52         |
| 122 | VMID_R[1]      | -8730 | -335 | 30x52         | 178 | TEST_VDD_O[2]  | -6210 | -335 | 30x52         |
| 123 | AVDD           | -8685 | -335 | 30x52         | 179 | TEST_VDD_O[3]  | -6165 | -335 | 30x52         |
| 124 | AVDD           | -8640 | -335 | 30x52         | 180 | VPP_GEN        | -6120 | -335 | 30x52         |
| 125 | AVDD           | -8595 | -335 | 30x52         | 181 | VPP_GEN        | -6075 | -335 | 30x52         |
| 126 | AVDD           | -8550 | -335 | 30x52         | 182 | VOTP           | -6030 | -335 | 30x52         |
| 127 | AVDD           | -8505 | -335 | 30x52         | 183 | VOTP           | -5985 | -335 | 30x52         |
| 128 | AVDD           | -8460 | -335 | 30x52         | 184 | Shielding_VSSA | -5940 | -335 | 30x52         |
| 129 | VSS            | -8415 | -335 | 30x52         | 185 | Shielding_VSSA | -5895 | -335 | 30x52         |
| 130 | VSS            | -8370 | -335 | 30x52         | 186 | Shielding_VSSA | -5850 | -335 | 30x52         |
| 131 | VSS            | -8325 | -335 | 30x52         | 187 | Shielding_VSSA | -5805 | -335 | 30x52         |
| 132 | VSS            | -8280 | -335 | 30x52         | 188 | Shielding_VSSA | -5760 | -335 | 30x52         |
| 133 | VSS            | -8235 | -335 | 30x52         | 189 | Shielding_VSSA | -5715 | -335 | 30x52         |
| 134 | VSS            | -8190 | -335 | 30x52         | 190 | Shielding_VSSA | -5670 | -335 | 30x52         |
| 135 | TEST_O[0]      | -8145 | -335 | 30x52         | 191 | Shielding_VSSA | -5625 | -335 | 30x52         |
| 136 | TEST_O[1]      | -8100 | -335 | 30x52         | 192 | Shielding_VSSA | -5580 | -335 | 30x52         |
| 137 | TEST_O[2]      | -8055 | -335 | 30x52         | 193 | Shielding_VSSA | -5535 | -335 | 30x52         |
| 138 | TEST_O[3]      | -8010 | -335 | 30x52         | 194 | Shielding_VSSA | -5490 | -335 | 30x52         |
| 139 | TEST_O[4]      | -7965 | -335 | 30x52         | 195 | Shielding_VSSA | -5445 | -335 | 30x52         |
| 140 | TEST_O[5]      | -7920 | -335 | 30x52         | 196 | VSSA           | -5400 | -335 | 30x52         |
| 141 | TEST_O[6]      | -7875 | -335 | 30x52         | 197 | VSSA           | -5355 | -335 | 30x52         |
| 142 | TEST_O[7]      | -7830 | -335 | 30x52         | 198 | VSSA           | -5310 | -335 | 30x52         |
| 143 | TEST_O[8]      | -7785 | -335 | 30x52         | 199 | VSSA           | -5265 | -335 | 30x52         |
| 144 | TEST_O[9]      | -7740 | -335 | 30x52         | 200 | VSSA           | -5220 | -335 | 30x52         |
| 145 | TEST_O[10]     | -7695 | -335 | 30x52         | 201 | VSSA           | -5175 | -335 | 30x52         |
| 146 | TEST_O[11]     | -7650 | -335 | 30x52         | 202 | VMID_R[0]      | -5130 | -335 | 30x52         |
| 147 | TEST_O[12]     | -7605 | -335 | 30x52         | 203 | VMID_R[0]      | -5085 | -335 | 30x52         |
| 148 | TEST_O[13]     | -7560 | -335 | 30x52         | 204 | VMID_R[0]      | -5040 | -335 | 30x52         |
| 149 | TEST_O[14]     | -7515 | -335 | 30x52         | 205 | VMID_R[0]      | -4995 | -335 | 30x52         |
| 150 | TEST_O[15]     | -7470 | -335 | 30x52         | 206 | VMID_R[0]      | -4950 | -335 | 30x52         |
| 151 | Shielding_VSSA | -7425 | -335 | 30x52         | 207 | VMID_R[0]      | -4905 | -335 | 30x52         |
| 152 | TEST_HV_I[0]   | -7380 | -335 | 30x52         | 208 | AVDD           | -4860 | -335 | 30x52         |
| 153 | TEST_HV_I[1]   | -7335 | -335 | 30x52         | 209 | AVDD           | -4815 | -335 | 30x52         |
| 154 | TEST_MV_I[0]   | -7290 | -335 | 30x52         | 210 | AVDD           | -4770 | -335 | 30x52         |
| 155 | TEST_MV_I[1]   | -7245 | -335 | 30x52         | 211 | AVDD           | -4725 | -335 | 30x52         |
| 156 | Shielding_VSSA | -7200 | -335 | 30x52         | 212 | AVDD           | -4680 | -335 | 30x52         |
| 157 | TEST_VDD_I[0]  | -7155 | -335 | 30x52         | 213 | AVDD           | -4635 | -335 | 30x52         |
| 158 | Shielding_VSSA | -7110 | -335 | 30x52         | 214 | VSS            | -4590 | -335 | 30x52         |
| 159 | Shielding_VSSA | -7065 | -335 | 30x52         | 215 | VSS            | -4545 | -335 | 30x52         |
| 160 | Shielding_VSSA | -7020 | -335 | 30x52         | 216 | VSS            | -4500 | -335 | 30x52         |
| 161 | Shielding_VSSA | -6975 | -335 | 30x52         | 217 | VSS            | -4455 | -335 | 30x52         |
| 162 | Shielding_VSSA | -6930 | -335 | 30x52         | 218 | VSS            | -4410 | -335 | 30x52         |
| 163 | Shielding_VSSA | -6885 | -335 | 30x52         | 219 | VSS            | -4365 | -335 | 30x52         |
| 164 | Shielding_VSSA | -6840 | -335 | 30x52         | 220 | VDD_15V        | -4320 | -335 | 30x52         |
| 165 | Shielding_VSSA | -6795 | -335 | 30x52         | 221 | VDD_15V        | -4275 | -335 | 30x52         |
| 166 | Shielding_VSSA | -6750 | -335 | 30x52         | 222 | VDD_15V        | -4230 | -335 | 30x52         |
| 167 | Shielding_VSSA | -6705 | -335 | 30x52         | 223 | VDD_15V        | -4185 | -335 | 30x52         |
| 168 | Shielding_VSSA | -6660 | -335 | 30x52         | 224 | VDD_15V        | -4140 | -335 | 30x52         |

| No. | Name       | X     | Y    | Bump size(um) | No. | Name       | X     | Y    | Bump size(um) |
|-----|------------|-------|------|---------------|-----|------------|-------|------|---------------|
| 225 | VDD_15V    | -4095 | -335 | 30x52         | 281 | CKN        | -1575 | -335 | 30x52         |
| 226 | VDD_R      | -4050 | -335 | 30x52         | 282 | CKN        | -1530 | -335 | 30x52         |
| 227 | VDD_R      | -4005 | -335 | 30x52         | 283 | CKN        | -1485 | -335 | 30x52         |
| 228 | VDD_R      | -3960 | -335 | 30x52         | 284 | CKN        | -1440 | -335 | 30x52         |
| 229 | VDD_R      | -3915 | -335 | 30x52         | 285 | CKN        | -1395 | -335 | 30x52         |
| 230 | VDD_R      | -3870 | -335 | 30x52         | 286 | VSS_IF     | -1350 | -335 | 30x52         |
| 231 | VDD_R      | -3825 | -335 | 30x52         | 287 | DP[2]      | -1305 | -335 | 30x52         |
| 232 | VDD_15V_IF | -3780 | -335 | 30x52         | 288 | DP[2]      | -1260 | -335 | 30x52         |
| 233 | VDD_15V_IF | -3735 | -335 | 30x52         | 289 | DP[2]      | -1215 | -335 | 30x52         |
| 234 | VDD_15V_IF | -3690 | -335 | 30x52         | 290 | DP[2]      | -1170 | -335 | 30x52         |
| 235 | VDD_15V_IF | -3645 | -335 | 30x52         | 291 | DP[2]      | -1125 | -335 | 30x52         |
| 236 | VDD_15V_IF | -3600 | -335 | 30x52         | 292 | DP[2]      | -1080 | -335 | 30x52         |
| 237 | VDD_15V_IF | -3555 | -335 | 30x52         | 293 | DN[2]      | -1035 | -335 | 30x52         |
| 238 | VDD_12V_IF | -3510 | -335 | 30x52         | 294 | DN[2]      | -990  | -335 | 30x52         |
| 239 | VDD_12V_IF | -3465 | -335 | 30x52         | 295 | DN[2]      | -945  | -335 | 30x52         |
| 240 | VSS_IF     | -3420 | -335 | 30x52         | 296 | DN[2]      | -900  | -335 | 30x52         |
| 241 | VSS_IF     | -3375 | -335 | 30x52         | 297 | DN[2]      | -855  | -335 | 30x52         |
| 242 | VSS_IF     | -3330 | -335 | 30x52         | 298 | DN[2]      | -810  | -335 | 30x52         |
| 243 | VSS_IF     | -3285 | -335 | 30x52         | 299 | VSS_IF     | -765  | -335 | 30x52         |
| 244 | VSS_IF     | -3240 | -335 | 30x52         | 300 | DP[3]      | -720  | -335 | 30x52         |
| 245 | VSS_IF     | -3195 | -335 | 30x52         | 301 | DP[3]      | -675  | -335 | 30x52         |
| 246 | T_IRX      | -3150 | -335 | 30x52         | 302 | DP[3]      | -630  | -335 | 30x52         |
| 247 | VSS_IF     | -3105 | -335 | 30x52         | 303 | DP[3]      | -585  | -335 | 30x52         |
| 248 | DP[0]      | -3060 | -335 | 30x52         | 304 | DP[3]      | -540  | -335 | 30x52         |
| 249 | DP[0]      | -3015 | -335 | 30x52         | 305 | DP[3]      | -495  | -335 | 30x52         |
| 250 | DP[0]      | -2970 | -335 | 30x52         | 306 | DN[3]      | -450  | -335 | 30x52         |
| 251 | DP[0]      | -2925 | -335 | 30x52         | 307 | DN[3]      | -405  | -335 | 30x52         |
| 252 | DP[0]      | -2880 | -335 | 30x52         | 308 | DN[3]      | -360  | -335 | 30x52         |
| 253 | DP[0]      | -2835 | -335 | 30x52         | 309 | DN[3]      | -315  | -335 | 30x52         |
| 254 | DN[0]      | -2790 | -335 | 30x52         | 310 | DN[3]      | -270  | -335 | 30x52         |
| 255 | DN[0]      | -2745 | -335 | 30x52         | 311 | DN[3]      | -225  | -335 | 30x52         |
| 256 | DN[0]      | -2700 | -335 | 30x52         | 312 | VSS_IF     | -180  | -335 | 30x52         |
| 257 | DN[0]      | -2655 | -335 | 30x52         | 313 | VSS_IF     | -135  | -335 | 30x52         |
| 258 | DN[0]      | -2610 | -335 | 30x52         | 314 | VDD_15V_IF | -90   | -335 | 30x52         |
| 259 | DN[0]      | -2565 | -335 | 30x52         | 315 | VDD_15V_IF | -45   | -335 | 30x52         |
| 260 | VSS_IF     | -2520 | -335 | 30x52         | 316 | VDDIO      | 0     | -335 | 30x52         |
| 261 | DP[1]      | -2475 | -335 | 30x52         | 317 | VDDIO      | 45    | -335 | 30x52         |
| 262 | DP[1]      | -2430 | -335 | 30x52         | 318 | VSS        | 90    | -335 | 30x52         |
| 263 | DP[1]      | -2385 | -335 | 30x52         | 319 | VSS        | 135   | -335 | 30x52         |
| 264 | DP[1]      | -2340 | -335 | 30x52         | 320 | VSD        | 180   | -335 | 30x52         |
| 265 | DP[1]      | -2295 | -335 | 30x52         | 321 | VSD        | 225   | -335 | 30x52         |
| 266 | DP[1]      | -2250 | -335 | 30x52         | 322 | VSS        | 270   | -335 | 30x52         |
| 267 | DN[1]      | -2205 | -335 | 30x52         | 323 | HSD        | 315   | -335 | 30x52         |
| 268 | DN[1]      | -2160 | -335 | 30x52         | 324 | HSD        | 360   | -335 | 30x52         |
| 269 | DN[1]      | -2115 | -335 | 30x52         | 325 | VSS        | 405   | -335 | 30x52         |
| 270 | DN[1]      | -2070 | -335 | 30x52         | 326 | DEN        | 450   | -335 | 30x52         |
| 271 | DN[1]      | -2025 | -335 | 30x52         | 327 | DEN        | 495   | -335 | 30x52         |
| 272 | DN[1]      | -1980 | -335 | 30x52         | 328 | VSS        | 540   | -335 | 30x52         |
| 273 | VSS_IF     | -1935 | -335 | 30x52         | 329 | CLKIN      | 585   | -335 | 30x52         |
| 274 | CKP        | -1890 | -335 | 30x52         | 330 | CLKIN      | 630   | -335 | 30x52         |
| 275 | CKP        | -1845 | -335 | 30x52         | 331 | VSS        | 675   | -335 | 30x52         |
| 276 | CKP        | -1800 | -335 | 30x52         | 332 | DB[7]      | 720   | -335 | 30x52         |
| 277 | CKP        | -1755 | -335 | 30x52         | 333 | DB[7]      | 765   | -335 | 30x52         |
| 278 | CKP        | -1710 | -335 | 30x52         | 334 | DB[6]      | 810   | -335 | 30x52         |
| 279 | CKP        | -1665 | -335 | 30x52         | 335 | DB[6]      | 855   | -335 | 30x52         |
| 280 | CKN        | -1620 | -335 | 30x52         | 336 | VSS        | 900   | -335 | 30x52         |

| No. | Name  | X    | Y    | Bump size(um) | No. | Name           | X    | Y    | Bump size(um) |
|-----|-------|------|------|---------------|-----|----------------|------|------|---------------|
| 337 | DB[5] | 945  | -335 | 30x52         | 393 | VDDIO          | 3465 | -335 | 30x52         |
| 338 | DB[5] | 990  | -335 | 30x52         | 394 | VDDIO          | 3510 | -335 | 30x52         |
| 339 | DB[4] | 1035 | -335 | 30x52         | 395 | VDDIO          | 3555 | -335 | 30x52         |
| 340 | DB[4] | 1080 | -335 | 30x52         | 396 | VSS            | 3600 | -335 | 30x52         |
| 341 | VSS   | 1125 | -335 | 30x52         | 397 | VSS            | 3645 | -335 | 30x52         |
| 342 | DB[3] | 1170 | -335 | 30x52         | 398 | VSS            | 3690 | -335 | 30x52         |
| 343 | DB[3] | 1215 | -335 | 30x52         | 399 | VSS            | 3735 | -335 | 30x52         |
| 344 | DB[2] | 1260 | -335 | 30x52         | 400 | VSS            | 3780 | -335 | 30x52         |
| 345 | DB[2] | 1305 | -335 | 30x52         | 401 | VSS            | 3825 | -335 | 30x52         |
| 346 | VSS   | 1350 | -335 | 30x52         | 402 | VDD_15V        | 3870 | -335 | 30x52         |
| 347 | DB[1] | 1395 | -335 | 30x52         | 403 | VDD_15V        | 3915 | -335 | 30x52         |
| 348 | DB[1] | 1440 | -335 | 30x52         | 404 | VDD_15V        | 3960 | -335 | 30x52         |
| 349 | DB[0] | 1485 | -335 | 30x52         | 405 | VDD_15V        | 4005 | -335 | 30x52         |
| 350 | DB[0] | 1530 | -335 | 30x52         | 406 | VDD_L          | 4050 | -335 | 30x52         |
| 351 | VSS   | 1575 | -335 | 30x52         | 407 | VDD_L          | 4095 | -335 | 30x52         |
| 352 | DG[7] | 1620 | -335 | 30x52         | 408 | VDD_L          | 4140 | -335 | 30x52         |
| 353 | DG[7] | 1665 | -335 | 30x52         | 409 | VDD_L          | 4185 | -335 | 30x52         |
| 354 | DG[6] | 1710 | -335 | 30x52         | 410 | AVDD           | 4230 | -335 | 30x52         |
| 355 | DG[6] | 1755 | -335 | 30x52         | 411 | AVDD           | 4275 | -335 | 30x52         |
| 356 | VSS   | 1800 | -335 | 30x52         | 412 | AVDD           | 4320 | -335 | 30x52         |
| 357 | DG[5] | 1845 | -335 | 30x52         | 413 | AVDD           | 4365 | -335 | 30x52         |
| 358 | DG[5] | 1890 | -335 | 30x52         | 414 | AVDD           | 4410 | -335 | 30x52         |
| 359 | DG[4] | 1935 | -335 | 30x52         | 415 | AVDD           | 4455 | -335 | 30x52         |
| 360 | DG[4] | 1980 | -335 | 30x52         | 416 | VMID_L[0]      | 4500 | -335 | 30x52         |
| 361 | VSS   | 2025 | -335 | 30x52         | 417 | VMID_L[0]      | 4545 | -335 | 30x52         |
| 362 | DG[3] | 2070 | -335 | 30x52         | 418 | VMID_L[0]      | 4590 | -335 | 30x52         |
| 363 | DG[3] | 2115 | -335 | 30x52         | 419 | VMID_L[0]      | 4635 | -335 | 30x52         |
| 364 | DG[2] | 2160 | -335 | 30x52         | 420 | VMID_L[0]      | 4680 | -335 | 30x52         |
| 365 | DG[2] | 2205 | -335 | 30x52         | 421 | VMID_L[0]      | 4725 | -335 | 30x52         |
| 366 | VSS   | 2250 | -335 | 30x52         | 422 | VSSA           | 4770 | -335 | 30x52         |
| 367 | DG[1] | 2295 | -335 | 30x52         | 423 | VSSA           | 4815 | -335 | 30x52         |
| 368 | DG[1] | 2340 | -335 | 30x52         | 424 | VSSA           | 4860 | -335 | 30x52         |
| 369 | DG[0] | 2385 | -335 | 30x52         | 425 | VSSA           | 4905 | -335 | 30x52         |
| 370 | DG[0] | 2430 | -335 | 30x52         | 426 | VSSA           | 4950 | -335 | 30x52         |
| 371 | VSS   | 2475 | -335 | 30x52         | 427 | VSSA           | 4995 | -335 | 30x52         |
| 372 | DR[7] | 2520 | -335 | 30x52         | 428 | Shielding_VSSA | 5040 | -335 | 30x52         |
| 373 | DR[7] | 2565 | -335 | 30x52         | 429 | Shielding_VSSA | 5085 | -335 | 30x52         |
| 374 | DR[6] | 2610 | -335 | 30x52         | 430 | TEST_MV_O[0]   | 5130 | -335 | 30x52         |
| 375 | DR[6] | 2655 | -335 | 30x52         | 431 | TEST_MV_O[1]   | 5175 | -335 | 30x52         |
| 376 | VSS   | 2700 | -335 | 30x52         | 432 | Shielding_VSSA | 5220 | -335 | 30x52         |
| 377 | DR[5] | 2745 | -335 | 30x52         | 433 | Shielding_VSSA | 5265 | -335 | 30x52         |
| 378 | DR[5] | 2790 | -335 | 30x52         | 434 | Shielding_VSSA | 5310 | -335 | 30x52         |
| 379 | DR[4] | 2835 | -335 | 30x52         | 435 | Shielding_VSSA | 5355 | -335 | 30x52         |
| 380 | DR[4] | 2880 | -335 | 30x52         | 436 | Shielding_VSSA | 5400 | -335 | 30x52         |
| 381 | VSS   | 2925 | -335 | 30x52         | 437 | Shielding_VSSA | 5445 | -335 | 30x52         |
| 382 | DR[3] | 2970 | -335 | 30x52         | 438 | Shielding_VSSA | 5490 | -335 | 30x52         |
| 383 | DR[3] | 3015 | -335 | 30x52         | 439 | Shielding_VSSA | 5535 | -335 | 30x52         |
| 384 | DR[2] | 3060 | -335 | 30x52         | 440 | Shielding_VSSA | 5580 | -335 | 30x52         |
| 385 | DR[2] | 3105 | -335 | 30x52         | 441 | Shielding_VSSA | 5625 | -335 | 30x52         |
| 386 | VSS   | 3150 | -335 | 30x52         | 442 | Shielding_VSSA | 5670 | -335 | 30x52         |
| 387 | DR[1] | 3195 | -335 | 30x52         | 443 | Shielding_VSSA | 5715 | -335 | 30x52         |
| 388 | DR[1] | 3240 | -335 | 30x52         | 444 | Shielding_VSSA | 5760 | -335 | 30x52         |
| 389 | DR[0] | 3285 | -335 | 30x52         | 445 | Shielding_VSSA | 5805 | -335 | 30x52         |
| 390 | DR[0] | 3330 | -335 | 30x52         | 446 | TEST_HV_O[0]   | 5850 | -335 | 30x52         |
| 391 | VSS   | 3375 | -335 | 30x52         | 447 | TEST_HV_O[1]   | 5895 | -335 | 30x52         |
| 392 | VSS   | 3420 | -335 | 30x52         | 448 | Shielding_VSSA | 5940 | -335 | 30x52         |

| No. | Name           | X    | Y    | Bump size(um) | No. | Name           | X     | Y    | Bump size(um) |
|-----|----------------|------|------|---------------|-----|----------------|-------|------|---------------|
| 449 | Shielding_VSSA | 5985 | -335 | 30x52         | 505 | VMIDOP_EN      | 8505  | -335 | 30x52         |
| 450 | Shielding_VSSA | 6030 | -335 | 30x52         | 506 | Shielding_VSSA | 8550  | -335 | 30x52         |
| 451 | Shielding_VSSA | 6075 | -335 | 30x52         | 507 | VQLI_L         | 8595  | -335 | 30x52         |
| 452 | SCL_I2C        | 6120 | -335 | 30x52         | 508 | VQLI_L         | 8640  | -335 | 30x52         |
| 453 | SCL_I2C        | 6165 | -335 | 30x52         | 509 | VQHI_L         | 8685  | -335 | 30x52         |
| 454 | SDA_I2C        | 6210 | -335 | 30x52         | 510 | VQHI_L         | 8730  | -335 | 30x52         |
| 455 | SDA_I2C        | 6255 | -335 | 30x52         | 511 | Shielding_VSSA | 8775  | -335 | 30x52         |
| 456 | Shielding_VSSA | 6300 | -335 | 30x52         | 512 | VSS            | 8820  | -335 | 30x52         |
| 457 | Shielding_VSSA | 6345 | -335 | 30x52         | 513 | VSS            | 8865  | -335 | 30x52         |
| 458 | Shielding_VSSA | 6390 | -335 | 30x52         | 514 | VSS            | 8910  | -335 | 30x52         |
| 459 | CSB            | 6435 | -335 | 30x52         | 515 | VSS            | 8955  | -335 | 30x52         |
| 460 | CSB            | 6480 | -335 | 30x52         | 516 | VSS            | 9000  | -335 | 30x52         |
| 461 | SCL            | 6525 | -335 | 30x52         | 517 | VSS            | 9045  | -335 | 30x52         |
| 462 | SCL            | 6570 | -335 | 30x52         | 518 | AVDD           | 9090  | -335 | 30x52         |
| 463 | SDI            | 6615 | -335 | 30x52         | 519 | AVDD           | 9135  | -335 | 30x52         |
| 464 | SDI            | 6660 | -335 | 30x52         | 520 | AVDD           | 9180  | -335 | 30x52         |
| 465 | SDO            | 6705 | -335 | 30x52         | 521 | AVDD           | 9225  | -335 | 30x52         |
| 466 | SDO            | 6750 | -335 | 30x52         | 522 | AVDD           | 9270  | -335 | 30x52         |
| 467 | TP_SYNC        | 6795 | -335 | 30x52         | 523 | AVDD           | 9315  | -335 | 30x52         |
| 468 | LEDPWM         | 6840 | -335 | 30x52         | 524 | VMID_L[1]      | 9360  | -335 | 30x52         |
| 469 | LEDPWM         | 6885 | -335 | 30x52         | 525 | VMID_L[1]      | 9405  | -335 | 30x52         |
| 470 | TEST_[24]      | 6930 | -335 | 30x52         | 526 | VMID_L[1]      | 9450  | -335 | 30x52         |
| 471 | TEST_[25]      | 6975 | -335 | 30x52         | 527 | VMID_L[1]      | 9495  | -335 | 30x52         |
| 472 | RESX           | 7020 | -335 | 30x52         | 528 | VMID_L[1]      | 9540  | -335 | 30x52         |
| 473 | RESX           | 7065 | -335 | 30x52         | 529 | VMID_L[1]      | 9585  | -335 | 30x52         |
| 474 | STBYB          | 7110 | -335 | 30x52         | 530 | VSSA           | 9630  | -335 | 30x52         |
| 475 | STBYB          | 7155 | -335 | 30x52         | 531 | VSSA           | 9675  | -335 | 30x52         |
| 476 | BIST_EN        | 7200 | -335 | 30x52         | 532 | VSSA           | 9720  | -335 | 30x52         |
| 477 | BIST_EN        | 7245 | -335 | 30x52         | 533 | VSSA           | 9765  | -335 | 30x52         |
| 478 | SHLR           | 7290 | -335 | 30x52         | 534 | VSSA           | 9810  | -335 | 30x52         |
| 479 | SHLR           | 7335 | -335 | 30x52         | 535 | VSSA           | 9855  | -335 | 30x52         |
| 480 | UPDN           | 7380 | -335 | 30x52         | 536 | Shielding_VSSA | 9900  | -335 | 30x52         |
| 481 | UPDN           | 7425 | -335 | 30x52         | 537 | VCOMIN         | 9945  | -335 | 30x52         |
| 482 | MODE           | 7470 | -335 | 30x52         | 538 | VCOMIN         | 9990  | -335 | 30x52         |
| 483 | PNSWAP         | 7515 | -335 | 30x52         | 539 | Shielding_VSSA | 10035 | -335 | 30x52         |
| 484 | DSWAP[1]       | 7560 | -335 | 30x52         | 540 | VCOM_OP        | 10080 | -335 | 30x52         |
| 485 | DSWAP[0]       | 7605 | -335 | 30x52         | 541 | VCOM_OP        | 10125 | -335 | 30x52         |
| 486 | LANSEL[1]      | 7650 | -335 | 30x52         | 542 | VCOM_OP        | 10170 | -335 | 30x52         |
| 487 | LANSEL[0]      | 7695 | -335 | 30x52         | 543 | VCOM_OP        | 10215 | -335 | 30x52         |
| 488 | IFSEL[1]       | 7740 | -335 | 30x52         | 544 | VCOM_OP        | 10260 | -335 | 30x52         |
| 489 | IFSEL[0]       | 7785 | -335 | 30x52         | 545 | VCOM_OP        | 10305 | -335 | 30x52         |
| 490 | ZIGZAG_ENB     | 7830 | -335 | 30x52         | 546 | Shielding_VSSA | 10350 | -335 | 30x52         |
| 491 | PTYPE[1]       | 7875 | -335 | 30x52         | 547 | Shielding_VSSA | 10395 | -335 | 30x52         |
| 492 | PTYPE[0]       | 7920 | -335 | 30x52         | 548 | Shielding_VSSA | 10440 | -335 | 30x52         |
| 493 | DCLKPOL        | 7965 | -335 | 30x52         | 549 | Shielding_VSSA | 10485 | -335 | 30x52         |
| 494 | INVSEL         | 8010 | -335 | 30x52         | 550 | Shielding_VSSA | 10530 | -335 | 30x52         |
| 495 | NBW            | 8055 | -335 | 30x52         | 551 | VQLO           | 10575 | -335 | 30x52         |
| 496 | LVFMT          | 8100 | -335 | 30x52         | 552 | VQLO           | 10620 | -335 | 30x52         |
| 497 | LVBIT          | 8145 | -335 | 30x52         | 553 | VQLO           | 10665 | -335 | 30x52         |
| 498 | CMD_SEL[1]     | 8190 | -335 | 30x52         | 554 | VQLO           | 10710 | -335 | 30x52         |
| 499 | CMD_SEL[0]     | 8235 | -335 | 30x52         | 555 | VMID_OP        | 10755 | -335 | 30x52         |
| 500 | VCOMIN_SEL     | 8280 | -335 | 30x52         | 556 | VMID_OP        | 10800 | -335 | 30x52         |
| 501 | VCOMIN_SEL     | 8325 | -335 | 30x52         | 557 | VMID_OP        | 10845 | -335 | 30x52         |
| 502 | VCOM_EN        | 8370 | -335 | 30x52         | 558 | VMID_OP        | 10890 | -335 | 30x52         |
| 503 | VCOM_EN        | 8415 | -335 | 30x52         | 559 | VMID_OP        | 10935 | -335 | 30x52         |
| 504 | VMIDOP_EN      | 8460 | -335 | 30x52         | 560 | VMID_OP        | 10980 | -335 | 30x52         |



| No. | Name           | X     | Y    | Bump size(um) | No. | Name      | X     | Y    | Bump size(um) |
|-----|----------------|-------|------|---------------|-----|-----------|-------|------|---------------|
| 561 | VQHO           | 11025 | -335 | 30x52         | 617 | GOUT_L[6] | 13545 | -335 | 30x52         |
| 562 | VQHO           | 11070 | -335 | 30x52         | 618 | GOUT_L[6] | 13590 | -335 | 30x52         |
| 563 | VQHO           | 11115 | -335 | 30x52         | 619 | GOUT_L[5] | 13635 | -335 | 30x52         |
| 564 | VQHO           | 11160 | -335 | 30x52         | 620 | GOUT_L[5] | 13680 | -335 | 30x52         |
| 565 | Shielding_VSSA | 11205 | -335 | 30x52         | 621 | GOUT_L[4] | 13725 | -335 | 30x52         |
| 566 | Shielding_VSSA | 11250 | -335 | 30x52         | 622 | GOUT_L[4] | 13770 | -335 | 30x52         |
| 567 | Shielding_VSSA | 11295 | -335 | 30x52         | 623 | GOUT_L[3] | 13815 | -335 | 30x52         |
| 568 | Shielding_VSSA | 11340 | -335 | 30x52         | 624 | GOUT_L[3] | 13860 | -335 | 30x52         |
| 569 | Shielding_VSSA | 11385 | -335 | 30x52         | 625 | GOUT_L[2] | 13905 | -335 | 30x52         |
| 570 | PASS1_L        | 11430 | -335 | 30x52         | 626 | GOUT_L[2] | 13950 | -335 | 30x52         |
| 571 | PASS1_L        | 11475 | -335 | 30x52         | 627 | GOUT_L[1] | 13995 | -335 | 30x52         |
| 572 | Shielding_VSSA | 11520 | -335 | 30x52         | 628 | GOUT_L[1] | 14040 | -335 | 30x52         |
| 573 | VGH_L          | 11565 | -335 | 30x52         | 629 | DUMMY[36] | 14085 | -335 | 30x52         |
| 574 | VGH_L          | 11610 | -335 | 30x52         | 630 | PASS2_L   | 14130 | -335 | 30x52         |
| 575 | VGH_L          | 11655 | -335 | 30x52         | 631 | PASS2_L   | 14175 | -335 | 30x52         |
| 576 | VGH_L          | 11700 | -335 | 30x52         | 632 | DUMMY     | 13864 | 330  | 18x60         |
| 577 | VGH_L          | 11745 | -335 | 30x52         | 633 | PASS2_L   | 13828 | 330  | 18x60         |
| 578 | VGH_L          | 11790 | -335 | 30x52         | 634 | PASS2_L   | 13792 | 330  | 18x60         |
| 579 | VGL            | 11835 | -335 | 30x52         | 635 | PASS2_L   | 13756 | 330  | 18x60         |
| 580 | VGL            | 11880 | -335 | 30x52         | 636 | PASS2_L   | 13720 | 330  | 18x60         |
| 581 | VGL            | 11925 | -335 | 30x52         | 637 | DUMMY     | 13684 | 330  | 18x60         |
| 582 | VGL            | 11970 | -335 | 30x52         | 638 | PASS1_L   | 13648 | 330  | 18x60         |
| 583 | VGL            | 12015 | -335 | 30x52         | 639 | PASS1_L   | 13612 | 330  | 18x60         |
| 584 | VGL            | 12060 | -335 | 30x52         | 640 | PASS1_L   | 13576 | 330  | 18x60         |
| 585 | GOUT_L[22]     | 12105 | -335 | 30x52         | 641 | PASS1_L   | 13540 | 330  | 18x60         |
| 586 | GOUT_L[22]     | 12150 | -335 | 30x52         | 642 | DUMMY     | 13504 | 330  | 18x60         |
| 587 | GOUT_L[21]     | 12195 | -335 | 30x52         | 643 | DUMMY     | 13468 | 330  | 18x60         |
| 588 | GOUT_L[21]     | 12240 | -335 | 30x52         | 644 | DUMMY     | 13432 | 330  | 18x60         |
| 589 | GOUT_L[20]     | 12285 | -335 | 30x52         | 645 | DUMMY     | 13396 | 330  | 18x60         |
| 590 | GOUT_L[20]     | 12330 | -335 | 30x52         | 646 | DUMMY     | 13360 | 330  | 18x60         |
| 591 | GOUT_L[19]     | 12375 | -335 | 30x52         | 647 | DUMMY     | 13324 | 330  | 18x60         |
| 592 | GOUT_L[19]     | 12420 | -335 | 30x52         | 648 | DUMMY     | 13288 | 330  | 18x60         |
| 593 | GOUT_L[18]     | 12465 | -335 | 30x52         | 649 | DUMMY     | 13252 | 330  | 18x60         |
| 594 | GOUT_L[18]     | 12510 | -335 | 30x52         | 650 | DUMMY     | 13216 | 330  | 18x60         |
| 595 | GOUT_L[17]     | 12555 | -335 | 30x52         | 651 | DUMMY     | 13180 | 330  | 18x60         |
| 596 | GOUT_L[17]     | 12600 | -335 | 30x52         | 652 | S[0]      | 13168 | 240  | 18x60         |
| 597 | GOUT_L[16]     | 12645 | -335 | 30x52         | 653 | S[1]      | 13156 | 150  | 18x60         |
| 598 | GOUT_L[16]     | 12690 | -335 | 30x52         | 654 | S[2]      | 13144 | 330  | 18x60         |
| 599 | GOUT_L[15]     | 12735 | -335 | 30x52         | 655 | S[3]      | 13132 | 240  | 18x60         |
| 600 | GOUT_L[15]     | 12780 | -335 | 30x52         | 656 | S[4]      | 13120 | 150  | 18x60         |
| 601 | GOUT_L[14]     | 12825 | -335 | 30x52         | 657 | S[5]      | 13108 | 330  | 18x60         |
| 602 | GOUT_L[14]     | 12870 | -335 | 30x52         | 658 | S[6]      | 13096 | 240  | 18x60         |
| 603 | GOUT_L[13]     | 12915 | -335 | 30x52         | 659 | S[7]      | 13084 | 150  | 18x60         |
| 604 | GOUT_L[13]     | 12960 | -335 | 30x52         | 660 | S[8]      | 13072 | 330  | 18x60         |
| 605 | GOUT_L[12]     | 13005 | -335 | 30x52         | 661 | S[9]      | 13060 | 240  | 18x60         |
| 606 | GOUT_L[12]     | 13050 | -335 | 30x52         | 662 | S[10]     | 13048 | 150  | 18x60         |
| 607 | GOUT_L[11]     | 13095 | -335 | 30x52         | 663 | S[11]     | 13036 | 330  | 18x60         |
| 608 | GOUT_L[11]     | 13140 | -335 | 30x52         | 664 | S[12]     | 13024 | 240  | 18x60         |
| 609 | GOUT_L[10]     | 13185 | -335 | 30x52         | 665 | S[13]     | 13012 | 150  | 18x60         |
| 610 | GOUT_L[10]     | 13230 | -335 | 30x52         | 666 | S[14]     | 13000 | 330  | 18x60         |
| 611 | GOUT_L[9]      | 13275 | -335 | 30x52         | 667 | S[15]     | 12988 | 240  | 18x60         |
| 612 | GOUT_L[9]      | 13320 | -335 | 30x52         | 668 | S[16]     | 12976 | 150  | 18x60         |
| 613 | GOUT_L[8]      | 13365 | -335 | 30x52         | 669 | S[17]     | 12964 | 330  | 18x60         |
| 614 | GOUT_L[8]      | 13410 | -335 | 30x52         | 670 | S[18]     | 12952 | 240  | 18x60         |
| 615 | GOUT_L[7]      | 13455 | -335 | 30x52         | 671 | S[19]     | 12940 | 150  | 18x60         |
| 616 | GOUT_L[7]      | 13500 | -335 | 30x52         | 672 | S[20]     | 12928 | 330  | 18x60         |

| No. | Name  | X     | Y   | Bump size(um) | No. | Name   | X     | Y   | Bump size(um) |
|-----|-------|-------|-----|---------------|-----|--------|-------|-----|---------------|
| 673 | S[21] | 12916 | 240 | 18x60         | 729 | S[77]  | 12244 | 330 | 18x60         |
| 674 | S[22] | 12904 | 150 | 18x60         | 730 | S[78]  | 12232 | 240 | 18x60         |
| 675 | S[23] | 12892 | 330 | 18x60         | 731 | S[79]  | 12220 | 150 | 18x60         |
| 676 | S[24] | 12880 | 240 | 18x60         | 732 | S[80]  | 12208 | 330 | 18x60         |
| 677 | S[25] | 12868 | 150 | 18x60         | 733 | S[81]  | 12196 | 240 | 18x60         |
| 678 | S[26] | 12856 | 330 | 18x60         | 734 | S[82]  | 12184 | 150 | 18x60         |
| 679 | S[27] | 12844 | 240 | 18x60         | 735 | S[83]  | 12172 | 330 | 18x60         |
| 680 | S[28] | 12832 | 150 | 18x60         | 736 | S[84]  | 12160 | 240 | 18x60         |
| 681 | S[29] | 12820 | 330 | 18x60         | 737 | S[85]  | 12148 | 150 | 18x60         |
| 682 | S[30] | 12808 | 240 | 18x60         | 738 | S[86]  | 12136 | 330 | 18x60         |
| 683 | S[31] | 12796 | 150 | 18x60         | 739 | S[87]  | 12124 | 240 | 18x60         |
| 684 | S[32] | 12784 | 330 | 18x60         | 740 | S[88]  | 12112 | 150 | 18x60         |
| 685 | S[33] | 12772 | 240 | 18x60         | 741 | S[89]  | 12100 | 330 | 18x60         |
| 686 | S[34] | 12760 | 150 | 18x60         | 742 | S[90]  | 12088 | 240 | 18x60         |
| 687 | S[35] | 12748 | 330 | 18x60         | 743 | S[91]  | 12076 | 150 | 18x60         |
| 688 | S[36] | 12736 | 240 | 18x60         | 744 | S[92]  | 12064 | 330 | 18x60         |
| 689 | S[37] | 12724 | 150 | 18x60         | 745 | S[93]  | 12052 | 240 | 18x60         |
| 690 | S[38] | 12712 | 330 | 18x60         | 746 | S[94]  | 12040 | 150 | 18x60         |
| 691 | S[39] | 12700 | 240 | 18x60         | 747 | S[95]  | 12028 | 330 | 18x60         |
| 692 | S[40] | 12688 | 150 | 18x60         | 748 | S[96]  | 12016 | 240 | 18x60         |
| 693 | S[41] | 12676 | 330 | 18x60         | 749 | S[97]  | 12004 | 150 | 18x60         |
| 694 | S[42] | 12664 | 240 | 18x60         | 750 | S[98]  | 11992 | 330 | 18x60         |
| 695 | S[43] | 12652 | 150 | 18x60         | 751 | S[99]  | 11980 | 240 | 18x60         |
| 696 | S[44] | 12640 | 330 | 18x60         | 752 | S[100] | 11968 | 150 | 18x60         |
| 697 | S[45] | 12628 | 240 | 18x60         | 753 | S[101] | 11956 | 330 | 18x60         |
| 698 | S[46] | 12616 | 150 | 18x60         | 754 | S[102] | 11944 | 240 | 18x60         |
| 699 | S[47] | 12604 | 330 | 18x60         | 755 | S[103] | 11932 | 150 | 18x60         |
| 700 | S[48] | 12592 | 240 | 18x60         | 756 | S[104] | 11920 | 330 | 18x60         |
| 701 | S[49] | 12580 | 150 | 18x60         | 757 | S[105] | 11908 | 240 | 18x60         |
| 702 | S[50] | 12568 | 330 | 18x60         | 758 | S[106] | 11896 | 150 | 18x60         |
| 703 | S[51] | 12556 | 240 | 18x60         | 759 | S[107] | 11884 | 330 | 18x60         |
| 704 | S[52] | 12544 | 150 | 18x60         | 760 | S[108] | 11872 | 240 | 18x60         |
| 705 | S[53] | 12532 | 330 | 18x60         | 761 | S[109] | 11860 | 150 | 18x60         |
| 706 | S[54] | 12520 | 240 | 18x60         | 762 | S[110] | 11848 | 330 | 18x60         |
| 707 | S[55] | 12508 | 150 | 18x60         | 763 | S[111] | 11836 | 240 | 18x60         |
| 708 | S[56] | 12496 | 330 | 18x60         | 764 | S[112] | 11824 | 150 | 18x60         |
| 709 | S[57] | 12484 | 240 | 18x60         | 765 | S[113] | 11812 | 330 | 18x60         |
| 710 | S[58] | 12472 | 150 | 18x60         | 766 | S[114] | 11800 | 240 | 18x60         |
| 711 | S[59] | 12460 | 330 | 18x60         | 767 | S[115] | 11788 | 150 | 18x60         |
| 712 | S[60] | 12448 | 240 | 18x60         | 768 | S[116] | 11776 | 330 | 18x60         |
| 713 | S[61] | 12436 | 150 | 18x60         | 769 | S[117] | 11764 | 240 | 18x60         |
| 714 | S[62] | 12424 | 330 | 18x60         | 770 | S[118] | 11752 | 150 | 18x60         |
| 715 | S[63] | 12412 | 240 | 18x60         | 771 | S[119] | 11740 | 330 | 18x60         |
| 716 | S[64] | 12400 | 150 | 18x60         | 772 | S[120] | 11728 | 240 | 18x60         |
| 717 | S[65] | 12388 | 330 | 18x60         | 773 | S[121] | 11716 | 150 | 18x60         |
| 718 | S[66] | 12376 | 240 | 18x60         | 774 | S[122] | 11704 | 330 | 18x60         |
| 719 | S[67] | 12364 | 150 | 18x60         | 775 | S[123] | 11692 | 240 | 18x60         |
| 720 | S[68] | 12352 | 330 | 18x60         | 776 | S[124] | 11680 | 150 | 18x60         |
| 721 | S[69] | 12340 | 240 | 18x60         | 777 | S[125] | 11668 | 330 | 18x60         |
| 722 | S[70] | 12328 | 150 | 18x60         | 778 | S[126] | 11656 | 240 | 18x60         |
| 723 | S[71] | 12316 | 330 | 18x60         | 779 | S[127] | 11644 | 150 | 18x60         |
| 724 | S[72] | 12304 | 240 | 18x60         | 780 | S[128] | 11632 | 330 | 18x60         |
| 725 | S[73] | 12292 | 150 | 18x60         | 781 | S[129] | 11620 | 240 | 18x60         |
| 726 | S[74] | 12280 | 330 | 18x60         | 782 | S[130] | 11608 | 150 | 18x60         |
| 727 | S[75] | 12268 | 240 | 18x60         | 783 | S[131] | 11596 | 330 | 18x60         |
| 728 | S[76] | 12256 | 150 | 18x60         | 784 | S[132] | 11584 | 240 | 18x60         |

| No. | Name   | X     | Y   | Bump size(um) | No. | Name   | X     | Y   | Bump size(um) |
|-----|--------|-------|-----|---------------|-----|--------|-------|-----|---------------|
| 785 | S[133] | 11572 | 150 | 18x60         | 841 | S[189] | 10900 | 240 | 18x60         |
| 786 | S[134] | 11560 | 330 | 18x60         | 842 | S[190] | 10888 | 150 | 18x60         |
| 787 | S[135] | 11548 | 240 | 18x60         | 843 | S[191] | 10876 | 330 | 18x60         |
| 788 | S[136] | 11536 | 150 | 18x60         | 844 | S[192] | 10864 | 240 | 18x60         |
| 789 | S[137] | 11524 | 330 | 18x60         | 845 | S[193] | 10852 | 150 | 18x60         |
| 790 | S[138] | 11512 | 240 | 18x60         | 846 | S[194] | 10840 | 330 | 18x60         |
| 791 | S[139] | 11500 | 150 | 18x60         | 847 | S[195] | 10828 | 240 | 18x60         |
| 792 | S[140] | 11488 | 330 | 18x60         | 848 | S[196] | 10816 | 150 | 18x60         |
| 793 | S[141] | 11476 | 240 | 18x60         | 849 | S[197] | 10804 | 330 | 18x60         |
| 794 | S[142] | 11464 | 150 | 18x60         | 850 | S[198] | 10792 | 240 | 18x60         |
| 795 | S[143] | 11452 | 330 | 18x60         | 851 | S[199] | 10780 | 150 | 18x60         |
| 796 | S[144] | 11440 | 240 | 18x60         | 852 | S[200] | 10768 | 330 | 18x60         |
| 797 | S[145] | 11428 | 150 | 18x60         | 853 | S[201] | 10756 | 240 | 18x60         |
| 798 | S[146] | 11416 | 330 | 18x60         | 854 | S[202] | 10744 | 150 | 18x60         |
| 799 | S[147] | 11404 | 240 | 18x60         | 855 | S[203] | 10732 | 330 | 18x60         |
| 800 | S[148] | 11392 | 150 | 18x60         | 856 | S[204] | 10720 | 240 | 18x60         |
| 801 | S[149] | 11380 | 330 | 18x60         | 857 | S[205] | 10708 | 150 | 18x60         |
| 802 | S[150] | 11368 | 240 | 18x60         | 858 | S[206] | 10696 | 330 | 18x60         |
| 803 | S[151] | 11356 | 150 | 18x60         | 859 | S[207] | 10684 | 240 | 18x60         |
| 804 | S[152] | 11344 | 330 | 18x60         | 860 | S[208] | 10672 | 150 | 18x60         |
| 805 | S[153] | 11332 | 240 | 18x60         | 861 | S[209] | 10660 | 330 | 18x60         |
| 806 | S[154] | 11320 | 150 | 18x60         | 862 | S[210] | 10648 | 240 | 18x60         |
| 807 | S[155] | 11308 | 330 | 18x60         | 863 | S[211] | 10636 | 150 | 18x60         |
| 808 | S[156] | 11296 | 240 | 18x60         | 864 | S[212] | 10624 | 330 | 18x60         |
| 809 | S[157] | 11284 | 150 | 18x60         | 865 | S[213] | 10612 | 240 | 18x60         |
| 810 | S[158] | 11272 | 330 | 18x60         | 866 | S[214] | 10600 | 150 | 18x60         |
| 811 | S[159] | 11260 | 240 | 18x60         | 867 | S[215] | 10588 | 330 | 18x60         |
| 812 | S[160] | 11248 | 150 | 18x60         | 868 | S[216] | 10576 | 240 | 18x60         |
| 813 | S[161] | 11236 | 330 | 18x60         | 869 | S[217] | 10564 | 150 | 18x60         |
| 814 | S[162] | 11224 | 240 | 18x60         | 870 | S[218] | 10552 | 330 | 18x60         |
| 815 | S[163] | 11212 | 150 | 18x60         | 871 | S[219] | 10540 | 240 | 18x60         |
| 816 | S[164] | 11200 | 330 | 18x60         | 872 | S[220] | 10528 | 150 | 18x60         |
| 817 | S[165] | 11188 | 240 | 18x60         | 873 | S[221] | 10516 | 330 | 18x60         |
| 818 | S[166] | 11176 | 150 | 18x60         | 874 | S[222] | 10504 | 240 | 18x60         |
| 819 | S[167] | 11164 | 330 | 18x60         | 875 | S[223] | 10492 | 150 | 18x60         |
| 820 | S[168] | 11152 | 240 | 18x60         | 876 | S[224] | 10480 | 330 | 18x60         |
| 821 | S[169] | 11140 | 150 | 18x60         | 877 | S[225] | 10468 | 240 | 18x60         |
| 822 | S[170] | 11128 | 330 | 18x60         | 878 | S[226] | 10456 | 150 | 18x60         |
| 823 | S[171] | 11116 | 240 | 18x60         | 879 | S[227] | 10444 | 330 | 18x60         |
| 824 | S[172] | 11104 | 150 | 18x60         | 880 | S[228] | 10432 | 240 | 18x60         |
| 825 | S[173] | 11092 | 330 | 18x60         | 881 | S[229] | 10420 | 150 | 18x60         |
| 826 | S[174] | 11080 | 240 | 18x60         | 882 | S[230] | 10408 | 330 | 18x60         |
| 827 | S[175] | 11068 | 150 | 18x60         | 883 | S[231] | 10396 | 240 | 18x60         |
| 828 | S[176] | 11056 | 330 | 18x60         | 884 | S[232] | 10384 | 150 | 18x60         |
| 829 | S[177] | 11044 | 240 | 18x60         | 885 | S[233] | 10372 | 330 | 18x60         |
| 830 | S[178] | 11032 | 150 | 18x60         | 886 | S[234] | 10360 | 240 | 18x60         |
| 831 | S[179] | 11020 | 330 | 18x60         | 887 | S[235] | 10348 | 150 | 18x60         |
| 832 | S[180] | 11008 | 240 | 18x60         | 888 | S[236] | 10336 | 330 | 18x60         |
| 833 | S[181] | 10996 | 150 | 18x60         | 889 | S[237] | 10324 | 240 | 18x60         |
| 834 | S[182] | 10984 | 330 | 18x60         | 890 | S[238] | 10312 | 150 | 18x60         |
| 835 | S[183] | 10972 | 240 | 18x60         | 891 | S[239] | 10300 | 330 | 18x60         |
| 836 | S[184] | 10960 | 150 | 18x60         | 892 | S[240] | 10288 | 240 | 18x60         |
| 837 | S[185] | 10948 | 330 | 18x60         | 893 | S[241] | 10276 | 150 | 18x60         |
| 838 | S[186] | 10936 | 240 | 18x60         | 894 | S[242] | 10264 | 330 | 18x60         |
| 839 | S[187] | 10924 | 150 | 18x60         | 895 | S[243] | 10252 | 240 | 18x60         |
| 840 | S[188] | 10912 | 330 | 18x60         | 896 | S[244] | 10240 | 150 | 18x60         |

| No. | Name   | X     | Y   | Bump size(um) | No.  | Name   | X    | Y   | Bump size(um) |
|-----|--------|-------|-----|---------------|------|--------|------|-----|---------------|
| 897 | S[245] | 10228 | 330 | 18x60         | 953  | S[301] | 9556 | 150 | 18x60         |
| 898 | S[246] | 10216 | 240 | 18x60         | 954  | S[302] | 9544 | 330 | 18x60         |
| 899 | S[247] | 10204 | 150 | 18x60         | 955  | S[303] | 9532 | 240 | 18x60         |
| 900 | S[248] | 10192 | 330 | 18x60         | 956  | S[304] | 9520 | 150 | 18x60         |
| 901 | S[249] | 10180 | 240 | 18x60         | 957  | S[305] | 9508 | 330 | 18x60         |
| 902 | S[250] | 10168 | 150 | 18x60         | 958  | S[306] | 9496 | 240 | 18x60         |
| 903 | S[251] | 10156 | 330 | 18x60         | 959  | S[307] | 9484 | 150 | 18x60         |
| 904 | S[252] | 10144 | 240 | 18x60         | 960  | S[308] | 9472 | 330 | 18x60         |
| 905 | S[253] | 10132 | 150 | 18x60         | 961  | S[309] | 9460 | 240 | 18x60         |
| 906 | S[254] | 10120 | 330 | 18x60         | 962  | S[310] | 9448 | 150 | 18x60         |
| 907 | S[255] | 10108 | 240 | 18x60         | 963  | S[311] | 9436 | 330 | 18x60         |
| 908 | S[256] | 10096 | 150 | 18x60         | 964  | S[312] | 9424 | 240 | 18x60         |
| 909 | S[257] | 10084 | 330 | 18x60         | 965  | S[313] | 9412 | 150 | 18x60         |
| 910 | S[258] | 10072 | 240 | 18x60         | 966  | S[314] | 9400 | 330 | 18x60         |
| 911 | S[259] | 10060 | 150 | 18x60         | 967  | S[315] | 9388 | 240 | 18x60         |
| 912 | S[260] | 10048 | 330 | 18x60         | 968  | S[316] | 9376 | 150 | 18x60         |
| 913 | S[261] | 10036 | 240 | 18x60         | 969  | S[317] | 9364 | 330 | 18x60         |
| 914 | S[262] | 10024 | 150 | 18x60         | 970  | S[318] | 9352 | 240 | 18x60         |
| 915 | S[263] | 10012 | 330 | 18x60         | 971  | S[319] | 9340 | 150 | 18x60         |
| 916 | S[264] | 10000 | 240 | 18x60         | 972  | S[320] | 9328 | 330 | 18x60         |
| 917 | S[265] | 9988  | 150 | 18x60         | 973  | S[321] | 9316 | 240 | 18x60         |
| 918 | S[266] | 9976  | 330 | 18x60         | 974  | S[322] | 9304 | 150 | 18x60         |
| 919 | S[267] | 9964  | 240 | 18x60         | 975  | S[323] | 9292 | 330 | 18x60         |
| 920 | S[268] | 9952  | 150 | 18x60         | 976  | S[324] | 9280 | 240 | 18x60         |
| 921 | S[269] | 9940  | 330 | 18x60         | 977  | S[325] | 9268 | 150 | 18x60         |
| 922 | S[270] | 9928  | 240 | 18x60         | 978  | S[326] | 9256 | 330 | 18x60         |
| 923 | S[271] | 9916  | 150 | 18x60         | 979  | S[327] | 9244 | 240 | 18x60         |
| 924 | S[272] | 9904  | 330 | 18x60         | 980  | S[328] | 9232 | 150 | 18x60         |
| 925 | S[273] | 9892  | 240 | 18x60         | 981  | S[329] | 9220 | 330 | 18x60         |
| 926 | S[274] | 9880  | 150 | 18x60         | 982  | S[330] | 9208 | 240 | 18x60         |
| 927 | S[275] | 9868  | 330 | 18x60         | 983  | S[331] | 9196 | 150 | 18x60         |
| 928 | S[276] | 9856  | 240 | 18x60         | 984  | S[332] | 9184 | 330 | 18x60         |
| 929 | S[277] | 9844  | 150 | 18x60         | 985  | S[333] | 9172 | 240 | 18x60         |
| 930 | S[278] | 9832  | 330 | 18x60         | 986  | S[334] | 9160 | 150 | 18x60         |
| 931 | S[279] | 9820  | 240 | 18x60         | 987  | S[335] | 9148 | 330 | 18x60         |
| 932 | S[280] | 9808  | 150 | 18x60         | 988  | S[336] | 9136 | 240 | 18x60         |
| 933 | S[281] | 9796  | 330 | 18x60         | 989  | S[337] | 9124 | 150 | 18x60         |
| 934 | S[282] | 9784  | 240 | 18x60         | 990  | S[338] | 9112 | 330 | 18x60         |
| 935 | S[283] | 9772  | 150 | 18x60         | 991  | S[339] | 9100 | 240 | 18x60         |
| 936 | S[284] | 9760  | 330 | 18x60         | 992  | S[340] | 9088 | 150 | 18x60         |
| 937 | S[285] | 9748  | 240 | 18x60         | 993  | S[341] | 9076 | 330 | 18x60         |
| 938 | S[286] | 9736  | 150 | 18x60         | 994  | S[342] | 9064 | 240 | 18x60         |
| 939 | S[287] | 9724  | 330 | 18x60         | 995  | S[343] | 9052 | 150 | 18x60         |
| 940 | S[288] | 9712  | 240 | 18x60         | 996  | S[344] | 9040 | 330 | 18x60         |
| 941 | S[289] | 9700  | 150 | 18x60         | 997  | S[345] | 9028 | 240 | 18x60         |
| 942 | S[290] | 9688  | 330 | 18x60         | 998  | S[346] | 9016 | 150 | 18x60         |
| 943 | S[291] | 9676  | 240 | 18x60         | 999  | S[347] | 9004 | 330 | 18x60         |
| 944 | S[292] | 9664  | 150 | 18x60         | 1000 | S[348] | 8992 | 240 | 18x60         |
| 945 | S[293] | 9652  | 330 | 18x60         | 1001 | S[349] | 8980 | 150 | 18x60         |
| 946 | S[294] | 9640  | 240 | 18x60         | 1002 | S[350] | 8968 | 330 | 18x60         |
| 947 | S[295] | 9628  | 150 | 18x60         | 1003 | S[351] | 8956 | 240 | 18x60         |
| 948 | S[296] | 9616  | 330 | 18x60         | 1004 | S[352] | 8944 | 150 | 18x60         |
| 949 | S[297] | 9604  | 240 | 18x60         | 1005 | S[353] | 8932 | 330 | 18x60         |
| 950 | S[298] | 9592  | 150 | 18x60         | 1006 | S[354] | 8920 | 240 | 18x60         |
| 951 | S[299] | 9580  | 330 | 18x60         | 1007 | S[355] | 8908 | 150 | 18x60         |
| 952 | S[300] | 9568  | 240 | 18x60         | 1008 | S[356] | 8896 | 330 | 18x60         |

| No.  | Name   | X    | Y   | Bump size(um) | No.  | Name   | X    | Y   | Bump size(um) |
|------|--------|------|-----|---------------|------|--------|------|-----|---------------|
| 1009 | S[357] | 8884 | 240 | 18x60         | 1065 | S[413] | 8212 | 330 | 18x60         |
| 1010 | S[358] | 8872 | 150 | 18x60         | 1066 | S[414] | 8200 | 240 | 18x60         |
| 1011 | S[359] | 8860 | 330 | 18x60         | 1067 | S[415] | 8188 | 150 | 18x60         |
| 1012 | S[360] | 8848 | 240 | 18x60         | 1068 | S[416] | 8176 | 330 | 18x60         |
| 1013 | S[361] | 8836 | 150 | 18x60         | 1069 | S[417] | 8164 | 240 | 18x60         |
| 1014 | S[362] | 8824 | 330 | 18x60         | 1070 | S[418] | 8152 | 150 | 18x60         |
| 1015 | S[363] | 8812 | 240 | 18x60         | 1071 | S[419] | 8140 | 330 | 18x60         |
| 1016 | S[364] | 8800 | 150 | 18x60         | 1072 | S[420] | 8128 | 240 | 18x60         |
| 1017 | S[365] | 8788 | 330 | 18x60         | 1073 | S[421] | 8116 | 150 | 18x60         |
| 1018 | S[366] | 8776 | 240 | 18x60         | 1074 | S[422] | 8104 | 330 | 18x60         |
| 1019 | S[367] | 8764 | 150 | 18x60         | 1075 | S[423] | 8092 | 240 | 18x60         |
| 1020 | S[368] | 8752 | 330 | 18x60         | 1076 | S[424] | 8080 | 150 | 18x60         |
| 1021 | S[369] | 8740 | 240 | 18x60         | 1077 | S[425] | 8068 | 330 | 18x60         |
| 1022 | S[370] | 8728 | 150 | 18x60         | 1078 | S[426] | 8056 | 240 | 18x60         |
| 1023 | S[371] | 8716 | 330 | 18x60         | 1079 | S[427] | 8044 | 150 | 18x60         |
| 1024 | S[372] | 8704 | 240 | 18x60         | 1080 | S[428] | 8032 | 330 | 18x60         |
| 1025 | S[373] | 8692 | 150 | 18x60         | 1081 | S[429] | 8020 | 240 | 18x60         |
| 1026 | S[374] | 8680 | 330 | 18x60         | 1082 | S[430] | 8008 | 150 | 18x60         |
| 1027 | S[375] | 8668 | 240 | 18x60         | 1083 | S[431] | 7996 | 330 | 18x60         |
| 1028 | S[376] | 8656 | 150 | 18x60         | 1084 | S[432] | 7984 | 240 | 18x60         |
| 1029 | S[377] | 8644 | 330 | 18x60         | 1085 | S[433] | 7972 | 150 | 18x60         |
| 1030 | S[378] | 8632 | 240 | 18x60         | 1086 | S[434] | 7960 | 330 | 18x60         |
| 1031 | S[379] | 8620 | 150 | 18x60         | 1087 | S[435] | 7948 | 240 | 18x60         |
| 1032 | S[380] | 8608 | 330 | 18x60         | 1088 | S[436] | 7936 | 150 | 18x60         |
| 1033 | S[381] | 8596 | 240 | 18x60         | 1089 | S[437] | 7924 | 330 | 18x60         |
| 1034 | S[382] | 8584 | 150 | 18x60         | 1090 | S[438] | 7912 | 240 | 18x60         |
| 1035 | S[383] | 8572 | 330 | 18x60         | 1091 | S[439] | 7900 | 150 | 18x60         |
| 1036 | S[384] | 8560 | 240 | 18x60         | 1092 | S[440] | 7888 | 330 | 18x60         |
| 1037 | S[385] | 8548 | 150 | 18x60         | 1093 | S[441] | 7876 | 240 | 18x60         |
| 1038 | S[386] | 8536 | 330 | 18x60         | 1094 | S[442] | 7864 | 150 | 18x60         |
| 1039 | S[387] | 8524 | 240 | 18x60         | 1095 | S[443] | 7852 | 330 | 18x60         |
| 1040 | S[388] | 8512 | 150 | 18x60         | 1096 | S[444] | 7840 | 240 | 18x60         |
| 1041 | S[389] | 8500 | 330 | 18x60         | 1097 | S[445] | 7828 | 150 | 18x60         |
| 1042 | S[390] | 8488 | 240 | 18x60         | 1098 | S[446] | 7816 | 330 | 18x60         |
| 1043 | S[391] | 8476 | 150 | 18x60         | 1099 | S[447] | 7804 | 240 | 18x60         |
| 1044 | S[392] | 8464 | 330 | 18x60         | 1100 | S[448] | 7792 | 150 | 18x60         |
| 1045 | S[393] | 8452 | 240 | 18x60         | 1101 | S[449] | 7780 | 330 | 18x60         |
| 1046 | S[394] | 8440 | 150 | 18x60         | 1102 | S[450] | 7768 | 240 | 18x60         |
| 1047 | S[395] | 8428 | 330 | 18x60         | 1103 | S[451] | 7756 | 150 | 18x60         |
| 1048 | S[396] | 8416 | 240 | 18x60         | 1104 | S[452] | 7744 | 330 | 18x60         |
| 1049 | S[397] | 8404 | 150 | 18x60         | 1105 | S[453] | 7732 | 240 | 18x60         |
| 1050 | S[398] | 8392 | 330 | 18x60         | 1106 | S[454] | 7720 | 150 | 18x60         |
| 1051 | S[399] | 8380 | 240 | 18x60         | 1107 | S[455] | 7708 | 330 | 18x60         |
| 1052 | S[400] | 8368 | 150 | 18x60         | 1108 | S[456] | 7696 | 240 | 18x60         |
| 1053 | S[401] | 8356 | 330 | 18x60         | 1109 | S[457] | 7684 | 150 | 18x60         |
| 1054 | S[402] | 8344 | 240 | 18x60         | 1110 | S[458] | 7672 | 330 | 18x60         |
| 1055 | S[403] | 8332 | 150 | 18x60         | 1111 | S[459] | 7660 | 240 | 18x60         |
| 1056 | S[404] | 8320 | 330 | 18x60         | 1112 | S[460] | 7648 | 150 | 18x60         |
| 1057 | S[405] | 8308 | 240 | 18x60         | 1113 | S[461] | 7636 | 330 | 18x60         |
| 1058 | S[406] | 8296 | 150 | 18x60         | 1114 | S[462] | 7624 | 240 | 18x60         |
| 1059 | S[407] | 8284 | 330 | 18x60         | 1115 | S[463] | 7612 | 150 | 18x60         |
| 1060 | S[408] | 8272 | 240 | 18x60         | 1116 | S[464] | 7600 | 330 | 18x60         |
| 1061 | S[409] | 8260 | 150 | 18x60         | 1117 | S[465] | 7588 | 240 | 18x60         |
| 1062 | S[410] | 8248 | 330 | 18x60         | 1118 | S[466] | 7576 | 150 | 18x60         |
| 1063 | S[411] | 8236 | 240 | 18x60         | 1119 | S[467] | 7564 | 330 | 18x60         |
| 1064 | S[412] | 8224 | 150 | 18x60         | 1120 | S[468] | 7552 | 240 | 18x60         |

| No.  | Name   | X    | Y   | Bump size(um) | No.  | Name   | X    | Y   | Bump size(um) |
|------|--------|------|-----|---------------|------|--------|------|-----|---------------|
| 1121 | S[469] | 7540 | 150 | 18x60         | 1177 | S[525] | 6868 | 240 | 18x60         |
| 1122 | S[470] | 7528 | 330 | 18x60         | 1178 | S[526] | 6856 | 150 | 18x60         |
| 1123 | S[471] | 7516 | 240 | 18x60         | 1179 | S[527] | 6844 | 330 | 18x60         |
| 1124 | S[472] | 7504 | 150 | 18x60         | 1180 | S[528] | 6832 | 240 | 18x60         |
| 1125 | S[473] | 7492 | 330 | 18x60         | 1181 | S[529] | 6820 | 150 | 18x60         |
| 1126 | S[474] | 7480 | 240 | 18x60         | 1182 | S[530] | 6808 | 330 | 18x60         |
| 1127 | S[475] | 7468 | 150 | 18x60         | 1183 | S[531] | 6796 | 240 | 18x60         |
| 1128 | S[476] | 7456 | 330 | 18x60         | 1184 | S[532] | 6784 | 150 | 18x60         |
| 1129 | S[477] | 7444 | 240 | 18x60         | 1185 | S[533] | 6772 | 330 | 18x60         |
| 1130 | S[478] | 7432 | 150 | 18x60         | 1186 | S[534] | 6760 | 240 | 18x60         |
| 1131 | S[479] | 7420 | 330 | 18x60         | 1187 | S[535] | 6748 | 150 | 18x60         |
| 1132 | S[480] | 7408 | 240 | 18x60         | 1188 | S[536] | 6736 | 330 | 18x60         |
| 1133 | S[481] | 7396 | 150 | 18x60         | 1189 | S[537] | 6724 | 240 | 18x60         |
| 1134 | S[482] | 7384 | 330 | 18x60         | 1190 | S[538] | 6712 | 150 | 18x60         |
| 1135 | S[483] | 7372 | 240 | 18x60         | 1191 | S[539] | 6700 | 330 | 18x60         |
| 1136 | S[484] | 7360 | 150 | 18x60         | 1192 | S[540] | 6688 | 240 | 18x60         |
| 1137 | S[485] | 7348 | 330 | 18x60         | 1193 | S[541] | 6676 | 150 | 18x60         |
| 1138 | S[486] | 7336 | 240 | 18x60         | 1194 | S[542] | 6664 | 330 | 18x60         |
| 1139 | S[487] | 7324 | 150 | 18x60         | 1195 | S[543] | 6652 | 240 | 18x60         |
| 1140 | S[488] | 7312 | 330 | 18x60         | 1196 | S[544] | 6640 | 150 | 18x60         |
| 1141 | S[489] | 7300 | 240 | 18x60         | 1197 | S[545] | 6628 | 330 | 18x60         |
| 1142 | S[490] | 7288 | 150 | 18x60         | 1198 | S[546] | 6616 | 240 | 18x60         |
| 1143 | S[491] | 7276 | 330 | 18x60         | 1199 | S[547] | 6604 | 150 | 18x60         |
| 1144 | S[492] | 7264 | 240 | 18x60         | 1200 | S[548] | 6592 | 330 | 18x60         |
| 1145 | S[493] | 7252 | 150 | 18x60         | 1201 | S[549] | 6580 | 240 | 18x60         |
| 1146 | S[494] | 7240 | 330 | 18x60         | 1202 | S[550] | 6568 | 150 | 18x60         |
| 1147 | S[495] | 7228 | 240 | 18x60         | 1203 | S[551] | 6556 | 330 | 18x60         |
| 1148 | S[496] | 7216 | 150 | 18x60         | 1204 | S[552] | 6544 | 240 | 18x60         |
| 1149 | S[497] | 7204 | 330 | 18x60         | 1205 | S[553] | 6532 | 150 | 18x60         |
| 1150 | S[498] | 7192 | 240 | 18x60         | 1206 | S[554] | 6520 | 330 | 18x60         |
| 1151 | S[499] | 7180 | 150 | 18x60         | 1207 | S[555] | 6508 | 240 | 18x60         |
| 1152 | S[500] | 7168 | 330 | 18x60         | 1208 | S[556] | 6496 | 150 | 18x60         |
| 1153 | S[501] | 7156 | 240 | 18x60         | 1209 | S[557] | 6484 | 330 | 18x60         |
| 1154 | S[502] | 7144 | 150 | 18x60         | 1210 | S[558] | 6472 | 240 | 18x60         |
| 1155 | S[503] | 7132 | 330 | 18x60         | 1211 | S[559] | 6460 | 150 | 18x60         |
| 1156 | S[504] | 7120 | 240 | 18x60         | 1212 | S[560] | 6448 | 330 | 18x60         |
| 1157 | S[505] | 7108 | 150 | 18x60         | 1213 | S[561] | 6436 | 240 | 18x60         |
| 1158 | S[506] | 7096 | 330 | 18x60         | 1214 | S[562] | 6424 | 150 | 18x60         |
| 1159 | S[507] | 7084 | 240 | 18x60         | 1215 | S[563] | 6412 | 330 | 18x60         |
| 1160 | S[508] | 7072 | 150 | 18x60         | 1216 | S[564] | 6400 | 240 | 18x60         |
| 1161 | S[509] | 7060 | 330 | 18x60         | 1217 | S[565] | 6388 | 150 | 18x60         |
| 1162 | S[510] | 7048 | 240 | 18x60         | 1218 | S[566] | 6376 | 330 | 18x60         |
| 1163 | S[511] | 7036 | 150 | 18x60         | 1219 | S[567] | 6364 | 240 | 18x60         |
| 1164 | S[512] | 7024 | 330 | 18x60         | 1220 | S[568] | 6352 | 150 | 18x60         |
| 1165 | S[513] | 7012 | 240 | 18x60         | 1221 | S[569] | 6340 | 330 | 18x60         |
| 1166 | S[514] | 7000 | 150 | 18x60         | 1222 | S[570] | 6328 | 240 | 18x60         |
| 1167 | S[515] | 6988 | 330 | 18x60         | 1223 | S[571] | 6316 | 150 | 18x60         |
| 1168 | S[516] | 6976 | 240 | 18x60         | 1224 | S[572] | 6304 | 330 | 18x60         |
| 1169 | S[517] | 6964 | 150 | 18x60         | 1225 | S[573] | 6292 | 240 | 18x60         |
| 1170 | S[518] | 6952 | 330 | 18x60         | 1226 | S[574] | 6280 | 150 | 18x60         |
| 1171 | S[519] | 6940 | 240 | 18x60         | 1227 | S[575] | 6268 | 330 | 18x60         |
| 1172 | S[520] | 6928 | 150 | 18x60         | 1228 | S[576] | 6256 | 240 | 18x60         |
| 1173 | S[521] | 6916 | 330 | 18x60         | 1229 | S[577] | 6244 | 150 | 18x60         |
| 1174 | S[522] | 6904 | 240 | 18x60         | 1230 | S[578] | 6232 | 330 | 18x60         |
| 1175 | S[523] | 6892 | 150 | 18x60         | 1231 | S[579] | 6220 | 240 | 18x60         |
| 1176 | S[524] | 6880 | 330 | 18x60         | 1232 | S[580] | 6208 | 150 | 18x60         |

| No.  | Name   | X    | Y   | Bump size(um) | No.  | Name   | X    | Y   | Bump size(um) |
|------|--------|------|-----|---------------|------|--------|------|-----|---------------|
| 1233 | S[581] | 6196 | 330 | 18x60         | 1289 | S[637] | 5524 | 150 | 18x60         |
| 1234 | S[582] | 6184 | 240 | 18x60         | 1290 | S[638] | 5512 | 330 | 18x60         |
| 1235 | S[583] | 6172 | 150 | 18x60         | 1291 | S[639] | 5500 | 240 | 18x60         |
| 1236 | S[584] | 6160 | 330 | 18x60         | 1292 | S[640] | 5488 | 150 | 18x60         |
| 1237 | S[585] | 6148 | 240 | 18x60         | 1293 | S[641] | 5476 | 330 | 18x60         |
| 1238 | S[586] | 6136 | 150 | 18x60         | 1294 | S[642] | 5464 | 240 | 18x60         |
| 1239 | S[587] | 6124 | 330 | 18x60         | 1295 | S[643] | 5452 | 150 | 18x60         |
| 1240 | S[588] | 6112 | 240 | 18x60         | 1296 | S[644] | 5440 | 330 | 18x60         |
| 1241 | S[589] | 6100 | 150 | 18x60         | 1297 | S[645] | 5428 | 240 | 18x60         |
| 1242 | S[590] | 6088 | 330 | 18x60         | 1298 | S[646] | 5416 | 150 | 18x60         |
| 1243 | S[591] | 6076 | 240 | 18x60         | 1299 | S[647] | 5404 | 330 | 18x60         |
| 1244 | S[592] | 6064 | 150 | 18x60         | 1300 | S[648] | 5392 | 240 | 18x60         |
| 1245 | S[593] | 6052 | 330 | 18x60         | 1301 | S[649] | 5380 | 150 | 18x60         |
| 1246 | S[594] | 6040 | 240 | 18x60         | 1302 | S[650] | 5368 | 330 | 18x60         |
| 1247 | S[595] | 6028 | 150 | 18x60         | 1303 | S[651] | 5356 | 240 | 18x60         |
| 1248 | S[596] | 6016 | 330 | 18x60         | 1304 | S[652] | 5344 | 150 | 18x60         |
| 1249 | S[597] | 6004 | 240 | 18x60         | 1305 | S[653] | 5332 | 330 | 18x60         |
| 1250 | S[598] | 5992 | 150 | 18x60         | 1306 | S[654] | 5320 | 240 | 18x60         |
| 1251 | S[599] | 5980 | 330 | 18x60         | 1307 | S[655] | 5308 | 150 | 18x60         |
| 1252 | S[600] | 5968 | 240 | 18x60         | 1308 | S[656] | 5296 | 330 | 18x60         |
| 1253 | S[601] | 5956 | 150 | 18x60         | 1309 | S[657] | 5284 | 240 | 18x60         |
| 1254 | S[602] | 5944 | 330 | 18x60         | 1310 | S[658] | 5272 | 150 | 18x60         |
| 1255 | S[603] | 5932 | 240 | 18x60         | 1311 | S[659] | 5260 | 330 | 18x60         |
| 1256 | S[604] | 5920 | 150 | 18x60         | 1312 | S[660] | 5248 | 240 | 18x60         |
| 1257 | S[605] | 5908 | 330 | 18x60         | 1313 | S[661] | 5236 | 150 | 18x60         |
| 1258 | S[606] | 5896 | 240 | 18x60         | 1314 | S[662] | 5224 | 330 | 18x60         |
| 1259 | S[607] | 5884 | 150 | 18x60         | 1315 | S[663] | 5212 | 240 | 18x60         |
| 1260 | S[608] | 5872 | 330 | 18x60         | 1316 | S[664] | 5200 | 150 | 18x60         |
| 1261 | S[609] | 5860 | 240 | 18x60         | 1317 | S[665] | 5188 | 330 | 18x60         |
| 1262 | S[610] | 5848 | 150 | 18x60         | 1318 | S[666] | 5176 | 240 | 18x60         |
| 1263 | S[611] | 5836 | 330 | 18x60         | 1319 | S[667] | 5164 | 150 | 18x60         |
| 1264 | S[612] | 5824 | 240 | 18x60         | 1320 | S[668] | 5152 | 330 | 18x60         |
| 1265 | S[613] | 5812 | 150 | 18x60         | 1321 | S[669] | 5140 | 240 | 18x60         |
| 1266 | S[614] | 5800 | 330 | 18x60         | 1322 | S[670] | 5128 | 150 | 18x60         |
| 1267 | S[615] | 5788 | 240 | 18x60         | 1323 | S[671] | 5116 | 330 | 18x60         |
| 1268 | S[616] | 5776 | 150 | 18x60         | 1324 | S[672] | 5104 | 240 | 18x60         |
| 1269 | S[617] | 5764 | 330 | 18x60         | 1325 | S[673] | 5092 | 150 | 18x60         |
| 1270 | S[618] | 5752 | 240 | 18x60         | 1326 | S[674] | 5080 | 330 | 18x60         |
| 1271 | S[619] | 5740 | 150 | 18x60         | 1327 | S[675] | 5068 | 240 | 18x60         |
| 1272 | S[620] | 5728 | 330 | 18x60         | 1328 | S[676] | 5056 | 150 | 18x60         |
| 1273 | S[621] | 5716 | 240 | 18x60         | 1329 | S[677] | 5044 | 330 | 18x60         |
| 1274 | S[622] | 5704 | 150 | 18x60         | 1330 | S[678] | 5032 | 240 | 18x60         |
| 1275 | S[623] | 5692 | 330 | 18x60         | 1331 | S[679] | 5020 | 150 | 18x60         |
| 1276 | S[624] | 5680 | 240 | 18x60         | 1332 | S[680] | 5008 | 330 | 18x60         |
| 1277 | S[625] | 5668 | 150 | 18x60         | 1333 | S[681] | 4996 | 240 | 18x60         |
| 1278 | S[626] | 5656 | 330 | 18x60         | 1334 | S[682] | 4984 | 150 | 18x60         |
| 1279 | S[627] | 5644 | 240 | 18x60         | 1335 | S[683] | 4972 | 330 | 18x60         |
| 1280 | S[628] | 5632 | 150 | 18x60         | 1336 | S[684] | 4960 | 240 | 18x60         |
| 1281 | S[629] | 5620 | 330 | 18x60         | 1337 | S[685] | 4948 | 150 | 18x60         |
| 1282 | S[630] | 5608 | 240 | 18x60         | 1338 | S[686] | 4936 | 330 | 18x60         |
| 1283 | S[631] | 5596 | 150 | 18x60         | 1339 | S[687] | 4924 | 240 | 18x60         |
| 1284 | S[632] | 5584 | 330 | 18x60         | 1340 | S[688] | 4912 | 150 | 18x60         |
| 1285 | S[633] | 5572 | 240 | 18x60         | 1341 | S[689] | 4900 | 330 | 18x60         |
| 1286 | S[634] | 5560 | 150 | 18x60         | 1342 | S[690] | 4888 | 240 | 18x60         |
| 1287 | S[635] | 5548 | 330 | 18x60         | 1343 | S[691] | 4876 | 150 | 18x60         |
| 1288 | S[636] | 5536 | 240 | 18x60         | 1344 | S[692] | 4864 | 330 | 18x60         |



| No.  | Name   | X    | Y   | Bump size(um) | No.  | Name           | X    | Y   | Bump size(um) |
|------|--------|------|-----|---------------|------|----------------|------|-----|---------------|
| 1345 | S[693] | 4852 | 240 | 18x60         | 1401 | S[749]         | 4180 | 330 | 18x60         |
| 1346 | S[694] | 4840 | 150 | 18x60         | 1402 | S[750]         | 4168 | 240 | 18x60         |
| 1347 | S[695] | 4828 | 330 | 18x60         | 1403 | S[751]         | 4156 | 150 | 18x60         |
| 1348 | S[696] | 4816 | 240 | 18x60         | 1404 | S[752]         | 4144 | 330 | 18x60         |
| 1349 | S[697] | 4804 | 150 | 18x60         | 1405 | S[753]         | 4132 | 240 | 18x60         |
| 1350 | S[698] | 4792 | 330 | 18x60         | 1406 | S[754]         | 4120 | 150 | 18x60         |
| 1351 | S[699] | 4780 | 240 | 18x60         | 1407 | S[755]         | 4108 | 330 | 18x60         |
| 1352 | S[700] | 4768 | 150 | 18x60         | 1408 | S[756]         | 4096 | 240 | 18x60         |
| 1353 | S[701] | 4756 | 330 | 18x60         | 1409 | S[757]         | 4084 | 150 | 18x60         |
| 1354 | S[702] | 4744 | 240 | 18x60         | 1410 | S[758]         | 4072 | 330 | 18x60         |
| 1355 | S[703] | 4732 | 150 | 18x60         | 1411 | S[759]         | 4060 | 240 | 18x60         |
| 1356 | S[704] | 4720 | 330 | 18x60         | 1412 | S[760]         | 4048 | 150 | 18x60         |
| 1357 | S[705] | 4708 | 240 | 18x60         | 1413 | S[761]         | 4036 | 330 | 18x60         |
| 1358 | S[706] | 4696 | 150 | 18x60         | 1414 | S[762]         | 4024 | 240 | 18x60         |
| 1359 | S[707] | 4684 | 330 | 18x60         | 1415 | S[763]         | 4012 | 150 | 18x60         |
| 1360 | S[708] | 4672 | 240 | 18x60         | 1416 | S[764]         | 4000 | 330 | 18x60         |
| 1361 | S[709] | 4660 | 150 | 18x60         | 1417 | S[765]         | 3988 | 240 | 18x60         |
| 1362 | S[710] | 4648 | 330 | 18x60         | 1418 | S[766]         | 3976 | 150 | 18x60         |
| 1363 | S[711] | 4636 | 240 | 18x60         | 1419 | S[767]         | 3964 | 330 | 18x60         |
| 1364 | S[712] | 4624 | 150 | 18x60         | 1420 | S[768]         | 3952 | 240 | 18x60         |
| 1365 | S[713] | 4612 | 330 | 18x60         | 1421 | Shielding_VSSA | 3888 | 330 | 18x60         |
| 1366 | S[714] | 4600 | 240 | 18x60         | 1422 | Shielding_VSSA | 3816 | 330 | 18x60         |
| 1367 | S[715] | 4588 | 150 | 18x60         | 1423 | Shielding_VSSA | 3744 | 330 | 18x60         |
| 1368 | S[716] | 4576 | 330 | 18x60         | 1424 | Shielding_VSSA | 3672 | 330 | 18x60         |
| 1369 | S[717] | 4564 | 240 | 18x60         | 1425 | Shielding_VSSA | 3600 | 330 | 18x60         |
| 1370 | S[718] | 4552 | 150 | 18x60         | 1426 | Shielding_VSSA | 3528 | 330 | 18x60         |
| 1371 | S[719] | 4540 | 330 | 18x60         | 1427 | Shielding_VSSA | 3456 | 330 | 18x60         |
| 1372 | S[720] | 4528 | 240 | 18x60         | 1428 | Shielding_VSSA | 3384 | 330 | 18x60         |
| 1373 | S[721] | 4516 | 150 | 18x60         | 1429 | Shielding_VSSA | 3312 | 330 | 18x60         |
| 1374 | S[722] | 4504 | 330 | 18x60         | 1430 | Shielding_VSSA | 3240 | 330 | 18x60         |
| 1375 | S[723] | 4492 | 240 | 18x60         | 1431 | Shielding_VSSA | 3168 | 330 | 18x60         |
| 1376 | S[724] | 4480 | 150 | 18x60         | 1432 | Shielding_VSSA | 3096 | 330 | 18x60         |
| 1377 | S[725] | 4468 | 330 | 18x60         | 1433 | Shielding_VSSA | 3024 | 330 | 18x60         |
| 1378 | S[726] | 4456 | 240 | 18x60         | 1434 | Shielding_VSSA | 2952 | 330 | 18x60         |
| 1379 | S[727] | 4444 | 150 | 18x60         | 1435 | Shielding_VSSA | 2880 | 330 | 18x60         |
| 1380 | S[728] | 4432 | 330 | 18x60         | 1436 | Shielding_VSSA | 2808 | 330 | 18x60         |
| 1381 | S[729] | 4420 | 240 | 18x60         | 1437 | Shielding_VSSA | 2736 | 330 | 18x60         |
| 1382 | S[730] | 4408 | 150 | 18x60         | 1438 | Shielding_VSSA | 2664 | 330 | 18x60         |
| 1383 | S[731] | 4396 | 330 | 18x60         | 1439 | Shielding_VSSA | 2592 | 330 | 18x60         |
| 1384 | S[732] | 4384 | 240 | 18x60         | 1440 | Shielding_VSSA | 2520 | 330 | 18x60         |
| 1385 | S[733] | 4372 | 150 | 18x60         | 1441 | Shielding_VSSA | 2448 | 330 | 18x60         |
| 1386 | S[734] | 4360 | 330 | 18x60         | 1442 | Shielding_VSSA | 2376 | 330 | 18x60         |
| 1387 | S[735] | 4348 | 240 | 18x60         | 1443 | Shielding_VSSA | 2304 | 330 | 18x60         |
| 1388 | S[736] | 4336 | 150 | 18x60         | 1444 | Shielding_VSSA | 2232 | 330 | 18x60         |
| 1389 | S[737] | 4324 | 330 | 18x60         | 1445 | Shielding_VSSA | 2160 | 330 | 18x60         |
| 1390 | S[738] | 4312 | 240 | 18x60         | 1446 | Shielding_VSSA | 2088 | 330 | 18x60         |
| 1391 | S[739] | 4300 | 150 | 18x60         | 1447 | Shielding_VSSA | 2016 | 330 | 18x60         |
| 1392 | S[740] | 4288 | 330 | 18x60         | 1448 | Shielding_VSSA | 1944 | 330 | 18x60         |
| 1393 | S[741] | 4276 | 240 | 18x60         | 1449 | Shielding_VSSA | 1872 | 330 | 18x60         |
| 1394 | S[742] | 4264 | 150 | 18x60         | 1450 | Shielding_VSSA | 1800 | 330 | 18x60         |
| 1395 | S[743] | 4252 | 330 | 18x60         | 1451 | Shielding_VSSA | 1728 | 330 | 18x60         |
| 1396 | S[744] | 4240 | 240 | 18x60         | 1452 | Shielding_VSSA | 1656 | 330 | 18x60         |
| 1397 | S[745] | 4228 | 150 | 18x60         | 1453 | Shielding_VSSA | 1584 | 330 | 18x60         |
| 1398 | S[746] | 4216 | 330 | 18x60         | 1454 | Shielding_VSSA | 1512 | 330 | 18x60         |
| 1399 | S[747] | 4204 | 240 | 18x60         | 1455 | Shielding_VSSA | 1440 | 330 | 18x60         |
| 1400 | S[748] | 4192 | 150 | 18x60         | 1456 | Shielding_VSSA | 1368 | 330 | 18x60         |



| No.  | Name           | X     | Y   | Bump size(um) | No.  | Name           | X     | Y   | Bump size(um) |
|------|----------------|-------|-----|---------------|------|----------------|-------|-----|---------------|
| 1457 | Shielding_VSSA | 1296  | 330 | 18x60         | 1513 | Shielding_VSSA | -2736 | 330 | 18x60         |
| 1458 | Shielding_VSSA | 1224  | 330 | 18x60         | 1514 | Shielding_VSSA | -2808 | 330 | 18x60         |
| 1459 | Shielding_VSSA | 1152  | 330 | 18x60         | 1515 | Shielding_VSSA | -2880 | 330 | 18x60         |
| 1460 | Shielding_VSSA | 1080  | 330 | 18x60         | 1516 | Shielding_VSSA | -2952 | 330 | 18x60         |
| 1461 | Shielding_VSSA | 1008  | 330 | 18x60         | 1517 | Shielding_VSSA | -3024 | 330 | 18x60         |
| 1462 | Shielding_VSSA | 936   | 330 | 18x60         | 1518 | Shielding_VSSA | -3096 | 330 | 18x60         |
| 1463 | Shielding_VSSA | 864   | 330 | 18x60         | 1519 | Shielding_VSSA | -3168 | 330 | 18x60         |
| 1464 | Shielding_VSSA | 792   | 330 | 18x60         | 1520 | Shielding_VSSA | -3240 | 330 | 18x60         |
| 1465 | Shielding_VSSA | 720   | 330 | 18x60         | 1521 | Shielding_VSSA | -3312 | 330 | 18x60         |
| 1466 | Shielding_VSSA | 648   | 330 | 18x60         | 1522 | Shielding_VSSA | -3384 | 330 | 18x60         |
| 1467 | Shielding_VSSA | 576   | 330 | 18x60         | 1523 | Shielding_VSSA | -3456 | 330 | 18x60         |
| 1468 | Shielding_VSSA | 504   | 330 | 18x60         | 1524 | Shielding_VSSA | -3528 | 330 | 18x60         |
| 1469 | Shielding_VSSA | 432   | 330 | 18x60         | 1525 | Shielding_VSSA | -3600 | 330 | 18x60         |
| 1470 | Shielding_VSSA | 360   | 330 | 18x60         | 1526 | Shielding_VSSA | -3672 | 330 | 18x60         |
| 1471 | Shielding_VSSA | 288   | 330 | 18x60         | 1527 | Shielding_VSSA | -3744 | 330 | 18x60         |
| 1472 | Shielding_VSSA | 216   | 330 | 18x60         | 1528 | Shielding_VSSA | -3816 | 330 | 18x60         |
| 1473 | Shielding_VSSA | 144   | 330 | 18x60         | 1529 | Shielding_VSSA | -3888 | 330 | 18x60         |
| 1474 | Shielding_VSSA | 72    | 330 | 18x60         | 1530 | S[769]         | -4029 | 150 | 18x60         |
| 1475 | Shielding_VSSA | 0     | 330 | 18x60         | 1531 | S[770]         | -4041 | 330 | 18x60         |
| 1476 | Shielding_VSSA | -72   | 330 | 18x60         | 1532 | S[771]         | -4053 | 240 | 18x60         |
| 1477 | Shielding_VSSA | -144  | 330 | 18x60         | 1533 | S[772]         | -4065 | 150 | 18x60         |
| 1478 | Shielding_VSSA | -216  | 330 | 18x60         | 1534 | S[773]         | -4077 | 330 | 18x60         |
| 1479 | Shielding_VSSA | -288  | 330 | 18x60         | 1535 | S[774]         | -4089 | 240 | 18x60         |
| 1480 | Shielding_VSSA | -360  | 330 | 18x60         | 1536 | S[775]         | -4101 | 150 | 18x60         |
| 1481 | Shielding_VSSA | -432  | 330 | 18x60         | 1537 | S[776]         | -4113 | 330 | 18x60         |
| 1482 | Shielding_VSSA | -504  | 330 | 18x60         | 1538 | S[777]         | -4125 | 240 | 18x60         |
| 1483 | Shielding_VSSA | -576  | 330 | 18x60         | 1539 | S[778]         | -4137 | 150 | 18x60         |
| 1484 | Shielding_VSSA | -648  | 330 | 18x60         | 1540 | S[779]         | -4149 | 330 | 18x60         |
| 1485 | Shielding_VSSA | -720  | 330 | 18x60         | 1541 | S[780]         | -4161 | 240 | 18x60         |
| 1486 | Shielding_VSSA | -792  | 330 | 18x60         | 1542 | S[781]         | -4173 | 150 | 18x60         |
| 1487 | Shielding_VSSA | -864  | 330 | 18x60         | 1543 | S[782]         | -4185 | 330 | 18x60         |
| 1488 | Shielding_VSSA | -936  | 330 | 18x60         | 1544 | S[783]         | -4197 | 240 | 18x60         |
| 1489 | Shielding_VSSA | -1008 | 330 | 18x60         | 1545 | S[784]         | -4209 | 150 | 18x60         |
| 1490 | Shielding_VSSA | -1080 | 330 | 18x60         | 1546 | S[785]         | -4221 | 330 | 18x60         |
| 1491 | Shielding_VSSA | -1152 | 330 | 18x60         | 1547 | S[786]         | -4233 | 240 | 18x60         |
| 1492 | Shielding_VSSA | -1224 | 330 | 18x60         | 1548 | S[787]         | -4245 | 150 | 18x60         |
| 1493 | Shielding_VSSA | -1296 | 330 | 18x60         | 1549 | S[788]         | -4257 | 330 | 18x60         |
| 1494 | Shielding_VSSA | -1368 | 330 | 18x60         | 1550 | S[789]         | -4269 | 240 | 18x60         |
| 1495 | Shielding_VSSA | -1440 | 330 | 18x60         | 1551 | S[790]         | -4281 | 150 | 18x60         |
| 1496 | Shielding_VSSA | -1512 | 330 | 18x60         | 1552 | S[791]         | -4293 | 330 | 18x60         |
| 1497 | Shielding_VSSA | -1584 | 330 | 18x60         | 1553 | S[792]         | -4305 | 240 | 18x60         |
| 1498 | Shielding_VSSA | -1656 | 330 | 18x60         | 1554 | S[793]         | -4317 | 150 | 18x60         |
| 1499 | Shielding_VSSA | -1728 | 330 | 18x60         | 1555 | S[794]         | -4329 | 330 | 18x60         |
| 1500 | Shielding_VSSA | -1800 | 330 | 18x60         | 1556 | S[795]         | -4341 | 240 | 18x60         |
| 1501 | Shielding_VSSA | -1872 | 330 | 18x60         | 1557 | S[796]         | -4353 | 150 | 18x60         |
| 1502 | Shielding_VSSA | -1944 | 330 | 18x60         | 1558 | S[797]         | -4365 | 330 | 18x60         |
| 1503 | Shielding_VSSA | -2016 | 330 | 18x60         | 1559 | S[798]         | -4377 | 240 | 18x60         |
| 1504 | Shielding_VSSA | -2088 | 330 | 18x60         | 1560 | S[799]         | -4389 | 150 | 18x60         |
| 1505 | Shielding_VSSA | -2160 | 330 | 18x60         | 1561 | S[800]         | -4401 | 330 | 18x60         |
| 1506 | Shielding_VSSA | -2232 | 330 | 18x60         | 1562 | S[801]         | -4413 | 240 | 18x60         |
| 1507 | Shielding_VSSA | -2304 | 330 | 18x60         | 1563 | S[802]         | -4425 | 150 | 18x60         |
| 1508 | Shielding_VSSA | -2376 | 330 | 18x60         | 1564 | S[803]         | -4437 | 330 | 18x60         |
| 1509 | Shielding_VSSA | -2448 | 330 | 18x60         | 1565 | S[804]         | -4449 | 240 | 18x60         |
| 1510 | Shielding_VSSA | -2520 | 330 | 18x60         | 1566 | S[805]         | -4461 | 150 | 18x60         |
| 1511 | Shielding_VSSA | -2592 | 330 | 18x60         | 1567 | S[806]         | -4473 | 330 | 18x60         |
| 1512 | Shielding_VSSA | -2664 | 330 | 18x60         | 1568 | S[807]         | -4485 | 240 | 18x60         |

| No.  | Name   | X     | Y   | Bump size(um) | No.  | Name   | X     | Y   | Bump size(um) |
|------|--------|-------|-----|---------------|------|--------|-------|-----|---------------|
| 1569 | S[808] | -4497 | 150 | 18x60         | 1625 | S[864] | -5169 | 240 | 18x60         |
| 1570 | S[809] | -4509 | 330 | 18x60         | 1626 | S[865] | -5181 | 150 | 18x60         |
| 1571 | S[810] | -4521 | 240 | 18x60         | 1627 | S[866] | -5193 | 330 | 18x60         |
| 1572 | S[811] | -4533 | 150 | 18x60         | 1628 | S[867] | -5205 | 240 | 18x60         |
| 1573 | S[812] | -4545 | 330 | 18x60         | 1629 | S[868] | -5217 | 150 | 18x60         |
| 1574 | S[813] | -4557 | 240 | 18x60         | 1630 | S[869] | -5229 | 330 | 18x60         |
| 1575 | S[814] | -4569 | 150 | 18x60         | 1631 | S[870] | -5241 | 240 | 18x60         |
| 1576 | S[815] | -4581 | 330 | 18x60         | 1632 | S[871] | -5253 | 150 | 18x60         |
| 1577 | S[816] | -4593 | 240 | 18x60         | 1633 | S[872] | -5265 | 330 | 18x60         |
| 1578 | S[817] | -4605 | 150 | 18x60         | 1634 | S[873] | -5277 | 240 | 18x60         |
| 1579 | S[818] | -4617 | 330 | 18x60         | 1635 | S[874] | -5289 | 150 | 18x60         |
| 1580 | S[819] | -4629 | 240 | 18x60         | 1636 | S[875] | -5301 | 330 | 18x60         |
| 1581 | S[820] | -4641 | 150 | 18x60         | 1637 | S[876] | -5313 | 240 | 18x60         |
| 1582 | S[821] | -4653 | 330 | 18x60         | 1638 | S[877] | -5325 | 150 | 18x60         |
| 1583 | S[822] | -4665 | 240 | 18x60         | 1639 | S[878] | -5337 | 330 | 18x60         |
| 1584 | S[823] | -4677 | 150 | 18x60         | 1640 | S[879] | -5349 | 240 | 18x60         |
| 1585 | S[824] | -4689 | 330 | 18x60         | 1641 | S[880] | -5361 | 150 | 18x60         |
| 1586 | S[825] | -4701 | 240 | 18x60         | 1642 | S[881] | -5373 | 330 | 18x60         |
| 1587 | S[826] | -4713 | 150 | 18x60         | 1643 | S[882] | -5385 | 240 | 18x60         |
| 1588 | S[827] | -4725 | 330 | 18x60         | 1644 | S[883] | -5397 | 150 | 18x60         |
| 1589 | S[828] | -4737 | 240 | 18x60         | 1645 | S[884] | -5409 | 330 | 18x60         |
| 1590 | S[829] | -4749 | 150 | 18x60         | 1646 | S[885] | -5421 | 240 | 18x60         |
| 1591 | S[830] | -4761 | 330 | 18x60         | 1647 | S[886] | -5433 | 150 | 18x60         |
| 1592 | S[831] | -4773 | 240 | 18x60         | 1648 | S[887] | -5445 | 330 | 18x60         |
| 1593 | S[832] | -4785 | 150 | 18x60         | 1649 | S[888] | -5457 | 240 | 18x60         |
| 1594 | S[833] | -4797 | 330 | 18x60         | 1650 | S[889] | -5469 | 150 | 18x60         |
| 1595 | S[834] | -4809 | 240 | 18x60         | 1651 | S[890] | -5481 | 330 | 18x60         |
| 1596 | S[835] | -4821 | 150 | 18x60         | 1652 | S[891] | -5493 | 240 | 18x60         |
| 1597 | S[836] | -4833 | 330 | 18x60         | 1653 | S[892] | -5505 | 150 | 18x60         |
| 1598 | S[837] | -4845 | 240 | 18x60         | 1654 | S[893] | -5517 | 330 | 18x60         |
| 1599 | S[838] | -4857 | 150 | 18x60         | 1655 | S[894] | -5529 | 240 | 18x60         |
| 1600 | S[839] | -4869 | 330 | 18x60         | 1656 | S[895] | -5541 | 150 | 18x60         |
| 1601 | S[840] | -4881 | 240 | 18x60         | 1657 | S[896] | -5553 | 330 | 18x60         |
| 1602 | S[841] | -4893 | 150 | 18x60         | 1658 | S[897] | -5565 | 240 | 18x60         |
| 1603 | S[842] | -4905 | 330 | 18x60         | 1659 | S[898] | -5577 | 150 | 18x60         |
| 1604 | S[843] | -4917 | 240 | 18x60         | 1660 | S[899] | -5589 | 330 | 18x60         |
| 1605 | S[844] | -4929 | 150 | 18x60         | 1661 | S[900] | -5601 | 240 | 18x60         |
| 1606 | S[845] | -4941 | 330 | 18x60         | 1662 | S[901] | -5613 | 150 | 18x60         |
| 1607 | S[846] | -4953 | 240 | 18x60         | 1663 | S[902] | -5625 | 330 | 18x60         |
| 1608 | S[847] | -4965 | 150 | 18x60         | 1664 | S[903] | -5637 | 240 | 18x60         |
| 1609 | S[848] | -4977 | 330 | 18x60         | 1665 | S[904] | -5649 | 150 | 18x60         |
| 1610 | S[849] | -4989 | 240 | 18x60         | 1666 | S[905] | -5661 | 330 | 18x60         |
| 1611 | S[850] | -5001 | 150 | 18x60         | 1667 | S[906] | -5673 | 240 | 18x60         |
| 1612 | S[851] | -5013 | 330 | 18x60         | 1668 | S[907] | -5685 | 150 | 18x60         |
| 1613 | S[852] | -5025 | 240 | 18x60         | 1669 | S[908] | -5697 | 330 | 18x60         |
| 1614 | S[853] | -5037 | 150 | 18x60         | 1670 | S[909] | -5709 | 240 | 18x60         |
| 1615 | S[854] | -5049 | 330 | 18x60         | 1671 | S[910] | -5721 | 150 | 18x60         |
| 1616 | S[855] | -5061 | 240 | 18x60         | 1672 | S[911] | -5733 | 330 | 18x60         |
| 1617 | S[856] | -5073 | 150 | 18x60         | 1673 | S[912] | -5745 | 240 | 18x60         |
| 1618 | S[857] | -5085 | 330 | 18x60         | 1674 | S[913] | -5757 | 150 | 18x60         |
| 1619 | S[858] | -5097 | 240 | 18x60         | 1675 | S[914] | -5769 | 330 | 18x60         |
| 1620 | S[859] | -5109 | 150 | 18x60         | 1676 | S[915] | -5781 | 240 | 18x60         |
| 1621 | S[860] | -5121 | 330 | 18x60         | 1677 | S[916] | -5793 | 150 | 18x60         |
| 1622 | S[861] | -5133 | 240 | 18x60         | 1678 | S[917] | -5805 | 330 | 18x60         |
| 1623 | S[862] | -5145 | 150 | 18x60         | 1679 | S[918] | -5817 | 240 | 18x60         |
| 1624 | S[863] | -5157 | 330 | 18x60         | 1680 | S[919] | -5829 | 150 | 18x60         |

| No.  | Name   | X     | Y   | Bump size(um) | No.  | Name    | X     | Y   | Bump size(um) |
|------|--------|-------|-----|---------------|------|---------|-------|-----|---------------|
| 1681 | S[920] | -5841 | 330 | 18x60         | 1737 | S[976]  | -6513 | 150 | 18x60         |
| 1682 | S[921] | -5853 | 240 | 18x60         | 1738 | S[977]  | -6525 | 330 | 18x60         |
| 1683 | S[922] | -5865 | 150 | 18x60         | 1739 | S[978]  | -6537 | 240 | 18x60         |
| 1684 | S[923] | -5877 | 330 | 18x60         | 1740 | S[979]  | -6549 | 150 | 18x60         |
| 1685 | S[924] | -5889 | 240 | 18x60         | 1741 | S[980]  | -6561 | 330 | 18x60         |
| 1686 | S[925] | -5901 | 150 | 18x60         | 1742 | S[981]  | -6573 | 240 | 18x60         |
| 1687 | S[926] | -5913 | 330 | 18x60         | 1743 | S[982]  | -6585 | 150 | 18x60         |
| 1688 | S[927] | -5925 | 240 | 18x60         | 1744 | S[983]  | -6597 | 330 | 18x60         |
| 1689 | S[928] | -5937 | 150 | 18x60         | 1745 | S[984]  | -6609 | 240 | 18x60         |
| 1690 | S[929] | -5949 | 330 | 18x60         | 1746 | S[985]  | -6621 | 150 | 18x60         |
| 1691 | S[930] | -5961 | 240 | 18x60         | 1747 | S[986]  | -6633 | 330 | 18x60         |
| 1692 | S[931] | -5973 | 150 | 18x60         | 1748 | S[987]  | -6645 | 240 | 18x60         |
| 1693 | S[932] | -5985 | 330 | 18x60         | 1749 | S[988]  | -6657 | 150 | 18x60         |
| 1694 | S[933] | -5997 | 240 | 18x60         | 1750 | S[989]  | -6669 | 330 | 18x60         |
| 1695 | S[934] | -6009 | 150 | 18x60         | 1751 | S[990]  | -6681 | 240 | 18x60         |
| 1696 | S[935] | -6021 | 330 | 18x60         | 1752 | S[991]  | -6693 | 150 | 18x60         |
| 1697 | S[936] | -6033 | 240 | 18x60         | 1753 | S[992]  | -6705 | 330 | 18x60         |
| 1698 | S[937] | -6045 | 150 | 18x60         | 1754 | S[993]  | -6717 | 240 | 18x60         |
| 1699 | S[938] | -6057 | 330 | 18x60         | 1755 | S[994]  | -6729 | 150 | 18x60         |
| 1700 | S[939] | -6069 | 240 | 18x60         | 1756 | S[995]  | -6741 | 330 | 18x60         |
| 1701 | S[940] | -6081 | 150 | 18x60         | 1757 | S[996]  | -6753 | 240 | 18x60         |
| 1702 | S[941] | -6093 | 330 | 18x60         | 1758 | S[997]  | -6765 | 150 | 18x60         |
| 1703 | S[942] | -6105 | 240 | 18x60         | 1759 | S[998]  | -6777 | 330 | 18x60         |
| 1704 | S[943] | -6117 | 150 | 18x60         | 1760 | S[999]  | -6789 | 240 | 18x60         |
| 1705 | S[944] | -6129 | 330 | 18x60         | 1761 | S[1000] | -6801 | 150 | 18x60         |
| 1706 | S[945] | -6141 | 240 | 18x60         | 1762 | S[1001] | -6813 | 330 | 18x60         |
| 1707 | S[946] | -6153 | 150 | 18x60         | 1763 | S[1002] | -6825 | 240 | 18x60         |
| 1708 | S[947] | -6165 | 330 | 18x60         | 1764 | S[1003] | -6837 | 150 | 18x60         |
| 1709 | S[948] | -6177 | 240 | 18x60         | 1765 | S[1004] | -6849 | 330 | 18x60         |
| 1710 | S[949] | -6189 | 150 | 18x60         | 1766 | S[1005] | -6861 | 240 | 18x60         |
| 1711 | S[950] | -6201 | 330 | 18x60         | 1767 | S[1006] | -6873 | 150 | 18x60         |
| 1712 | S[951] | -6213 | 240 | 18x60         | 1768 | S[1007] | -6885 | 330 | 18x60         |
| 1713 | S[952] | -6225 | 150 | 18x60         | 1769 | S[1008] | -6897 | 240 | 18x60         |
| 1714 | S[953] | -6237 | 330 | 18x60         | 1770 | S[1009] | -6909 | 150 | 18x60         |
| 1715 | S[954] | -6249 | 240 | 18x60         | 1771 | S[1010] | -6921 | 330 | 18x60         |
| 1716 | S[955] | -6261 | 150 | 18x60         | 1772 | S[1011] | -6933 | 240 | 18x60         |
| 1717 | S[956] | -6273 | 330 | 18x60         | 1773 | S[1012] | -6945 | 150 | 18x60         |
| 1718 | S[957] | -6285 | 240 | 18x60         | 1774 | S[1013] | -6957 | 330 | 18x60         |
| 1719 | S[958] | -6297 | 150 | 18x60         | 1775 | S[1014] | -6969 | 240 | 18x60         |
| 1720 | S[959] | -6309 | 330 | 18x60         | 1776 | S[1015] | -6981 | 150 | 18x60         |
| 1721 | S[960] | -6321 | 240 | 18x60         | 1777 | S[1016] | -6993 | 330 | 18x60         |
| 1722 | S[961] | -6333 | 150 | 18x60         | 1778 | S[1017] | -7005 | 240 | 18x60         |
| 1723 | S[962] | -6345 | 330 | 18x60         | 1779 | S[1018] | -7017 | 150 | 18x60         |
| 1724 | S[963] | -6357 | 240 | 18x60         | 1780 | S[1019] | -7029 | 330 | 18x60         |
| 1725 | S[964] | -6369 | 150 | 18x60         | 1781 | S[1020] | -7041 | 240 | 18x60         |
| 1726 | S[965] | -6381 | 330 | 18x60         | 1782 | S[1021] | -7053 | 150 | 18x60         |
| 1727 | S[966] | -6393 | 240 | 18x60         | 1783 | S[1022] | -7065 | 330 | 18x60         |
| 1728 | S[967] | -6405 | 150 | 18x60         | 1784 | S[1023] | -7077 | 240 | 18x60         |
| 1729 | S[968] | -6417 | 330 | 18x60         | 1785 | S[1024] | -7089 | 150 | 18x60         |
| 1730 | S[969] | -6429 | 240 | 18x60         | 1786 | S[1025] | -7101 | 330 | 18x60         |
| 1731 | S[970] | -6441 | 150 | 18x60         | 1787 | S[1026] | -7113 | 240 | 18x60         |
| 1732 | S[971] | -6453 | 330 | 18x60         | 1788 | S[1027] | -7125 | 150 | 18x60         |
| 1733 | S[972] | -6465 | 240 | 18x60         | 1789 | S[1028] | -7137 | 330 | 18x60         |
| 1734 | S[973] | -6477 | 150 | 18x60         | 1790 | S[1029] | -7149 | 240 | 18x60         |
| 1735 | S[974] | -6489 | 330 | 18x60         | 1791 | S[1030] | -7161 | 150 | 18x60         |
| 1736 | S[975] | -6501 | 240 | 18x60         | 1792 | S[1031] | -7173 | 330 | 18x60         |

| No.  | Name    | X     | Y   | Bump size(um) | No.  | Name    | X     | Y   | Bump size(um) |
|------|---------|-------|-----|---------------|------|---------|-------|-----|---------------|
| 1793 | S[1032] | -7185 | 240 | 18x60         | 1849 | S[1088] | -7857 | 330 | 18x60         |
| 1794 | S[1033] | -7197 | 150 | 18x60         | 1850 | S[1089] | -7869 | 240 | 18x60         |
| 1795 | S[1034] | -7209 | 330 | 18x60         | 1851 | S[1090] | -7881 | 150 | 18x60         |
| 1796 | S[1035] | -7221 | 240 | 18x60         | 1852 | S[1091] | -7893 | 330 | 18x60         |
| 1797 | S[1036] | -7233 | 150 | 18x60         | 1853 | S[1092] | -7905 | 240 | 18x60         |
| 1798 | S[1037] | -7245 | 330 | 18x60         | 1854 | S[1093] | -7917 | 150 | 18x60         |
| 1799 | S[1038] | -7257 | 240 | 18x60         | 1855 | S[1094] | -7929 | 330 | 18x60         |
| 1800 | S[1039] | -7269 | 150 | 18x60         | 1856 | S[1095] | -7941 | 240 | 18x60         |
| 1801 | S[1040] | -7281 | 330 | 18x60         | 1857 | S[1096] | -7953 | 150 | 18x60         |
| 1802 | S[1041] | -7293 | 240 | 18x60         | 1858 | S[1097] | -7965 | 330 | 18x60         |
| 1803 | S[1042] | -7305 | 150 | 18x60         | 1859 | S[1098] | -7977 | 240 | 18x60         |
| 1804 | S[1043] | -7317 | 330 | 18x60         | 1860 | S[1099] | -7989 | 150 | 18x60         |
| 1805 | S[1044] | -7329 | 240 | 18x60         | 1861 | S[1100] | -8001 | 330 | 18x60         |
| 1806 | S[1045] | -7341 | 150 | 18x60         | 1862 | S[1101] | -8013 | 240 | 18x60         |
| 1807 | S[1046] | -7353 | 330 | 18x60         | 1863 | S[1102] | -8025 | 150 | 18x60         |
| 1808 | S[1047] | -7365 | 240 | 18x60         | 1864 | S[1103] | -8037 | 330 | 18x60         |
| 1809 | S[1048] | -7377 | 150 | 18x60         | 1865 | S[1104] | -8049 | 240 | 18x60         |
| 1810 | S[1049] | -7389 | 330 | 18x60         | 1866 | S[1105] | -8061 | 150 | 18x60         |
| 1811 | S[1050] | -7401 | 240 | 18x60         | 1867 | S[1106] | -8073 | 330 | 18x60         |
| 1812 | S[1051] | -7413 | 150 | 18x60         | 1868 | S[1107] | -8085 | 240 | 18x60         |
| 1813 | S[1052] | -7425 | 330 | 18x60         | 1869 | S[1108] | -8097 | 150 | 18x60         |
| 1814 | S[1053] | -7437 | 240 | 18x60         | 1870 | S[1109] | -8109 | 330 | 18x60         |
| 1815 | S[1054] | -7449 | 150 | 18x60         | 1871 | S[1110] | -8121 | 240 | 18x60         |
| 1816 | S[1055] | -7461 | 330 | 18x60         | 1872 | S[1111] | -8133 | 150 | 18x60         |
| 1817 | S[1056] | -7473 | 240 | 18x60         | 1873 | S[1112] | -8145 | 330 | 18x60         |
| 1818 | S[1057] | -7485 | 150 | 18x60         | 1874 | S[1113] | -8157 | 240 | 18x60         |
| 1819 | S[1058] | -7497 | 330 | 18x60         | 1875 | S[1114] | -8169 | 150 | 18x60         |
| 1820 | S[1059] | -7509 | 240 | 18x60         | 1876 | S[1115] | -8181 | 330 | 18x60         |
| 1821 | S[1060] | -7521 | 150 | 18x60         | 1877 | S[1116] | -8193 | 240 | 18x60         |
| 1822 | S[1061] | -7533 | 330 | 18x60         | 1878 | S[1117] | -8205 | 150 | 18x60         |
| 1823 | S[1062] | -7545 | 240 | 18x60         | 1879 | S[1118] | -8217 | 330 | 18x60         |
| 1824 | S[1063] | -7557 | 150 | 18x60         | 1880 | S[1119] | -8229 | 240 | 18x60         |
| 1825 | S[1064] | -7569 | 330 | 18x60         | 1881 | S[1120] | -8241 | 150 | 18x60         |
| 1826 | S[1065] | -7581 | 240 | 18x60         | 1882 | S[1121] | -8253 | 330 | 18x60         |
| 1827 | S[1066] | -7593 | 150 | 18x60         | 1883 | S[1122] | -8265 | 240 | 18x60         |
| 1828 | S[1067] | -7605 | 330 | 18x60         | 1884 | S[1123] | -8277 | 150 | 18x60         |
| 1829 | S[1068] | -7617 | 240 | 18x60         | 1885 | S[1124] | -8289 | 330 | 18x60         |
| 1830 | S[1069] | -7629 | 150 | 18x60         | 1886 | S[1125] | -8301 | 240 | 18x60         |
| 1831 | S[1070] | -7641 | 330 | 18x60         | 1887 | S[1126] | -8313 | 150 | 18x60         |
| 1832 | S[1071] | -7653 | 240 | 18x60         | 1888 | S[1127] | -8325 | 330 | 18x60         |
| 1833 | S[1072] | -7665 | 150 | 18x60         | 1889 | S[1128] | -8337 | 240 | 18x60         |
| 1834 | S[1073] | -7677 | 330 | 18x60         | 1890 | S[1129] | -8349 | 150 | 18x60         |
| 1835 | S[1074] | -7689 | 240 | 18x60         | 1891 | S[1130] | -8361 | 330 | 18x60         |
| 1836 | S[1075] | -7701 | 150 | 18x60         | 1892 | S[1131] | -8373 | 240 | 18x60         |
| 1837 | S[1076] | -7713 | 330 | 18x60         | 1893 | S[1132] | -8385 | 150 | 18x60         |
| 1838 | S[1077] | -7725 | 240 | 18x60         | 1894 | S[1133] | -8397 | 330 | 18x60         |
| 1839 | S[1078] | -7737 | 150 | 18x60         | 1895 | S[1134] | -8409 | 240 | 18x60         |
| 1840 | S[1079] | -7749 | 330 | 18x60         | 1896 | S[1135] | -8421 | 150 | 18x60         |
| 1841 | S[1080] | -7761 | 240 | 18x60         | 1897 | S[1136] | -8433 | 330 | 18x60         |
| 1842 | S[1081] | -7773 | 150 | 18x60         | 1898 | S[1137] | -8445 | 240 | 18x60         |
| 1843 | S[1082] | -7785 | 330 | 18x60         | 1899 | S[1138] | -8457 | 150 | 18x60         |
| 1844 | S[1083] | -7797 | 240 | 18x60         | 1900 | S[1139] | -8469 | 330 | 18x60         |
| 1845 | S[1084] | -7809 | 150 | 18x60         | 1901 | S[1140] | -8481 | 240 | 18x60         |
| 1846 | S[1085] | -7821 | 330 | 18x60         | 1902 | S[1141] | -8493 | 150 | 18x60         |
| 1847 | S[1086] | -7833 | 240 | 18x60         | 1903 | S[1142] | -8505 | 330 | 18x60         |
| 1848 | S[1087] | -7845 | 150 | 18x60         | 1904 | S[1143] | -8517 | 240 | 18x60         |

| No.  | Name    | X     | Y   | Bump size(um) | No.  | Name    | X     | Y   | Bump size(um) |
|------|---------|-------|-----|---------------|------|---------|-------|-----|---------------|
| 1905 | S[1144] | -8529 | 150 | 18x60         | 1961 | S[1200] | -9201 | 240 | 18x60         |
| 1906 | S[1145] | -8541 | 330 | 18x60         | 1962 | S[1201] | -9213 | 150 | 18x60         |
| 1907 | S[1146] | -8553 | 240 | 18x60         | 1963 | S[1202] | -9225 | 330 | 18x60         |
| 1908 | S[1147] | -8565 | 150 | 18x60         | 1964 | S[1203] | -9237 | 240 | 18x60         |
| 1909 | S[1148] | -8577 | 330 | 18x60         | 1965 | S[1204] | -9249 | 150 | 18x60         |
| 1910 | S[1149] | -8589 | 240 | 18x60         | 1966 | S[1205] | -9261 | 330 | 18x60         |
| 1911 | S[1150] | -8601 | 150 | 18x60         | 1967 | S[1206] | -9273 | 240 | 18x60         |
| 1912 | S[1151] | -8613 | 330 | 18x60         | 1968 | S[1207] | -9285 | 150 | 18x60         |
| 1913 | S[1152] | -8625 | 240 | 18x60         | 1969 | S[1208] | -9297 | 330 | 18x60         |
| 1914 | S[1153] | -8637 | 150 | 18x60         | 1970 | S[1209] | -9309 | 240 | 18x60         |
| 1915 | S[1154] | -8649 | 330 | 18x60         | 1971 | S[1210] | -9321 | 150 | 18x60         |
| 1916 | S[1155] | -8661 | 240 | 18x60         | 1972 | S[1211] | -9333 | 330 | 18x60         |
| 1917 | S[1156] | -8673 | 150 | 18x60         | 1973 | S[1212] | -9345 | 240 | 18x60         |
| 1918 | S[1157] | -8685 | 330 | 18x60         | 1974 | S[1213] | -9357 | 150 | 18x60         |
| 1919 | S[1158] | -8697 | 240 | 18x60         | 1975 | S[1214] | -9369 | 330 | 18x60         |
| 1920 | S[1159] | -8709 | 150 | 18x60         | 1976 | S[1215] | -9381 | 240 | 18x60         |
| 1921 | S[1160] | -8721 | 330 | 18x60         | 1977 | S[1216] | -9393 | 150 | 18x60         |
| 1922 | S[1161] | -8733 | 240 | 18x60         | 1978 | S[1217] | -9405 | 330 | 18x60         |
| 1923 | S[1162] | -8745 | 150 | 18x60         | 1979 | S[1218] | -9417 | 240 | 18x60         |
| 1924 | S[1163] | -8757 | 330 | 18x60         | 1980 | S[1219] | -9429 | 150 | 18x60         |
| 1925 | S[1164] | -8769 | 240 | 18x60         | 1981 | S[1220] | -9441 | 330 | 18x60         |
| 1926 | S[1165] | -8781 | 150 | 18x60         | 1982 | S[1221] | -9453 | 240 | 18x60         |
| 1927 | S[1166] | -8793 | 330 | 18x60         | 1983 | S[1222] | -9465 | 150 | 18x60         |
| 1928 | S[1167] | -8805 | 240 | 18x60         | 1984 | S[1223] | -9477 | 330 | 18x60         |
| 1929 | S[1168] | -8817 | 150 | 18x60         | 1985 | S[1224] | -9489 | 240 | 18x60         |
| 1930 | S[1169] | -8829 | 330 | 18x60         | 1986 | S[1225] | -9501 | 150 | 18x60         |
| 1931 | S[1170] | -8841 | 240 | 18x60         | 1987 | S[1226] | -9513 | 330 | 18x60         |
| 1932 | S[1171] | -8853 | 150 | 18x60         | 1988 | S[1227] | -9525 | 240 | 18x60         |
| 1933 | S[1172] | -8865 | 330 | 18x60         | 1989 | S[1228] | -9537 | 150 | 18x60         |
| 1934 | S[1173] | -8877 | 240 | 18x60         | 1990 | S[1229] | -9549 | 330 | 18x60         |
| 1935 | S[1174] | -8889 | 150 | 18x60         | 1991 | S[1230] | -9561 | 240 | 18x60         |
| 1936 | S[1175] | -8901 | 330 | 18x60         | 1992 | S[1231] | -9573 | 150 | 18x60         |
| 1937 | S[1176] | -8913 | 240 | 18x60         | 1993 | S[1232] | -9585 | 330 | 18x60         |
| 1938 | S[1177] | -8925 | 150 | 18x60         | 1994 | S[1233] | -9597 | 240 | 18x60         |
| 1939 | S[1178] | -8937 | 330 | 18x60         | 1995 | S[1234] | -9609 | 150 | 18x60         |
| 1940 | S[1179] | -8949 | 240 | 18x60         | 1996 | S[1235] | -9621 | 330 | 18x60         |
| 1941 | S[1180] | -8961 | 150 | 18x60         | 1997 | S[1236] | -9633 | 240 | 18x60         |
| 1942 | S[1181] | -8973 | 330 | 18x60         | 1998 | S[1237] | -9645 | 150 | 18x60         |
| 1943 | S[1182] | -8985 | 240 | 18x60         | 1999 | S[1238] | -9657 | 330 | 18x60         |
| 1944 | S[1183] | -8997 | 150 | 18x60         | 2000 | S[1239] | -9669 | 240 | 18x60         |
| 1945 | S[1184] | -9009 | 330 | 18x60         | 2001 | S[1240] | -9681 | 150 | 18x60         |
| 1946 | S[1185] | -9021 | 240 | 18x60         | 2002 | S[1241] | -9693 | 330 | 18x60         |
| 1947 | S[1186] | -9033 | 150 | 18x60         | 2003 | S[1242] | -9705 | 240 | 18x60         |
| 1948 | S[1187] | -9045 | 330 | 18x60         | 2004 | S[1243] | -9717 | 150 | 18x60         |
| 1949 | S[1188] | -9057 | 240 | 18x60         | 2005 | S[1244] | -9729 | 330 | 18x60         |
| 1950 | S[1189] | -9069 | 150 | 18x60         | 2006 | S[1245] | -9741 | 240 | 18x60         |
| 1951 | S[1190] | -9081 | 330 | 18x60         | 2007 | S[1246] | -9753 | 150 | 18x60         |
| 1952 | S[1191] | -9093 | 240 | 18x60         | 2008 | S[1247] | -9765 | 330 | 18x60         |
| 1953 | S[1192] | -9105 | 150 | 18x60         | 2009 | S[1248] | -9777 | 240 | 18x60         |
| 1954 | S[1193] | -9117 | 330 | 18x60         | 2010 | S[1249] | -9789 | 150 | 18x60         |
| 1955 | S[1194] | -9129 | 240 | 18x60         | 2011 | S[1250] | -9801 | 330 | 18x60         |
| 1956 | S[1195] | -9141 | 150 | 18x60         | 2012 | S[1251] | -9813 | 240 | 18x60         |
| 1957 | S[1196] | -9153 | 330 | 18x60         | 2013 | S[1252] | -9825 | 150 | 18x60         |
| 1958 | S[1197] | -9165 | 240 | 18x60         | 2014 | S[1253] | -9837 | 330 | 18x60         |
| 1959 | S[1198] | -9177 | 150 | 18x60         | 2015 | S[1254] | -9849 | 240 | 18x60         |
| 1960 | S[1199] | -9189 | 330 | 18x60         | 2016 | S[1255] | -9861 | 150 | 18x60         |

| No.  | Name    | X      | Y   | Bump size(um) | No.  | Name    | X      | Y   | Bump size(um) |
|------|---------|--------|-----|---------------|------|---------|--------|-----|---------------|
| 2017 | S[1256] | -9873  | 330 | 18x60         | 2073 | S[1312] | -10545 | 150 | 18x60         |
| 2018 | S[1257] | -9885  | 240 | 18x60         | 2074 | S[1313] | -10557 | 330 | 18x60         |
| 2019 | S[1258] | -9897  | 150 | 18x60         | 2075 | S[1314] | -10569 | 240 | 18x60         |
| 2020 | S[1259] | -9909  | 330 | 18x60         | 2076 | S[1315] | -10581 | 150 | 18x60         |
| 2021 | S[1260] | -9921  | 240 | 18x60         | 2077 | S[1316] | -10593 | 330 | 18x60         |
| 2022 | S[1261] | -9933  | 150 | 18x60         | 2078 | S[1317] | -10605 | 240 | 18x60         |
| 2023 | S[1262] | -9945  | 330 | 18x60         | 2079 | S[1318] | -10617 | 150 | 18x60         |
| 2024 | S[1263] | -9957  | 240 | 18x60         | 2080 | S[1319] | -10629 | 330 | 18x60         |
| 2025 | S[1264] | -9969  | 150 | 18x60         | 2081 | S[1320] | -10641 | 240 | 18x60         |
| 2026 | S[1265] | -9981  | 330 | 18x60         | 2082 | S[1321] | -10653 | 150 | 18x60         |
| 2027 | S[1266] | -9993  | 240 | 18x60         | 2083 | S[1322] | -10665 | 330 | 18x60         |
| 2028 | S[1267] | -10005 | 150 | 18x60         | 2084 | S[1323] | -10677 | 240 | 18x60         |
| 2029 | S[1268] | -10017 | 330 | 18x60         | 2085 | S[1324] | -10689 | 150 | 18x60         |
| 2030 | S[1269] | -10029 | 240 | 18x60         | 2086 | S[1325] | -10701 | 330 | 18x60         |
| 2031 | S[1270] | -10041 | 150 | 18x60         | 2087 | S[1326] | -10713 | 240 | 18x60         |
| 2032 | S[1271] | -10053 | 330 | 18x60         | 2088 | S[1327] | -10725 | 150 | 18x60         |
| 2033 | S[1272] | -10065 | 240 | 18x60         | 2089 | S[1328] | -10737 | 330 | 18x60         |
| 2034 | S[1273] | -10077 | 150 | 18x60         | 2090 | S[1329] | -10749 | 240 | 18x60         |
| 2035 | S[1274] | -10089 | 330 | 18x60         | 2091 | S[1330] | -10761 | 150 | 18x60         |
| 2036 | S[1275] | -10101 | 240 | 18x60         | 2092 | S[1331] | -10773 | 330 | 18x60         |
| 2037 | S[1276] | -10113 | 150 | 18x60         | 2093 | S[1332] | -10785 | 240 | 18x60         |
| 2038 | S[1277] | -10125 | 330 | 18x60         | 2094 | S[1333] | -10797 | 150 | 18x60         |
| 2039 | S[1278] | -10137 | 240 | 18x60         | 2095 | S[1334] | -10809 | 330 | 18x60         |
| 2040 | S[1279] | -10149 | 150 | 18x60         | 2096 | S[1335] | -10821 | 240 | 18x60         |
| 2041 | S[1280] | -10161 | 330 | 18x60         | 2097 | S[1336] | -10833 | 150 | 18x60         |
| 2042 | S[1281] | -10173 | 240 | 18x60         | 2098 | S[1337] | -10845 | 330 | 18x60         |
| 2043 | S[1282] | -10185 | 150 | 18x60         | 2099 | S[1338] | -10857 | 240 | 18x60         |
| 2044 | S[1283] | -10197 | 330 | 18x60         | 2100 | S[1339] | -10869 | 150 | 18x60         |
| 2045 | S[1284] | -10209 | 240 | 18x60         | 2101 | S[1340] | -10881 | 330 | 18x60         |
| 2046 | S[1285] | -10221 | 150 | 18x60         | 2102 | S[1341] | -10893 | 240 | 18x60         |
| 2047 | S[1286] | -10233 | 330 | 18x60         | 2103 | S[1342] | -10905 | 150 | 18x60         |
| 2048 | S[1287] | -10245 | 240 | 18x60         | 2104 | S[1343] | -10917 | 330 | 18x60         |
| 2049 | S[1288] | -10257 | 150 | 18x60         | 2105 | S[1344] | -10929 | 240 | 18x60         |
| 2050 | S[1289] | -10269 | 330 | 18x60         | 2106 | S[1345] | -10941 | 150 | 18x60         |
| 2051 | S[1290] | -10281 | 240 | 18x60         | 2107 | S[1346] | -10953 | 330 | 18x60         |
| 2052 | S[1291] | -10293 | 150 | 18x60         | 2108 | S[1347] | -10965 | 240 | 18x60         |
| 2053 | S[1292] | -10305 | 330 | 18x60         | 2109 | S[1348] | -10977 | 150 | 18x60         |
| 2054 | S[1293] | -10317 | 240 | 18x60         | 2110 | S[1349] | -10989 | 330 | 18x60         |
| 2055 | S[1294] | -10329 | 150 | 18x60         | 2111 | S[1350] | -11001 | 240 | 18x60         |
| 2056 | S[1295] | -10341 | 330 | 18x60         | 2112 | S[1351] | -11013 | 150 | 18x60         |
| 2057 | S[1296] | -10353 | 240 | 18x60         | 2113 | S[1352] | -11025 | 330 | 18x60         |
| 2058 | S[1297] | -10365 | 150 | 18x60         | 2114 | S[1353] | -11037 | 240 | 18x60         |
| 2059 | S[1298] | -10377 | 330 | 18x60         | 2115 | S[1354] | -11049 | 150 | 18x60         |
| 2060 | S[1299] | -10389 | 240 | 18x60         | 2116 | S[1355] | -11061 | 330 | 18x60         |
| 2061 | S[1300] | -10401 | 150 | 18x60         | 2117 | S[1356] | -11073 | 240 | 18x60         |
| 2062 | S[1301] | -10413 | 330 | 18x60         | 2118 | S[1357] | -11085 | 150 | 18x60         |
| 2063 | S[1302] | -10425 | 240 | 18x60         | 2119 | S[1358] | -11097 | 330 | 18x60         |
| 2064 | S[1303] | -10437 | 150 | 18x60         | 2120 | S[1359] | -11109 | 240 | 18x60         |
| 2065 | S[1304] | -10449 | 330 | 18x60         | 2121 | S[1360] | -11121 | 150 | 18x60         |
| 2066 | S[1305] | -10461 | 240 | 18x60         | 2122 | S[1361] | -11133 | 330 | 18x60         |
| 2067 | S[1306] | -10473 | 150 | 18x60         | 2123 | S[1362] | -11145 | 240 | 18x60         |
| 2068 | S[1307] | -10485 | 330 | 18x60         | 2124 | S[1363] | -11157 | 150 | 18x60         |
| 2069 | S[1308] | -10497 | 240 | 18x60         | 2125 | S[1364] | -11169 | 330 | 18x60         |
| 2070 | S[1309] | -10509 | 150 | 18x60         | 2126 | S[1365] | -11181 | 240 | 18x60         |
| 2071 | S[1310] | -10521 | 330 | 18x60         | 2127 | S[1366] | -11193 | 150 | 18x60         |
| 2072 | S[1311] | -10533 | 240 | 18x60         | 2128 | S[1367] | -11205 | 330 | 18x60         |



| No.  | Name    | X      | Y   | Bump size(um) | No.  | Name    | X      | Y   | Bump size(um) |
|------|---------|--------|-----|---------------|------|---------|--------|-----|---------------|
| 2129 | S[1368] | -11217 | 240 | 18x60         | 2185 | S[1424] | -11889 | 330 | 18x60         |
| 2130 | S[1369] | -11229 | 150 | 18x60         | 2186 | S[1425] | -11901 | 240 | 18x60         |
| 2131 | S[1370] | -11241 | 330 | 18x60         | 2187 | S[1426] | -11913 | 150 | 18x60         |
| 2132 | S[1371] | -11253 | 240 | 18x60         | 2188 | S[1427] | -11925 | 330 | 18x60         |
| 2133 | S[1372] | -11265 | 150 | 18x60         | 2189 | S[1428] | -11937 | 240 | 18x60         |
| 2134 | S[1373] | -11277 | 330 | 18x60         | 2190 | S[1429] | -11949 | 150 | 18x60         |
| 2135 | S[1374] | -11289 | 240 | 18x60         | 2191 | S[1430] | -11961 | 330 | 18x60         |
| 2136 | S[1375] | -11301 | 150 | 18x60         | 2192 | S[1431] | -11973 | 240 | 18x60         |
| 2137 | S[1376] | -11313 | 330 | 18x60         | 2193 | S[1432] | -11985 | 150 | 18x60         |
| 2138 | S[1377] | -11325 | 240 | 18x60         | 2194 | S[1433] | -11997 | 330 | 18x60         |
| 2139 | S[1378] | -11337 | 150 | 18x60         | 2195 | S[1434] | -12009 | 240 | 18x60         |
| 2140 | S[1379] | -11349 | 330 | 18x60         | 2196 | S[1435] | -12021 | 150 | 18x60         |
| 2141 | S[1380] | -11361 | 240 | 18x60         | 2197 | S[1436] | -12033 | 330 | 18x60         |
| 2142 | S[1381] | -11373 | 150 | 18x60         | 2198 | S[1437] | -12045 | 240 | 18x60         |
| 2143 | S[1382] | -11385 | 330 | 18x60         | 2199 | S[1438] | -12057 | 150 | 18x60         |
| 2144 | S[1383] | -11397 | 240 | 18x60         | 2200 | S[1439] | -12069 | 330 | 18x60         |
| 2145 | S[1384] | -11409 | 150 | 18x60         | 2201 | S[1440] | -12081 | 240 | 18x60         |
| 2146 | S[1385] | -11421 | 330 | 18x60         | 2202 | S[1441] | -12093 | 150 | 18x60         |
| 2147 | S[1386] | -11433 | 240 | 18x60         | 2203 | S[1442] | -12105 | 330 | 18x60         |
| 2148 | S[1387] | -11445 | 150 | 18x60         | 2204 | S[1443] | -12117 | 240 | 18x60         |
| 2149 | S[1388] | -11457 | 330 | 18x60         | 2205 | S[1444] | -12129 | 150 | 18x60         |
| 2150 | S[1389] | -11469 | 240 | 18x60         | 2206 | S[1445] | -12141 | 330 | 18x60         |
| 2151 | S[1390] | -11481 | 150 | 18x60         | 2207 | S[1446] | -12153 | 240 | 18x60         |
| 2152 | S[1391] | -11493 | 330 | 18x60         | 2208 | S[1447] | -12165 | 150 | 18x60         |
| 2153 | S[1392] | -11505 | 240 | 18x60         | 2209 | S[1448] | -12177 | 330 | 18x60         |
| 2154 | S[1393] | -11517 | 150 | 18x60         | 2210 | S[1449] | -12189 | 240 | 18x60         |
| 2155 | S[1394] | -11529 | 330 | 18x60         | 2211 | S[1450] | -12201 | 150 | 18x60         |
| 2156 | S[1395] | -11541 | 240 | 18x60         | 2212 | S[1451] | -12213 | 330 | 18x60         |
| 2157 | S[1396] | -11553 | 150 | 18x60         | 2213 | S[1452] | -12225 | 240 | 18x60         |
| 2158 | S[1397] | -11565 | 330 | 18x60         | 2214 | S[1453] | -12237 | 150 | 18x60         |
| 2159 | S[1398] | -11577 | 240 | 18x60         | 2215 | S[1454] | -12249 | 330 | 18x60         |
| 2160 | S[1399] | -11589 | 150 | 18x60         | 2216 | S[1455] | -12261 | 240 | 18x60         |
| 2161 | S[1400] | -11601 | 330 | 18x60         | 2217 | S[1456] | -12273 | 150 | 18x60         |
| 2162 | S[1401] | -11613 | 240 | 18x60         | 2218 | S[1457] | -12285 | 330 | 18x60         |
| 2163 | S[1402] | -11625 | 150 | 18x60         | 2219 | S[1458] | -12297 | 240 | 18x60         |
| 2164 | S[1403] | -11637 | 330 | 18x60         | 2220 | S[1459] | -12309 | 150 | 18x60         |
| 2165 | S[1404] | -11649 | 240 | 18x60         | 2221 | S[1460] | -12321 | 330 | 18x60         |
| 2166 | S[1405] | -11661 | 150 | 18x60         | 2222 | S[1461] | -12333 | 240 | 18x60         |
| 2167 | S[1406] | -11673 | 330 | 18x60         | 2223 | S[1462] | -12345 | 150 | 18x60         |
| 2168 | S[1407] | -11685 | 240 | 18x60         | 2224 | S[1463] | -12357 | 330 | 18x60         |
| 2169 | S[1408] | -11697 | 150 | 18x60         | 2225 | S[1464] | -12369 | 240 | 18x60         |
| 2170 | S[1409] | -11709 | 330 | 18x60         | 2226 | S[1465] | -12381 | 150 | 18x60         |
| 2171 | S[1410] | -11721 | 240 | 18x60         | 2227 | S[1466] | -12393 | 330 | 18x60         |
| 2172 | S[1411] | -11733 | 150 | 18x60         | 2228 | S[1467] | -12405 | 240 | 18x60         |
| 2173 | S[1412] | -11745 | 330 | 18x60         | 2229 | S[1468] | -12417 | 150 | 18x60         |
| 2174 | S[1413] | -11757 | 240 | 18x60         | 2230 | S[1469] | -12429 | 330 | 18x60         |
| 2175 | S[1414] | -11769 | 150 | 18x60         | 2231 | S[1470] | -12441 | 240 | 18x60         |
| 2176 | S[1415] | -11781 | 330 | 18x60         | 2232 | S[1471] | -12453 | 150 | 18x60         |
| 2177 | S[1416] | -11793 | 240 | 18x60         | 2233 | S[1472] | -12465 | 330 | 18x60         |
| 2178 | S[1417] | -11805 | 150 | 18x60         | 2234 | S[1473] | -12477 | 240 | 18x60         |
| 2179 | S[1418] | -11817 | 330 | 18x60         | 2235 | S[1474] | -12489 | 150 | 18x60         |
| 2180 | S[1419] | -11829 | 240 | 18x60         | 2236 | S[1475] | -12501 | 330 | 18x60         |
| 2181 | S[1420] | -11841 | 150 | 18x60         | 2237 | S[1476] | -12513 | 240 | 18x60         |
| 2182 | S[1421] | -11853 | 330 | 18x60         | 2238 | S[1477] | -12525 | 150 | 18x60         |
| 2183 | S[1422] | -11865 | 240 | 18x60         | 2239 | S[1478] | -12537 | 330 | 18x60         |
| 2184 | S[1423] | -11877 | 150 | 18x60         | 2240 | S[1479] | -12549 | 240 | 18x60         |

| No.  | Name    | X      | Y   | Bump size(um) | No.  | Name    | X      | Y   | Bump size(um) |
|------|---------|--------|-----|---------------|------|---------|--------|-----|---------------|
| 2241 | S[1480] | -12561 | 150 | 18x60         | 2297 | S[1536] | -13233 | 240 | 18x60         |
| 2242 | S[1481] | -12573 | 330 | 18x60         | 2298 | S[1537] | -13245 | 150 | 18x60         |
| 2243 | S[1482] | -12585 | 240 | 18x60         | 2299 | DUMMY   | -13257 | 330 | 18x60         |
| 2244 | S[1483] | -12597 | 150 | 18x60         | 2300 | DUMMY   | -13293 | 330 | 18x60         |
| 2245 | S[1484] | -12609 | 330 | 18x60         | 2301 | DUMMY   | -13329 | 330 | 18x60         |
| 2246 | S[1485] | -12621 | 240 | 18x60         | 2302 | DUMMY   | -13365 | 330 | 18x60         |
| 2247 | S[1486] | -12633 | 150 | 18x60         | 2303 | DUMMY   | -13401 | 330 | 18x60         |
| 2248 | S[1487] | -12645 | 330 | 18x60         | 2304 | DUMMY   | -13437 | 330 | 18x60         |
| 2249 | S[1488] | -12657 | 240 | 18x60         | 2305 | DUMMY   | -13473 | 330 | 18x60         |
| 2250 | S[1489] | -12669 | 150 | 18x60         | 2306 | DUMMY   | -13509 | 330 | 18x60         |
| 2251 | S[1490] | -12681 | 330 | 18x60         | 2307 | DUMMY   | -13545 | 330 | 18x60         |
| 2252 | S[1491] | -12693 | 240 | 18x60         | 2308 | DUMMY   | -13581 | 330 | 18x60         |
| 2253 | S[1492] | -12705 | 150 | 18x60         | 2309 | PASS1_R | -13617 | 330 | 18x60         |
| 2254 | S[1493] | -12717 | 330 | 18x60         | 2310 | PASS1_R | -13653 | 330 | 18x60         |
| 2255 | S[1494] | -12729 | 240 | 18x60         | 2311 | PASS1_R | -13689 | 330 | 18x60         |
| 2256 | S[1495] | -12741 | 150 | 18x60         | 2312 | PASS1_R | -13725 | 330 | 18x60         |
| 2257 | S[1496] | -12753 | 330 | 18x60         | 2313 | DUMMY   | -13761 | 330 | 18x60         |
| 2258 | S[1497] | -12765 | 240 | 18x60         | 2314 | PASS2_R | -13797 | 330 | 18x60         |
| 2259 | S[1498] | -12777 | 150 | 18x60         | 2315 | PASS2_R | -13833 | 330 | 18x60         |
| 2260 | S[1499] | -12789 | 330 | 18x60         | 2316 | PASS2_R | -13869 | 330 | 18x60         |
| 2261 | S[1500] | -12801 | 240 | 18x60         | 2317 | PASS2_R | -13905 | 330 | 18x60         |
| 2262 | S[1501] | -12813 | 150 | 18x60         | 2318 | DUMMY   | -13941 | 330 | 18x60         |
| 2263 | S[1502] | -12825 | 330 | 18x60         |      |         |        |     |               |
| 2264 | S[1503] | -12837 | 240 | 18x60         |      |         |        |     |               |
| 2265 | S[1504] | -12849 | 150 | 18x60         |      |         |        |     |               |
| 2266 | S[1505] | -12861 | 330 | 18x60         |      |         |        |     |               |
| 2267 | S[1506] | -12873 | 240 | 18x60         |      |         |        |     |               |
| 2268 | S[1507] | -12885 | 150 | 18x60         |      |         |        |     |               |
| 2269 | S[1508] | -12897 | 330 | 18x60         |      |         |        |     |               |
| 2270 | S[1509] | -12909 | 240 | 18x60         |      |         |        |     |               |
| 2271 | S[1510] | -12921 | 150 | 18x60         |      |         |        |     |               |
| 2272 | S[1511] | -12933 | 330 | 18x60         |      |         |        |     |               |
| 2273 | S[1512] | -12945 | 240 | 18x60         |      |         |        |     |               |
| 2274 | S[1513] | -12957 | 150 | 18x60         |      |         |        |     |               |
| 2275 | S[1514] | -12969 | 330 | 18x60         |      |         |        |     |               |
| 2276 | S[1515] | -12981 | 240 | 18x60         |      |         |        |     |               |
| 2277 | S[1516] | -12993 | 150 | 18x60         |      |         |        |     |               |
| 2278 | S[1517] | -13005 | 330 | 18x60         |      |         |        |     |               |
| 2279 | S[1518] | -13017 | 240 | 18x60         |      |         |        |     |               |
| 2280 | S[1519] | -13029 | 150 | 18x60         |      |         |        |     |               |
| 2281 | S[1520] | -13041 | 330 | 18x60         |      |         |        |     |               |
| 2282 | S[1521] | -13053 | 240 | 18x60         |      |         |        |     |               |
| 2283 | S[1522] | -13065 | 150 | 18x60         |      |         |        |     |               |
| 2284 | S[1523] | -13077 | 330 | 18x60         |      |         |        |     |               |
| 2285 | S[1524] | -13089 | 240 | 18x60         |      |         |        |     |               |
| 2286 | S[1525] | -13101 | 150 | 18x60         |      |         |        |     |               |
| 2287 | S[1526] | -13113 | 330 | 18x60         |      |         |        |     |               |
| 2288 | S[1527] | -13125 | 240 | 18x60         |      |         |        |     |               |
| 2289 | S[1528] | -13137 | 150 | 18x60         |      |         |        |     |               |
| 2290 | S[1529] | -13149 | 330 | 18x60         |      |         |        |     |               |
| 2291 | S[1530] | -13161 | 240 | 18x60         |      |         |        |     |               |
| 2292 | S[1531] | -13173 | 150 | 18x60         |      |         |        |     |               |
| 2293 | S[1532] | -13185 | 330 | 18x60         |      |         |        |     |               |
| 2294 | S[1533] | -13197 | 240 | 18x60         |      |         |        |     |               |
| 2295 | S[1534] | -13209 | 150 | 18x60         |      |         |        |     |               |
| 2296 | S[1535] | -13221 | 330 | 18x60         |      |         |        |     |               |



## 14. Revision History

| Reversion | Content                                           | Date       |
|-----------|---------------------------------------------------|------------|
| 0.0.1     | New SPEC                                          | 2023/11/27 |
| 0.0.2     | Remove VDD18V_BPS function(p12)                   | 2024/04/02 |
| 0.0.3     | Modify VCOM voltage operation range(p6/p125/p152) | 2024/04/18 |
| 0.0.3     | Modify flexible GIP output mapping table(p134)    | 2024/04/18 |
| 0.0.3     | Modify chip information(p154~p176)                | 2024/04/18 |